M^{3}LB

A modular lattice Boltzmann solver on Kokkos

Scheme description, methodology and validation

Abstract

This document describes the numerical schemes, the software architecture and the complete validation record of `M3LB`, a portable lattice Boltzmann solver written in C++20 on top of Kokkos. The code targets NVIDIA GPUs and CPU clusters from a single source. It supports four lattices, two streaming schemes (including the in-place Esoteric Pull), two population storage conventions, four collision operators for the fluid (BGK, TRT, raw multiple relaxation time and central moments), no-slip, regularised and free-slip walls, a passive-scalar module for heat transfer a magnetohydrodynamic module built on a vector-valued distribution, and a multiphase module in which a conservative Allen–Cahn phase field drives a pressure-based fluid formulation at a density ratio, with rigid bodies that fall, float and roll under the flow's own reaction, and an electrohydrodynamic module coupling charge transport, a Poisson potential and the Coulomb force. Every scheme documented here is accompanied by the quantitative test that establishes it, and every stated limitation is one that has been measured rather than assumed.

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1 Scope and design

1.1 What the code is

`M3LB` solves the weakly compressible Navier–Stokes equations by the lattice Boltzmann method, optionally coupled to a passive scalar (temperature), to a magnetic field, or to a phase field carrying an immiscible interface between fluids of different density. It is written once and compiled for any Kokkos backend: Serial, Threads, OpenMP, CUDA, HIP or SYCL. Scalar precision is selected at configure time.

1.2 Governing equations

The fluid obeys

$$\partial_t \rho + \nabla\cdot(\rho \boldsymbol{u}) = 0, \tag{1}$$

$$\partial_t(\rho \boldsymbol{u}) + \nabla\cdot(\rho \boldsymbol{u}\boldsymbol{u}) = -\nabla p + \nabla\cdot\boldsymbol{\tau} + \boldsymbol{F}, \tag{2}$$

with $p = \rho c_s^2$ and $\boldsymbol{\tau}$ the viscous stress. When the thermal module is active a scalar T is transported by

$$\partial_t T + \nabla\cdot(\boldsymbol{u}T) = D\,\nabla^2 T, \tag{3}$$

and couples back to the flow through the Boussinesq body force. When the magnetohydrodynamic module is active the magnetic field $\boldsymbol{b}$ obeys the induction equation

$$\partial_t \boldsymbol{b} = \nabla\times(\boldsymbol{u}\times\boldsymbol{b}) + \eta\,\nabla^2\boldsymbol{b}, \qquad \nabla\cdot\boldsymbol{b} = 0, \tag{4}$$

and acts on the flow through the Lorentz force $\boldsymbol{j}\times\boldsymbol{b}$ with $\boldsymbol{j} = \nabla\times\boldsymbol{b}$. When the multiphase module is active an order parameter ϕ marks the two fluids and obeys the conservative Allen–Cahn equation

$$\partial_t\phi + \nabla\cdot(\boldsymbol{u}\phi) = \nabla\cdot\Big[M\Big(\nabla\phi - \frac{4}{W}\,\phi(1-\phi)\,\boldsymbol{n}\Big)\Big], \qquad \boldsymbol{n} = \frac{\nabla\phi}{|\nabla\phi|}, \tag{5}$$

with M a mobility and W the interface width; density and viscosity are interpolated in ϕ, and the flow gains a capillary force. The fluid equation of state changes with it, and §10 is where that change is paid for.

1.3 Architectural rules

Seven rules govern the implementation. They are stated here because most of the design decisions in later sections follow from them rather than from the physics.

1. **No virtual functions in device code.** Lattice, streaming scheme, equilibrium, forcing, storage convention and collision operator are all compile-time policies that inline into a single kernel. Dynamic dispatch exists only in a host-side factory above the solver. A virtual call inside a Kokkos kernel runs, but it blocks inlining of the entire collision operator.

2. **Never pin the Kokkos memory layout.** Population fields are declared `View<Real**>` with extents `(node, q)` and the *default* layout. Kokkos then selects `LayoutLeft` on CUDA/HIP — node index fastest, hence a structure of arrays with coalesced access — and `LayoutRight` on CPU backends, giving an array of structures with good cache locality per node. One declaration is correct on both; hardcoding either costs performance on the other.

3. **Streaming and collision are one fused kernel.** The in-place streaming schemes cannot have a separate streaming pass, so nothing in the solver is written in a way that would assume one.

4. **Parity is a compile-time parameter.** Esoteric Pull addresses different storage slots on even and odd time steps. That decision must fold away at compile time, so `FluidSolver::step` dispatches to `run_step<0>` or `run_step<1>`.
5. **Periodicity is index wrapping, not halo copying.** See §3.3.
6. **Boundary conditions are alternative collision operators** applied to flagged cells, giving one branch per node rather than one per direction.
7. **The reference implementation is never deleted.** The two-lattice streaming scheme is retained permanently and cross-checked against Esoteric Pull after every step.

2 Lattices

2.1 Definition and constraints

A lattice is a set of Q velocities $\boldsymbol{c}_i$ with weights w_i and a lattice sound speed c_s. To recover the Navier–Stokes equations at second order the set must satisfy

$$\sum_i w_i = 1, \qquad \sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}, \tag{6}$$

$$\sum_i w_i c_{i\alpha} c_{i\beta} c_{i\gamma} = 0, \qquad \sum_i w_i c_{i\alpha} c_{i\beta} c_{i\gamma} c_{i\delta} = c_s^4 (\delta_{\alpha\beta}\delta_{\gamma\delta} + \delta_{\alpha\gamma}\delta_{\beta\delta} + \delta_{\alpha\delta}\delta_{\beta\gamma}). \tag{7}$$

Four lattices are provided. **D2Q5** and **D3Q7** satisfy (6) but *not* (7): they cannot support Navier–Stokes and are used only for scalar transport and for the magnetic field, where only the second moment is required.

lattice	Q	D	c_s^2	weights	Navier–Stokes
D2Q5	5	2	1/3	1/3; 1/6	no
D2Q9	9	2	1/3	4/9; 1/9; 1/36	yes
D3Q7	7	3	1/4	1/4; 1/8	no
D3Q27	27	3	1/3	8/27; 2/27; 1/54; 1/216	yes

Table 1: The four lattices. Note that D3Q7 has $c_s^2 = 1/4$, not 1/3.

2.2 Velocity sets

The complete velocity sets follow, in the Esoteric Pull ordering of §2.3. These tables are generated from the same data the code compiles against, and the identities in the captions are re-verified in exact rational arithmetic at generation time.

2.3 The direction-ordering contract

Esoteric Pull walks opposite pairs as `for (i = 1; i < Q; i += 2)`. This requires that the rest velocity occupies index 0 and that opposite directions occupy *adjacent* indices, so that

$$\mathrm{opp}(0) = 0, \qquad \mathrm{opp}(i) = \begin{cases} i+1 & i \text{ odd} \\ i-1 & i \text{ even} \end{cases} \tag{8}$$

is pure integer arithmetic, with no lookup table in the kernel. This is a *contract*: every moment matrix in the code is generated against it.

Reordering the directions of a lattice is a column permutation of the moment transform matrices together with the matching permutation of f_i and f_i^{eq}. Every moment, raw or central, is therefore invariant, and only the per-direction post-collision expressions are permuted. This was verified numerically in exact rational arithmetic before the orderings were adopted.

i	$\boldsymbol{c}_i$	w_i	opp	i	$\boldsymbol{c}_i$	w_i	opp
0	$(0,0)$	$\frac{1}{3}$	0	3	$(0,1)$	$\frac{1}{6}$	4
1	$(1,0)$	$\frac{1}{6}$	2	4	$(0,-1)$	$\frac{1}{6}$	3
2	$(-1,0)$	$\frac{1}{6}$	1				

Table 2: D2Q5: velocity set in the Esoteric Pull ordering (§2.3), weights, and opp(i). $c_s^2 = \frac{1}{3}$. Verified: $\sum_i w_i = 1$ and $\sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}$ in exact rational arithmetic.

i	$\boldsymbol{c}_i$	w_i	opp	i	$\boldsymbol{c}_i$	w_i	opp
0	$(0,0)$	$\frac{4}{9}$	0	5	$(1,1)$	$\frac{1}{36}$	6
1	$(1,0)$	$\frac{1}{9}$	2	6	$(-1,-1)$	$\frac{1}{36}$	5
2	$(-1,0)$	$\frac{1}{9}$	1	7	$(1,-1)$	$\frac{1}{36}$	8
3	$(0,1)$	$\frac{1}{9}$	4	8	$(-1,1)$	$\frac{1}{36}$	7
4	$(0,-1)$	$\frac{1}{9}$	3				

Table 3: D2Q9: velocity set in the Esoteric Pull ordering (§2.3), weights, and opp(i). $c_s^2 = \frac{1}{3}$. Verified: $\sum_i w_i = 1$ and $\sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}$ in exact rational arithmetic.

2.4 Compile-time verification

Weights are stored as exact rationals ($w_i = \texttt{w_num}[i]/\texttt{w_den}$) and c_s^2 as `cs2_num/cs2_den`, so all the identities above are checked at compile time in exact integer arithmetic rather than in floating point:

```
static_assert(check_pairing<L>(),        "opposite directions not adjacent");
static_assert(check_unique<L>(),         "duplicated velocity");
static_assert(check_w_sum<L>(),          "weights do not sum to 1");
static_assert(check_second_moment<L>(), "<cc> != cs2 I");
static_assert(check_third_moment<L>(),  "<ccc> != 0");
static_assert(check_fourth_moment<L>() == L::supports_navier_stokes, "");
```

The last line is an equality rather than a truth test: D2Q5 and D3Q7 are *required* to fail fourth-order isotropy, and a lattice whose flag disagrees with its actual moments will not compile.

3 Streaming

3.1 Two-lattice (reference)

Two arrays, $2Q$ reals per node. Semantics are *pull*: the kernel gathers the populations that have arrived at node n,

$$f_i^{\text{in}}(n) = f_i^{\text{src}}(n - \boldsymbol{c}_i), \tag{9}$$

collides, and writes locally to the destination array, which is then swapped. This scheme is obviously race-free and obviously correct. It is not the production scheme, but it is retained permanently: when an exotic collision operator produces incorrect results, being able to switch to a streaming scheme that cannot be wrong is what makes the fault bisectable.

3.2 Esoteric Pull

Esoteric Pull [4] stores a single copy of the populations and updates them in place. Write $A[n][s]$ for storage slot s at node n, and consider one opposite pair $(i, i+1)$ with i odd. The partner node is *always* $n + \boldsymbol{c}_i$, at both parities; only which of the two slots lives remotely alternates.

i	$\boldsymbol{c}_i$	w_i	opp	i	$\boldsymbol{c}_i$	w_i	opp
0	$(0,0,0)$	$\frac{1}{4}$	0	4	$(0,-1,0)$	$\frac{1}{8}$	3
1	$(1,0,0)$	$\frac{1}{8}$	2	5	$(0,0,1)$	$\frac{1}{8}$	6
2	$(-1,0,0)$	$\frac{1}{8}$	1	6	$(0,0,-1)$	$\frac{1}{8}$	5
3	$(0,1,0)$	$\frac{1}{8}$	4				

Table 4: D3Q7: velocity set in the Esoteric Pull ordering (§2.3), weights, and opp(i). $c_s^2 = \frac{1}{4}$. Verified: $\sum_i w_i = 1$ and $\sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}$ in exact rational arithmetic.

i	$\boldsymbol{c}_i$	w_i	opp	i	$\boldsymbol{c}_i$	w_i	opp	i	$\boldsymbol{c}_i$	w_i	opp
0	$(0,0,0)$	$\frac{8}{27}$	0	9	$(1,-1,0)$	$\frac{1}{54}$	10	18	$(0,-1,1)$	$\frac{1}{54}$	17
1	$(1,0,0)$	$\frac{2}{27}$	2	10	$(-1,1,0)$	$\frac{1}{54}$	9	19	$(1,1,1)$	$\frac{1}{216}$	20
2	$(-1,0,0)$	$\frac{2}{27}$	1	11	$(1,0,1)$	$\frac{1}{54}$	12	20	$(-1,-1,-1)$	$\frac{1}{216}$	19
3	$(0,1,0)$	$\frac{2}{27}$	4	12	$(-1,0,-1)$	$\frac{1}{54}$	11	21	$(1,1,-1)$	$\frac{1}{216}$	22
4	$(0,-1,0)$	$\frac{2}{27}$	3	13	$(1,0,-1)$	$\frac{1}{54}$	14	22	$(-1,-1,1)$	$\frac{1}{216}$	21
5	$(0,0,1)$	$\frac{2}{27}$	6	14	$(-1,0,1)$	$\frac{1}{54}$	13	23	$(1,-1,1)$	$\frac{1}{216}$	24
6	$(0,0,-1)$	$\frac{2}{27}$	5	15	$(0,1,1)$	$\frac{1}{54}$	16	24	$(-1,1,-1)$	$\frac{1}{216}$	23
7	$(1,1,0)$	$\frac{1}{54}$	8	16	$(0,-1,-1)$	$\frac{1}{54}$	15	25	$(-1,1,1)$	$\frac{1}{216}$	26
8	$(-1,-1,0)$	$\frac{1}{54}$	7	17	$(0,1,-1)$	$\frac{1}{54}$	18	26	$(1,-1,-1)$	$\frac{1}{216}$	25

Table 5: D3Q27: velocity set in the Esoteric Pull ordering (§2.3), weights, and opp(i). $c_s^2 = \frac{1}{3}$. Verified: $\sum_i w_i = 1$ and $\sum_i w_i c_{i\alpha} c_{i\beta} = c_s^2 \delta_{\alpha\beta}$ in exact rational arithmetic.

Consistency. At an even step node n writes its outgoing f_i into $A[n + \boldsymbol{c}_i][i{+}1]$; at the next (odd) step node $m = n + \boldsymbol{c}_i$ reads f_i from $A[m][i{+}1]$, which is exactly the odd-step rule. The scheme closes.

Race freedom. Every slot has exactly one reader and one writer, and they are the same node. No temporary buffer and no synchronisation within a step are required.

Implicit bounce-back. Halfway bounce-back sets $f_i^{\text{out}} = f_{i+1}^{\text{in}}$ and $f_{i+1}^{\text{out}} = f_i^{\text{in}}$. Substituting into Table 6, every write lands back in the slot it was read from: the entire update is the identity. *Solid cells are therefore skipped outright* — no load, no store, no arithmetic. The population a fluid node emits toward a wall sits untouched for one step and is read back as the opposite direction on the next, placing the wall exactly midway between the two cells.

Memory. Q reals per node against $2Q$ for two-lattice. At 512^3 with D3Q27 in FP64 this is the difference between 29 GB and 58 GB, i.e. between fitting and not fitting on a 40 GB device.

3.3 Periodicity by index wrapping

Esoteric Pull writes into its neighbours' storage. A halo-copy treatment of periodicity would therefore require a *two-way* exchange whose set of exchanged slots depends on the parity. The code instead resolves periodicity by wrapping the neighbour index, which removes that entire class of bug:

$$\text{shift}(v) = \begin{cases} v + n & v < h \\ v - n & v \geq h + n \\ v & \text{otherwise} \end{cases} \quad \text{(periodic)}, \qquad \text{clamp}(v, 0, s-1) \quad \text{(otherwise)}. \tag{10}$$

even step t	odd step t
$f_i \leftarrow A[n][i]$	$f_i \leftarrow A[n][i{+}1]$
$f_{i+1} \leftarrow A[n+\boldsymbol{c}_i][i{+}1]$	$f_{i+1} \leftarrow A[n+\boldsymbol{c}_i][i]$
$A[n+\boldsymbol{c}_i][i{+}1] \leftarrow f_i^{\text{out}}$	$A[n+\boldsymbol{c}_i][i] \leftarrow f_i^{\text{out}}$
$A[n][i] \leftarrow f_{i+1}^{\text{out}}$	$A[n][i{+}1] \leftarrow f_{i+1}^{\text{out}}$

Table 6: The Esoteric Pull access pattern, one opposite pair. Each parity reads exactly two slots and writes those same two, crossed.

A halo layer is allocated only in non-periodic directions, where edge cells genuinely need somewhere out-of-domain to read from. A fast path skips the wrapping entirely for nodes away from every face.

4 Population storage

Two conventions are supported.

Raw the arrays hold f_i.

Shifted the arrays hold $g_i = f_i - w_i$.

The shift exists for conditioning. Populations are $O(w_i) \sim 10^{-1}$, while the collision operates on differences $O(10^{-6})$; forming $f_i - f_i^{\text{eq}}$ in FP32 therefore discards most of the mantissa. The momentum sum is worse, since $\sum_i \boldsymbol{c}_i f_i$ cancels terms of order 10^{-1} down to 10^{-2}. Storing g_i removes both cancellations, because $\sum_i \boldsymbol{c}_i w_i = 0$ exactly and hence

$$\sum_i \boldsymbol{c}_i \, g_i \;\equiv\; \sum_i \boldsymbol{c}_i \, f_i \tag{11}$$

identically, while every summand is small. The density follows from $\rho = 1 + \sum_i g_i$.

Almost nothing else changes. Streaming is a pure copy and is untouched. Halfway bounce-back is $g_i \leftarrow g_{\text{opp}(i)}$, untouched, because $w_i = w_{\text{opp}(i)}$. The Guo source term is already an $O(F)$ quantity and is untouched. Only the equilibrium and the density reconstruction differ.

Measured effect. Poiseuille flow at the magic relaxation time, $H = 32$, in FP32:

storage	FP64	FP32
raw	1.433×10^{-13}	5.969×10^{-4}
shifted	1.626×10^{-14}	$\mathbf{2.778 \times 10^{-5}}$
gain	8.8×	**21.5×**

The validation suite asserts that gain rather than merely tolerating its absence: a regression to raw storage fails the build.

For the passive scalar the same argument applies but there is *no universal reference*. Density is always near unity; a temperature scale is whatever the problem says it is. The scalar collision therefore exposes T_{ref} as a runtime parameter, defaulting to 0, which reproduces unshifted storage exactly.

5 Equilibria

This section states, for every lattice, the equilibrium that the code actually evaluates. The distinction matters in one place in particular: the moment-space operators of §6.3 *never form equilibrium populations at all.* Their equilibrium is defined directly in moment space, and the corresponding populations exist only implicitly, as the inverse transform of those moments.

5.1 The ϕ formulation

Each populations-space equilibrium policy supplies only $\phi_i(\boldsymbol{u})$, defined by

$$f_i^{\mathrm{eq}} = w_i\,\rho\,(1+\phi_i), \tag{12}$$

from which both storage conventions of §4 follow once:

$$\text{raw:}\quad f_i^{\mathrm{eq}} = w_i\,\rho\,(1+\phi_i), \tag{13}$$

$$\text{shifted:}\quad f_i^{\mathrm{eq}} - w_i = w_i(\delta\rho + (1+\delta\rho)\,\phi_i), \qquad \delta\rho = \rho - 1. \tag{14}$$

Form (14) is written that way deliberately: every term is small, so it never forms the difference of two $O(w_i)$ quantities. Writing it as $f_i^{\mathrm{eq}} - w_i$ would be algebraically identical and numerically useless.

5.2 Second-order (truncated Hermite) — D2Q9, D3Q27

The default for BGK, TRT and the MHD fluid operator, on both Navier–Stokes lattices:

$$\boxed{f_i^{\mathrm{eq}} = w_i\,\rho\left[1 + \frac{\boldsymbol{c}_i\cdot\boldsymbol{u}}{c_s^2} + \frac{(\boldsymbol{c}_i\cdot\boldsymbol{u})^2}{2c_s^4} - \frac{\boldsymbol{u}\cdot\boldsymbol{u}}{2c_s^2}\right]} \tag{15}$$

With $c_s^2 = 1/3$ on both, this is $f_i^{\mathrm{eq}} = w_i\rho\big[1 + 3(\boldsymbol{c}_i\cdot\boldsymbol{u}) + \frac{9}{2}(\boldsymbol{c}_i\cdot\boldsymbol{u})^2 - \frac{3}{2}\boldsymbol{u}^2\big]$. It is correct to $O(u^2)$; its central moments carry $O(u^3)$ defects, which is what the Galilean-invariance test of §12.1.4 measures and what the moment operators remove.

5.3 Product form — D2Q9 and D3Q27

The complete (fourth-order) equilibrium, equivalent to `index_equilibrium = 4` of the symbolic generator. Writing $a_\alpha = c_{i\alpha}^2 - c_s^2$,

$$\boxed{\begin{aligned} f_i^{\mathrm{eq}} = w_i\rho\Big[1 &+ \frac{c_{ix}u_x + c_{iy}u_y}{c_s^2} + \frac{a_x u_x^2 + a_y u_y^2 + 2c_{ix}c_{iy}u_x u_y}{2c_s^4} \\ &+ \frac{a_x c_{iy} u_x^2 u_y + a_y c_{ix} u_x u_y^2}{2c_s^6} + \frac{a_x a_y\, u_x^2 u_y^2}{4c_s^8}\Big] \end{aligned}} \tag{16}$$

Its central moments are *exactly* Maxwellian for all nine representable k_{pq} — verified symbolically, 0 of 9 deviating — so it is fully Galilean invariant on this lattice.

D3Q27: sixth order. D3Q27 is a product lattice, so the same construction runs to sixth order — the highest its 27 velocities admit. The symbolic generator writes it as a sum of Hermite tensors of orders one to six. That sum was checked, in exact rational arithmetic, against the factorised form

$$\boxed{f_i^{\mathrm{eq}} = \rho\,\psi(c_{ix},u_x)\,\psi(c_{iy},u_y)\,\psi(c_{iz},u_z), \qquad \begin{aligned} \psi(0,u) &= 1 - c_s^2 - u^2, \\ \psi(\pm1,u) &= \tfrac{1}{2}(c_s^2 + u^2 \pm u), \end{aligned}} \tag{17}$$

and the two are *identical, term for term.* The one-dimensional factor ψ is simply the three-point distribution carrying the exact moments $\{1,\, u,\, c_s^2 + u^2\}$. Equation (17) is what the code evaluates: three multiplications instead of six Hermite tensors, bit-for-bit the same polynomial. Dividing each factor by its one-dimensional weight $\{\frac{1}{6}, \frac{2}{3}, \frac{1}{6}\}$ puts it in the ϕ form of §5.1, since $w_i = w^{\mathrm{1D}}(c_{ix})w^{\mathrm{1D}}(c_{iy})w^{\mathrm{1D}}(c_{iz})$ on a product lattice:

$$\phi_i = \chi(c_{ix}, u_x)\,\chi(c_{iy}, u_y)\,\chi(c_{iz}, u_z) - 1, \qquad \chi(0,u) = 1 - \tfrac{3}{2}u^2, \quad \chi(\pm 1, u) = 1 + 3u(\pm 1 + u). \tag{18}$$

The defining property. The central moments of (16) and (17) are *exactly* Maxwellian in every representable moment:

$$k^{\mathrm{eq}}_{pqr} = \begin{cases} \rho\,(c_s^2)^{\,n}, & p,q,r \text{ all even},\ n = \#\{p,q,r \text{ equal to } 2\}, \\ 0, & \text{otherwise}, \end{cases} \tag{19}$$

so 8 of the 27 D3Q27 moments are nonzero and the remaining 19 vanish identically. Every Galilean-invariance defect of the truncation (15) is absent.

Where these are used. (16) and (17) are collected as `HighOrderEquilibrium<L>`, which selects the highest order each lattice admits. This is the default for the MHD operators of §6.4. The BGK and TRT defaults remain (15), and the moment-space operators of §5.4 reach the same equilibrium without evaluating any of these forms explicitly.

5.4 Equilibria in moment space — D2Q9, D3Q27

The MRT and central-moment operators specify their equilibrium as *moments*, in closed form. Writing $\boldsymbol{u}_b$ for the basis velocity ($\mathbf{0}$ for raw moments, $\boldsymbol{u}$ for central) and $\delta\boldsymbol{u} = \boldsymbol{u} - \boldsymbol{u}_b$, the equilibrium moment in the stored variable is a product of one-dimensional factors,

$$\boxed{E_{pqr} = \rho\, Q_p(\delta u_x)\, Q_q(\delta u_y)\, Q_r(\delta u_z) \;-\; [\text{shifted}]\; A_p(u_{bx})\, A_q(u_{by})\, A_r(u_{bz})} \tag{20}$$

with the triples below. A second, c_s^2-carrying set belonged to the monomial basis, which went with D3Q19 on 2026-09-18; `SelectBasis` now always returns `ProductBasis`, so only the product triples remain reachable:

lattice	basis	$Q_{0,1,2}$	$A_{0,1,2}$
D2Q9, D3Q27	product, $\varphi_2 = C^2 - c_s^2$	$1,\ \delta u,\ \delta u^2$	$1,\ -u_b,\ u_b^2$

Two consequences are worth stating explicitly.

Central moments on D2Q9 and D3Q27. Here $\delta\boldsymbol{u} = \mathbf{0}$, so $Q = (1, 0, 0)$ and (20) collapses to

$$E_{0\ldots0} = \rho, \qquad E_{pqr} = 0 \ \text{ otherwise} \qquad \text{(raw storage)}. \tag{21}$$

One nonzero equilibrium moment. This is the property that makes the operator cheap: no equilibrium populations, no stored K^{eq} vector, no $Q \times Q$ matrices. It was verified symbolically — 0 of 9 deviating on D2Q9, 0 of 27 on D3Q27 — and it is equivalent to relaxing toward (16) and its D3Q27 counterpart.

Raw MRT. Here $\boldsymbol{u}_b = \mathbf{0}$ and $\delta\boldsymbol{u} = \boldsymbol{u}$, so $Q = (1, u, u^2)$ on the product basis — the raw moments of a Maxwellian centred on $\boldsymbol{u}$. (The monomial form was $(1, u, c_s^2 + u^2)$; that basis was removed with D3Q19 on 2026-09-18.) Note this gives raw MRT the *complete* equilibrium, not a truncated one, which is why it recovers most of the Galilean-invariance gain on its own (§12.1.4).

5.5 Scalar transport — D2Q5, D3Q7

The passive scalar carries only its zeroth moment, $T = \sum_i g_i$, and is advected by a velocity supplied from outside:

$$\boxed{g_i^{\mathrm{eq}} = w_i\, T\left(1 + \frac{\boldsymbol{c}_i\cdot\boldsymbol{u}}{c_s^2}\right)} \qquad D = c_s^2\Big(\frac{1}{\omega} - \frac{1}{2}\Big). \tag{22}$$

Explicitly, with $c_s^2 = 1/3$ on D2Q5 and $c_s^2 = 1/4$ on D3Q7,

$$\text{D2Q5:}\quad g_i^{\mathrm{eq}} = w_i T(1 + 3\,\boldsymbol{c}_i\cdot\boldsymbol{u}), \qquad w = (\tfrac{1}{3};\ \tfrac{1}{6}, \tfrac{1}{6}, \tfrac{1}{6}, \tfrac{1}{6}), \tag{23}$$

$$\text{D3Q7:}\quad g_i^{\mathrm{eq}} = w_i T(1 + 4\,\boldsymbol{c}_i\cdot\boldsymbol{u}), \qquad w = (\tfrac{1}{4};\ \tfrac{1}{8}\ \text{(six times)}). \tag{24}$$

In the shifted storage of §4 the code evaluates

$$h_i^{\mathrm{eq}} = w_i\left[\delta T + (T_{\mathrm{ref}} + \delta T)\,\frac{\boldsymbol{c}_i\cdot\boldsymbol{u}}{c_s^2}\right], \qquad \delta T = T - T_{\mathrm{ref}}, \tag{25}$$

which is (22) minus $w_i T_{\mathrm{ref}}$, written so that no term is the difference of two $O(w_i T)$ quantities.

Why first order. The truncation in (22) is deliberate. D2Q5 and D3Q7 have an isotropic second moment but not an isotropic fourth-order one, and on D3Q7 every velocity has a single nonzero component, so $(\boldsymbol{c}_i\cdot\boldsymbol{u})^2$ reduces to $c_\alpha^2 u_\alpha^2$ and the cross terms of the $\boldsymbol{uu}$ tensor cannot be represented at all. A second-order term would add an anisotropic error rather than accuracy. The cost is an $O(u^2)$ defect in the advection term, which is why these lattices suit low-Mach transport.

5.6 Magnetic induction — D2Q5, D3Q7

The magnetic field is carried by a distribution that is a *vector* at every link, with $b_\alpha = \sum_i g_i^\alpha$:

$$\boxed{g_i^{\mathrm{eq},\alpha} = w_i\left[b_\alpha + \frac{c_{i\beta}}{c_s^2}(u_\beta b_\alpha - b_\beta u_\alpha)\right]} \qquad \eta = c_s^2\Big(\frac{1}{\omega_\eta} - \frac{1}{2}\Big). \tag{26}$$

Its first two moments are $\sum_i g_i^{\mathrm{eq},\alpha} = b_\alpha$ and $\sum_i c_{i\beta} g_i^{\mathrm{eq},\alpha} = u_\beta b_\alpha - b_\beta u_\alpha$, which is exactly the antisymmetric flux of the induction equation. Only that first moment is required, which is why a five- or seven-velocity lattice suffices for the field while the flow runs on nine or twenty-seven.

5.7 MHD fluid equilibrium — D2Q9, D3Q27

The Lorentz force enters through the equilibrium's second moment rather than as a body force. To (15) is added

$$\boxed{\delta f_i = \frac{w_i}{2c_s^4}\, M_{\alpha\beta}(c_{i\alpha}c_{i\beta} - c_s^2\delta_{\alpha\beta}), \qquad M_{\alpha\beta} = \tfrac{1}{2}|\boldsymbol{b}|^2\delta_{\alpha\beta} - b_\alpha b_\beta} \tag{27}$$

so that $\sum_i c_{i\alpha}c_{i\beta} f_i^{\mathrm{eq}} = \rho u_\alpha u_\beta + (p + \tfrac{1}{2}|\boldsymbol{b}|^2)\delta_{\alpha\beta} - b_\alpha b_\beta$, while $\sum_i \delta f_i = 0$ and $\sum_i \boldsymbol{c}_i\,\delta f_i = \boldsymbol{0}$ leave mass and momentum untouched.

Dimensional form. In two dimensions $M_{\alpha\beta}\delta_{\alpha\beta} = |\boldsymbol{b}|^2 - |\boldsymbol{b}|^2 = 0$, so (27) reduces to the familiar

$$\delta f_i = \frac{w_i}{2c_s^4}\Big[\tfrac{1}{2}|\boldsymbol{c}_i|^2|\boldsymbol{b}|^2 - (\boldsymbol{c}_i\cdot\boldsymbol{b})^2\Big] \qquad \text{(2D only)}. \tag{28}$$

In three dimensions $M_{\alpha\beta}\delta_{\alpha\beta} = \tfrac{1}{2}|\boldsymbol{b}|^2 \neq 0$ and the trace subtraction is retained. This is not cosmetic: it is precisely the term whose projection onto the plane governs the quasi-two-dimensional reduction test of §12.4.2.

5.8 Summary: which equilibrium is used where

operator	lattices	equilibrium	where
BGK, TRT	D2Q9, D3Q27	second order, (15)	§5.2
MRT (raw moments)	D2Q9, D3Q27	moment space, (20), $\boldsymbol{u}_b = \mathbf{0}$	§5.4
central moments	D2Q9, D3Q27	moment space, (20), $\boldsymbol{u}_b = \boldsymbol{u}$	§5.4
MHD fluid (BGK)	D2Q9, D3Q27	(15) + (27)	§5.7
MHD central moments	D2Q9	(16) + (27); CMs (42)	§6.4
	D3Q27	(17) + (27); CMs (45)	§6.5
	(`-eq2`)	second order (15); CMs (41)	§6.4
scalar transport	D2Q5, D3Q7	first order, (22)	§5.5
magnetic induction	D2Q5, D3Q7	Dellar, (26)	§5.6

highest order each lattice admits: (16) D2Q9, (17) D3Q27
regularised walls rebuild f_i from the same equilibrium, §8.1

6 Collision operators

6.1 BGK

$$f_i^\star = f_i + \omega\,(f_i^{\mathrm{eq}} - f_i) + S_i, \qquad \nu = c_s^2\Big(\frac{1}{\omega} - \frac{1}{2}\Big). \tag{29}$$

6.2 TRT

Split each population into parts symmetric and antisymmetric about the opposite direction,

$$f_i^{\pm} = \tfrac{1}{2}(f_i \pm f_{\mathrm{opp}(i)}), \tag{30}$$

and relax them at different rates ω^+ (which sets the viscosity) and ω^- (free). The free rate buys the *magic parameter*

$$\Lambda = \Big(\frac{1}{\omega^+} - \frac{1}{2}\Big)\Big(\frac{1}{\omega^-} - \frac{1}{2}\Big). \tag{31}$$

At $\Lambda = 3/16$ the halfway bounce-back wall sits exactly midway between the fluid and solid nodes at *any* viscosity. BGK manages this only at the single value $\tau = \frac{1}{2} + \frac{\sqrt{3}}{4}$, because BGK is TRT with $\omega^- = \omega^+$.

The direction-ordering contract (8) pays off here: the pair loop is `for (i = 1; i < Q; i += 2)` with no lookup table. The equilibrium is evaluated in a separate flat loop first, because the pair loop reads f_i and f_{i+1} together and does not vectorise; leaving the equilibrium inside it cost more than the temporary array does (§17).

6.3 Moment-space collision: raw MRT and central moments

Both are one implementation. They differ only in the velocity at which the moment basis is constructed:

$$\boldsymbol{u}_b = \mathbf{0} \ \text{(raw moments, MRT)}, \qquad \boldsymbol{u}_b = \boldsymbol{u} \ \text{(central moments)}, \tag{32}$$

which is precisely the `central_moments` toggle of the symbolic generator.

6.3.1 Product basis (D2Q9, D3Q27)

A *product lattice* contains every velocity in $\{-1, 0, 1\}^D$ exactly once. For these the basis is a product of the same three one-dimensional functions on each axis,

$$\varphi_0(C) = 1, \qquad \varphi_1(C) = C, \qquad \varphi_2(C) = C^2 - c_s^2, \qquad C = c - u_b, \tag{33}$$

so the transform factorises into D successive one-dimensional passes rather than one $Q \times Q$ contraction:

$$\text{D3Q27: } 1458 \to 324 \text{ multiply–adds } (4.5\times), \qquad \text{D2Q9: } 162 \to 72. \tag{34}$$

Each pass, on values at $c = -1, 0, +1$, is

$$m_0 = g_- + g_0 + g_+, \qquad m_1 = (g_+ - g_-) - u\, m_0, \tag{35}$$

$$m_2 = (g_+ + g_-) - 2u(g_+ - g_-) + (u^2 - c_s^2)\, m_0, \tag{36}$$

with an exactly invertible counterpart. The factorised transform was checked against the dense contraction in exact rational arithmetic.

Why this basis. Basis (33) diagonalises the Maxwellian. The product-form equilibrium has *exactly one* nonzero moment in it:

$$k^{\text{eq}}_{00\ldots} = \rho, \qquad k^{\text{eq}}_{pqr} = 0 \;\; \text{otherwise}, \tag{37}$$

verified symbolically for D2Q9 (0 of 9 deviating) and D3Q27 (0 of 27). No equilibrium populations are ever formed and no K^{eq} vector is stored, which is what collapses the collision to a handful of lines.

6.3.2 Relaxation and forcing

Writing $\delta\boldsymbol{u} = \boldsymbol{u} - \boldsymbol{u}_b$, the equilibrium moments in the *stored* variable are

$$E_{pqr} = \rho\, Q_p(\delta u_x) Q_q(\delta u_y) Q_r(\delta u_z) - [\text{shifted}]\; A_p(u_{bx}) A_q(u_{by}) A_r(u_{bz}), \tag{38}$$

with $Q = \{1, \delta u, \delta u^2\}$ and $A = \{1, -u_b, u_b^2\}$ for the product basis, which since 2026-09-18 is the only basis in the tree. The relaxation is then

$$\begin{aligned} \text{order } 0: &\quad \text{conserved} \\ \text{order } 1: &\quad k^\star = k + F_\alpha \\ \text{order } 2: &\quad \text{trace at } \omega_{\text{bulk}},\ \text{deviatoric and shear at } \omega \\ \text{order } \geq 3: &\quad k^\star = E. \end{aligned} \tag{39}$$

The Guo source has moments $\sum_i c_{i\alpha} F_i = F_\alpha$ and $\sum_i c_{i\alpha} c_{i\beta} F_i = u_\alpha F_\beta + u_\beta F_\alpha$. Transformed to the basis at $\boldsymbol{u}_b$ these become F_α and $(F_\beta\, \delta u_\alpha + F_\alpha\, \delta u_\beta)$. *For central moments $\delta\boldsymbol{u} = \boldsymbol{0}$, so the second-order force contribution vanishes identically* and the force enters only through the first moments — the well-known elegance of collision in the co-moving frame. For raw MRT it does not vanish and is applied with the usual $(1 - \omega/2)$ prefactor.

Bulk viscosity is an explicit `omega_bulk`, defaulting to ω.

6.4 MHD central moments

The hybrid scheme of De Rosis, Lévêque and Chahine [6] is implemented separately, because its equilibria are *not* the Maxwellian ones. Central moments are taken in that paper's non-orthogonal basis

$$\begin{aligned} &t_0 = 1, && t_1 = C_x, && t_2 = C_y, && t_3 = C_x^2 + C_y^2, \\ &t_4 = C_x^2 - C_y^2, && t_5 = C_x C_y, && t_6 = C_x^2 C_y, && t_7 = C_x C_y^2, && t_8 = C_x^2 C_y^2, \end{aligned} \tag{40}$$

with $\boldsymbol{C} = \boldsymbol{c}_i - \boldsymbol{u}$, and relaxed toward

$$\begin{aligned} k_0^{\mathrm{eq}} &= \rho, & k_1^{\mathrm{eq}} &= k_2^{\mathrm{eq}} = 0, & k_3^{\mathrm{eq}} &= \tfrac{2}{3}\rho, \\ k_4^{\mathrm{eq}} &= b_y^2 - b_x^2, & k_5^{\mathrm{eq}} &= -b_x b_y, & & \\ k_6^{\mathrm{eq}} &= -\rho u_x^2 u_y + \tfrac{u_y}{2}(b_x^2 - b_y^2) + 2u_x b_x b_y, & & & & \\ k_7^{\mathrm{eq}} &= -\rho u_x u_y^2 + \tfrac{u_x}{2}(b_y^2 - b_x^2) + 2u_y b_x b_y, & & & & \\ k_8^{\mathrm{eq}} &= \tfrac{\rho}{9}(27 u_x^2 u_y^2 + 1) + \tfrac{u_x^2 - u_y^2}{2}(b_x^2 - b_y^2) - 4u_x u_y b_x b_y, & & & & \end{aligned} \tag{41}$$

with $\omega_4 = \omega_5 = \omega_\nu$, ω_3 setting the bulk viscosity and $\omega_6 = \omega_7 = \omega_8 = 1$. k_0, k_1, k_2 are collision invariants.

Raising the equilibrium order in two dimensions. Substituting the product form (16) for (15) leaves k_0–k_5 of (41) untouched and removes precisely the ρ-dependent terms from the rest:

$$k_6^{\mathrm{eq}} = \tfrac{u_y}{2}(b_x^2 - b_y^2) + 2u_x b_x b_y, \qquad k_7^{\mathrm{eq}} = \tfrac{u_x}{2}(b_y^2 - b_x^2) + 2u_y b_x b_y, \qquad k_8^{\mathrm{eq}} = \tfrac{\rho}{9} + \tfrac{u_x^2 - u_y^2}{2}(b_x^2 - b_y^2) - 4u_x u_y b_x b_y. \tag{42}$$

This is the default; (41) is selected by `-eq2` and is what the validation of §12.4.2 against the published Table 1 uses.

These equilibria are deliberately non-Maxwellian. They are the central moments of the *second-order truncated* equilibrium plus the Maxwell term, not of a product-form equilibrium. The terms $-\rho u_x^2 u_y$ in k_6 and $3\rho u_x^2 u_y^2$ in k_8 are exactly the Galilean defects of the truncation. Reproducing the published scheme means reproducing them rather than improving them away, which is why this is a separate operator rather than a configuration of §6.3.

Verification. Equation (41) was not taken on trust. The test suite constructs the paper's equilibrium (73) directly, contracts its central moments by summation, and compares: all nine agree to 1.1×10^{-16}, which also confirms the sign convention on $\boldsymbol{b}$.

6.5 Three dimensions: D3Q27, and the effect of equilibrium order

Reference [6] prescribes D3Q27 for the velocity field (with D3Q7 for the magnetic field) in three dimensions. The operator is extended accordingly. Writing $\boldsymbol{C} = \boldsymbol{c}_i - \boldsymbol{u}$ and

$$k_{pqr} = \sum_i f_i \, C_x^p \, C_y^q \, C_z^r, \qquad M_{\alpha\beta} = \tfrac{1}{2}|\boldsymbol{b}|^2 \delta_{\alpha\beta} - b_\alpha b_\beta, \tag{43}$$

the equilibrium central moments follow from contracting the three-dimensional equilibrium — hydrodynamic part plus the Maxwell term (27) — against the monomial basis. They were obtained symbolically in exact rational arithmetic, and depend on which hydrodynamic equilibrium is used.

With the second-order truncation (15)**.** Orders two and three close in a form that holds for every index combination,

$$k^{\mathrm{eq}}_{\alpha\beta} = \rho c_s^2 \delta_{\alpha\beta} + M_{\alpha\beta}, \qquad k^{\mathrm{eq}}_{\alpha\beta\gamma} = -\rho\, u_\alpha u_\beta u_\gamma - \big(u_\alpha M_{\beta\gamma} + u_\beta M_{\alpha\gamma} + u_\gamma M_{\alpha\beta}\big), \tag{44}$$

and 24 of the 27 moments are nonzero. Setting $u_z = b_z = 0$ reduces (44) and its fourth-order companions to (41) exactly, moment for moment — the three-dimensional scheme is a genuine extension of the published one, not a different model that happens to agree in the bulk.

With the sixth-order equilibrium (17)**.** Because (19) holds, the hydrodynamic contribution collapses to the Maxwellian and *every ρ-dependent term above second order disappears*:

$$\begin{aligned}
k^{\mathrm{eq}}_{000} &= \rho, \qquad k^{\mathrm{eq}}_{\alpha} = 0,\\
k^{\mathrm{eq}}_{\alpha\beta} &= \rho c_s^2 \delta_{\alpha\beta} + M_{\alpha\beta},\\
k^{\mathrm{eq}}_{\alpha\beta\gamma} &= -\big(u_\alpha M_{\beta\gamma} + u_\beta M_{\alpha\gamma} + u_\gamma M_{\alpha\beta}\big),\\
k^{\mathrm{eq}}_{aacc} &= \tfrac{\rho}{9} + \tfrac{1}{3} b_d^2 + u_c^2 M_{aa} + u_a^2 M_{cc} + 4 u_a u_c M_{ac},\\
k^{\mathrm{eq}}_{aacd} &= u_c u_d M_{aa} + \big(\tfrac{1}{3} + u_a^2\big) M_{cd} + 2 u_a u_d M_{ac} + 2 u_a u_c M_{ad},\\
k^{\mathrm{eq}}_{acc\,dd} &= -\tfrac{1}{3} u_a b_a^2 - \tfrac{2}{3}\big(u_c M_{ac} + u_d M_{ad}\big) - 2 u_c u_d^2 M_{ac} - 2 u_c^2 u_d M_{ad}\\
&\quad - 4 u_a u_c u_d M_{cd} - u_a u_d^2 M_{cc} - u_a u_c^2 M_{dd},\\
k^{\mathrm{eq}}_{222} &= \tfrac{\rho}{27} + \tfrac{1}{18}|\boldsymbol{b}|^2 + \tfrac{1}{3}\sum_\alpha u_\alpha^2 b_\alpha^2 + \tfrac{4}{3}\sum_{\alpha<\beta} u_\alpha u_\beta M_{\alpha\beta}\\
&\quad + 4\big(u_x u_y u_z^2 M_{xy} + u_x u_y^2 u_z M_{xz} + u_x^2 u_y u_z M_{yz}\big)\\
&\quad + u_y^2 u_z^2 M_{xx} + u_x^2 u_z^2 M_{yy} + u_x^2 u_y^2 M_{zz},
\end{aligned} \tag{45}$$

where (a, c, d) denotes a permutation of (x, y, z). Comparing with the second-order case, the terms lost are exactly $-\rho u_\alpha u_\beta u_\gamma$ at order three, $3\rho u_a^2 u_c^2$ and $3\rho u_a^2 u_c u_d$ at order four, and so on: the Galilean defects, and nothing else. What remains is the Maxwell stress, which is not a defect but the physics.

Implementation and verification. Rather than evaluate 27 closed forms — orders five and six do not factorise compactly — the operator builds the equilibrium populations and transforms them with the same monomial central-moment transform used for the populations themselves. The order of the hydrodynamic part is a compile-time switch, defaulting to (17); the second-order form is retained so the published results of [6] stay reproducible. The test suite checks, at *both* orders: the monomial transform against a direct contraction (2.2×10^{-16}); the closed forms (44) and (45) (5.6×10^{-17}); the reduction to the two-dimensional scheme at $u_z = b_z = 0$; and conservation of mass and all three momentum components.

Relaxation. As in two dimensions, the trace $k_{200} + k_{020} + k_{002}$ relaxes at ω_{bulk}, its five deviatoric partners at ω_ν, and everything of order three and above goes straight to equilibrium. Note that in three dimensions the trace carries a magnetic contribution that is absent in two,

$$k^{\mathrm{eq}}_{200} + k^{\mathrm{eq}}_{020} + k^{\mathrm{eq}}_{002} = \rho + \tfrac{1}{2}|\boldsymbol{b}|^2, \tag{46}$$

because $M_{\alpha\alpha} = \frac{1}{2}|\boldsymbol{b}|^2$ in 3D but vanishes in 2D — the same trace term discussed in §5.7.

7 Forcing

7.1 Guo

$$\boldsymbol{u} = \frac{1}{\rho}\Big(\sum_i \boldsymbol{c}_i f_i + \tfrac{1}{2}\boldsymbol{F}\Big), \qquad S_i = \Big(1 - \frac{\omega}{2}\Big) w_i \left[\frac{\boldsymbol{c}_i - \boldsymbol{u}}{c_s^2} + \frac{(\boldsymbol{c}_i \cdot \boldsymbol{u})\,\boldsymbol{c}_i}{c_s^4}\right] \cdot \boldsymbol{F}. \tag{47}$$

The $(1-\omega/2)$ prefactor is applied *per relaxed mode*, not once: TRT and the moment operators relax different modes at different rates, so only BGK can fold a single prefactor in. The raw source is exposed separately for that reason.

7.2 High-order Hermite forcing, and why it is already here

The Guo term (47) is the body force expanded in Hermite polynomials and truncated at second order. Nothing stops the expansion being carried further, and on these lattices it can be taken as far as the equilibrium goes: fourth order on D2Q9, sixth on D3Q27. Writing $\mathcal{H}^{(n)}$ for the Hermite tensors in $\boldsymbol{c}_i$ and building the coefficients by the product rule — replacing one $\boldsymbol{u}$ slot by $\boldsymbol{F}$ at a time,

$$a^{(1)}_{\alpha} = F_\alpha, \quad a^{(2)}_{\alpha\beta} = F_\alpha u_\beta + u_\alpha F_\beta, \quad a^{(3)}_{\alpha\beta\gamma} = F_\alpha u_\beta u_\gamma + u_\alpha F_\beta u_\gamma + u_\alpha u_\beta F_\gamma, \ \ldots \tag{48}$$

— the source term is

$$F_i = w_i \sum_{n\geq 1} \frac{1}{n!\, c_s^{2n}}\, a^{(n)}_{\alpha_1\ldots\alpha_n}\, \mathcal{H}^{(n)}_{\alpha_1\ldots\alpha_n}(\boldsymbol{c}_i). \tag{49}$$

Which components survive. Not every Hermite component may be kept. Two rules, both forced by the lattice rather than chosen:

- Components with an odd exponent ≥ 3 on one axis *vanish identically*. On D2Q9, $c_x^3 = c_x$ and $1/c_s^2 = 3$ give $\mathcal{H}^{(3)}_{xxx} = \hat{c}_x^3 - 3\hat{c}_x \equiv 0$. They may be kept or dropped freely.
- Components with an *even* exponent ≥ 4 on one axis are nonzero but *not independent*: $\mathcal{H}^{(4)}_{xxxx} = -3\,\mathcal{H}^{(2)}_{xx}$. Including them injects a spurious $O(Fu^3)$ term into the second moment and destroys $k_{20} = k_{02} = 0$. They must be omitted.

What remains is exactly the set of exponent triples in $\{0,1,2\}$ — the same representable set the equilibrium uses.

The result, and it is simple. Transforming (49) to *monomial* central moments $k_{pqr} = \sum_i F_i\, C_x^p C_y^q C_z^r$ collapses to one rule. If exactly one axis a carries exponent 1 and m other axes carry exponent 2, then

$$\boxed{k^{\text{force}} = c_s^{2m} F_a} \qquad \text{and every other moment is exactly zero.} \tag{50}$$

On D2Q9 that is $[0, F_x, F_y, 0, 0, 0, c_s^2 F_y, c_s^2 F_x, 0]$ in the ordering $(00, 10, 01, 20, 02, 11, 21, 12, 22)$. On D3Q27 it is three entries F_a, six at $c_s^2 F_a$, three at $c_s^4 F_a$, and fifteen exact zeros. In the co-moving frame the force touches only the moments that are odd in exactly one axis — which is the whole reason for working there.

And the code already does this. It is tempting to read (50) as saying that an operator which adds $\boldsymbol{F}$ only to the first-order moments, as §6.3 does, is missing the c_s^{2m} entries. It is not. `ProductBasis` stores $\varphi_2(C) = C^2 - c_s^2$ per axis, so expanding $C^2 = \varphi_2 + c_s^2$,

$$k^{\text{monomial}}_{122} = \tilde{k}_{122} + c_s^2 \tilde{k}_{120} + c_s^2 \tilde{k}_{102} + c_s^4 \tilde{k}_{100}, \tag{51}$$

and with only $\tilde{k}_{100} = F_x$ populated the monomial value is $c_s^4 F_x$, exactly as required — one factor of c_s^2 per squared axis, delivered by the basis function itself. The momentum source alone reproduces all of (50).

This was established the hard way. The third-order terms were added explicitly on top, and the operator then injected $1.5\, c_s^2 F_y$ where $c_s^2 F_y$ is correct — a clean 50% overshoot from double

counting. No change to the collision operator was needed or kept. `validation/forcing_cm.cpp` checks both representations of the same populations on both lattices: monomial must show the full c_s^{2m} pattern, Hermite must show the force in the first-order slots and nowhere else. Worst deviation 5.6×10^{-13}.

7.3 Boussinesq buoyancy

$$\boldsymbol{F}(n) = \rho_0 \, \boldsymbol{g} \, \beta \, (T(n) - T_0), \tag{52}$$

delivered through the same Guo machinery. Because this force varies in space, the collision interface carries a node index; that hook is also what the Lorentz force uses.

7.4 Density-anomaly buoyancy: water near 4 °C

Water is densest at $T_m \approx 4\,^\circ\mathrm{C}$, and Gebhart & Mollendorf (1977) fit pure water at one atmosphere by $\rho(T) = \rho_m(1 - w_{GM}\,|T - T_m|^q)$ with $\rho_m = 999.972\,\mathrm{kg\,m^{-3}}$, $w_{GM} = 9.297173 \times 10^{-6}\,(^\circ\mathrm{C})^{-q}$, $q = 1.894816$ and $T_m = 4.029325\,^\circ\mathrm{C}$. `DensityAnomalyGuo` delivers the corresponding buoyancy through the Guo machinery,

$$\boldsymbol{F}(n) = \rho_0 \, \hat{\boldsymbol{e}}_\uparrow \, w \, \left(|T(n) - T_m|^q - |T_0 - T_m|^q\right), \tag{53}$$

with $w = g\, w_{GM}\, \Delta T^q$ in lattice units when the scalar carries $\theta = (T - T_{\mathrm{ref}})/\Delta T$. Over 0–8 °C the density moves by 1.2×10^{-4} of itself, so the approximation is Boussinesq exactly as in Sec. 7.3; only the shape of the buoyancy changes, and it is not monotone: water between 0 and 4 °C heated from below is unstable, and above 4 °C stable. For $q = 1$ with every temperature above T_m the policy is `BoussinesqGuo` with $\beta = w$, and the two agree on a growing Rayleigh–Bénard mode to 1.7×10^{-15} in T and 7.5×10^{-14} in v (not bit for bit, because $|T - T_m| - |T_0 - T_m|$ and $T - T_0$ round differently).

Validation. `validation/density_anomaly` brackets the onset of penetrative convection in a rigid–rigid layer with $b(\theta) = |\theta - \lambda|^q$ against an exact reference: `tools/penetrative_onset.py` solves the marginal equations by shooting, on a mesh split and graded at the kink $z = 1 - \lambda$, and reproduces the classical $\mathrm{Ra}_c = 1707.762$, $k_c H = 3.117$ before it reports anything. At the Gebhart–Mollendorf q the references are $\mathrm{Ra} = 16764.642$ at $kH = 4.0212$ for $\lambda = 0.5$ (maximum mid-layer, upper half stably stratified; the minimum is $\mathrm{Ra}_c = 16763.68$ at $k_c H = 3.995$, a factor 9.8 above the classical layer) and $\mathrm{Ra} = 1674.292$ at $kH = \pi$ for $\lambda = 0$. On D3Q27 central moments with a D3Q7 scalar at $n_z = 1$, halfway walls:

	$H = 16$	$H = 32$	$H = 64$	order
$\lambda = 0.5$	+4.665%	+1.059%	+0.250%	2.14, 2.08
$\lambda = 0$	+2.133%	+0.481%	+0.113%	2.15, 2.09

Richardson extrapolation of the last pair lands 0.019% and 0.010% from the references. Nothing nonlinear is validated with this law yet: no Nusselt number and no sidewall flow has been compared with a published result.

7.5 Implicit Darcy drag: the mushy-zone sink

An enthalpy-porosity model holds the solid still with a Carman–Kozeny drag $\boldsymbol{F} = -A\,\boldsymbol{u}$, $A = A_{\mathrm{lat}}\,\epsilon\,(1 - f_l)^2/(f_l^3 + \epsilon)$. Built by a driver from the velocity the solver last *reported*, the force lags a step; with Guo's half shift that is the two-step recurrence $u^{n+1} = (1 - \frac{3}{2}A)\,u^n + \frac{1}{2}A\,u^{n-1}$, whose spectral radius reaches one at $A = 1$, and the solid keeps a leakage floor of about 2% of

the peak velocity. `DarcyGuo` instead closes the half shift with the drag, which a linear local force allows in one line,

$$\rho \boldsymbol{u} = \boldsymbol{m} + \tfrac{1}{2}(\boldsymbol{F}_{\text{ext}} - A\boldsymbol{u}) \quad \Longrightarrow \quad \boldsymbol{u} = \frac{\boldsymbol{m} + \boldsymbol{F}_{\text{ext}}/2}{\rho + A/2}, \tag{54}$$

and evaluates the force at that $\boldsymbol{u}$ in the source. A uniform state then decays by $g = (2-A)/(2+A)$ per step, $|g| < 1$ for every $A > 0$. Operators that cannot be handed the velocity refuse the policy at compile time. On the reduced channel of `validation/mushy_sink` the implicit and lagged profiles agree to 2.5×10^{-14} wherever the lagged one converges (a steady state cannot see a lag), every A up to 10^6 is stable on BGK and on central moments, and the leakage falls as $1/A$: 2.79×10^{-6} of the peak at $A = 10^4$. The stiff sink's no-slip plane is *not* on the first solid node: it lies 0.570/0.736/0.904 cells past the last liquid node on BGK and 0.672/0.809/0.904 on central moments at $\tau = 0.6/0.8/1.0$, so the relaxation time also places the solid surface, to within 0.43 of a cell.

7.6 Ice on water at equilibrium

`validation/ice_equilibrium` couples the three pieces above — `EnthalpyBGK`, `DarcyGuo` and the density-anomaly buoyancy — in the configuration of Wang *et al.* (PRFluids 6, L091501, 2021): ice grown down from a plate at $T_t < 0$ over water heated from below at T_b. Their $T_b = 10\,^\circ\text{C}$ comparison convects at $\text{Ra} \sim 10^8$ and needs days of physical time; this case takes the limit in which the answer is exact. With $T_b \le 4\,^\circ\text{C}$ every water temperature lies below the density maximum, the layer is stably stratified, and the equilibrium is two-slab conduction, $h/H = k_i|T_t|/(k_i|T_t| + k_w T_b)$, with water's $k_i/k_w = 3.944$ and $(\rho c_p)_i/(\rho c_p)_w = 0.4824$. The buoyancy is sized so that the water would sit at $\text{Ra} \approx 5 \times 10^3$ if its stratification were unstable, and it is seeded. At every resolution the flow contributes nothing (a frozen-fluid twin agrees to 1.2×10^{-8} of the temperature span), the residual velocity is below 2.5×10^{-9} of the would-be free-fall velocity, and the seed decays by at least 10^9.

The isothermal front ($T_s = T_l$) has no unique discrete equilibrium: any link whose two-slab interface temperature lies in $[-\frac{1}{2}s_w, \frac{1}{2}s_i]$, with s the per-cell temperature drop on each side, is steady, and growing ice stops within one cell of the window's thin edge. The predicted edges are 0.906 and 1.222 cells; the fronts land at 0.903 and 1.241 at $H = 64$. The error in cells does not refine away. A mushy band $[-d, d]$ makes f_l a function of T and converges onto its own exact equilibrium, obtained in closed form from $K(T) = \int k\,dT$: within 0.09 cells at $H = 64$, with the pointwise temperature error falling from 1.8% to 0.18% and 0.094% over $H = 16$–64. The band is a model change, though: $d = 0.2\,^\circ\text{C}$ moves the equilibrium by 0.87 cells here, so it must be sized against the gradient at the front.

8 Boundary conditions

Boundary conditions are alternative collision operators applied to flagged cells. This gives one branch per *node* rather than one per direction, keeps the population loads unconditional and coalesced, and generalises to any wall model.

condition	rule	wall location
no-slip (halfway bounce-back)	$f_i^{\text{out}} = f_{\text{opp}(i)}^{\text{in}}$	midway
adiabatic scalar	$g_i^{\text{out}} = g_{\text{opp}(i)}^{\text{in}}$	midway
Dirichlet scalar (anti-bounce-back)	$g_i^{\text{out}} = -g_{\text{opp}(i)}^{\text{in}} + 2w_i\,(T_w - T_{\text{ref}})$	midway

There is an asymmetry worth stating. Bounce-back is the *identity* on Esoteric Pull's storage (§3.2), so those cells are skipped entirely. Anti-bounce-back flips a sign and adds a source, so

those cells must be processed at every step. Both still write back into the same two slots they read, so neither disturbs the in-place scheme.

All three conditions above put the wall *midway* between nodes. The three that follow put it *on* the node. That difference is not a detail of accuracy but of geometry: a channel of N nodes is N lattice units wide under bounce-back and $N-1$ under an on-node condition, and mixing the two families in one problem means the fluid and the field disagree about where the wall is.

8.1 Regularised velocity walls

Boundary condition BC3 of Latt, Chopard, Malaspinas, Deville and Michler [1]. *Every* population on the node is replaced, using only ρ, the imposed $\boldsymbol{u}$, and the non-equilibrium stress:

$$\boxed{g_i = f_i^{\mathrm{eq}}(\rho, \boldsymbol{u}) + \frac{w_i}{2c_s^4}\,\boldsymbol{Q}_i : \Pi^{(1)}, \qquad \boldsymbol{Q}_i = \boldsymbol{c}_i\boldsymbol{c}_i - c_s^2\boldsymbol{I}.} \tag{55}$$

Note what (55) does *not* contain: ω. The reconstruction is a projection of a given tensor onto the lattice, so it serves BGK, TRT, MRT and the central-moment operators without change.

Density. The paper's closure is written for one particular figure and velocity numbering. It is implemented here geometrically, which is numbering independent and therefore survives the Esoteric Pull ordering: with $\hat{\boldsymbol{n}}$ the *outward* normal, the known populations are those with $\boldsymbol{c}_i \cdot \hat{\boldsymbol{n}} > 0$, and

$$\rho = \frac{1}{1 + \boldsymbol{u}\cdot\hat{\boldsymbol{n}}}\Big(2\sum_{\boldsymbol{c}_i\cdot\hat{\boldsymbol{n}}>0} f_i + \sum_{\boldsymbol{c}_i\cdot\hat{\boldsymbol{n}}=0} f_i\Big). \tag{56}$$

Stress. The unknown populations are given, *temporarily*, a bounce-back of their off-equilibrium part, and $\Pi^{(1)} = \sum_i \boldsymbol{c}_i\boldsymbol{c}_i(f_i - f_i^{\mathrm{eq}})$ is evaluated from that. The paper is explicit that this may not be used as the boundary condition itself — it cannot enforce $\boldsymbol{u}$ exactly — and it exists here only to give $\Pi^{(1)}$ a value.

Which populations are unknown. Not a half-space. A dot-product test $\boldsymbol{c}_i\cdot\hat{\boldsymbol{n}} < 0$ is correct on a straight wall and *wrong* on a corner: at the corner $(0,0)$ of a box, both $(1,-1)$ and $(-1,1)$ are unknown, each having its source node outside a different wall. The code therefore carries a per-node bitmask, built once from the geometry: bit i is set when the source $\boldsymbol{p} - \boldsymbol{c}_i$ lies outside the fluid. Where *both* members of a pair are unknown — possible only on corners and edges — those populations are set to equilibrium, contributing nothing to $\Pi^{(1)}$, which is the only choice that invents no stress.

Corners and edges. Equation (56) needs a single normal, which a corner does not have, so ρ is extrapolated to second order along one wall from two straight-wall neighbours — never reaching into the bulk, so no density field is required. For $\Pi^{(1)}$ there are two routes, and the default is the second:

- *local* — the same closure as a straight wall. Operator-agnostic, but a 2D corner offers only three streamed directions to build a stress from.
- *finite-difference* — the route of [1], Sec. V: $\Pi^{(1)}$ from second-order one-sided velocity gradients through

$$\Pi^{(1)} = -\frac{2c_s^2}{\omega}\,\rho\,\boldsymbol{S}, \qquad \boldsymbol{S} = \tfrac{1}{2}\big[\nabla\boldsymbol{u} + (\nabla\boldsymbol{u})^T\big]. \tag{57}$$

Neighbour velocities are taken from the imposed value when the neighbour is itself a wall node — its own unknowns are not yet fixed, so reading its populations would be wrong — and from its populations otherwise.

This is where the condition acquires a collision dependence. Equation (57) is a constitutive law, and its coefficient is the viscosity $\nu = c_s^2(1/\omega - 1/2)$. Inferring stress from strain rate therefore *requires* naming the shear relaxation rate, whereas measuring it from the populations does not: the operator's effect is already carried by the data. The rate is read from the operator (`omega`, or `omega_p` for TRT, whose even rate sets the viscosity), never supplied by the caller; an operator exposing none cannot select this route at all, which removes by construction the possibility of silently evaluating (57) with a placeholder $\omega = 1$ and imposing the wrong viscosity at the wall.

Body forces. Guo forcing defines $\rho\boldsymbol{u} = \sum_i f_i\boldsymbol{c}_i + \boldsymbol{F}/2$, so a reconstruction must satisfy $\sum_i \boldsymbol{c}_i g_i = \rho\boldsymbol{u}_w - \boldsymbol{F}/2$. Equation (55) delivers $\rho\boldsymbol{u}_w$, because $\sum_i w_i\boldsymbol{c}_i\boldsymbol{Q}_i = \boldsymbol{0}$. Two corrections follow from the forced Chapman–Enskog expansion and both are applied by default.

First, the expansion gives (dropping $O(\mathrm{Ma}^3)$)

$$\sum_i \boldsymbol{c}_i f_i^{(1)} = -\frac{\boldsymbol{F}}{2}, \qquad \Pi^{(1)} = -\frac{2\rho c_s^2}{\omega}\boldsymbol{S} - \frac{1}{2}(\boldsymbol{u}\boldsymbol{F} + \boldsymbol{F}\boldsymbol{u}), \tag{58}$$

the second term vanishing at a no-slip wall. The second-order Hermite reconstruction of $f^{(1)}$ consistent with (58) therefore needs its own first-moment term, which $\boldsymbol{Q}_i{:}\Pi$ cannot supply:

$$g_i \;\to\; g_i \;-\; \frac{w_i}{2c_s^2}\,\boldsymbol{c}_i{\cdot}\boldsymbol{F}. \tag{59}$$

Second — and this is the larger effect — the *measurement* of $\Pi^{(1)}$ is biased. Since $f^{(1)}$ carries the odd part $-\frac{w_i}{2c_s^2}\boldsymbol{c}_i\cdot\boldsymbol{F}$, bounce-back reverses its sign, so each unknown is wrong by $(w_i/c_s^2)\,\boldsymbol{c}_i\cdot\boldsymbol{F}$. Contracting $\boldsymbol{c}_i\boldsymbol{c}_i$ against that error would cancel over the full velocity set, but the unknowns form a *half-space* and it does not. On D2Q9 at a flat wall:

force direction	$\delta\Pi_{xx}$	$\delta\Pi_{yy}$	$\delta\Pi_{xy}$
tangential	0	0	$F/6$
normal	$F/6$	$F/2$	0

The tensors are different, which is why no correction of the form $\boldsymbol{u}_w \mp \boldsymbol{F}/2\rho$ can serve both cases: a tangential force produces spurious *shear*, a normal force spurious *normal* stress. The scaffold is therefore given the correct odd part rather than the reversed one.

What remains is not a force effect. After (59) and the scaffold repair, a residual wall offset survives for $\omega \neq 1$, obeying

$$\text{offset} \times W = \frac{4}{3}\,\frac{\omega - 1}{\omega^2} \qquad \Longleftrightarrow \qquad \delta u_{\text{slip}} = -\frac{2}{3}\,\frac{\omega - 1}{\omega^2}\,\frac{\partial^2 u_\parallel}{\partial n^2}. \tag{60}$$

This is a *curvature*-induced slip, not a force term. It is independent of $\boldsymbol{F}$ to five digits across an eightfold range of forcing; it vanishes identically for a linear profile, so Couette flow is exact to 4×10^{-15} at every τ from 0.6 to 2.0; and it vanishes at $\omega = 1$, consistent with the factor $(1-\omega)$ by which collision erases any error in the emitted stress. It is a pre-existing consequence of truncating $f^{(1)}$ at second Hermite order, which is what "regularised" means, and no correction for it is applied. The law (60) is empirical: it was fitted at two values of τ and then *predicted* three held-out values to within 0.3%, but the coefficient 4/3 has not been derived.

Departure from the paper. Reference [1] uses BC5, Eq. (48), on corners and edges; the form implemented here is BC4, Eq. (46). The two differ only by terms nonlinear in $\boldsymbol{u}$, which — as [1] states — "cancel for symmetry reasons during the evaluation of the tensor $\Pi^{(1)}$", and the asymptotic dynamics depends on $\Pi^{(1)}$ alone. The two therefore impose the same stress and differ only in ghost-population detail.

8.2 Specular reflection: a free-slip wall

A population arriving at the wall leaves with its wall-normal component reversed and its tangential components untouched,

$$f_i^{\text{out}} = f_{m(i)}^{\text{in}}, \qquad \boldsymbol{c}_{m(i)} = \boldsymbol{c}_i \text{ with component } \alpha \text{ negated}, \tag{61}$$

which imposes $\boldsymbol{u}\cdot\boldsymbol{n} = 0$ and zero tangential stress. Compare bounce-back, $f_i^{\text{out}} = f_{\text{opp}(i)}^{\text{in}}$, which reverses *all* components and imposes $\boldsymbol{u} = \boldsymbol{0}$.

Why this was thought impossible here, and why it is not. An earlier version of §12.5.2 asserted that specular reflection could not be added to Esoteric Pull: bounce-back is cheap there only because opposite directions occupy adjacent slots, and a mirror about a wall normal lands in a different pair. The second clause is true and the conclusion does not follow. Read the storage table of §3.2: at node n, even parity, for odd i, the scheme *reads* $A[n][i]$ and $A[n + \boldsymbol{c}_i][i{+}1]$ and *writes those same two slots*, crossed. Over all pairs the read set and the write set are the same Q slots. A node therefore holds its complete incoming population vector and writes back only into slots it alone owns, so *any* permutation applied between the load and the store is race-free and needs no buffer. Equation (61) is exactly such a permutation.

What the wrong claim was conflated with is real: bounce-back is the *identity* on the storage, so a `Solid` cell is skipped outright — no load, no store, no arithmetic. Specular is not the identity, so a free-slip cell has to run. That is a cost and a cell type, not an obstruction.

The test is an identity, not a rate. Poiseuille flow in a channel of width $2H$ has a steady state exactly mirror-symmetric about its mid-plane. Replace the upper half by a specular wall on that plane and the half-domain must reproduce the lower half of the full solution to *round-off*.

	half vs. full	no-slip plane	symmetry plane	mass drift
D2Q9, $\tau = 0.8$	5.5×10^{-14}	0.5041	16.5000	-1.5×10^{-16}
D2Q9, $\tau = 1.2$	1.2×10^{-15}	0.4874	16.5000	-1.5×10^{-16}
D3Q27, $\tau = 0.8$	2.0×10^{-13}	0.5041	16.5000	0

The reflecting plane lands halfway, at $H + 0.5 = 16.5$, for the same reason bounce-back's does. Mass is conserved exactly because a permutation cannot change a sum — unlike the regularised walls of §8.1, which overwrite populations. A free-slip wall that leaked a little momentum would pass a convergence test and fail this one. `validation/specular.cpp`.

What it does not do. Axis-aligned normals only: an oblique free-slip surface has no exact specular permutation on a lattice and is not faked. Two specular walls meeting at an edge would need both mirrors composed and are rejected at setup rather than approximated — which is affordable here only because the cell is a ghost, and stops being affordable once the plane moves on-node (§8.4).

8.3 The on-node zero-flux scalar wall

The mirror of §8.2 is a *fluid* boundary and it is a ghost cell: the plane falls half a cell outside the last fluid node. A scalar in this solver is frequently on-node instead — `ScalarMoment` puts a Dirichlet value exactly *at* a node, for the reason §12.5.1 measures, namely that $\boldsymbol{E} = -\nabla\varphi$ is a *derivative* of the field carrying the boundary value and a halfway stencil never sees the imposed potential. A zero-flux condition half a cell away from an on-node Dirichlet one mixes two wall families, so the scalar needs its own variant, and `ScalarSpecular` (`mirror_unknowns`) is it.

It acts on the unknowns only. After streaming, a wall node holds real data in every direction that arrived from inside the domain and nothing usable in every direction that would have arrived from outside. Mirror symmetry about the node says the missing population equals its normal-reflected partner, and that partner is one of the arrived ones:

$$q(-1) = q(+1) \quad \Longrightarrow \quad g_i \leftarrow g_{m(i)} \ \text{ for every } i \text{ with } \boldsymbol{c}_i \cdot \boldsymbol{n} < 0, \tag{62}$$

with $m(\cdot)$ the same mirror permutation as (61). Everything else is left alone and the node then *collides* like any other. Three properties follow, and each is the reason one of the pre-existing conditions fails where this one is needed.

- **It reports the real value.** $\sum_i g_i$ at the node is the field. `ScalarAdiabatic` reports a structural *zero*, which is harmless at a node nothing reads and fatal in a case that differentiates φ or integrates q across the wall column.
- **It reverses only the normal component.** Bounce-back reverses all of them, and at $\omega \to 2$ a full reversal re-injects an odd–even mode that never damps. A mirror does not.
- **The normal flux vanishes identically**, $\sum_i g_i \boldsymbol{c}_i \cdot \boldsymbol{n} = 0$ by the symmetry of the mirrored set — exactly, not to $O(\Delta x)$.

Checked as an identity, not as a rate. A box of width M closed by two of these mirrors is not merely similar to the periodic box of width $2M$ whose mirror it is; it is the same problem. `validation/scalar_specular.cpp` advects a scalar by a solenoidal, mirror-symmetric velocity field and compares node for node:

lattice	ω	$\max \lvert q_{\text{half}} - q_{\text{full}} \rvert$	relative
D2Q9	1.99	8.9×10^{-15}	5.3×10^{-15}
D2Q5	1.99	7.3×10^{-15}	4.3×10^{-15}
D2Q5	1.00	4.4×10^{-16}	3.1×10^{-16}
D3Q7	1.00	4.4×10^{-16}	3.0×10^{-16}

The hard case is run first and deliberately: $\omega = 1.99$ is where the two existing conditions fail. The same test prints two guards, because an identity is easy to satisfy vacuously. The reference field must still carry structure — two nearly-uniform fields agree to round-off whatever the wall does — and the same half box closed by `ScalarAdiabatic` is reported as a control, where the disagreement is the size of the field itself.

A cell type is either prescribed or closed, and the source list is what will not tell you. `ScalarSolver`'s source kernel visited `ScalarBulk` only. That is right for Dirichlet, Moment, Outflow and Adiabatic, all of which are overwritten downstream so a source added to them would be discarded — and wrong for `ScalarSpecular`, which is a bulk node carrying a mirror closure. The Poisson source $\beta q/\varepsilon$ was therefore absent in the wall columns, which quietly solved $\nabla^2 \varphi = 0$ instead. Measured on §12.5.4's hydrostatic reference at $N = 41$: I_0 came out $+8.87\,\%$ against $+0.51\,\%$ for the boundary-free box, and the convergence fell from second order to first. Nothing failed; it merely looked like a mediocre boundary condition.

8.4 The on-node specular fluid wall

§8.2's mirror is exact and its plane sits half a cell outside the last fluid node; §8.3's is exact and its plane sits *on* the node, because the Dirichlet family it must pair with is on-node. So the two were exact about planes half a cell apart, and a case whose walls are all on-node —

§12.5.4 is one — had no matching free-slip option for the fluid. `SpecNode` (`set_specular_nodes`, `mirror_unknowns_faces`) is that option: the same construction as (62), applied to the fluid populations, so the node mirrors its *unknown* directions and then collides like any other. It reports a real ρ and a real $\boldsymbol{u}$, where `SpecWall` is a ghost and reports neither.

It is a new cell type rather than a mode of the old one on purpose: `validation/specular.cpp` asserts the ghost plane at 16.5000 and §12.5.2's free-slip geometry is built on it, so moving that plane would silently reinterpret both.

A face mask, not a normal. A `NormalCode` names one outward direction. An on-node wall node can sit on an edge or a corner of a closed box, where two or three mirror planes meet, and unlike the ghost version it cannot refuse them — a ghost outside the domain is never read diagonally, whereas an on-node node in the corner of a box is a real fluid node and there is nowhere else to put it. So the condition takes a bitmask over the six faces. Reusing the normal code would also collide outright, `NrmYp` = 3 reading as `SpecXp|SpecXm`. For a direction $\boldsymbol{c}_i$ the mirror negates every component whose sign points *into* the domain, i.e. composes the axis mirrors over exactly that subset. The image then points outward on every masked axis — it is a direction that actually arrived — and the map is many-to-one by construction: at a two-face edge the three inward quadrants all take the value of the single outward one, which is what reflection in two planes says. A mask carrying both faces of one axis is rejected at setup, since two mirror planes one cell apart would map an unknown onto another unknown.

Checked as an identity, and against the wall it is not. A box closed by two on-node mirrors is the same problem as the periodic box of twice the width whose mirror it is, node for node. `validation/specular_node.cpp` runs both and compares, in two geometries:

geometry	what it exercises	worst relative
half box, D2Q9, 17×20 vs 32×20	single-face masks	1.9×10^{-15}
quarter box, D3Q27, $17 \times 20 \times 13$	four two-face edge lines	3.7×10^{-15}

The initial state is even about every mirror plane, drawn from a streamfunction so that it is solenoidal, and every plane falls on a node. Two guards accompany the number, because an identity is easy to satisfy vacuously. The reference must still carry structure: at $\tau = 0.8$ the seeded mode decays by e^{-4} over the run and arrives at $|\boldsymbol{u}| = 6 \times 10^{-4}$, close enough to nothing that the comparison would pass for any wall at all, which is why the case runs at $\tau = 0.6$. And the same box closed by the *halfway* mirror of §8.2 is reported as a control: it is out by 16 % of the field. Both walls are exact; they are exact about different planes, and that control is what makes the distinction measurable rather than asserted.

What the mirror imposes, checked without a solver. A symmetric setup can hide a wall that is symmetric but not zero-flux, so the permutation is also applied to a random population vector and its moments read off directly. On every masked axis $\sum_i f_i \boldsymbol{c}_i \cdot \boldsymbol{n} = 0$ and $\sum_i f_i (\boldsymbol{c}_i \cdot \boldsymbol{n})(\boldsymbol{c}_i \cdot \boldsymbol{t}) = 0$, both to 0 exactly, so there is no normal flux and no tangential stress. The density is left *free* — the mirror imposes no value on it — and the sharper statement, which is what the test asserts, is that an already-symmetric state is a fixed point: an equilibrium at $(\rho, \boldsymbol{u})$ with vanishing normal components is returned bit for bit. Momentum *along* a two-face edge line survives, which is correct rather than a leak, the edge being the intersection of two symmetry planes and constrained by neither; the test asserts that it survives, which is what distinguishes the mirror from bounce-back — bounce-back would kill every component and would still pass both flux tests on a symmetric setup.

What it buys, measured. §12.5.2's `-fsnode` runs the pairing with every plane on a node. Against the boundary-free doubled box at $T = 190$ it reduces the lateral gap from $+1.55\,\%$ to $-0.56\,\%$ at the finest grid, i.e. by a factor 2.8 — and does *not* close it. The full ladder, the coarse-grid branch trap, and the separate finding that the on-node *scalar* wall is meanwhile twice as large as the halfway one, are in §12.5.2. `GPU/` carries the same fluid condition (`specular.cuh`), asserted the same way in `test/host_physics.cpp`, at 1.6×10^{-15} and 1.4×10^{-15} in FP64 and 8×10^{-7} in FP32.

8.5 Outflow: constant back-pressure

An open outlet needs a different closure from a wall: the velocity there is not known, it is an output of the interior flow. The condition implemented reuses the regularised reconstruction of §8.1 and differs only in where $(\rho, \boldsymbol{u})$ come from.

Why the density must be imposed. The obvious choice — copy both ρ and $\boldsymbol{u}$ from the upstream neighbour, a zero-gradient outlet — is *ill posed* in this scheme, and not marginally so. The regularised nodes overwrite every population from $(\rho, \boldsymbol{u}, \Pi^{(1)})$, so they are not mass conserving; with a velocity inlet injecting mass and no condition fixing the pressure anywhere, the total mass has no equilibrium. Run on the channel of §12.1.12 it settles at $\rho \approx 181$ with a density ramp along x and a centreline velocity a factor two below the analytic profile. The outlet must therefore fix the pressure:

$$\rho(\text{outlet}) = \rho_0. \tag{63}$$

Normal velocity. With ρ known, the wall closure (56) inverts. Partitioning the populations by the sign of $\boldsymbol{c}_i \cdot \hat{\boldsymbol{n}}$ with $\hat{\boldsymbol{n}}$ the *outward* normal, and writing $\Sigma_+ = \sum_{\boldsymbol{c}_i \cdot \hat{\boldsymbol{n}} > 0} f_i$ (streamed from the interior, hence known) and $\Sigma_0 = \sum_{\boldsymbol{c}_i \cdot \hat{\boldsymbol{n}} = 0} f_i$,

$$\rho\,(1 + \boldsymbol{u} \cdot \hat{\boldsymbol{n}}) = 2\Sigma_+ + \Sigma_0 \qquad \Longrightarrow \qquad \boxed{\boldsymbol{u} \cdot \hat{\boldsymbol{n}} = \frac{2\Sigma_+ + \Sigma_0}{\rho} - 1} \tag{64}$$

which is the same relation as the wall's, read in the other direction. The outlet therefore passes whatever flux the interior sends it rather than prescribing one, and the condition is self-regulating.

Tangential velocity. The node's own populations carry no inflow directions, so the tangential components are taken from the upstream neighbour (zero gradient, first order). A second-order variant — linear extrapolation from two upstream nodes, `set_outflow_order(2)` — is implemented and is *not* used, because it was measured to be no better: worse on coarse grids (1.388×10^{-3} against 1.160×10^{-3} at $L_y = 41$) and indistinguishable on the finest (6.184×10^{-6} against 6.192×10^{-6} at $L_y = 641$). §12.1.12 shows why — the outlet is not what limits the order there.

Only the $+x$ face is implemented (`NrmOutXp`). Nothing in (64) is specific to that direction; the other five are a matter of extending `upstream_of` and `normal_of`.

The magnetic counterpart, and its defect. The same idea is available on the induction lattices (`MagOutXp`): the node takes $\boldsymbol{b}$ from its upstream neighbour instead of an imposed value, and is then closed by the moment condition (68) exactly as a Dirichlet magnetic wall is. It reuses the field pass rather than the collision pass – the upstream neighbour's populations are read where $\boldsymbol{b}$ is computed, which is race free because that kernel only reads populations, and the node's own $\boldsymbol{b}$ then serves as the target when the moment is imposed.

This condition is not validated and has a measured defect. On the inlet-driven Hartmann flow of §12.1.5 — whose exact solution is independent of the streamwise coordinate, so that any x variation in the answer is boundary error and nothing else – it drives $\boldsymbol{b}$ about 6% high over the last ten or so nodes, and the overshoot decays upstream only slowly:

L_x	b/b_{exact} along the channel at $L_y = 65$
21	runs $1.000 \to 1.054$ across the *whole* domain
121	flat plateau at 1.005 in the bulk, rising to 1.061 at the end
241	plateau reached well before the sampling station

The effect on a score is large: at $L_y = 129$ the error in $\boldsymbol{b}$ falls from 3.367×10^{-3} to 9.793×10^{-4} purely by moving the outlet from $L_x = 121$ to $L_x = 241$. It is specifically the magnetic outlet and not the fluid one — pinning $\boldsymbol{b}$ analytically at the outlet instead gives errors agreeing to three significant figures with the zero-gradient case at $L_y = 17, 33, 65$, so the two differ only through that contamination. The cause is not identified. The two natural suspects are the interaction with $\nabla \cdot \boldsymbol{b} = 0$, which this code does not preserve to machine precision anyway, and the fact that a pure Neumann condition does not constrain the field level at all — the same class of ill-posedness that made a zero-gradient *fluid* outlet drive ρ to 181. Until it is understood, keep the outlet far from anything being measured.

A race that does not fire, and two repairs that made it worse. The arbitrary-face closure reads its donor's populations from inside the main `stream_collide` kernel. Under Esoteric Pull the two slots a node reads are exactly the two it writes, so the donor — an ordinary fluid cell, updated in the same pass — is rewriting the slots the outflow node is reading. Whether the pre- or post-collision state is obtained is not determined by the program.

Measured on the aorta of §13, 4000 steps with 2000 outlet nodes, the output is **byte-identical on 1, 2, 4 and 8 threads**. The reason is layout, not correctness: `rebuild_lists` builds the active list by scanning node indices in ascending order, and a donor is an immediate spatial neighbour, so an outflow node and its donor fall inside one thread's contiguous chunk. Only a pair straddling a chunk boundary can race, and with eight threads there are seven boundaries against two thousand outlet nodes. **That guarantee is a property of contiguous CPU chunking and will not survive a GPU**, where every node is its own work item.

Both obvious repairs are worse than the race. The natural fix is the one the scalar's open boundary uses (§8.8): read what cannot be read safely in a separate pass, so the ordering is defined. Two versions were built and measured against the documented steady case of §13.3, 13 500 steps at $\text{Re} = 50$:

donor read	$\lvert\boldsymbol{u}\rvert_{\max}$	$\lvert\Delta m\rvert$	ρ_{out}	behaviour
in-kernel (current)	0.0720	$\sim 10^{-5}$	0.9934	converges by step 7000
pre-collision, prefetched	0.0400	2×10^{-3}	oscillating	**limit cycle**
post-collision, gathered	0.0335	7.3×10^{-4}	0.9929	converges

Reading the donor *pre*-collision replaces a fixed point with a limit cycle, which is unambiguously a regression. Reading it *post*-collision is the correct phase — the result is written through `store_pair`, which stores post-collision populations — and it does remove the limit cycle and bring ρ_{out} within 0.05% of the documented value. But it still leaves mass conservation seventy-three times worse and the peak speed at half. Mass conservation is the only quantity here with a known correct value, so the in-kernel read stands.

Why defining the read should cost mass conservation is not understood. One untested story: the closure rescales the donor's distribution to a fixed density, and under the

racy mixed read adjacent outlet cells received slightly different treatments whose errors partly cancelled, where a uniform read makes the error systematic. That is speculation, recorded as such.

The practical conclusion is not "leave it alone" but a sharper one: the GPU port needs a closure that is *right*, not merely one that is *ordered*. Imposing an order on the existing closure is exactly what was tried, twice, and it costs one to two orders of magnitude of mass conservation.

8.6 Forcing: every component, every axis, and a fully 3D case

§12.1.1 drives its channel along x with $\boldsymbol{F} = (G, 0, 0)$. That exercises $k_{12} = c_s^2 F_x$ from (50) and never touches $k_{21} = c_s^2 F_y$, so on its own it validates half the forcing. Three tests close that.

Rotating the channel, D2Q9. Walls x-normal and the force along $+y$ swaps which component is live. Both orientations are exact at the central-moment magic point, and — since D2Q9 is invariant under a quarter turn — they must agree with each other, which no single orientation can check.

	$\tau = 0.6$	$\tau = 7/8$	$\tau = 1.2$
along x (tests k_{12})	3.9225×10^{-3}	3.06×10^{-14}	4.6356×10^{-3}
along y (tests k_{21})	3.9225×10^{-3}	3.14×10^{-14}	4.6356×10^{-3}
|difference|	5.3×10^{-14}	6.1×10^{-15}	1.1×10^{-14}

$H = 16$. Off-magic the two agree to 10^{-13}, which is rotational isotropy of the forcing, equilibrium and streaming together.

All three axes, D3Q27. Walls normal to one axis, force along the next, periodic in the third; the three cyclic combinations put every axis in both roles. At $H = 16$ they give 2.99×10^{-14}, 3.68×10^{-14}, 3.12×10^{-14} at $\tau = 7/8$, and off-magic all three are *bit-identical* — 3.9225×10^{-3} at $\tau = 0.6$ and 4.6356×10^{-3} at $\tau = 1.2$, the same values D2Q9 gives, since a plane channel is quasi-one-dimensional and D3Q27 reduces to D2Q9 exactly on it.

A fully three-dimensional forced flow. A plane channel varies along one axis only, so it never excites a moment with two squared axes and barely touches the c_s^4 group. This case does. Periodic cube, no walls, $F_x = F_0 \sin(ky)\sin(kz)$ with $k = 2\pi/N$, whose steady solution is *exact* for the full Navier–Stokes equations,

$$u_x = \frac{F_0 \sin(ky)\sin(kz)}{2\rho\nu k^2}, \qquad u_y = u_z = 0, \tag{65}$$

because u_x does not depend on x, so $(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}$ vanishes identically and the pressure stays uniform. The nonlinear term is not small here; it is zero.

N	$\tau = 0.8$		$\tau = 1.2$	
	rel L_2	order	rel L_2	order
8	7.4948×10^{-2}	—	8.1933×10^{-2}	—
16	2.0117×10^{-2}	1.897	2.0529×10^{-2}	1.997
32	5.1128×10^{-3}	1.976	5.1381×10^{-3}	1.998
64	1.2834×10^{-3}	1.994	1.2850×10^{-3}	2.000

Second order, and the L_2 error *equals* the peak error at every entry: the profile shape is right to round-off and all the error is amplitude.

Two magic points, and they are not the same one. That amplitude error changes sign with τ, so some viscosity makes it vanish. Scanning τ from 0.6 to 1.4 at $N = 32$ locates the crossing at $\tau = 0.9994$, and the whole family collapses onto

$$\frac{\Delta A}{A} = 2c_s^2 k^2(\tau - 1) + O(k^4), \tag{66}$$

with measured-to-model ratios running $0.9946 \to 0.9999$ as the grid refines, across $\tau \in [0.6, 1.4]$ and $N \in [8, 64]$. So the forced scheme has two distinct magic points and no single τ gives both: $\tau = 7/8$ puts the *wall* exactly on the node, a boundary property; $\tau = 1$ makes the *bulk* amplitude exact, a dispersion property. Which one limits a given run depends on whether it is wall-dominated or bulk-dominated.

The tests can fail. An exactness claim is only worth as much as its sensitivity, so the third-order force moments were deliberately perturbed by +50% and the tests re-run. The D2Q9 channel degrades from 8.44×10^{-14} to 6.69×10^{-4} and its wall slip from 7.4×10^{-14} to -3.9×10^{-3}; the transverse case, where $F_x = 0$ so only k_{21} is live, degrades from 3.14×10^{-14} to 2.67×10^{-3}. Ten orders of magnitude in both. These are not tests that pass regardless.

8.7 Moment-based walls for the scalar and magnetic fields

Dellar's moment conditions [2], Eqs. (13a)–(13b). The fields carried by the advection–diffusion lattices are *zeroth moments*,

$$B_\alpha = \sum_i g_{\alpha i}, \qquad T = \sum_i h_i, \tag{67}$$

so imposing a boundary value is one linear equation in the unknown populations. On a cross lattice a straight wall leaves *exactly one* unknown — the direction pointing into the domain along the normal — and that single equation closes the system uniquely and exactly:

$$\boxed{g_{x1} = B_0 - (g_{x0} + g_{x2} + g_{x3} + g_{x4}), \qquad g_{y1} = -(g_{y0} + g_{y2} + g_{y3} + g_{y4}).} \tag{68}$$

There is no closure assumption and no free parameter. The scalar case is identical with $\sum_i h_i = T_w - T_{\mathrm{ref}}$, the shift being the storage reference of §5.5.

Why B may simply be imposed. Maxwell's equations make both the normal and the tangential components of $\boldsymbol{b}$ continuous across the wall, so $\boldsymbol{b}$ takes its external applied value there. The velocity enjoys no such freedom, which is why $\boldsymbol{u}$ needs the machinery of §8.1 and $\boldsymbol{b}$ does not.

Corners. More than one unknown, and the single moment no longer closes the system. The deficit is then shared in proportion to the lattice weights: the moment is still reproduced exactly, only its distribution among the unknowns is a choice, and the weight-proportional one introduces no directional preference. This is a fallback, not part of the published method.

A trap worth recording. The field kernels compute B_α and T as $\sum_i$ of the streamed populations. At a moment wall those still contain the *unfixed* inward direction, so their sum is not the field. Reading it anyway feeds a wrong $\boldsymbol{b}$ into the magnetic equilibrium and the computation diverges — observed here at 10^{150} within a few hundred steps. The field at such a node is the imposed value by construction and is reported as such. The same applies to ρ and $\boldsymbol{u}$ at a regularised wall.

condition	fields	rule	wall location
regularised	$\rho, \boldsymbol{u}$	(55) with (56)	on the node
moment (magnetic)	$\boldsymbol{b}$	(68)	on the node
moment (scalar)	T	$\sum_i h_i = T_w - T_{\mathrm{ref}}$	on the node

8.8 Open boundaries for the scalar

The two scalar wall conditions above are both closed: adiabatic reflects everything, Dirichlet pins a value. Neither lets a plume *leave*. A transported scalar on an open domain needs a third condition, zero-gradient on C, so that what the flow carries to the boundary passes through it instead of accumulating behind it.

The condition implemented is equilibrium extrapolation. Each cell marked `ScalarOutflow` is assigned a *donor*, an interior neighbour, and every outgoing population is set to

$$h_i^{\mathrm{out}} = h_i^{\mathrm{eq}}(C_{\mathrm{donor}},\, \boldsymbol{u}_{\mathrm{node}})\,, \tag{69}$$

so the node carries its neighbour's concentration at its own velocity. The node is fully prescribed: whatever streamed into it is discarded, and its inward directions carry the interior's value back into the domain. Dropping the non-equilibrium part slightly damps the *diffusive* flux at the exit, which is negligible when the exit is advection-dominated — and is measured rather than assumed in §12.3.2.

Why this needs a second kernel. The obvious implementation reads the donor's populations inside the main kernel, as the arbitrary-face outflow of §8.5 does for the fluid. Under Esoteric Pull that is a genuine read/write race: the two slots a node reads are exactly the two it writes, so a donor being updated concurrently is rewriting the very slots the outflow node is reading, and which state is obtained depends on thread scheduling. Outflow cells are therefore skipped by the main kernel and handled by a second one after a fence, and that pass reads the *macroscopic field* rather than populations: the main kernel has just written C at every bulk cell, donors are required to be bulk cells, and so nothing the second pass reads is written by the second pass. Every storage slot still has exactly one writer, because the main pass skips these cells entirely and the outflow pass writes only the slots they own. The result is byte-identical on 1, 2 and 4 threads.

Donor selection. The outward directions of a cell are the axes whose neighbour lies outside the field; the donor is the neighbour one step inward along all of them at once. A face cell therefore takes its axis neighbour, an edge cell the diagonal and a corner cell the body diagonal. Without that, a box edge has no purely axial interior neighbour and every edge and corner of the domain would be inert. A cell with no bulk neighbour at all is left pointing at itself, which makes the condition a no-op there — it falls back to bounce-back — and is counted and reported at setup rather than silently reading a neighbour's garbage.

9 Coupled modules

9.1 Thermal transport

A passive scalar is carried by a second distribution set on its own lattice, with

$$T = \sum_i g_i, \qquad g_i^{\mathrm{eq}} = w_i\, T\Big(1 + \frac{\boldsymbol{c}_i \cdot \boldsymbol{u}}{c_s^2}\Big), \qquad D = c_s^2\Big(\frac{1}{\omega} - \frac{1}{2}\Big). \tag{70}$$

The velocity is an *input*, not something recovered from these populations, which is why the scalar has its own collision concept.

The first-order truncation in (70) is deliberate. D2Q5 and D3Q7 have an isotropic second moment but not an isotropic fourth-order one, and on D3Q7 every velocity has a single nonzero component, so $(\boldsymbol{c}_i\cdot\boldsymbol{u})^2$ reduces to $c_\alpha^2 u_\alpha^2$ and the cross terms of the $\boldsymbol{uu}$ tensor cannot be represented at all. A second-order term would add an anisotropic error rather than accuracy. The cost is an $O(u^2)$ defect in the advection term, which is why these lattices suit low-Mach transport — exactly the regime Boussinesq convection occupies.

Architectural significance. This module was the first test of the composition the design was built around. D3Q7 has seven populations to the fluid's twenty-seven, a different sound speed, a different collision concept and different boundary conditions — and nothing in `FluidSolver`, the lattices, the streaming schemes or the moment operators required modification. The only interface addition was the node index on the collision.

9.2 Magnetohydrodynamics

Induction. Following Dellar [5], the magnetic field is carried by a *vector-valued* distribution g_i^α, with

$$b_\alpha = \sum_i g_i^\alpha, \qquad g_i^{\mathrm{eq},\alpha} = W_i\left[b_\alpha + \frac{c_{i\beta}}{\theta^2}(u_\beta b_\alpha - b_\beta u_\alpha)\right], \qquad \eta = \theta^2\Big(\frac{1}{\omega_\eta} - \frac{1}{2}\Big). \tag{71}$$

Taking moments gives $\sum_i g_i^{\mathrm{eq}} = b_\alpha$ and $\sum_i c_{i\beta} g_i^{\mathrm{eq},\alpha} = u_\beta b_\alpha - b_\beta u_\alpha$, which is exactly the antisymmetric flux of (4). Only that first moment is needed, so the magnetic lattice can be much smaller than the fluid's: D2Q5 or D3Q7 suffices. Running the field on a different lattice from the flow is the point, not an economy.

Lorentz force. The momentum equation gains $-\nabla\cdot(\boldsymbol{bb} - \frac{1}{2}|\boldsymbol{b}|^2 I)$. This is *not* applied as a body force, which would require derivatives of $\boldsymbol{b}$ and lose an order. The fluid equilibrium is instead given the correct second moment directly,

$$\sum_i c_{i\alpha}c_{i\beta} f_i^{\mathrm{eq}} = \rho u_\alpha u_\beta + \Big(p + \tfrac{1}{2}|\boldsymbol{b}|^2\Big)\delta_{\alpha\beta} - b_\alpha b_\beta, \tag{72}$$

achieved by adding to the hydrodynamic equilibrium

$$\delta f_i = \frac{w_i}{2c_s^4} M_{\alpha\beta}(c_{i\alpha}c_{i\beta} - c_s^2\delta_{\alpha\beta}), \qquad M_{\alpha\beta} = \tfrac{1}{2}|\boldsymbol{b}|^2\delta_{\alpha\beta} - b_\alpha b_\beta. \tag{73}$$

Since $\sum_i \delta f_i = 0$ and $\sum_i \boldsymbol{c}_i\,\delta f_i = \boldsymbol{0}$, this term perturbs only the stress and leaves the shifted-storage bookkeeping untouched. In two dimensions $M_{\alpha\beta}\delta_{\alpha\beta} = 0$, so (73) reduces to the familiar $\frac{w_i}{2c_s^4}\big[\frac{1}{2}|\boldsymbol{c}_i|^2|\boldsymbol{b}|^2 - (\boldsymbol{c}_i\cdot\boldsymbol{b})^2\big]$; in three dimensions the trace subtraction matters and is retained.

9.3 Electrohydrodynamics

Unipolar charge injected into a dielectric liquid between two plane electrodes. Four fields, each feeding the next [21]:

$$\nabla^2\varphi = -\frac{q}{\varepsilon}, \qquad \boldsymbol{E} = -\nabla\varphi, \qquad \partial_t q + \nabla\cdot\big[q(\boldsymbol{u} + K\boldsymbol{E})\big] = D\nabla^2 q, \tag{74}$$

with the Coulomb force $q\boldsymbol{E}$ entering the momentum equation. The dimensionless groups are the electric Rayleigh number $T = \varepsilon\Delta\varphi/(\rho_0\nu K)$, the injection strength $C = q_0 L_y^2/(\varepsilon\Delta\varphi)$, the mobility parameter $M = \sqrt{\varepsilon/\rho_0}/K$ and the charge diffusion number $\alpha = D/(K\Delta\varphi)$. Note $\mathrm{Re} = T/M^2$: at $T = 190$, $M = 10$ that is 1.9, so this is a creeping flow and a large τ is the correct consequence of the groups rather than a symptom of a bad setup.

The charge, and why it is not `ScalarBGK`. `ChargeCentralMoments` carries q on D3Q27 or D2Q9. Two things separate it from the passive scalar. The advecting velocity is the *drift* velocity $\boldsymbol{u} + K\boldsymbol{E}$, not the fluid velocity — `ScalarSolver` already takes three arbitrary velocity views, so the plumbing existed, but the operator takes its central moments *about that* velocity. And the equilibrium is full order rather than the first-order truncation of (70): the argument that justifies truncating on D3Q7 — no isotropic fourth moment, and every velocity with a single nonzero component — simply does not apply on a product lattice.

The collision is then almost nothing. In the shifted product basis (§5.3) the equilibrium has exactly one nonzero moment, $k_{0\ldots0} = q$, and taking moments about the drift velocity makes the first-order equilibrium vanish too. With the generator's relaxation schedule $\boldsymbol{\Lambda}_q = \mathrm{diag}(1, \omega_q, \omega_q, \omega_q, 1, \ldots, 1)$ the whole operator reads: keep q, decay the three flux moments by $(1-\omega_q)$, annihilate everything of order two and above. That is the structure of `ScalarRegularised` — relax the flux, kill the ghosts — which is why that operator can be read as this one's D3Q7 special case.

A basis trap that does not announce itself. The reference implementation tabulates $k_9 = Q$, $k_{17} = Qc_s^2$, $k_{18} = Qc_s^4$, $k_{26} = Qc_s^6$. Those are *monomial* central moments. Here the same states are identically zero, because `ProductBasis` is shifted and carries the c_s^2 inside the basis function. Transcribing that list slot-for-slot double-counts, and the result still conserves charge and still runs. This is the trap of §10.5 in a second module, and it fired while the operator was being written: the test asserted $q\,c_s^{2n}$, the operator correctly produced zero, and the disagreement was the basis rather than the physics.

The potential needs no new solver. Poisson is solved as a pseudo-transient diffusion, $\partial_t\varphi = \beta\nabla^2\varphi + \beta q/\varepsilon$, whose steady state is (74) for any β; the code is `ScalarBGK` with no velocity set — so its equilibrium collapses to $w_i\varphi$ — plus a source $w_i\beta[\,\frac{3}{2}q(t) - \frac{1}{2}q(t-1)]/\varepsilon$. Following the reference, $\beta = 0.3$. The relaxation time to the elliptic solution is $L_y^2/(\pi^2\beta)$, which is of the same order as the flow's own t_0 at the finer grids: the potential lags, and that is a property of the pseudo-transient approach rather than of this implementation.

***E* by finite difference.** The reference compares three ways of obtaining the field and finds the finite-difference route (its model C) beats reconstructing $\boldsymbol{E}$ from the distribution's own moments at every C — 1.11% against 2.49% at $C = 10$ — because the moment reconstruction is sensitive to grid-scale oscillation. Model C is used here.

Architectural significance, and a second confirmation. Like the thermal module, this one required no new solver: the Coulomb force is `FieldGuo`, the per-node forcing policy written for the penalised rigid body, which does not care that the force is electric; the drift velocity is one addition in the kernel that already builds $\boldsymbol{E}$; and the charge operator satisfies `ScalarSolver`'s existing collision interface. The whole of the two-way coupling is a force field and one added term.

An initial condition that cannot be taken literally. The reference states $\varphi(\boldsymbol{x}, 0) = 0$. Poisson is *elliptic*: φ has no time derivative, and that zero describes where a relaxation solver starts its iteration, not a state of the system. Against $\varphi = \Delta\varphi$ imposed on the plate it puts the entire potential difference across one cell, so $\boldsymbol{E} = -\nabla\varphi$ reads $1.5\,\Delta\varphi$ at the injector instead of $\Delta\varphi/L_y$ — and the drift $K\boldsymbol{E}$ is then larger by the factor L_y. Measured at $n_y = 163$: a drift of 2.43 lattice units, comfortably supersonic, and the run went non-finite inside 810 steps. Starting φ at its charge-free linear profile removes a transient that was never physical.

9.4 Coupling order

Two-way coupling must be evaluated *simultaneously*. If the fluid is stepped before the magnetic field is refreshed, it collides against $\boldsymbol{b}(t-1)$. That is a first-order splitting error, and — crucially — it does *not* vanish under refinement: with diffusive scaling the ratio

$$\frac{\omega^2\,\Delta t}{\nu k^2} \tag{75}$$

is independent of N, so it appears as a damping offset that survives every grid refinement.

This was found in practice. The shear Alfvén damping error *grew* with resolution, $1.55\times10^{-2}\to2.79\times10^{-2}\to3.16\times10^{-2}$, while the phase speed converged cleanly at order 2; the amplitude history showed a beat. Refreshing $\boldsymbol{b}$ before stepping the fluid removed both. `MagneticSolver::step` takes a `field_is_current` flag so a coupled driver does not pay for the field pass twice.

A non-converging error sitting beside a converging one is the signature of a coupling-order problem, and any future two-way coupling should be checked the same way.

10 Multiphase flow

Every module before this one leaves the fluid solver alone. The scalar and the magnetic field ride on their own distributions, the Lorentz force enters through a second moment, and a single-phase run is bit-for-bit what it was. The multiphase module is different: it changes the equation of state, and that change propagates through the macroscopic definitions, the forcing, the boundary treatment and the initial condition.

The reason is that a two-fluid flow at a density ratio cannot recover p from ρ. Elsewhere in `M3LB` the zeroth moment of f_i is the density and $p=\rho c_s^2$ follows; here ρ is a material property of whichever fluid occupies the node, not a thermodynamic response to the pressure, and a scheme that ties the two together makes the sound speed and the acoustic time step functions of the phase. The populations therefore carry a normalised pressure instead. That single substitution is what puts a density ratio of 800 within reach — and it is also the origin of the conditioning problem in §10.7, which is this module's principal open issue and is stated there rather than buried.

The composition rule of §1.3 still holds. The multiphase operators are separate compile-time collision policies and the phase field is a separate solver on its own lattice; nothing single-phase was modified to admit them, and a single-component case takes none of this code. `FluidSolver`, `EsotericPull` and the lattices are unchanged.

The scheme follows De Rosis and Enan (2021), Sec. II.B, for the phase-field populations and Sec. II.C for the central-moment fluid operator; the rigid-body coupling of §10.8 does not, and §10.8 says why.

Most of this section is that model. §10.9 is a *second* immiscible model on a different principle, from a different paper, sharing no code with the first beyond the lattices and the moment transform — it is here because the two fail differently, and §16.12 measures both on one case. A third route, which resolves only the liquid, is §11.

10.1 Interface capture: conservative Allen–Cahn

The interface is carried by a *third* distribution set h_i, on its own lattice, with the velocity as an *input* exactly as the thermal scalar has it:

$$\phi=\sum_i h_i,\qquad h_i^{*}=h_i+\omega\big(h_i^{\mathrm{eq}}-h_i\big)+S_i, \tag{76}$$

with $\phi = 1$ in the heavy phase and 0 in the light one, obeying (5). The reference throughout this section is De Rosis & Enan, *Physics of Fluids* **33**, 043315 (2021), Sec. II.B; the relaxation, the equilibrium, the relation $\tau_\phi = M/c_s^2$, the source and the gradient stencil are its Eqs. (9), (11), (13), (15)–(16) and (20). Structurally (76) is the passive-scalar operator of (70) plus one source term, and the only moment carried is the zeroth.

The conservative form earns its name from an exact one-dimensional equilibrium,

$$\phi(x) = \tfrac{1}{2}\Big[1 + \tanh\frac{2x}{W}\Big], \qquad |\nabla\phi| = \frac{4}{W}\,\phi(1-\phi) \equiv \theta, \tag{77}$$

the identity that makes the diffusive flux $M\nabla\phi$ and the anti-diffusive flux $M\theta\,\boldsymbol{n}$ of (5) cancel term for term. The interface neither spreads nor sharpens, and W is a prescribed number of lattice units rather than something that emerges from a balance and has to be measured afterwards. The pointwise vanishing of the total flux at a static flat interface is the cheapest check on the whole construction, and the one the implementation is tested against first.

The equilibrium is a policy, because the right truncation is the lattice's. On a Navier–Stokes lattice the second-order form (15) is used unchanged, with ϕ in place of ρ — that is Eq. (11) of the reference exactly, and the fluid's own equilibrium serves without being retyped. On D2Q5 and D3Q7 it drops to first order,

$$h_i^{\mathrm{eq}} = w_i\,\phi\Big(1 + \frac{\boldsymbol{c}_i\cdot\boldsymbol{u}}{c_s^2}\Big), \tag{78}$$

for the reason given for the scalar in §5.5: those lattices have no isotropic fourth-order moment, and on D3Q7 every velocity has a single nonzero component, so $(\boldsymbol{c}_i\cdot\boldsymbol{u})^2$ collapses to $c_\alpha^2 u_\alpha^2$ and the cross terms of $\boldsymbol{uu}$ cannot be represented at all. A second-order term there adds anisotropy, not accuracy. The trade is cost against advection accuracy: the reduced lattice moves 5 or 7 populations against 9 or 27 and pays an $O(u^2)$ defect in the advection term. A static interface cannot see that difference; a sheared one can.

The source, and why its prefactor is not Guo's. Chapman–Enskog on the advection–diffusion LBE with a source obeying $\sum_i S_i = 0$ and $\sum_i \boldsymbol{c}_i S_i = \boldsymbol{A}_S$ gives

$$\partial_t\phi + \nabla\cdot(\phi\boldsymbol{u}) = M\nabla^2\phi - \frac{1}{\omega}\nabla\cdot\boldsymbol{A}_S, \tag{79}$$

so matching the anti-diffusion term of (5) requires $\boldsymbol{A}_S = \omega M\,\theta\,\boldsymbol{n} = c_s^2(1-\omega/2)\,\theta\,\boldsymbol{n}$, using $\omega M = c_s^2(1-\omega/2)$. With $S_i = w_i(\boldsymbol{c}_i\cdot\boldsymbol{A}_S)/c_s^2$ the c_s^2 cancels and the source is simply

$$\boxed{\;S_i = \Big(1 - \frac{\omega}{2}\Big)\,w_i\,\theta\,(\boldsymbol{c}_i\cdot\boldsymbol{n}), \qquad \theta = \frac{4}{W}\,\phi(1-\phi)\;} \tag{80}$$

carrying no mobility and no speed of sound at all. Since $\sum_i S_i = 0$, ϕ is conserved exactly by the collision, and streaming conserves it too — that is the "conservative" in conservative Allen–Cahn, and it is verified rather than assumed: ϕ drifts by 8.4×10^{-13} over 10 000 steps.

The coefficient in the recovered equation is $1/\omega$, *not* the $1/(1-\omega/2)$ that Guo's force term (47) compensates, and the difference is structural rather than a convention. A body force enters the *first*-moment equation directly, where the half-step correction is absorbed by redefining the velocity. This source enters the *zeroth*-moment equation through the divergence of the first moment, so it contributes twice — once through $h_i^{(1)}$ and once through the second-order Taylor term — and $(1/\omega - \tfrac{1}{2}) + \tfrac{1}{2} = 1/\omega$ is where that lands.

It was written with Guo's prefactor and a $1/c_s^2$ first. The source then comes out a factor $M/c_s^2 = 1/\omega - \tfrac{1}{2}$ too small, the anti-diffusion cannot balance the diffusion, and the interface

spreads as though the term were absent: a prescribed $W = 4$ measured 14.1 after 600 steps, against the $\sqrt{4Mt} = 11$ of pure diffusion. With (80) the same measurement returns 4.0011, a relative error of 2.8×10^{-4}. Nothing else misbehaves in the wrong case, which is what makes the failure worth recording: it is loud in its effect and silent in its cause.

Mobility. $M = c_s^2(1/\omega - \frac{1}{2})$, exactly as the diffusivity in (70), with $c_s^2 = 1/3$ on D2Q5 and $1/4$ on D3Q7. It is a numerical parameter and not a physical one: too small and the interface cannot heal after being advected through the grid, too large and it diffuses faster than the flow deforms it. $M \in [0.01, 0.1]$ is the working range.

Gradients are taken from the field, not from neighbouring populations. $\nabla\phi$ is formed from the ϕ *field* by a lattice-weighted stencil in a pass of its own. Reading a neighbour's populations inside the fused kernel would be a genuine read/write race under Esoteric Pull (§3.2) — the two slots a node reads are exactly the two it writes — which is the same defect the scalar outflow condition avoids in the same way (§8.5). `compute_field()` ends in a fence and the derivative kernel reads only what that kernel wrote, so every slot still has exactly one writer. The stencils are Eqs. (20) and (21) of the reference,

$$\nabla\phi(\boldsymbol{x}) = \frac{1}{c_s^2}\sum_i w_i\,\boldsymbol{c}_i\,\phi(\boldsymbol{x}+\boldsymbol{c}_i), \qquad \nabla^2\phi(\boldsymbol{x}) = \frac{2}{c_s^2}\sum_i w_i\big[\phi(\boldsymbol{x}+\boldsymbol{c}_i) - \phi(\boldsymbol{x})\big], \tag{81}$$

evaluated on a *gradient lattice* that defaults to the richest velocity set of the same dimension — D2Q9 in two dimensions, D3Q27 in three — rather than the phase field's own D2Q5 or D3Q7. Spurious currents around a static droplet are dominated by the isotropy of this stencil and the axis-only set is markedly worse, so the cost of a 27-neighbour gather is accepted in a kernel that does nothing else. Both derivatives come out of that one gather: the Laplacian needs exactly the neighbours the gradient has already loaded, so it is one accumulator and one store more, not a second pass. BOTH pressure-form fluid operators read it — each applies $\boldsymbol{F}_s = \mu_\phi\nabla\phi$ with μ_ϕ carrying $-\kappa\nabla^2\phi$. Only the matched-density capillary-stress form of §10.3 needs no second derivative at all.

The anti-diffusion vector $\boldsymbol{A} = \theta\,\boldsymbol{n}$ is assembled once per node rather than once per direction — it costs a square root and a divide and contains no i — and is guarded by $|\nabla\phi| > 1 \times 10^{-12}$. Away from the interface $\phi(1-\phi)$ vanishes anyway, so the guard is a second line of defence rather than the only one: it stops 0/0 in the bulk, where $\nabla\phi$ is round-off noise about zero and $\boldsymbol{n}$ is meaningless.

Cost. Five extra `Real` fields per node — ϕ, three gradient components and the Laplacian — on top of the Q populations. In FP32 on D3Q7 that is 28 bytes of populations against 20 of fields, so the fields are not a rounding error in the bandwidth budget. They are the price of not differentiating populations, and it is a price worth naming.

Coupling order. ϕ and $\nabla\phi$ must be refreshed *before* the fluid steps, so that the fluid collides against the interface at time t rather than $t-1$: `pf.refresh()`, then `fluid.step()`, then `pf.step()`. This is the same first-order splitting error the magnetic module documents in §9.4, and it does not refine away for the same reason. Here it is worse in kind rather than in size: the interface *moves*, so the error is a systematically misplaced interface rather than a damping offset. One refresh serves both consumers, and that is not an economy but a consistency requirement — the fluid's capillary stress and the phase field's interface normal are the same gradient, and evaluating them from one array at one time is what stops them disagreeing about where the interface is.

What is not implemented. `PhaseWall` is zero-flux on the populations, which is the right condition for the transport but sets no contact angle; that needs a wetting condition on $\nabla\phi$ at the wall. A zero-flux wall never collides, so its slots hold whatever a neighbour emitted toward it one step ago, and that sum is not an order parameter — the field kernel therefore leaves ϕ at its initialised value there rather than writing a meaningless one, and the gradient stencil reads that value. An interface should not be put against a wall. Open boundaries for ϕ are also absent; the donor machinery of §8.8 would port, with the same second pass and the same fence.

10.2 The fluid side: a pressure-based formulation

THREE fluid operators exist, and they are TWO PHYSICS TIERS rather than three collisions of one model. `MultiphaseBGK` is the matched-density one: ρ is a single field near 1, and ϕ enters the flow only through the capillary STRESS of §10.3. It is a different equation of state from the other two, not a cheaper version of them — it carries no ρ_L/ρ_H, no μ_L/μ_H and no Laplacian, so substituting it for either does not compile rather than running wrong, and `validation/laplace.cpp` is the one case that wants it. (It also exposes a per-node relaxation rate, `omega_n`, for a viscosity contrast; nothing in the tree ever binds it, so that path has never executed and the header says so.) `MultiphasePotentialBGK` and `MultiphaseCentralMoments` are the other tier: the same pressure-based model and the same field set, differing only in the collision, and both carry a density ratio. To do so they change what the zeroth moment means.

`MultiphaseCentralMoments` is the DEFAULT, and has been since 2026-09-18 — `DefaultMultiphaseCollision` names it, and every case in the tree that couples a phase field to a flow uses it. `MultiphasePotentialBGK` is what the `-op bgk` flag selects, and is there to attribute a difference to the collision. One caveat when using it for that: the two differ in EQUILIBRIUM as well, the central-moment form using the product (sixth-order) expansion and the BGK form the second-order truncation of (15), so a difference measured across that flag is both effects at $O(u^3)$, not the collision alone.

The reason is conditioning. At a ratio ρ is no longer a small perturbation about 1, and a density-carrying distribution stops being well posed for the shifted storage of §4, which assumes ρ_{ref} is exactly 1: at a ratio of 10 the stored g_i are not small and the shift buys nothing. The incompressible model removes the density from the distribution entirely. ρ becomes a function of ϕ, and the populations carry a normalised pressure:

$$\tilde{p} = \sum_i f_i, \qquad \boldsymbol{u} = \sum_i \boldsymbol{c}_i f_i + \frac{\boldsymbol{F}}{2\rho}, \qquad f_i^{\text{eq}} = w_i\big[\tilde{p} + \Phi_i(\boldsymbol{u})\big], \tag{82}$$

Eqs. (24) and (10) of the reference, with Φ_i the second-order Hermite polynomial that (15) multiplies by ρ — written Φ_i here rather than ϕ_i only because ϕ is taken. Its zeroth moment vanishes and its first is $\boldsymbol{u}$, so $\sum_i f_i^{\text{eq}} = \tilde{p}$ exactly and $\sum_i \boldsymbol{c}_i f_i^{\text{eq}} = \boldsymbol{u}$, *not* $\rho\boldsymbol{u}$. There is consequently no division by ρ in the momentum sum; only the half-force correction carries one. Getting that wrong is a viscosity error that presents as a bad boundary condition. A unit test asserts all three moment statements, and the momentum gain $\boldsymbol{F}/\rho$ under the full force, directly against the equilibrium and to round-off.

The physical pressure follows from $\tilde{p}$ by a factor that is itself a choice, and one that turns out to matter more than its algebra suggests; §10.7 is where that is settled. The shift, meanwhile, is easier here than anywhere else in the code: f_i^{eq} is *affine* in $\tilde{p}$, so storing $f_i - w_i$ merely replaces $\tilde{p}$ by $\tilde{p} - 1$ in the same expression, and one line serves both storage conventions.

Density, viscosity, and the clamp on the interpolant. With s the normalised order parameter,

$$s = \frac{\phi - \phi_L}{\phi_H - \phi_L}\bigg|_0^1, \qquad \rho = \rho_L + s\,(\rho_H - \rho_L), \qquad \mu = \mu_L + s\,(\mu_H - \mu_L), \qquad \tau = \frac{\mu}{\rho\, c_s^2}, \tag{83}$$

Eqs. (25) and (12) of the reference. Both interpolations are linear in ϕ and the prescribed viscosities are *dynamic*, so the relaxation rate $\omega = 1/(\tau + \frac{1}{2})$ varies from node to node through both μ and ρ; $\nu = \mu/\rho$ is recovered from the same single read of ϕ, for §10.4.

The clamp is on s, the interpolant, and not on ϕ itself. The conservative Allen–Cahn form keeps ϕ close to $[0, 1]$ but does not guarantee it — shear against a solid, or any under-resolved feature, overshoots locally — and interpolating off an unclamped ϕ puts the density outside the two phases it is meant to lie between. A large enough undershoot makes ρ *negative*, at which point the $1/\rho$ in (82) and in the force is unbounded. Clamping the interpolant leaves the transported field alone and bounds only the equation of state, which is the narrower repair of the two.

Four forces, one of which is not obvious. The collision is sourced by $\boldsymbol{F} = \boldsymbol{F}_s + \boldsymbol{F}_p + \boldsymbol{F}_\nu + \rho\,\boldsymbol{b}$: surface tension (§10.3), the pressure mismatch (§10.7), the viscous interface force (§10.4), and a body acceleration. The middle two are not optional extras — they are the difference between what the LBE recovers and what the variable-density, variable-viscosity momentum equation asks for — and both vanish identically at a matched density, because $\nabla\rho$ does. Under the default normalisation $\boldsymbol{F}_s$ and $\boldsymbol{F}_p$ are both multiples of $\nabla\phi$, one through the chemical potential and the other through $\nabla\rho = (\rho_H - \rho_L)/(\phi_H - \phi_L)\,\nabla\phi$ (Eq. 23), so they are assembled as a single coefficient on one vector rather than as two vectors added, and no second gradient field is needed for the density. An arbitrary external per-node force field occupies a separate slot from $\boldsymbol{F}_\nu$, so that a penalised solid and the viscous pass never have to agree about who clears the array.

The source itself is Guo's (47) in its first-order form, $F_i = (1 - \omega/2)\, w_i\, (\boldsymbol{c}_i \cdot \boldsymbol{F})/(\rho c_s^2)$, which is Eq. (14) of the reference. The central-moment version of this operator, and what it changes, is §10.5.

One cost of the fixed `Macro` shape. `Macro` carries four `Reals` and all four are spoken for, so the force — needed once to correct the velocity in `macroscopic()` and once to source the collision — is assembled twice per node instead of being carried between them, about eight extra view reads. Widening `Macro` by three defaulted fields would remove it, at the cost of three `Reals` in every other operator's hot loop. That trade is stated rather than taken, because it should be measured first.

10.3 Surface tension: two routes

Route one: a capillary stress in the equilibrium. The matched-density operator does not apply surface tension as a body force. It gives the equilibrium the right second moment directly, exactly as the MHD operator does for the Maxwell stress in (27). With $\boldsymbol{G} = \nabla\phi$,

$$\sum_i c_{i\alpha} c_{i\beta} f_i^{\mathrm{eq}} = \rho u_\alpha u_\beta + \Big(p - \tfrac{1}{2}\kappa|\boldsymbol{G}|^2\Big)\delta_{\alpha\beta} + \kappa\, G_\alpha G_\beta, \tag{84}$$

achieved by adding to the hydrodynamic equilibrium

$$\delta f_i = \frac{w_i}{2c_s^4}\, M_{\alpha\beta}\big(c_{i\alpha}c_{i\beta} - c_s^2\delta_{\alpha\beta}\big), \qquad M_{\alpha\beta} = \kappa\Big(G_\alpha G_\beta - \tfrac{1}{2}|\boldsymbol{G}|^2\delta_{\alpha\beta}\Big). \tag{85}$$

This is (27) with the magnetic field replaced by $\sqrt{\kappa}\,\nabla\phi$ and the sign reversed. The tensor structure is the same and so are the invariants — $\sum_i \delta f_i = 0$ and $\sum_i \boldsymbol{c}_i\,\delta f_i = \boldsymbol{0}$, so it perturbs only the stress and leaves mass, momentum and the shifted-storage bookkeeping untouched — but the reversal is physics, not convention. The Maxwell stress is a tension *along* $\boldsymbol{b}$ and pulls field lines together; surface tension is a tension along the *interface*, which is perpendicular to $\nabla\phi$. Relative to the vector that defines the tensor, the two therefore differ by a sign.

Taking the MHD expression over unchanged is the mistake that was made, and it is not a subtle one in its consequences: with the Maxwell sign the mechanical surface-tension integral across the profile comes out negative and the droplet develops a pressure *deficit*, measured -2.5×10^{-4} where $+8.3 \times 10^{-4}$ was expected. `capillary()` is thus $-\kappa$ times `maxwell()`. The twelve duplicated lines were kept rather than factored into a shared helper, on the grounds that duplicating them is cheaper than disturbing a validated MHD path; a third stress of this shape would change that judgement.

Route two: the chemical potential. The pressure-based operator uses the potential form instead, Eqs. (18) and (19) of the reference,

$$\boldsymbol{F}_s = \mu_\phi \, \nabla\phi, \qquad \mu_\phi = 4\beta\,(\phi-\phi_L)(\phi-\phi_H)(\phi-\phi_0) - \kappa\,\nabla^2\phi, \qquad \phi_0 = \tfrac{1}{2}(\phi_L+\phi_H). \tag{86}$$

The coefficients. Integrating $\kappa(\partial_x\phi)^2$ across the tanh profile (77) of width W, for $\phi \in [0,1]$, gives

$$\sigma = \frac{2\kappa}{3W} \quad\Longleftrightarrow\quad \kappa = \frac{3\sigma W}{2}, \qquad \text{and consistently} \qquad \beta = \frac{12\sigma}{W}. \tag{87}$$

`kappa_from_sigma` returns the first and `beta_from_sigma` the second. In the stress form the double-well strength β never appears at all: the conservative Allen–Cahn equation maintains the profile geometrically, from W and the normal, so the free energy is implicit and only κ is needed. In the potential form both appear, and (87) is the pair consistent with the same σ and the same W.

Which is used where. The two are algebraically equivalent, and the stress form is the cheaper and safer of the two: it needs only $\nabla\phi$, which `PhaseFieldSolver` computes anyway for the interface normal, so it is one field cheaper and one derivative order safer — the same argument Dellar's form of the Lorentz force makes in §5.7. The pressure-based operator nonetheless uses the potential form, following the reference, and pays a Laplacian for it. What the exchange buys is a density ratio the stress form cannot reach, and the Laplacian is drawn from the neighbour gather that (81) already performs, so the cost is one accumulator rather than one pass. At matched density the two routes agree, as they must: on a droplet of radius 20 the surface tension inferred from the Laplace jump differs between them by 3.58×10^{-3}, against a prescribed $\sigma = 1 \times 10^{-2}$.

10.4 The viscous interface force

The LBE recovers $\mu\nabla^2\boldsymbol{u}$, while the momentum equation at variable viscosity wants $\nabla\cdot\big[\mu(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^{\mathsf{T}})\big]$. The difference is Eq. (22) of the reference,

$$\boldsymbol{F}_\nu = \nu\big(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^{\mathsf{T}}\big)\cdot\nabla\rho. \tag{88}$$

It is identically zero wherever $\nabla\rho$ is — that is, everywhere except the interface — so at a matched density the whole object may be left unbound and the collision reads zero for it. At a density ratio it is not a correction one may drop.

It is the only one of the four forces that cannot be assembled from node-local data: it needs the velocity *gradient*, and therefore a pass of its own. That pass uses the same isotropic stencil as (81), on the same gradient lattice.

Only the contraction is stored. The velocity gradient is nine numbers per node; contracted against $\nabla\rho$ it is three. Since $\nabla\rho$ is available in the same kernel — it is $\nabla\phi$ times the constant $(\rho_H-\rho_L)/(\phi_H-\phi_L)$ — the contraction is performed where the tensor is still in registers and only the resulting vector is written. Nine fields per node become three. The normalisation

factors out of the bracket for the same reason: the accumulator holds the raw sum $\sum_i w_i c_{i\alpha} u_\beta$, and since it and its transpose need the same $1/c_s^2$, one factor multiplies the whole contraction rather than each term of it.

Coupling order, and why the pass is opt-in. The kernel reads the velocity *field*, which `FluidSolver` fills from the populations, so the order

`pf.refresh()` → `fl.compute_macroscopic()` → `vf.refresh(coll)` → `fl.step()`

builds the force from $\boldsymbol{u}(t)$ and $\phi(t)$ and consumes it in the collision at time t, with no lag anywhere — unlike the phase field itself, whose splitting error §10.1 names. The price is one extra pass over the populations per step, because the fused step writes $\boldsymbol{u}$ only as it goes, too late for a neighbour gather within the same step. That is a real cost, and it is why this pass is requested by the driver rather than performed automatically.

Not valid against a wall. The stencil reads $\boldsymbol{u}$ at every neighbour, including non-fluid ones, where the velocity field holds whatever `FluidSolver` left there, which is zero. At an interface touching a wall that is a one-sided derivative taken as though it were central. The gap has the same shape as the missing wetting condition, and the same remedy would be needed.

An earlier version of this paragraph added "every case that currently runs this operator is periodic", which was never a safe thing to assert and is now false: `floating_body`, `layered_poiseuille` and `water_entry` all refresh $\boldsymbol{F}_\nu$ against walls. Worse, two cases that LOOK like counter examples were not — `enan_rt` and `enan_wedge` each constructed a `ViscousInterfaceForce`, bound it into the collision, handed it its inputs and then never called `refresh`, so the array stayed at its zero-initialised value and $\boldsymbol{F}_\nu$ was identically zero for the whole run. Nothing failed, because the operator's guard asks whether the array is BOUND, not whether it has been COMPUTED. Both were fixed on 2026-09-19; every measurement in §12.2.4 taken before that date was made without this term.

It earns its place. For as long as the only case running the pressure form was a static droplet — where (88) is identically zero — $\boldsymbol{F}_\nu$ was implemented and unexercised, and was documented as such rather than glossed. Layered Poiseuille flow at both a viscosity and a density ratio is the case that excites it, and switching it off there gives:

H	$L_2(u)$ with $\boldsymbol{F}_\nu$	without	ratio
32	4.2967×10^{-2}	4.2837×10^{-1}	9.97
64	4.2896×10^{-2}	5.9350×10^{-1}	13.84

A term that is exactly zero in every case previously run is precisely the kind that can stay wrong indefinitely without anything failing. This is the measurement that says it is not.

10.5 Central moments for the pressure form

BGK relaxes every mode at the same rate, so at high Reynolds number, where ω approaches 2, the non-hydrodynamic modes are over-relaxed and ring instead of decaying. That is not a margin one tunes around: at $Re = 30000$ on a 128-wide box, $\omega = 1.99795$. The symptom that prompted a moment operator here was measured rather than anticipated — a Rayleigh–Taylor run at a density ratio of 999 completed under BGK, but its velocity field was dominated by a one-cell alternating mode, 70× stronger than the same case at a ratio of 3. The point of collision in moment space is that the modes carrying no physics can be sent straight to equilibrium at rate 1 while the shear modes alone carry the viscosity.

The equilibrium moments factorise. That is the one derivation worth writing down, because it is what keeps the operator as cheap as the single-phase one. The pressure-form equilibrium splits exactly:

$$f_i^{\mathrm{eq}} = w_i\big[\tilde{p} + \Phi_i(\boldsymbol{u})\big] = \underbrace{w_i\big(1 + \Phi_i(\boldsymbol{u})\big)}_{\text{Maxwellian at } \rho=1} + (\tilde{p}-1)\, w_i.$$

The first bracket has central moments $\{1, 0, 0, \ldots\}$ *exactly* — that is the product-form property verified symbolically in §6.3, 0 of 9 moments deviating on D2Q9 and 0 of 27 on D3Q27. The second is $(\tilde{p}-1)$ times the central moments of the *weights* about $\boldsymbol{u}$, and on a product lattice those factorise into the same one-dimensional triple that the shifted storage of §6.3 already carries. Hence the closed form

$$k^{\mathrm{eq}}_{pqr} = \big[\, pqr = 000 \,\big] + (\tilde{p}-1)\, A_p(u_x)\, A_q(u_y)\, A_r(u_z), \qquad A = \{1,\ -u,\ u^2\}, \tag{89}$$

with the third factor absent in two dimensions. No dense $Q \times Q$ matrix, no stored equilibrium vector, and the same factorised transform the single-phase operator uses. Equation (89) was checked against Eq. (34) of De Rosis and Enan on four independent moments: $k_5 = (u_x^2 - u_y^2)(\tilde{p}-1)$ and $k_7 = u_x u_y(\tilde{p}-1)$ agree directly, and after converting this Hermite-like basis to the paper's monomial one through $k_{21}^{\mathrm{mono}} = k_{21} + c_s^2 k_{01}$, their k_{10} and k_{16} come out identical too. The paper's odd-order moments are printed without the leading minus the derivation gives; the even ones agree outright, which is what identifies it as typography rather than a difference of scheme. The test suite does not check against the paper at all: it contracts the central moments of the equilibrium directly by summation, and all nine agree with (89) to 5.0×10^{-17}.

One deliberate departure. The reference builds its equilibrium central moments from the second-order truncated equilibrium of its Eq. (10). This operator uses the product form instead, whose central moments are exactly Maxwellian on D2Q9 and D3Q27, so the $O(u^3)$ Galilean defects of the truncation are absent rather than reproduced — the same choice, for the same reason, that §6.3 makes for single-phase flow. The gap between the two is measurable and is exactly of that size: run against BGK at $\omega = 1$, where the two operators must otherwise coincide, both recover the same $\tilde{p}$ and the same u_x to round-off, and their post-collision states differ by 1.977×10^{-4} at $|\boldsymbol{u}|^3 = 7.712 \times 10^{-4}$.

Relaxation. Following Eq. (35) of the reference, $K = \mathrm{diag}[1, 1, 1, 1, 1, \omega, \omega, \omega, \omega, \omega, 1, \ldots, 1]$:

$$\begin{aligned}
\text{order } 0: &\quad \text{conserved — } \tilde{p} \text{ is untouched} \\
\text{order } 1: &\quad \text{exact assignment, the force included} \\
\text{order } 2: &\quad \text{trace at } \omega_{\mathrm{bulk}} \text{ (the paper's 1), deviatoric and shear at } \omega \\
\text{order} \geq 3: &\quad k^{\star} = k^{\mathrm{eq}}.
\end{aligned} \tag{90}$$

ω_{bulk} is exposed as a parameter because it is the bulk viscosity and nothing else in the scheme pins it.

The force is exact in the first moment, not relaxed into it. With $\boldsymbol{u} = \sum_i \boldsymbol{c}_i f_i + \boldsymbol{F}/(2\rho)$, the pre-collision first moment is

$$k_1 = \sum_i f_i\,(c_{ix} - u_x) = k^{\mathrm{eq}}_{100} - \frac{F_x}{2\rho}, \tag{91}$$

and the post-collision one has to be $k^{\mathrm{eq}}_{100} + F_x/(2\rho)$ for the next step to recover the right velocity. The update is therefore an assignment rather than a relaxation, and the rate-1 entry in K

is that assignment. The source of Eq. (14), $F_i = w_i(\boldsymbol{c}_i \cdot \boldsymbol{F})/(\rho c_s^2)$, is odd in $\boldsymbol{c}_i$ and so has no second moment at all: unlike Guo's there is no second-order force term to add, in any basis, and the co-moving-frame cancellation that §6.3 relies on is not even needed here. The test suite checks the consequence rather than the algebra — collision conserves $\tilde{p}$ to 2.2×10^{-16} and the momentum gains exactly F_x/ρ and F_y/ρ, to 1.4×10^{-17}, with the full multiphase force assembled.

Two constraints on where this operator may be used. It requires a product lattice: D2Q9 or D3Q27, enforced by a static assertion. And it takes raw storage only. The shifted convention centres the stored variable on $\tilde{p} = 1$, which is precisely the pressure gauge that must be avoided at a density ratio — the subject of §10.7.

10.6 Central moments for the phase field

§10.1 gives this paper's *physics* for h_i — the equilibrium, $\tau_\phi = M/c_s^2$, the anti-diffusion source — and presents BGK as the collision. Their Sec. II.D is a different collision for the same physics, and it is implemented here as `PhaseFieldCentralMoments`. Since 2026-09-18 it is also the *default* on any product lattice: `PhaseFieldSolver`'s collision template argument defaults to `DefaultPhaseCollision`, which is this operator on D2Q9 and D3Q27 and `PhaseFieldBGK` on D3Q7 and D2Q5. The campaign of §12.2.4 runs it throughout.

Why it exists, and the ceiling is not the same one the fluid has. The mobility fixes the rate outright, $M = c_s^2(1/\omega_\phi - \frac{1}{2})$, so $\omega_\phi = (M/c_s^2 + \frac{1}{2})^{-1}$; and an interface-capture test fixes the mobility through its Péclet number. A high Péclet number is a low mobility is $\omega_\phi \to 2$, and under BGK there is nothing to trade against it, because BGK relaxes the ghost modes at that same rate. The moment form relaxes only the three first-order moments at ω_ϕ and sends everything else to equilibrium at 1.

A retraction, and it matters because it was the justification. An earlier version of this section reported that the operator completes the reference's Table III at Pe = 80, 400, 800 and 4000 where BGK reaches 10^{300} on the last three, and read that as a stability ceiling the central-moment form lifts. The Péclet number had been read as $\mathrm{Pe} = U_0\xi/M$, which is what their text says; their drivers all compute $M = U_{\mathrm{ref}}\, d/\mathrm{Pe}$ with d the *domain size*. On Table III at Pe = 80 that is $M = 0.0497$ and $\omega_\phi = 1.54$, where the interface-width reading gives $M = 7.5 \times 10^{-4}$ and $\omega_\phi = 1.991$. BGK was being run 66 times short of mobility, hard against its own stability edge, and it diverged for that reason rather than for anything intrinsic at their operating point — across every case in the paper their ω_ϕ never exceeds 1.988. So this operator has *no measured advantage over BGK recorded here*, and the mobility ladder that used to appear in this section described a regime the paper does not visit. The numbers were real measurements of the wrong thing.

The equilibrium is one slot. Not an approximation — in this basis it is exact:

$$k^{\mathrm{eq}} = (\phi,\, 0,\, 0,\, \ldots,\, 0). \tag{92}$$

The reference tabulates five nonzero entries in the *monomial* central moments (its Eq. 54): $k_0 = \phi$, $k_4 = \phi$ and $k_{16} = k_{17} = k_{18} = \phi c_s^4$. `ProductBasis` is shifted, $\varphi_2 = C^2 - c_s^2$, and all four of the non-zeroth ones collapse in it:

$$k_4' = k_4 - 3c_s^2 k_0 = \phi - \phi = 0, \qquad k_{16}' = \phi c_s^4 - \phi c_s^4 - \phi c_s^4 + \phi c_s^4 = 0.$$

So no equilibrium table is stored at all: `eq_populations` writes ϕ into the $(0,0,0)$ slot and inverse-transforms. Note what that inverse transform returns — the product-form (sixth-order)

equilibrium, not the second-order truncation (15) that §10.1 gives for BGK. The two differ at $O(u^3)$, and the test suite pins exactly that: the post-collision states of the two operators at matched ω agree to 1.084×10^{-4} against $|\boldsymbol{u}|^3 = 2.342 \times 10^{-4}$, while ϕ and the first moment agree to 2.2×10^{-16} and 1.0×10^{-17}.

Relaxation, and where the mobility lives. Eq. (55) of the reference is $K_\phi = \mathrm{diag}[1, \omega_\phi, \omega_\phi, \omega_\phi, 1, \ldots, 1]$:

$$\begin{aligned} \text{order } 0: &\quad k^\star = \phi \quad \text{— conserved} \\ \text{order } 1: &\quad k^\star = (1-\omega_\phi)\, k + (1-\tfrac{\omega_\phi}{2}) c_s^2 A_a \\ \text{order } \geq 2: &\quad k^\star = 0. \end{aligned} \tag{93}$$

Twenty-three of the twenty-seven moments are annihilated, against seventeen for the fluid operator of §10.5, and the difference is instructive: the fluid's viscosity lives at order 2, so its order-2 slots must be relaxed rather than zeroed, whereas the phase field's mobility lives at order 1 and leaves order 2 free. The other asymmetry is that the fluid's first-order update is an *assignment* (91) and the phase field's is a genuine relaxation — which is the same statement, from the other side.

The source occupies three slots, and the reference lists nine. With $\sum_i S_i = 0$ and $\sum_i \boldsymbol{c}_i S_i = c_s^2(1-\omega_\phi/2)\,\theta \boldsymbol{n}$, the source is first order in $\boldsymbol{c}_i$, so its central and raw first moments differ by $\boldsymbol{u}$ times its zeroth, which vanishes. It is therefore added to k_1, k_2, k_3 directly, with the same $(1-\omega_\phi/2)$ prefactor §10.1 derives and for the same reason.

The c_s^2 in (93) is not decoration. `PhaseFieldBGK` adds $S_i = (1-\omega_\phi/2)\, w_i(\boldsymbol{c}_i \cdot \boldsymbol{A})$ to the *populations*, whose first moment is $(1-\omega_\phi/2)c_s^2 \boldsymbol{A}$ because $\sum_i w_i c_{ia} c_{ib} = c_s^2 \delta_{ab}$. Writing $\boldsymbol{A}$ here without it makes the anti-diffusion three times too weak on a $c_s^2 = 1/3$ lattice, which neither blows up nor looks wrong — the interface simply spreads, the way it does when the prefactor is got wrong in the other direction.

The reference's Eq. (61) lists nine nonzero entries: $R_{1,2,3}$ and, at third order, $R_{10} = R_{15} = c_s^2 A_y$, $R_{11} = R_{13} = c_s^2 A_x$, $R_{12} = R_{14} = c_s^2 A_z$. Those six are not missing here; they are *zero in this basis*. Eq. (61) is written in monomial central moments, and in the shifted basis the (a, a, b) slot is $(C_a^2 - c_s^2)C_b$, to which the same source contributes $c_s^4 A_b - c_s^2 \cdot c_s^2 A_b = 0$. Transcribing the list slot-for-slot would double-count; deleting the entries from a monomial implementation would lose them. Neither fails visibly, so `tests/test_phase_field.cpp` pins *both* directions: the raw moments reproduce Eq. (61) exactly, and the shifted moments outside the first three come out at 4.1×10^{-20}.

That cancellation is at $\boldsymbol{u} = 0$, and the qualifier is the honest part. Transform S_i at a general velocity and 26 of the 27 shifted moments are nonzero, not 3: the sparsity is a property of the rest frame. The reference's drivers add a $\boldsymbol{u}$-independent source and so does this, so both truncate the source's central moments at $\boldsymbol{u} = 0$ — a shared approximation rather than an identity, and `MATLAB/D3Q27_CM_phase.m` prints both forms so the difference stays visible. The discarded terms are $O(|\boldsymbol{u}|\, A)$, the same order as the equilibrium gap against BGK above. A `full_source` switch transforms the source at the actual velocity; it is kept because the reasoning that says it should matter is sound, and defaulted off because measurement says it does not. On the advected flat slab of `validation/phase_flat` — 50 000 steps, $W = 3$, the case most able to see it — the width drift is -4.58×10^{-2} truncated against -4.52×10^{-2} transformed at $\omega_\phi = 1.9881$, and $+7.43 \times 10^{-1}$ against $+7.45 \times 10^{-1}$ at 1.9940: 1.3% and 0.3% apart. It is simply not what limits this operator.

Two constraints, the same two the fluid operator has. A product lattice — D2Q9 or D3Q27, enforced by a static assertion, so a D3Q7 phase field must use `PhaseFieldBGK` and a

request for central moments there is a compile error rather than a silent downgrade. And raw storage, for the reason §10.1 gives: ϕ is $O(1)$ on $[0, 1]$, there is no cancellation to protect against, and `RawPopulations` is the right answer rather than a placeholder.

10.7 The pressure gauge, and two normalisations

This is the most consequential finding in the module, and it is a conditioning result rather than a bug. The populations carry $\tilde{p} = p/(\rho c_s^2)$. The physical pressure p is continuous across an interface; ρ is not. So $\tilde{p}$ jumps by the whole density ratio wherever p is not near zero, and the term that permits it to, $F_p = -\tilde{p}\, c_s^2 \nabla\rho$, is proportional to the pressure *level*. It is cancelled, to whatever accuracy the two discrete operators allow, by the LBE's own $\rho c_s^2 \nabla\tilde{p}$. The physical answer is their difference; their individual sizes — and hence the size of the residual mismatch — are set by a gauge that changes no physics.

The layered channel makes it visible. Two immiscible layers driven by a uniform force G have an exact profile that does not contain ρ at all, so a density ratio alone must leave the answer unchanged. It does not. Turning the two contrasts on one at a time (D2Q9, central moments, FP64, interface width 4):

configuration	$L_2(u)$, H=32	H=64	order
single phase	2.6751×10^{-4}	6.6877×10^{-5}	2.00
viscosity ratio 5 only	3.6187×10^{-2}	2.3587×10^{-2}	0.62
density ratio 10 only	2.5078×10^{-2}	2.8995×10^{-2}	−0.21
both	4.2967×10^{-2}	4.2896×10^{-2}	0.00

The first row says the base scheme is exact and second order. The second is the diffuse interface smoothing a kink in $\mathrm{d}u/\mathrm{d}y$ over W cells, first order in W/h and converging as it should. The third should not happen at all, and extending it to $H = 128$ gives 2.4×10^{-2} — a few per cent that refinement does not remove.

It is the pressure force, and it is not spurious. Dumping the profile at a density ratio of 10 with uniform viscosity gives

$\tilde{p}$	heavy 2.07×10^{-3}	light 2.18×10^{-2}	(ratio 10.5)
$p = \rho c_s^2 \tilde{p}$	heavy 6.72×10^{-3}	light 7.27×10^{-3}	(nearly uniform),

which is exactly what the formulation demands. F_p reaches 220 times the driving force at $H = 32$ and 408 at $H = 64$, and the reported error is the discrete mismatch between it and $\rho c_s^2 \nabla\tilde{p}$. That also explains the sign of the order directly: G scales as $1/h^2$ to hold the peak velocity fixed, so F_p/G *grows* as h^2 under refinement while the mismatch stays a fixed fraction of F_p. Refining makes the cancellation harder, not easier.

A gauge experiment confirms it. Seeding a uniform offset in $\tilde{p}$ changes no physics, since only ∇p is physical. Measured $L_2(u)$ at $H = 32$:

$\tilde{p}$ offset	single phase	viscosity only	density only
0	2.675×10^{-4}	3.619×10^{-2}	2.51×10^{-2}
0.2	2.675×10^{-4}	3.630×10^{-2}	4.34×10^{-1}

The two configurations without a density ratio are exactly gauge invariant, as they must be. The one with a density ratio degrades seventeenfold, and F_p/G goes from 220 to 4650. The conditioning, not the physics, sets the error.

Two normalisations. The reference normalises by the local density; the alternative normalises by a constant:

$$p = \rho(\phi)\, c_s^2\, \tilde{p}, \qquad F_p = -\tilde{p}\, c_s^2\, \nabla\rho \qquad \text{(default)}, \tag{94}$$

$$p = \rho_0\, c_s^2\, \tilde{p}, \qquad F_p = c_s^2\, (\rho - \rho_0)\, \nabla\tilde{p}. \tag{95}$$

Under (95) $\tilde{p}$ is uniform wherever p is, so F_p tracks the pressure *gradient* instead of the level. It needs $\nabla\tilde{p}$, which is a field the phase solver does not carry, so a separate pass differentiates the fluid's own zeroth moment on the isotropic stencil

$$\nabla f(\boldsymbol{x}) = \frac{1}{c_s^2} \sum_i w_i\, \boldsymbol{c}_i\, f(\boldsymbol{x} + \boldsymbol{c}_i), \tag{96}$$

run after the macroscopic update. Isotropic rather than axis-only for the reason the phase gradient is: this gradient is differenced against the LBE's own pressure term, and an anisotropic stencil would leave a direction-dependent residual exactly where the two are meant to cancel.

On the density-only configuration the two forms measure as

normalisation	$L_2(u)$, H=32	H=64	order	$\max \lvert F_p \rvert / G$
local ρ (default)	2.5078×10^{-2}	2.8995×10^{-2}	-0.21	$220 \to 408$
constant ρ_0	1.8572×10^{-2}	9.6381×10^{-3}	0.95	$379 \to 747$

At an offset of 0.2 the same orders are -0.37 and 0.86, and the gauge sensitivity — the factor by which the error moves under a shift that changes no physics — is 17.3× and 19.4× for the local form against 1.93× and 2.04× for the constant one. Note what does *not* improve: F_p/G is larger, 379 against 220 at $H = 32$. The gain is not that the term shrinks but that it stops tracking the pressure level. The residual factor of about two is the discrete equilibrium's own $\tilde{p}$ dependence: $f_i^{\mathrm{eq}} = w_i[\tilde{p} + \Phi_i]$ carries $(\tilde{p} - 1)$ into its higher moments through (89), so a $\tilde{p}$ offset is not a pure gauge on the lattice even when it is one in the continuum.

Neither dominates. The recovered pressure term under (95) is $-(\rho_0/\rho)c_s^2\nabla\tilde{p}$, so one phase always pays an acoustic price. With $\rho_0 = \rho_L$ the heavy phase runs at an effective sound speed $c_s\sqrt{\rho_L/\rho_H}$: stable, but the pressure term there is a cancellation of ratio size, which is tolerable where ∇p is diffuse and is not where it is concentrated. At a Laplace jump that form diverges at a ratio of 10 and above, on a case the local-ρ form handles at a ratio of 100. The opposite choice is worse and was tried first: with $\rho_0 = \rho_H$ the light phase gets $c_s\sqrt{\rho_H/\rho_L}$, which at a ratio of 10 is 1.83 lattice units per step — past the lattice speed — and the run diverged inside 500 steps. That is why ρ_0, when used at all, must be the light phase. The default remains the local-ρ form of (94), because every validated result in this tree was obtained with it; (95) is available where the pressure field is real but its gradients are diffuse, as in a driven channel, and the layered case exercises both.

Where the zero goes. Within (94) the absolute level is still free in the continuum and still not free numerically, and this is the single setting that decides whether the scheme works at all. Adding a constant P_0 to p adds $P_0/(\rho c_s^2)$ to $\tilde{p}$, which is not a constant when ρ varies, and both large terms scale with it. Seeding $p = \rho_L c_s^2$ — a uniform physical pressure, the natural-looking choice — was measured on the static droplet to return a *negative* surface tension, in error by thousands of per cent at a density ratio; the acceptance test for that case is normalised by the prescribed σ rather than by the smallest measured one precisely because a sign flip would otherwise turn a gross failure into an apparent pass, which it did. The same gauge made a Rayleigh–Taylor run at $At = 0.5$ diverge inside 424 steps, with $|\boldsymbol{u}|$ at twice the free-fall velocity at step zero, before anything had moved.

Seeding $p = 0$ where the two fluids meet keeps both terms small and leaves only the physical jump in $\tilde{p}$. With that gauge and nothing else changed, the same droplet recovers Laplace's law across the whole ratio range, and the spurious currents *fall* as the ratio rises:

ρ_H/ρ_L	σ recovered	rel. err.	$\max \lvert\boldsymbol{u}\rvert$
1	9.27607×10^{-3}	7.24×10^{-2}	1.67×10^{-5}
10	9.71478×10^{-3}	2.85×10^{-2}	3.21×10^{-6}
100	9.83470×10^{-3}	1.65×10^{-2}	4.90×10^{-7}

σ varies by 5.59×10^{-2} of the prescribed value over a hundredfold ratio, and most of that spread is the matched-density row's ordinary resolution deficit rather than any dependence on the ratio. The rule the module follows is therefore to put the pressure zero at the interface — for a hydrostatic configuration, at the free surface — and it is a rule about conditioning, not about physics. Its immediate corollary is the storage restriction stated at the end of §10.5: the shifted convention centres the stored variable on $\tilde{p} = 1$, which is the one gauge that must not be chosen at a density ratio.

10.8 Rigid bodies by volume penalisation

A rigid body is coupled to the flow by volume penalisation: a smooth solid indicator, and a direct forcing that drives the fluid inside it toward the body's rigid velocity field. The two standard alternatives were considered and rejected for specific reasons. A Peskin-style immersed boundary — Lagrangian markers on the surface, interpolation and spreading kernels, a body force fed back into $\boldsymbol{F}$ — is the usual tool and it is more accurate *at the surface* than this, but it is also a great deal more machinery, and for a body whose *interior* must move rigidly, a solid square rather than a thin shell, a volumetric penalisation enforces the same condition with a fraction of it. De Rosis and Enan drive their wedge that way. Moving bounce-back geometry is the other route and is worse here on three counts: the solid mask must be rebuilt on the host every time the body advances, which is a full pass over the domain plus `rebuild_lists()`; the nodes the body uncovers need a refill scheme, their populations being meaningless; and the body's position is quantised to whole cells, so a smoothly falling square arrives in lattice-sized jumps and each one radiates a pressure pulse. Penalisation has none of these. Nothing about the geometry is ever rebuilt, the body's position *and angle* are continuous, and there are no fresh nodes. Rotation in particular is nearly free here and would be a rewrite in either of the other two.

The indicator and the force. `Rect::chi` is the product of two tanh profiles across the faces, evaluated after the point has been taken into the body frame, so the shape turns with the body and so does the slight rounding of the corners that the product form introduces. It is smoothed over about a cell and a half (`Rect::smooth`, default 1.5 cells) rather than being a sharp mask: a sharp mask is the staircase again, with the surface sitting on lattice planes and the effective shape changing as the body falls, or as it turns. The smoothing is what lets the body occupy a continuously varying pose. The forcing is

$$\boldsymbol{F}(\boldsymbol{x}) = \chi(\boldsymbol{x})\, 2\rho(\boldsymbol{x}) \Big[\boldsymbol{U} + \Omega\, \hat{\boldsymbol{z}} \times \boldsymbol{r} - \boldsymbol{u}^{*}(\boldsymbol{x}) \Big], \qquad \boldsymbol{r} = \boldsymbol{x} - \boldsymbol{x}_c, \tag{97}$$

which is direct forcing: applied through the same half-force convention as (47), one step of it takes $\boldsymbol{u}$ to the rigid field exactly where $\chi = 1$. It must run after `compute_macroscopic()` and before the fluid steps. The density is read node by node rather than taken as a constant, so the same body meets a thousandfold heavier medium when it reaches the water; a penalisation that ignored that would decelerate the body as though it were still in air.

$\boldsymbol{u}^*$ is not the stored velocity. This is the one thing that has to be right. The solver defines $\boldsymbol{u} = \left(\sum_i \boldsymbol{c}_i f_i + \frac{1}{2}\boldsymbol{F}\right)/\rho$ with $\boldsymbol{F}$ the *total* force, so the stored velocity field already contains this body's force from the previous step. Feeding it straight back into (97) applies the correction twice: the body over-shoots, the fluid over-shoots back, and the pair diverges within a few hundred steps — measured, the square left the domain at six million times its impact speed. The previous contribution is therefore removed first,

$$\boldsymbol{u}^* = \boldsymbol{u}_{\text{stored}} - \frac{\boldsymbol{F}_{\text{prev}}}{2\rho},$$

which leaves the velocity the fluid would have had from every *other* force, and that is what direct forcing is defined against. The force arrays hold $\boldsymbol{F}_{\text{prev}}$ already, so this costs one subtraction and no extra state.

Because the force is known at every node, the force and torque the fluid exerts on the body are minus its sum and minus the sum of $\boldsymbol{r} \times \boldsymbol{F}$. That closes Newton's equations, and it is the reason for doing it this way: the body falls, floats and rolls under gravity and buoyancy rather than being pushed at a prescribed speed.

The fictitious fluid, and why the obvious arrangement cannot float. The penalised region is not empty. It is full of fluid that the forcing drags along, and that fluid's inertia and weight are both already inside the reaction, so using the body's own mass alone would count them twice. Writing m_f for that fictitious fluid mass, the exact bookkeeping is

$$m_b \frac{d\boldsymbol{U}}{dt} = \boldsymbol{R} + \frac{d\boldsymbol{P}_{\text{in}}}{dt} + (m_b - m_f)\,\boldsymbol{g},$$

where $\boldsymbol{g}$ is the body force per unit mass — the same vector the collision operator is given, or the buoyancy term describes a different problem from the fluid it is meant to displace. Uhlmann's classical step approximates $d\boldsymbol{P}_{\text{in}}/dt$ as $m_f\, d\boldsymbol{U}/dt$, moves it across and divides:

$$\delta\boldsymbol{U} = \frac{\boldsymbol{R} + (m_b - m_f)\,\boldsymbol{g}}{m_b - m_f}.$$

That denominator is the whole problem. It vanishes for a neutrally buoyant body and changes sign for a light one, so a floating body — the only interesting kind, since floating is what a density ratio is *for* — is precisely the case the arrangement cannot express. This never announced itself as a failure. The old code did not crash and produced no NaN: the guard on that denominator silently zeroed the acceleration, so a body that could not be floated simply did not accelerate.

Targeting the velocity the body is about to have. The fix is not a different model. It is the observation that the reaction is *linear* in the body velocity that produces it,

$$\boldsymbol{R} = -\sum \chi\, 2\rho\,\left(\boldsymbol{U} + \Omega\,\hat{\boldsymbol{z}} \times \boldsymbol{r} - \boldsymbol{u}^*\right),$$

so if the force written into the arrays targets the velocity the body is *about* to have rather than the one it already has, $\boldsymbol{R}$ moves to the left-hand side and the same equation comes out with a different denominator,

$$(m_b + m_f)\,\delta\boldsymbol{U} = 2\,\delta\boldsymbol{P} + (m_b - m_f)\,\boldsymbol{g}, \tag{98}$$

with $\delta\boldsymbol{P} = \sum \chi\rho\,(\boldsymbol{u}^* - \boldsymbol{u}_{\text{rigid}})$ the inner fluid's momentum deficit relative to rigid motion — the hydrodynamic signal, and the only place the fluid enters. Algebraically (98) is Uhlmann's equation multiplied through: the same discrete equation, rearranged, with nothing added to it. The difference is entirely that the body and the fluid are now solved at the *same* instant

instead of the fluid lagging the body by a step — the lesson of §9.4 in another guise. The denominator $m_b + m_f$ is positive for every mass, the homogeneous amplification factor is $(m_b - m_f)/(m_b + m_f)$, whose modulus is below one for every mass, and the neutrally buoyant case, previously a division by zero, is now the best-conditioned point on the line. Nothing is tuned and no limit is imposed.

Rotation, and why it does not decouple. The same substitution with rotation gives a 3×3 system rather than a scalar. It does not split into a translation part and a roll part, because the first moments $\boldsymbol{S} = \sum \chi\rho\, \boldsymbol{r}$ of the fictitious mass about the body centre are non-zero the moment the body straddles an interface — half of it is standing in water and half in air — and those moments couple sway to roll. With $A = m_b + m_f$ and $B = I_b + I_f$,

$$\begin{bmatrix} A & 0 & -S_y \\ 0 & A & S_x \\ -S_y & S_x & B \end{bmatrix} \begin{bmatrix} \delta U_x \\ \delta U_y \\ \delta\Omega \end{bmatrix} = \begin{bmatrix} 2\,\delta P_x + (m_b - m_f)\, g_x \\ 2\,\delta P_y + (m_b - m_f)\, g_y \\ 2\,\delta L_z - (S_x g_y - S_y g_x) \end{bmatrix}. \tag{99}$$

The matrix is symmetric, and it is positive definite for every body with mass and inertia of its own: Cauchy–Schwarz on the measure $\chi\rho$ gives $|\boldsymbol{S}|^2 \le m_f I_f \le AB$, so the Schur complement $B - |\boldsymbol{S}|^2/A$ is positive. There is therefore no pivoting and no case in which the solve has to give up; it is done in closed form, on the host, once a step. All seven integrals — m_f, $\boldsymbol{S}$, I_f, $\delta\boldsymbol{P}$ and δL_z — come from a single sweep, and the force is then written in a plain `parallel_for`: one reduction, not two. Locking a degree of freedom removes the coupling rather than the row, so a body held against translation solves $B\,\delta\Omega$ against the third right-hand side alone, which is not the same answer as solving (99) and discarding two rows. Finally, m_b and I_b are taken from the integrals of the indicator itself rather than from $4h_x h_y$ and the textbook $(w^2 + h^2)/12$, so that they describe the body the force actually acts on: the smoothing makes the penalised body slightly larger than the nominal rectangle, and its second moment exceeds the sharp rectangle's by 2.55% at the default smoothing width.

The last term of the third row is the righting moment. The body's own weight acts at its centre and exerts no torque about it; the displaced fluid's weight does not, because the displaced fluid is heavier on the submerged side, and $-(\boldsymbol{S} \times \boldsymbol{g})$ is exactly the hydrostatic couple that metacentric theory writes as $\rho g V\, \overline{GM} \sin\theta$. It was not put there for that purpose. It falls out of the same subtraction that produced $(m_b - m_f)\boldsymbol{g}$ in the rows above, and it is the reason a raft comes back upright. It is reported separately in `Reaction::righting` because it is the only part of the roll balance with a closed form to check against, and nothing in the way it is computed knows that form exists; §12.2.3 makes the comparison.

The reaction is a derived diagnostic, not a measurement. Once the solve has produced $\delta\boldsymbol{U}$ and $\delta\Omega$, the force and torque on the body follow from the same integrals in closed form,

$$\boldsymbol{R} = 2\,\delta\boldsymbol{P} - 2\big[\, m_f\, \delta\boldsymbol{U} + \delta\Omega\, (\hat{\boldsymbol{z}} \times \boldsymbol{S}) \,\big], \qquad T = 2\,\delta L_z - 2\big[\, I_f\, \delta\Omega + \boldsymbol{S} \times \delta\boldsymbol{U} \,\big], \tag{100}$$

so nothing has to be summed a second time to report them. What is reported is consequently exact to round-off against $-\sum \boldsymbol{F}$ and $-\sum \boldsymbol{r} \times \boldsymbol{F}$, and is an identity rather than an independent measurement of them: the physics enters through $\delta\boldsymbol{P}$ and δL_z, and (100) is algebra on top. That identity is checked against the summed force array in §12.2.3.

What this does not model. There is no contact line. The phase field is advected by the fluid velocity and knows nothing about the body, so nothing sets the angle at which the free surface meets the solid; inside the body $\boldsymbol{u}$ is driven to the rigid field, so the interface is carried

with the body rather than through it, and the conservative Allen–Cahn form keeps the profile from simply diffusing in — but the wetting physics is absent, and water entry is a contact-line problem. Read the splash, not the meniscus. The same gap means a floating body gets no surface-tension contribution to its draft, so a body small enough for that to matter, at Bond number of order one, is out of scope. There is no sub-cell surface: penalisation resolves the body to the smoothing width, so a pressure read at the face of the square is a smoothed pressure over a cell or two, and the integrated force is far more trustworthy than the local pressure, which is the usual situation with this family of methods; the draft is likewise resolved only to the interface width. There is no collision model: two bodies, or a body meeting a wall, will simply interpenetrate, because nothing here computes a contact force, and the floor stop in `demonstrator/water_entry` is a clamp in the driver rather than physics.

Three solves, chosen by the shape. This used to be a planar scheme and is not one any more; an earlier version of this paragraph said "the scheme is two-dimensional, and says so in the compiler's voice — a `static_assert` on $D = 2$", and described the three-dimensional work as something that had not been done. The assertion is now `static_assert(L::D == 2 || Shape::three_d)`, which is the same defence pointed the other way: a planar shape on a 3-D lattice would silently model an infinite prism in z, so it is refused, and a three-dimensional shape is let through. Five indicators are provided, sorting into three solves:

shape	geometry	rigid-body solve	driver
`Rect`, `Wedge`	planar prism	3×3: x, y, θ_z	`floating_body`, `enan_wedge`
`Sphere`	3-D	3 translations, no rotation	`sphere_entry`
`Box`, `Disc`	3-D	full 6×6, quaternion pose	`cube_entry`, `stone_skip`

`Sphere` is the row worth reading twice: it is three-dimensional in geometry and *not* a 6-DOF body. It declares `six_dof = false` and carries three translations with no quaternion and no rotating inertia tensor, which is exact for an axial entry — there is no torque about any axis — and would not be for anything else. The full 6×6 with the inertia tensor held in the body frame and rotated out each step is `RigidBody3D.hpp`, reached by `Box` and `Disc`. The two paths are separate rather than layered: calling `refresh()` on a `Box` or `refresh6()` on a `Rect` is a compile error rather than a wrong answer. What has *not* changed is that the fluid side needed no modification for any of it.

Uniform density, which is still a real limit. The body's centre of mass is taken to be its geometric centre, which is exactly what lets its own weight drop out of the roll equation. Ballast low in the hull is the standard way to make a tall body float upright, and it is precisely what this cannot express: it would need a second centre, a $\overline{BG}$ that no longer follows from the draft, and a weight torque restored to the third row. The same assumption has a sharper edge on the 6-DOF path — `set_uniform_density6` measures the inertia in the world frame and stores it in the body frame, and using the planar `set_uniform_density` on a 6-DOF body fills the mass and leaves the tensor at *zero*, which makes the angular half singular and produces a plausible tumble rather than a failure. Anything carrying a signed distance function drops straight in as a sixth shape.

10.9 A second immiscible model: the colour gradient

Everything above carries the interface in a field with its own advection equation. The colour-gradient model, implemented in `ColourGradient.hpp` and `ColourGradientSolver.hpp`, reaches the same physics with no interface equation at all, and it is in the tree because the two fail differently.

The reference changed on 2026-09-19. The implementation followed Saito *et al.* (Phys. Rev. E **98**, 013305, 2018) and now follows Appendix D of De Rosis, Huang & Coreixas (Phys. Fluids **31**, 117102, 2019), whose equations are cited below as (Dn). Three things differ, and only the first is a change of physics: the equilibrium is the complete sixth-order Hermite set of Eq. (D5) rather than Saito's third-order truncation; the rest term's α is interpolated by Eq. (D13) rather than taken per colour (§10.9); and the perturbation carries $A/2$ colour-blind, Eq. (D14), so $\sigma = 2A\tau/9$ where Saito's Eq. (32) gives $4A\tau/9$ for the same symbol — a normalisation of A, reached through `A_from_sigma()` so that it never has to be compared across the two.

Two distribution sets f_i^r and f_i^b are carried. The interface is wherever both are non-zero, and it is held together by a *recolouring* step that redistributes the colour-blind post-collision state back into red and blue along the colour gradient. Nothing diffuses, nothing is reinitialised, and the interface width is an *outcome* rather than a parameter — the opposite of the conservative Allen–Cahn equation of §10.1, whose W is prescribed and whose profile is its exact one-dimensional equilibrium.

Three suboperators. Equation (10) of the paper reads right to left,

$$\Omega_i^k = \left(\Omega_i^k\right)^{(3)}\Big[\left(\Omega_i\right)^{(1)} + \left(\Omega_i\right)^{(2)}\Big]: \tag{101}$$

collide the colour-blind $f_i = f_i^r + f_i^b$, perturb it to make surface tension, then recolour. Only the third carries a colour index, which is the economy of the scheme: the expensive part, the central-moment collision, is done *once* for both fluids.

The single-phase collision is a general MRT in central moments with the relaxation grouping $s_0 = s_1 = 0$, five second-order moments at the shear rate, the trace on its own, and everything above second order straight to equilibrium. That is already what `MomentCollision.hpp` does, so `ProductBasis` is reused unchanged and only the *equilibrium* is new. The paper reaches the same operator through De Rosis's nonorthogonal basis and this code through a Hermite product basis; above second order the two cannot disagree, because when every moment of a subspace relaxes at one rate any basis of that subspace gives the same operator, and at second order the split into trace and traceless deviatoric is basis-independent for the same reason.

The density ratio lives in the rest weight. ϕ_i of Eq. (20) replaces w_i in the rest term of the equilibrium and carries a free parameter α,

$$\phi_0 = \alpha, \quad \phi_{|\boldsymbol{c}_i|^2=1} = \tfrac{2(1-\alpha)}{19}, \quad \phi_{|\boldsymbol{c}_i|^2=2} = \tfrac{1-\alpha}{38}, \quad \phi_{|\boldsymbol{c}_i|^2=3} = \tfrac{1-\alpha}{152}, \tag{102}$$

which sums to one for any α and whose second moment is $c_s^2 = 9(1-\alpha)/19$. *The sound speed is therefore a property of the phase.* Pressure continuity across the interface, $p = \rho c_s^2$, then fixes the density ratio without a single density appearing in the collision,

$$\gamma = \frac{\rho_r^0}{\rho_b^0} = \frac{1-\alpha_b}{1-\alpha_r}. \tag{103}$$

Taking $\alpha_b = 8/27$ recovers $c_s^2 = 1/3$ exactly for the blue fluid, which is the paper's choice and makes the blue phase an ordinary lattice. Note what it does *not* do: it does not make $c_s^2 = 1/3$ anywhere else, so the standard lattice sound speed is not available as a global constant here, and nothing in the operator assumes it. `ProductBasis`'s c_s^2 is a basis constant, not a physical one, and the two are deliberately different numbers.

The last term of the equilibrium, Φ_i of Eq. (21), restores Galilean invariance when the density varies, through the second moment, using $G = \frac{1}{48}[\boldsymbol{u}\otimes\nabla\rho + (\boldsymbol{u}\otimes\nabla\rho)^{\mathsf{T}}]$. It is the same physics that `ViscousInterfaceForce.hpp` adds to the phase-field module as an explicit body force $\nu(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^{\mathsf{T}})\cdot\nabla\rho$ — the same missing stress, entered by a different door, and here at no

extra field beyond $\nabla\rho$ because it goes into the equilibrium rather than into F. It is also the term most easily dropped by accident, and dropping it changes nothing that any static test checks.

A factor of two, and what it is and is not. The implementation now follows Eq. (D14), which is written *colour-blind* and carries $A/2$:

$$(\Omega_i)^{(2)} = \frac{A}{2}\,|\nabla\phi|\Big[w_i\frac{(\boldsymbol{c}_i\cdot\nabla\phi)^2}{|\nabla\phi|^2} - B_i\Big], \qquad \sigma = \int (S_{tt} - S_{nn})\,\mathrm{d}n = \frac{A\tau}{9}\,\Delta\phi = \frac{2A\tau}{9}, \tag{104}$$

since ϕ runs from -1 to $+1$. Saito *et al.* write the perturbation *per colour* in their Eq. (30) with $A_k/2$, and their Eq. (39) sums the capillary stress over k as well as over i; applied to the colour-blind population the two halves add, giving A and their Eq. (32), $\sigma = 4A\tau/9$. This document previously argued that reading at length and called the difference "not cosmetic".

It is a normalisation of the symbol A, and nothing computed depends on it. Both pairs are internally consistent; what a simulation is given is σ, and `A_from_sigma()` and `sigma_from_A()` convert with whichever pair is in force. The three drivers that set A by hand had it doubled when the convention changed, so their surface tension did not move and an old-versus-new comparison measures the equilibrium alone. What *would* be a real error is mixing the two — taking A from one paper and σ from the other — and that is exactly what routing every driver through the converters prevents.

The perturbation conserves mass and momentum exactly and acts only across the interface: $\sum_i w_i(\boldsymbol{c}_i\cdot\boldsymbol{n})^4 = 3c_s^4 = 1/3$ cancels $\sum_i B_i(\boldsymbol{c}_i\cdot\boldsymbol{n})^2 = 1/3$ identically, so the *normal* component of the capillary stress vanishes for every direction of $\boldsymbol{n}$ while the tangential component does not — both direction-independent, because D3Q27's fourth moment is isotropic.

Recolouring is a partition. Equations (33)–(34) split the post-collision population back,

$$f_i^r = \frac{\rho_r}{\rho}f_i + \beta\frac{\rho_r\rho_b}{\rho^2}\cos\theta_i\, f_i^{\mathrm{eq}}(\rho,\mathbf{0}), \tag{105}$$

$$f_i^b = \frac{\rho_b}{\rho}f_i - \beta\frac{\rho_r\rho_b}{\rho^2}\cos\theta_i\, f_i^{\mathrm{eq}}(\rho,\mathbf{0}), \tag{106}$$

with $\cos\theta_i$ the cosine of the angle between $\boldsymbol{c}_i$ and $\nabla\phi$. The two sum to f_i identically, so mass and momentum survive whatever β is; all β does is push colour up the gradient. $\beta = 0.7$ is the paper's value and the largest that keeps the interface smooth. Note that $f_i^{\mathrm{eq}}(\rho,\mathbf{0})$ is simply $\rho\phi_i$: at zero velocity every other term of the equilibrium vanishes, Φ_i included.

What the operator does not own. It does not stream, it does not hold the two distributions, and it does not compute a gradient. Under Esoteric Pull the neighbour populations inside a fused kernel are a mixture of two time levels, so a gradient taken from them is meaningless; $\nabla\phi$ and $\nabla\rho$ are node fields computed by `ColourGradientSolver` in a separate pass, for exactly the reason §10.1 gives for the phase field.

Which model to use is not settled here. The phase field has a conservative advection equation and a prescribed interface width; the colour gradient has neither. What the colour gradient has instead is the density ratio: it reaches 1000 in the paper's own static tests, where the pressure form of §10.7 is already fighting its conditioning problem at 100. §16.12 measures both on one case.

Which α the rest term sees, and the one mechanism that was diagnosed. Both readings of Eq. (D6) are the *same functional form.* φ_i is affine in α — its rest slot is α and every other slot a fixed multiple of $(1-\alpha)$ — so, identically, population by population,

$$\sum_k \rho_k \varphi_i(\alpha_k) \;\equiv\; \rho\,\varphi_i(\alpha_P), \qquad \alpha_P = 1 - \frac{19P}{9\rho}, \qquad P = \sum_k \rho_k c_{s,k}^2. \tag{107}$$

The per-colour reading is therefore Eq. (D6)'s own single-α form with α chosen to make the pressure continuous, and the entire difference between the two is the *interpolation rule*: linear in the order parameter, Eq. (D13), or α_P. `tests/test_colour_gradient.cpp` block 2 pins that identity at 2.6×10^{-15}.

The linear rule is not free. c_s^2 is linear in α while ρ is not correspondingly interpolated, so at a symmetric interface's midpoint ($\phi = 0$, $\bar{\alpha} = (\alpha_r + \alpha_b)/2$, $\rho = (\gamma+1)/2$) the pressure is

$$\frac{p_{\mathrm{mid}}}{p_{\mathrm{bulk}}} = \frac{(\gamma+1)^2}{4\gamma}, \tag{108}$$

i.e. 1.00, 3.03, 25.50 and 250.50 at $\gamma = 1, 10, 100, 1000$ against a bulk of 1/3. Measured on `validation/static_droplet` at 48^3, $R = 16$, $\tau = 1$, σ held fixed, *converged* (32 000 steps), the Laplace error is

rest term	$\gamma = 1$	10	20	100	1000
`PerColour` (default)	0.88%	0.68%	0.46%	3.21%	comes apart
`AlphaBar`, Eq. (D6)	0.88%	3.34%	1.51%	NaN	NaN

Run at a *matched* 1500 steps — unconverged, but the same protocol on both sides, which is what an A/B needs — Saito's *third-order* equilibrium with `PerColour` reads 0.23/0.95/493.49/3871.32 against the sixth-order one's 0.23/0.95/493.50/3871.32, agreeing to four or five significant figures at every ratio: **the equilibrium order is not what is at stake here, and an earlier version of this section said it was.** That wrong attribution came from an isolation run in which the collision's rest term was changed while the solver's seed and the recolouring were left on the other reading — a 3× pressure mismatch at the interface on step 0, which returned NaN and looked like the model failing. The operator now owns the split, so the solver cannot disagree with it.

The step count has to scale with γ. Relaxation slows as the ratio grows while τ is held at 1, so a ladder quoted at one fixed step count measures the transient. Here $\gamma \le 20$ is converged by 8000 steps (0.04 points of drift out to 32 000) while $\gamma = 100$ reads 2.12% at 8000, 3.11% at 16 000 and 3.21% at 32 000. An earlier version of this table carried 493% and 3871% from a 1500-step run, which was pure transient. At $\gamma = 1000$ it is not slow convergence but a qualitative failure: the measured Laplace jump crosses *zero* ($+4.84 \times 10^{-3}$, $+1.20 \times 10^{-3}$, -2.34×10^{-4} at 8000/16 000/32 000) while the interface widens from 4.92 to 5.47 cells. Nothing is quotable there, and the cause is not diagnosed. `ColourGradient::RestTerm` selects the reading and `static_droplet -rest bar|percolour` measures the gap. Note that at $\gamma = 1$ the two are algebraically identical, so no matched-density test can see any of this.

11 Free-surface flow

§10 resolves both fluids, and everything difficult about it follows from that: the pressure-gauge conditioning of §10.7, the α interpolation of §10.9, the recolouring. The third route in this code changes the bargain instead of the parameters. It does not resolve the gas at all. The gas is a *void* held at a prescribed pressure — no populations, no velocity, no dynamics — and the liquid meets it across a boundary condition rather than across a diffuse interface.

What that buys is the thing the other two fight for. There is no density ratio, because there is no second density: the ratio is infinite by construction and it costs nothing, because the expensive half of the problem was deleted rather than discretised.

module	interface	fluids resolved	density ratio
phase field (§10.1)	diffuse	both	1000 measured; the gauge still bites
colour gradient (§10.9)	diffuse	both	1000 in principle
free surface (this section)	sharp	liquid only	infinite, by construction

`FreeSurfaceSolver.hpp` follows the standard formulation of Körner, Thies, Hofmann, Theobald & Rüde, J. Stat. Phys. **121**, 179 (2005), with the refinements in Thürey's and Rüde's later work. Unlike §10.9 there is no paper in `SomeRefs/` to check equation by equation, so nothing below cites an equation number and every choice is justified from the mechanics.

11.1 The state, and the mass that is not a moment

Every cell is Gas, Interface or Fluid (or a wall, or halo). Fluid cells are ordinary LBM cells; Gas cells hold nothing. Interface cells are the entire method: they carry an explicit mass m alongside their populations, and a fill level

$$\varepsilon = m/\rho, \tag{109}$$

which is 1 when the cell is full and 0 when empty. The interface is a closed, one-cell-thick shell separating Fluid from Gas, and keeping it closed through arbitrary topology change is most of the work.

Mass is advected, not taken as a moment, and that is the sharpest difference from everything else in this tree. Elsewhere $\rho = \sum_i f_i$ and conservation is a property of the collision operator, checkable by algebra. Here mass moves because populations cross faces:

$$\mathrm{d}m_i(\boldsymbol{x}) = f_{\bar{\imath}}(\boldsymbol{x} + \boldsymbol{c}_i) - f_i(\boldsymbol{x}), \tag{110}$$

for both cells Fluid or Interface, scaled by the mean fill level when both are Interface, because a half-full cell can only exchange half as much. Nothing enforces conservation but the *antisymmetry* of (110): what $\boldsymbol{x}$ sends across a face is exactly what its neighbour receives. That is why the validation checks total mass to round-off first and everything else second — it is the one property that can be silently lost.

11.2 The free-surface condition

For every direction i whose source cell $\boldsymbol{x} - \boldsymbol{c}_i$ is Gas, the population that would have arrived is reconstructed as

$$f_i(\boldsymbol{x}) = f_i^{\mathrm{eq}}(\rho_G, \boldsymbol{u}) + f_{\bar{\imath}}^{\mathrm{eq}}(\rho_G, \boldsymbol{u}) - f_{\bar{\imath}}(\boldsymbol{x}), \tag{111}$$

with ρ_G fixing the gas pressure through $p = \rho_G c_s^2$. Its two moments are what a free surface *is*: the normal stress equals the gas pressure, and the tangential stress vanishes. Here $f_{\bar{\imath}}(\boldsymbol{x})$ is the post-collision population at $\boldsymbol{x}$ from the previous step, which the two-lattice scheme still holds in its source array, so no extra storage is needed.

The velocity in that reconstruction is one step old. $\boldsymbol{u}$ is needed to build the equilibria and $\boldsymbol{u}$ comes from a sum over the populations, some of which are the ones being reconstructed; the circle is broken with the previous step's velocity. That is first order in time and is what the literature does. It matters where the interface accelerates hard and is invisible in a standing wave.

11.3 Why this module alone uses two-lattice streaming

Everything else in `M3LB` streams by Esoteric Pull (§3). This solver `static_asserts` *against* it, and the reason is structural rather than a matter of effort. Three requirements, of which Esoteric Pull fails two outright:

1. Mass exchange needs $f_i(\boldsymbol{x})$ and $f_{\bar{i}}(\boldsymbol{x}+\boldsymbol{c}_i)$ as *post-collision* values, after the whole domain has collided. In place, those slots hold a mixture of two time levels, and which one depends on parity and on the direction's index. Under two-lattice they are simply the destination array, complete and unambiguous, once the collide pass has fenced.
2. A cell that fills up turns its gas neighbours into interface cells, which then need populations. In place, a cell's populations live in its neighbours' slots, so initialising a neighbour means writing storage that other cells are concurrently reading through their own aliases.
3. Reading a neighbour's post-collision state while writing one's own is the normal case here and impossible there.

Requirement 2 is dodged rather than met: every pass is a *gather*. A cell about to become interface initialises itself by averaging its neighbours, and a cell with excess mass publishes it in a field that its neighbours collect. Nothing ever writes another cell's memory, so there is no race to reason about and no atomics — a deliberate shape, worth keeping even if scatters were free. The cost is two population arrays instead of one, which for a method that deletes an entire phase is not the expensive part.

11.4 A standing gravity wave

The validation case is the linear dispersion relation $\omega^2 = gk\tanh(kh)$, and it is that rather than a dam break because a dam break's reference is experimental and comparing against a digitised curve tests the digitisation as much as the solver. The dispersion relation is exact, closed form, has no fitted constant, and — the part that matters — it is a statement about precisely the two conditions a free-surface scheme exists to impose: the *kinematic* condition, that the surface moves with the fluid, which here is the mass exchange and nothing else; and the *dynamic* condition, that the normal stress equals the gas pressure, which here is the population reconstruction and nothing else. Get either wrong and the period moves.

Two depths, on purpose. At $kh = 3.1$ the tanh is 0.9963 and the relation is indistinguishable from the deep-water limit $\omega^2 = gk$; at $kh = 0.79$ it is 0.657, and a scheme with the surface condition right but the depth coupling wrong would pass the first and fail the second. Testing one depth tests a number; testing two tests the relation.

The two residual errors are separate, and each was identified by making it move without moving the other. *Deep* is limited by the free surface's own discretisation: the surface is resolved to about a cell, so a wave of amplitude A carries a relative position error of order $1/A$, and refining at fixed kh took it from 6.15% at $L_x = 64$, $A = 1.5$ to 1.77% at $L_x = 128$, $A = 3$. *Shallow* is limited by the viscous boundary layer on the floor: at $h = 16$ cells the layer is $\sqrt{\nu T} \approx 4$ cells, a quarter of the depth, against a relation with no viscosity in it at all, and dropping ν fivefold took it from 8.44% to 4.23%, monotonically. Neither knob touches the other's number, which is what says these are two effects rather than one tolerance being chased.

11.5 What this does not model

- **No surface tension.** ρ_G is uniform, so the gas pressure carries no curvature term. Adding it means $p_G = p_{\text{atm}} - \sigma\kappa$ with κ from the fill-level field, and the machinery for a curvature is not here. Everything at a Bond number where surface tension competes with gravity is out of scope — which includes most droplet problems and none of the dam-break family.

- **No gas dynamics**, and this is the method rather than an omission. An enclosed bubble does not compress, because its pressure is prescribed rather than solved. Modelling that needs a volume-of-gas tracker with a bubble-merging graph.
- **The reconstruction is applied to gas-facing directions only.** The literature also reconstructs directions with $\boldsymbol{c}_i \cdot \boldsymbol{n} > 0$, which stabilises interfaces nearly tangential to the lattice. If a case shows interface roughening along a diagonal, this is the first thing to add.
- **Excess mass is redistributed uniformly** among eligible neighbours rather than weighted by the interface normal. Uniform redistribution conserves mass exactly — which matters most — but is more diffusive at a sharply curved interface.
- **No subgrid interface.** A film thinner than a cell either survives as a one-cell layer or vanishes.
- **A moving obstacle is not yet reliable**, and the reason is in the solver rather than in the caller. `mass_exchange()` conserves mass by the antisymmetry of (110), which assumes both cells agree on the mask for the whole pass. `move_obstacle()` changes the mask between passes, so every cell the body sweeps breaks the assumption. `transfer_covered_mass()` keeps the total right but does not restore the per-face antisymmetry, and the difference shows up wherever many mask cells change at once: a flat face landing on a flat surface, a corner entering, a rotated rectangle's staircase flipping cells as it turns. The fix is to make the mass exchange itself aware of the moving boundary; bounce-back, momentum exchange and refill are all in place and are not the problem.

12 Validation

Every scheme in this document is accompanied by the test that establishes it, and this section collects all of them. It runs in both FP64 and FP32; the numbers below are FP64 unless stated.

The tests come from three places, and the difference matters when reading a number. Most were written alongside the feature they exercise, and are chosen to isolate it: an exact solution where one exists, a convergence rate where it does not. Two blocks instead work through somebody else's published suite without modification, so that the numbers can be put beside theirs — the fluid-only campaign of De Rosis and Coreixas (§12.1.6) and the thermal suite of Zhou, De Rosis and Revell (§12.3.5). Where a test is one of those, its heading says so.

The organisation below is by physics rather than by provenance: single-phase flow, multiphase flow, thermal transport, magnetohydrodynamics. A few tests sit awkwardly in that scheme and are placed where the bulk of their content lives rather than split — the boundary-condition survey of §12.1.5 is mostly velocity walls and sits under single-phase flow, but its magnetic-wall part is the Hartmann validation and §12.4 points back to it.

On the field snapshots. The tables say whether a number is right. The figures say whether the *flow* is right, which is a different question and one a norm cannot answer: a scheme can hit an integral quantity while getting the structure wrong. Every field shown is the one the corresponding test actually converged to, dumped from the run that produced the numbers quoted for it. Signed quantities — vorticity, current — use a diverging map centred on zero; quantities with no meaningful zero — temperature, speed — use a sequential one. Where the colour saturates below the peak the caption says so: several of these fields are strongly localised, and scaling by the maximum leaves the structure that matters invisible.

12.1 Single-phase flows

12.1.1 Poiseuille flow: wall placement and the magic parameter

Force-driven plane Poiseuille flow, periodic in x, halfway bounce-back walls in y. With H fluid nodes the no-slip planes sit at $y = 0.5$ and $y = H + 0.5$, so

$$u(y) = \frac{G}{2\rho\nu}\,(y - 0.5)\,(H + 0.5 - y), \qquad u_{\max} = \frac{GH^2}{8\rho\nu}. \tag{112}$$

Which relaxation rates play the roles in (31) depends on the operator, so each has its own magic point. Measured wall slip (D2Q9, $H = 32$):

τ	BGK	TRT(3/16)	MRT	central moments
0.6000000	-7.40×10^{-3}	$+2.51 \times 10^{-14}$	-5.73×10^{-3}	-5.73×10^{-3}
0.8000000	-4.06×10^{-3}	$+7.53 \times 10^{-14}$	-1.56×10^{-3}	-1.56×10^{-3}
0.8750000	-1.95×10^{-3}	-1.92×10^{-14}	$\mathbf{+4.11 \times 10^{-14}}$	$\mathbf{+7.39 \times 10^{-14}}$
0.9330127	$\mathbf{-1.08 \times 10^{-14}}$	$+5.35 \times 10^{-14}$	$+1.21 \times 10^{-3}$	$+1.21 \times 10^{-3}$
1.2000000	$+1.26 \times 10^{-2}$	$+3.55 \times 10^{-15}$	$+6.77 \times 10^{-3}$	$+6.77 \times 10^{-3}$

TRT holds the wall at 0.5 to $\sim 10^{-14}$ at *every* viscosity. BGK manages it only at $\tau = \frac{1}{2} + \frac{\sqrt{3}}{4} = 0.9330127$. The moment operators relax third-order modes at unity, so their magic point is $(\frac{1}{\tau} - \frac{1}{2})\frac{1}{2} = \frac{3}{16}$, i.e. $\tau = 7/8$ exactly — *predicted before the run and confirmed to* 7×10^{-14}.

Resolution behaviour. At its magic point an operator is exact at every resolution. Away from it (FP64, D2Q9):

operator @ τ	L_2, H=16	H=32	H=64	ord(L_2)	ord(slip)
BGK @ 0.9330127	4.2e-14	1.2e-14	5.8e-13	exact	exact
TRT(3/16) @ 0.6	4.6e-14	1.1e-13	1.6e-13	exact	exact
MRT @ 0.875	3.2e-14	9.3e-14	3.1e-13	exact	exact
central moments @ 0.875	3.6e-14	8.4e-14	3.2e-13	exact	exact
BGK @ 0.6 (off-magic)	5.1e-03	1.3e-03	3.2e-04	**2.00**	**1.00**

Off-magic bounce-back is still *second* order: a lattice-aligned wall does not degrade the scheme. The pair of orders is what identifies the defect — the discrete error is a constant velocity offset $C \sim (\Lambda - 3/16)\,G/\nu$, which at fixed peak velocity scales as $1/H^2$, while the fitted wall *position* moves only as $1/H$ because the parabola's curvature shrinks with it. Seeing 2 and 1 together says "offset", not "displaced wall".

These two assertions are FP64-only by design: at $H = 64$ the discretisation error is 3×10^{-4}, which in FP32 lies below the accumulated round-off floor of about 10^{-3}. Loosening the tolerance would not fix that, it would only stop the test measuring anything.

12.1.2 Decaying flows with exact solutions

Three periodic flows, chosen so that each isolates something the others cannot.

Taylor–Green $\boldsymbol{u} = U(-\cos kx \sin ky,\ \sin kx \cos ky)\, e^{-2\nu k^2 t}$. Nonlinear and exact, but z-independent — so it is the *same 2D problem* on every lattice. It is a cross-lattice consistency check, not a 3D test.

ABC / Beltrami $\boldsymbol{u} = U(\sin kz + \cos ky,\ \sin kx + \cos kz,\ \sin ky + \cos kx)$. Here $\nabla \times \boldsymbol{u} = k\boldsymbol{u}$, so $\boldsymbol{u} \times (\nabla \times \boldsymbol{u}) = \boldsymbol{0}$ and the nonlinear term collapses to $\nabla(|\boldsymbol{u}|^2/2)$, absorbed by the pressure: $\boldsymbol{u}(t) = \boldsymbol{u}_0 e^{-\nu k^2 t}$ exactly. Every component varies in every direction. This is the nonlinear 3D convergence test.

Diagonal shear wave $\boldsymbol{u} = U\boldsymbol{e}\sin(\boldsymbol{K}\cdot\boldsymbol{x})e^{-\nu|K|^2 t}$ with $\boldsymbol{K} = \frac{2\pi}{L}(1,1,1)$ and $\boldsymbol{e}\cdot\boldsymbol{K} = 0$. Linear by construction, and off-axis, so it probes lattice isotropy where an axis-aligned wave cannot.

The three are complementary and none substitutes for another. The ABC flow is built from axis-aligned sinusoids — it tests nonlinear 3D convergence but says nothing about isotropy.

Interpreting the effective viscosity. ν_{eff} does *not* equal ν at finite resolution: a sinusoid decays at the rate set by the discrete Laplacian, which differs from νk^2 at $O(k^2)$. What is asserted is therefore that the error converges away at second order, not that it is small at any given N. Diffusive scaling $U \sim 1/N$ keeps the Mach-number error second order alongside the spatial error.

12.1.3 Grid convergence: all lattices, all operators

A dedicated sweep (`validation/grid_convergence`) covers every lattice × operator combination over a resolution ladder and reports 56 data points — 92 before D3Q19 was removed on 2026-09-18. Slopes are least-squares fits of log(error) against $\log N$ over the whole ladder, not pairwise ratios.

case	lattice	operator	slope ν err	slope $L_2(u)$	worst R^2
Taylor–Green	D2Q9	BGK, TRT, MRT, CM	−2.02	−1.99	
Taylor–Green	D3Q27	BGK, TRT, MRT, CM	−2.02	−1.99	
ABC/Beltrami	D3Q27	BGK, TRT, MRT, CM	−2.01	−2.01	
24 fitted slopes, range −2.023 **to** −1.994					$R^2 \geq 0.99982$

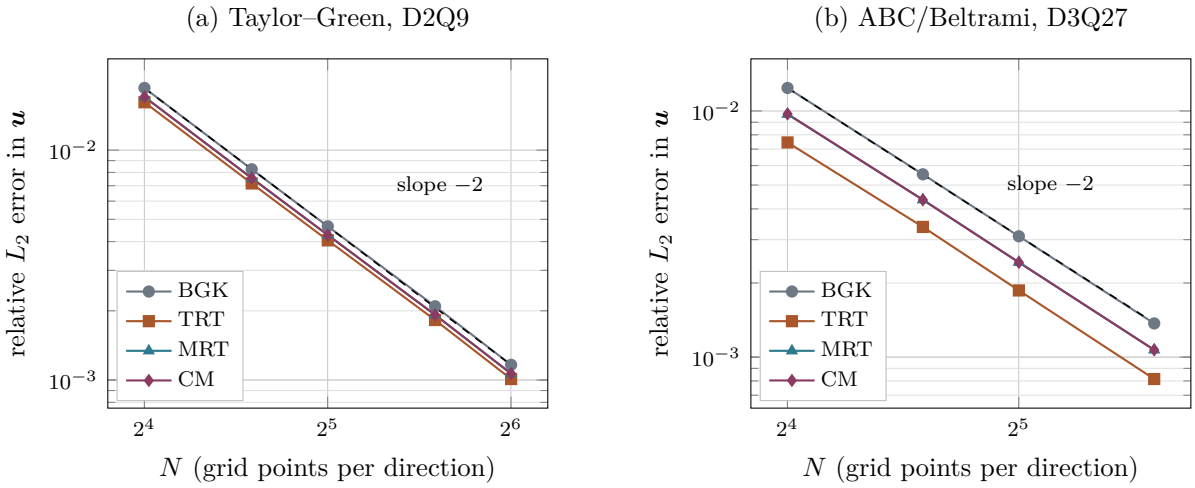


Figure 1: Grid convergence of the four collision operators, log–log, with a slope −2 reference (dashed). Diffusive scaling $U \sim 1/N$. Panel (a) is the Taylor–Green vortex on D2Q9; D3Q27 lies exactly on top of these points (spread 3.9×10^{-14}) because the field is z-independent. Panel (b) is the ABC/Beltrami flow on D3Q27. The lines are parallel: the operators differ in error *amplitude*, not in order. Fitted slopes over the whole ladder range from −1.994 to −2.023 across all 24 series.

No operator gains or loses an order. What differs is the error *amplitude*, and no operator dominates everywhere:

	ABC flow, ν error at $N = 48$	diagonal shear wave, $N = 64$
BGK	2.75×10^{-3}	1.00× (reference)
TRT(3/16)	$\mathbf{1.63 \times 10^{-3}}$	—
MRT	2.15×10^{-3}	—
central moments	2.15×10^{-3}	**1.38**× better (D3Q27)

TRT is the most accurate in a periodic bulk flow, because the magic parameter tunes the third-order modes that the moment operators send straight to equilibrium. On the off-axis wave the ordering reverses.

12.1.4 Galilean invariance

Superimposing a uniform velocity U_0 on a solution only translates it, so the recovered viscosity must not depend on U_0. ABC/Beltrami flow, $N = 32$, $\tau = 0.8$:

lattice	operator	U_0=0	0.05	0.10	0.15	drift vs BGK
D3Q27	BGK	0.100620	0.099864	0.097595	0.093812	—
D3Q27	TRT(3/16)	0.100368	0.099618	0.097368	0.093617	1×
D3Q27	MRT (raw)	0.100484	0.100485	0.100486	0.100490	1153×
D3Q27	central moments	0.100484	0.100484	0.100482	0.100479	**1439**×

Two findings deserve emphasis.

1. **TRT is no better than BGK here.** The magic parameter fixes where the *wall* sits, not which frame the collision prefers. These are independent defects and it takes both operators to address both.
2. **Most of the gain is the complete equilibrium, not the co-moving frame.** Raw MRT already recovers 1153×, because this implementation gives it the full product-form equilibrium; centring the basis adds a further $\sim 25\%$. The split may differ for flows whose nonlinearity is less balanced than a Beltrami field, but the effect should not be attributed wholly to centring.

Every residual scales as U_0^2 (fitted powers 2.00–2.10), so what remains is the weakly-compressible $O(\mathrm{Ma}^2)$ error of the method itself, not frame dependence of the collision.

12.1.5 Boundary conditions

Four tests, one per claim. All are second order, which is the accuracy of the scheme itself, so what distinguishes the conditions is *where the wall is*, and that is what each table measures rather than assumes.

Regularised walls: order and wall placement. Poiseuille flow with regularised walls on the boundary rows, $\tau = 0.8$. The wall position is obtained by fitting a parabola to the three interior nodes nearest the wall and taking its root; reading $\boldsymbol{u}$ back at the wall node would test nothing, since the solver reports the imposed value there by construction.

W	L_2 error	order	fitted wall position
8	1.708×10^{-2}	—	2.68×10^{-2}
16	4.419×10^{-3}	1.950	1.33×10^{-2}
32	1.123×10^{-3}	1.976	6.67×10^{-3}

The offset halves with each refinement: first order in lattice units, hence second order relative to the channel width. It is entirely the curvature slip (60): at $\tau = 0.8$, $\frac{4}{3}(\omega - 1)/\omega^2 = 0.2133$ against 0.2128 measured. At $\tau = 1$ the same run is exact to 1×10^{-15}.

These numbers are worse than they were before the force corrections of §8.1 were applied, and that is worth stating rather than hiding: without them the offset at $\tau = 0.8$ was 1.42×10^{-2}, because the force defect and the curvature slip carry opposite signs there and partially cancelled. Two errors cancelling is not accuracy, and removing the one that is understood is the right move even though a headline number rises. The order of convergence is unchanged. D2Q9 and D3Q27 agree to seven significant figures on this flow, the plane channel being quasi-one dimensional so that D3Q27 with periodic span reduces exactly.

Corners: the two closures compared. Lid-driven cavity, 129^2, against Ghia, Ghia and Shin [3].

Re	corner closure	$\max \lvert\Delta u\rvert$	$\max \lvert\Delta v\rvert$
100	local	0.0044	0.0071
100	finite diff.	0.0046	0.0065
1000	local	0.0187	0.0149
1000	finite diff.	**0.0123**	**0.0117**

At $Re = 100$ the two are indistinguishable — four corner nodes out of 16 641 do not move the centrelines. At $Re = 1000$ the finite-difference closure is better at *every* station of the u profile, by roughly a third in both the maximum and the rms. The improvement sits in the near-wall regions and vanishes in the core, which is the expected signature: the corner closure perturbs the boundary layers, and at $Re = 1000$ those are thin enough for the perturbation to reach the centreline. This is why the finite-difference route is the default.

Magnetic walls: Hartmann flow. The MHD analogue of Poiseuille flow, Ha = 10, $\nu = \eta$, against the exact solution [2]

$$b(\xi) = \frac{FL}{B_0}\left[\frac{\sinh(H\xi)}{\sinh H} - \xi\right], \qquad u(\xi) = \frac{FL}{B_0}\sqrt{\frac{\eta}{\nu}}\coth H\left[1 - \frac{\cosh(H\xi)}{\cosh H}\right], \tag{113}$$

with $\xi = x/L$ and $H = B_0 L/\sqrt{\nu\eta}$. Velocity uses regularised walls, $\boldsymbol{b}$ uses the moment condition (68); both put the wall on the node, so they agree about where it is.

n	$\ell_2(u)$	order	$\ell_2(b)$	order	$\lvert u\rvert$ wall	$\lvert b\rvert$ wall
17	3.873×10^{-4}	—	4.117×10^{-4}	—	0	0
33	8.831×10^{-5}	2.133	1.009×10^{-4}	2.029	0	0
65	2.136×10^{-5}	2.047	2.544×10^{-5}	1.988	0	0
129	5.476×10^{-6}	1.964	6.572×10^{-6}	1.953	0	0

Second order in both fields, and both vanish at the walls exactly rather than to $O(h^2)$ — the property the moment condition exists to provide. The computed profile reproduces the flat Hartmann core, the $O(L/H)$ boundary layers, and $b(0) = 1.1 \times 10^{-17}$, antisymmetry at round-off.

Inlet-driven Hartmann flow: the same problem without a body force. The test above drives the channel with a uniform force through a streamwise-periodic domain. Removing the force and driving it from an inlet instead changes what is being tested in three ways, and each is the reason for doing it:

- It is the only way to put `MhdCentralMoments` against an analytic profile at all. That operator hardcodes `NoForcing` and cannot be body-force driven, so the force-driven test covers `MhdBGK` and could never have covered it.
- It is free of the Guo half-force interaction and the curvature slip (60), exactly as the inlet Poiseuille of §12.1.12 is.
- It is the only test in the suite exercising the fluid and magnetic outflow conditions at once.

Dellar's axes are rotated so the streamwise direction is x: walls on the two y-normal planes, half-width $L = (L_y - 1)/2$, $\xi = (y - L)/L$, and the applied field B_0 wall-normal along y, stretched by the flow into a streamwise b_x. The exact solution is (113) with x and y interchanged. The velocity inlet imposes the analytic u, the outlet is the constant back-pressure condition of §8.5, and $\boldsymbol{b}$ is imposed at the inlet but left free at the outlet — pinning it at both ends would tell the induction equation the answer at both boundaries and make the interior test partly circular.

L_y	rel $L_2(u)$	order	rel $L_2(b)$	order	$\lvert u\rvert$ wall	$\lvert b\rvert$ wall
17	2.281×10^{-2}	—	6.499×10^{-2}	—	0	0
33	5.391×10^{-3}	2.081	1.590×10^{-2}	2.031	0	0
65	1.286×10^{-3}	2.068	3.916×10^{-3}	2.022	0	0
129	3.071×10^{-4}	2.066	9.793×10^{-4}	2.000	0	0

Ha $= 10$, $\nu = \eta = 0.1$, $u_{\max} = 0.005$, $L_x = 241$, $L_z = 1$. Second order in both fields, and both vanish at the wall node exactly. The Hartmann profile has a flat core with layers of thickness L/Ha, a tenth of the channel here, so this is a considerably harder test of the wall region than the parabolic Poiseuille profile.

Why L_x is so large. Not for physics: the magnetic outflow condition contaminates a long stretch upstream, as §8.7 records. With the $L_x = 21$ that the Poiseuille case uses, the contamination covers the entire domain and the measured order is meaningless. The Mach floor of §12.1.12 appears here too but is the smaller effect: at $u_{\max} = 0.02$ the last rate in u falls to 1.461, and dropping to $u_{\max} = 0.005$ recovers it.

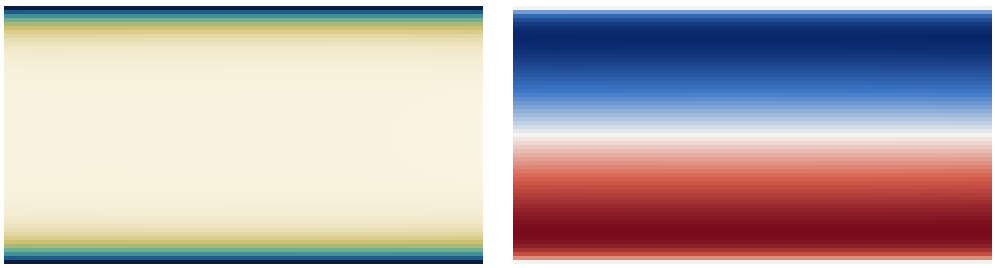
Scalar walls: placement, measured. Steady one-dimensional conduction between walls at $T = 0$ and $T = 1$. The profile is exactly linear, so the residual is round-off ($\sim 10^{-14}$ throughout) and the only meaningful measurement is where a least-squares line through the *interior* nodes reaches the wall values.

	anti-bounce-back		moment condition	
N	$T = 0$ at	$T = 1$ at	$T = 0$ at	$T = 1$ at
9	0.500000	7.500000	−0.000000	8.000000
17	0.500000	15.500000	−0.000000	16.000000
33	0.500000	31.500000	−0.000000	32.000000
65	0.500000	63.500000	−0.000000	64.000000

Anti-bounce-back places its wall half a spacing outside the node, the moment condition exactly on it, at every resolution and to machine precision.

12.1.6 A published campaign: De Rosis and Coreixas (2020)

The tests above were written alongside the features they exercise. The ones that follow are different: they work through a published test suite without modification, so that the numbers can be put beside someone else's.



(a) velocity u_x (b) induced field b_x

Figure 2: Inlet-driven Hartmann flow, Ha = 10, $L_y = 65$, D2Q9 fluid with D2Q5 induction, central moments (§12.1.5). The velocity shows the flat Hartmann core with boundary layers of thickness L/Ha; the induced field is antisymmetric about the centreline, vanishing there and at both walls. Both are visibly independent of the streamwise coordinate, which is the property the exact solution has and which the convergence table depends on — any x structure here would be boundary error, and it is how the magnetic outflow defect (§8.7) was found.

The suite is Sec. III of De Rosis and Coreixas [12], tests III.A to III.E, plus the inlet-driven Poiseuille flow of Sec. III.A of De Rosis [13]. Test III.F is out of scope: the campaign was specified to exclude applied body forces, and III.F has one.

Settings, and where they depart from the papers. Everything runs on the central-moments operator with the highest-order equilibrium each lattice admits (§5.8) — sixth order on D2Q9 and D3Q27. BGK and MRT were deliberately not run. The two-dimensional cases are run on D2Q9 and on D3Q27 with a single periodic point in z; III.E is genuinely three-dimensional and uses D3Q27. Fixed $u_0 = 0.02$ throughout rather than the papers' sweeps, and reduced resolutions where cost demanded: $D = 64$ rather than 128 in III.E, $D_{\mathrm{lb}} = 512$ only in III.D, $N = 129$ rather than 201 in III.C. These are stated per test below because in two cases they change what the number means.

12.1.7 Taylor–Green vortex

Periodic square of side 2π, N^2 points, Re = 1000, error measured at $t = T$ against the exact decaying solution.

N	8	16	32	64	128	256
D2Q9	1.106×10^{-1}	2.566×10^{-2}	6.259×10^{-3}	1.444×10^{-3}	2.427×10^{-4}	6.010×10^{-5}
D3Q27	1.106×10^{-1}	2.566×10^{-2}	6.259×10^{-3}	1.444×10^{-3}	2.427×10^{-4}	6.010×10^{-5}

Fitted rate 2.186 on both lattices. The paper reports an optimal rate of 2 at its lowest u_0.

A correction to the published initial condition. Reference [12] prints

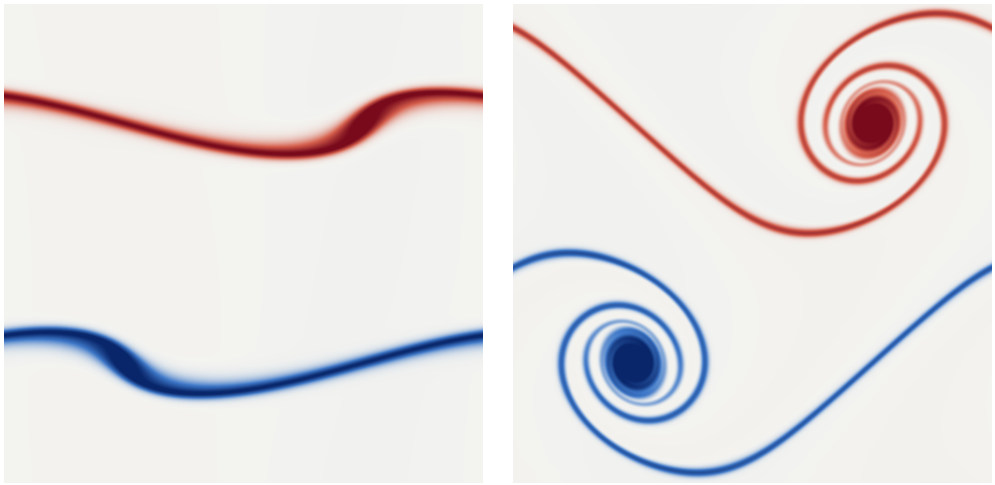
$$\boldsymbol{u}_0 = u_0[\cos\gamma x \, \sin\gamma y, \; \sin\gamma x \, \cos\gamma y, \; 0],$$

whose divergence is $-2u_0\gamma \sin\gamma x \sin\gamma y \neq 0$. Almost certainly a lost minus sign in typesetting; the standard solenoidal field, with $-\cos\gamma x \sin\gamma y$ in the first component, is used here. The printed density likewise carries a leading factor 3 that would give $\rho \approx 3$, and the standard form is used instead.

12.1.8 Double shear layer

$L = 256$, $\mathrm{Re} = 30\,000$, $\tau = 0.500512$ — a stability test rather than an accuracy one, run to $t/t_0 = 1$. Both lattices complete without divergence at that τ, and the normalised invariants agree closely:

	E/E_0	Ψ/Ψ_0
D2Q9	0.986436	0.622701
D3Q27	0.986436	0.622701



(a) $t/t_0 = 0.5$ (b) $t/t_0 = 1$

Figure 3: Double shear layer, vorticity, $L = 512$, $\mathrm{Re} = 30\,000$, D2Q9 central moments (§12.1.6). By (a) the perturbation has begun to distort the two layers; by (b) each has rolled into a vortex, and the two are joined by a single unbroken braid. That the braid stays thin and connected is the whole test: at $\tau = 0.50102$ the flow is barely resolved, and an under-dissipative or unstable scheme shows it here first, by fragmenting the braid into secondary vortices that are numerical rather than physical. Doubling the resolution from the $L = 256$ of the campaign table sharpens the structure and, more usefully, quantifies the numerical dissipation: the enstrophy retained at $t/t_0 = 1$ rises from 0.6227 to 0.6432, and the energy from 0.98644 to 0.98696. The physical dissipation at this Reynolds number is negligible over one turnover, so nearly all of that difference is the scheme's own, and it shrinks with resolution as it should. Colour saturates at the 99th percentile of $|\zeta|$.

12.1.9 Lid-driven cavity

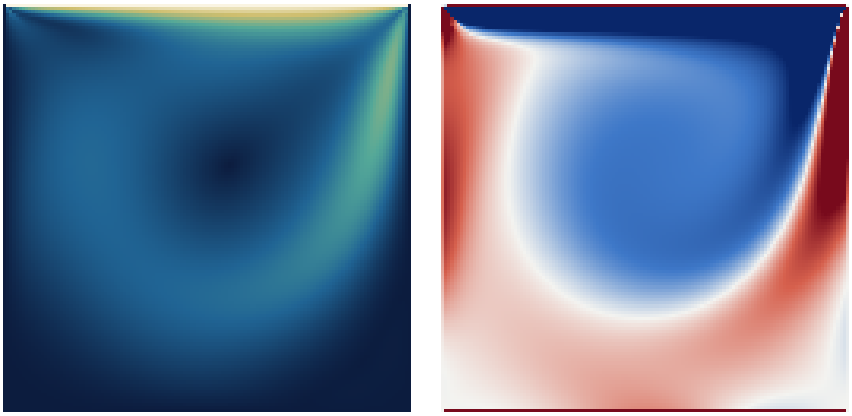
129^2, $u_{\mathrm{lid}} = 0.02$, against Ghia, Ghia and Shin [3]. Reported as the largest absolute deviation over the 17 tabulated stations, in lid-velocity units.

	$\mathrm{Re} = 100$		$\mathrm{Re} = 400$		$\mathrm{Re} = 1000$	
	$\max\lvert\Delta u\rvert$	$\max\lvert\Delta v\rvert$	$\max\lvert\Delta u\rvert$	$\max\lvert\Delta v\rvert$	$\max\lvert\Delta u\rvert$	$\max\lvert\Delta v\rvert$
D2Q9	0.0049	0.0067	0.0069	0.0071*	0.0094	0.0099
D3Q27	0.0049	0.0067	0.0069	0.0071*	0.0094	0.0099

* *One published reference value is anomalous, and it is Ghia's, not a transcription error here.* All three tables in `validation/cavity_cm.cpp` have been checked entry-for-entry against an independent digitisation of [3] and they match. But the v entry at $\mathrm{Re} = 400$, $x = 0.9063$ reads -0.23827, and three things say it cannot be right:

- Ghia's own extremum table gives $\min v = -0.44993$ at $x = 0.8594$ for this Reynolds number. The three tabulated neighbours are then -0.44993 (0.8594), -0.23827 (0.9063), -0.22847 (0.9453): essentially the whole recovery from the minimum happens in one interval, after which the profile is flat.
- The same three stations are smooth at $\mathrm{Re} = 100$ (-0.22445, -0.16914, -0.10313) and at $\mathrm{Re} = 1000$ (-0.42665, -0.51550, -0.39188). Only $\mathrm{Re} = 400$ kinks, and at exactly one point.
- Every other station agrees here to 0.0071 or better; this one disagrees by 0.1415, a factor of twenty. The value computed here, -0.37974, is what a smooth profile through Ghia's own minimum would give.

The entry is left in the table — it is what the paper prints, and quietly substituting a number nobody published would be worse — but it is excluded from the headline score above and reported separately by the run. It could not be checked against the original paper, which is not among the references to hand. An earlier draft of this document blamed the discrepancy on a faulty transcription on this side; that was wrong, and the correction is recorded here rather than silently applied.



(a) speed $|\boldsymbol{u}|/u_{\mathrm{lid}}$ (b) vorticity

Figure 4: Lid-driven cavity at $\mathrm{Re} = 400$, 129^2, D2Q9 central moments, the run scored against Ghia *et al.* in §12.1.6 ($\max|\Delta u| = 0.0069$). The primary vortex sits above and right of centre and both downstream corner eddies are resolved. The vorticity is clipped hard, at the 88th percentile against a peak twenty times larger, because it is concentrated in the lid boundary layer; without that the interior is blank.

12.1.10 Dipole–wall collision

$D_{\mathrm{lb}} = 512$ (a 1025^2 grid), walls regularised, corners on the finite-difference closure — the only test in the campaign where the corner treatment matters, since III.A, III.B and III.E are periodic. Against Table II of [12], which reproduces the lattice Boltzmann results of Mohammed, Graham and Reis [14] and the spectral reference of Clercx and Bruneau [15]:

		energy E			enstrophy Ψ		
Re	t	here	[14]	[15]	here	[14]	[15]
625	0.25	1.4976	1.494	1.501	466.3	467.2	472.1
	0.5	1.0167	1.010	1.013	387.2	374.0	382.6
	0.75	0.7702	0.765	0.767	251.3	244.8	256.0
2500	0.25	1.8355	1.838	1.848	689.6	705.3	725.6
	0.5	1.5379	1.534	1.540	889.1	898.1	917.6
	0.75	1.3274	1.320	1.325	790.3	790.2	809.9

Energy agrees to between 0.1% and 0.7% everywhere. Enstrophy is the more demanding invariant and the agreement is between 0.01% and 3.5%; in most rows the value computed here falls *between* the published lattice Boltzmann result and the spectral one, and at $\mathrm{Re} = 2500$, $t = 0.75$ it matches the published lattice Boltzmann enstrophy to four significant figures (790.3 against 790.2). D2Q9 and D3Q27 are identical to every digit shown.

An error of ours, and how it was caught. The published initial condition is written in physical units with $|\omega_e| = 299.56$, which makes $E(0) = 2$ and $\Psi(0) = 800$ exactly. It was first scaled so that the *peak* speed was u_0, which is wrong: the Reynolds number is built on the rms velocity, and for this field the peak-to-rms ratio is 11.019. The nominal Reynolds number and the elapsed physical time were both a factor eleven out, the flow dissipated far too slowly, and the discrepancy grew with t — a signature that pointed at the time scale rather than at the wall. Verifying Eq. (32) numerically ($E(0) = 2.0004$, $\Psi(0) = 799.96$, $U_{\mathrm{rms}} = 1.0001$) settled it. The results above are from the corrected normalisation.

12.1.11 Three-dimensional Taylor–Green vortex

$D = 64$, run to $t^* = tu_0/D = 10$. There is no exact solution; the comparison is between lattices. At the campaign's fixed $u_0 = 0.02$ the Mach number is 0.035 rather than the paper's 0.2–0.6, so a unit of physical time costs up to seventeen times more steps than in the paper.

	$\mathrm{Re} = 1600$	$\mathrm{Re} = 30\,000$
D3Q27	completes, $E/E_0 = 0.0145$ at $t^* = 10$	completes, $E/E_0 = 0.0399$ at $t^* = 10$

At $\mathrm{Re} = 30\,000$ and this Mach number $\tau = 0.50026$. That case was initially judged infeasible on the available hardware; the judgement was wrong, and D3Q27 runs it to $t^* = 10$ without difficulty.

12.1.12 Inlet-driven Poiseuille flow

Sec. III.A of [13]. Laminar flow along x; no-slip on the two y-normal planes; the parabolic profile $u_x(0, y) = 4U_0\,(y/L_y)(1 - y/L_y)$ with $U_0 = 0.005$ imposed at the west face; outflow opposite; $L_x = 21$; $\mathrm{Re} = U_0 L_y/\nu = 10$; started from rest and run to steady state. The error is the relative L_2 norm, Eq. (15) there, between the analytic profile — which equals the inlet profile, the flow being fully developed — and a y-aligned cross section at mid-domain. $L_z = 1$ rather than the paper's 21; the flow has no z dependence and z is periodic either way. The outflow is the constant back-pressure condition of §8.5.

This test is worth having precisely because it is driven by the inlet and not by a body force, so it is free of both the Guo half-force interaction and the curvature slip (60) that dominate the forced channel of §12.1.1.

L_y	41	81	161	321	641	fitted rate
D2Q9	1.160×10^{-3}	3.251×10^{-4}	7.261×10^{-5}	1.558×10^{-5}	6.192×10^{-6}	1.964
D3Q27	1.160×10^{-3}	3.251×10^{-4}	7.261×10^{-5}	1.558×10^{-5}	6.192×10^{-6}	1.964

Against the published 2.063 that is 4.8% low. The shortfall is entirely in the last refinement, where the pairwise rate drops from 2.231 to 1.334, and it is understood — see below. It is worth noting first that the paper states its error at the finest grid, $L_y = 1281$, is of order 10^{-5}; the error here at $L_y = 641$ is 6.2×10^{-6}, already below that.

What limits the order — three falsified hypotheses. The outlet was the obvious suspect, and it is not the cause.

1. *Per-station error.* Breaking the error down by x at $L_y = 641$ makes the outlet look guilty: against $L_y = 161$ the inlet station converges at 1.99 and the bulk minimum at 1.92, both clean second order, while the outlet station converges at 0.70.
2. *Moving the outlet away does not help.* Repeating at $L_x = 41$ and $L_x = 81$ makes the last pairwise rate *worse*, not better: $1.334 \to 0.885 \to 0.007$. At $L_x = 81$ the error stops responding to refinement altogether — 5.974×10^{-6} at $L_y = 321$ against 5.945×10^{-6} at $L_y = 641$.
3. *A second-order outlet does not help.* Replacing the zero-gradient tangential extrapolation by a linear one (§8.5) is worse on the coarse grids and indistinguishable on the finest.

What it actually is: a non-refining, velocity-dependent error. Sweeping U_0 at fixed $\mathrm{Re} = 10$ moves the onset of the degradation, and always toward *coarser* grids as U_0 rises.

U_0	Ma	fitted rate	pairwise rates			
0.0025	0.0043	**2.067**	1.720	2.026	2.205	2.263
0.005	0.0087	1.964	1.868	2.182	2.231	1.334
0.01	0.0173	1.523	2.091	2.179	0.974	0.812
0.02	0.0346	0.844	2.097	0.150	0.644	0.947

At $U_0 = 0.0025$ the velocity-dependent term stays below the discretisation error across the whole range and the fitted rate is 2.067, against the published 2.063 — agreement to 0.2%. The scheme is second order; the shortfall at $U_0 = 0.005$ is an artefact of the test's Mach number, not a defect of the wall or the outlet.

This is not a novel observation. Reference [12] reports the same behaviour on its own Taylor–Green convergence study — "for the lowest value of u_0, an optimal convergence value equal to 2 is found. As u_0 increases, the accuracy and convergence properties of the method deteriorate as well" — and attributes it to the cubic aliasing defect $c_{i\zeta}^3 = c_{i\zeta}$, which prevents the diagonal terms u_ζ^3 from entering the third-order equilibrium moments and can only be removed by explicit correction terms. The paper on this Poiseuille case [13] instead asserts that $U_0 = 0.005$ gives a Mach number "sufficiently low to annihilate detrimental compressibility effects"; at the accuracy reached here that assertion does not hold, which is unsurprising given that the error here at $L_y = 641$ is already below the error there at $L_y = 1281$.

Fitting $\varepsilon = A/L_y^2 + B$ to the two finest grids gives $B < 0$ at $U_0 = 0.0025$ — no floor reached — and $B = 3.1 \times 10^{-6}$, 9.2×10^{-6}, 2.1×10^{-5} at $U_0 = 0.005$, 0.01, 0.02. That is growth somewhere between linear and quadratic in U_0, but a two-point extraction is far too noisy to pin the exponent and none is claimed.

12.1.13 A finding that cuts across the campaign

D2Q9 and D3Q27 with one point in z are bit-identical. Not merely close: identical to every digit printed, in the Taylor–Green convergence, the shear layer, the dipole, the cavity and the inlet Poiseuille — five independent tests. The reduction of D3Q27 to D2Q9 under a single periodic z point is exact, which is a useful correctness check in both directions: it exercises the 3D code path against the 2D one on every one of these flows.

12.2 Multiphase flows

The immiscible models of §10 carry three tests here: the one law with a closed form (Laplace's), one flow with an exact profile across a viscosity and density jump, and one coupled fluid–body equilibrium. The free surface is validated in its own section (§11.4), because its one quantitative test is a dispersion relation rather than a field comparison.

12.2.1 Laplace's law

A static droplet is the first two-phase validation and deliberately so: it has an exact answer, it needs no reference table, and the two quantities it measures fail independently. The run below is `validation/laplace`, registered under `ctest`, on a triply periodic 96^2 box with $\sigma = 0.01$, $W = 4$, $\tau = 0.8$, mobility $M = 0.05$ and 10 000 steps, D2Q9 fluid against a D2Q9 phase field.

The flat interface, which must do nothing. A slab has zero curvature, so it is the control: the width must be maintained and no flow may be driven. The conservative Allen–Cahn equation (5) has the tanh profile of prescribed width as its exact one-dimensional equilibrium, and W is recovered from an integral rather than from a fit,

$$\int \phi\,(1-\phi)\,\mathrm{d}y = \frac{W}{4} \quad \text{per interface,}$$

because for $\phi = \frac{1}{2}[1 + \tanh(2y/W)]$ the integrand is $\frac{1}{4}\operatorname{sech}^2(2y/W)$, whose integral is $W/4$ exactly. An integral measure uses every node in the profile instead of the two or three near the steepest point, so it does not degrade as W shrinks towards the grid. Measured against a prescribed $W = 4.0000$: width 4.0011, relative error 2.79×10^{-4}; $\max|\boldsymbol{u}| = 2.224 \times 10^{-9}$; drift in $\sum\phi$ of 8.440×10^{-13}. The width is second-order accurate rather than exact — the *continuum* equilibrium is the tanh profile and the lattice converges to it — so round-off is not what is expected here, and is not what the tolerance asks for.

The pressure jump. Across a curved interface $\Delta p = \sigma(D-1)/R$, so in two dimensions $\Delta p = \sigma/R$ and measuring Δp at several radii recovers σ with no free parameter. What makes this a test of the capillary stress rather than of the initial condition is that σ is never handled as such anywhere in the code: it enters only through $\kappa = 3\sigma W/2$, a coefficient in a tensor added to the equilibrium's second moment, so the number coming back out is a statement about the whole chain. Far from the interface $\nabla\phi$ vanishes and the capillary stress with it, leaving the total normal stress at ρc_s^2; the jump is therefore read straight off the zeroth moment, averaged over $r < R - 2W$ against $r > R + 2W$, with the interface region excluded from both.

R	Δp (half)	Δp (final)	$\sigma = \Delta p\, R$	rel. err	$\max\lvert\boldsymbol{u}\rvert$
12.0	7.81376×10^{-4}	7.82300×10^{-4}	9.38760×10^{-3}	6.12×10^{-2}	5.05×10^{-5}
16.0	5.81377×10^{-4}	5.83552×10^{-4}	9.33683×10^{-3}	6.63×10^{-2}	5.33×10^{-5}
20.0	4.61264×10^{-4}	4.65468×10^{-4}	9.30935×10^{-3}	6.91×10^{-2}	5.34×10^{-5}
24.0	3.79953×10^{-4}	3.87025×10^{-4}	9.28860×10^{-3}	7.11×10^{-2}	5.65×10^{-5}

Convergence is shown rather than assumed: Δp is reported at half the run and at the end, and if those two disagreed the droplet would still be equilibrating and the fitted σ would mean nothing whatever it happened to equal. The droplet also has to still be the droplet that was asked for. The area-equivalent radius $\sqrt{\sum \phi/\pi}$ comes back at 12.136, 16.102, 20.082 and 24.068 against nominal 12, 16, 20 and 24, and the plateaus hold at ϕ_{in} between 0.999816 and 0.999931 against $\phi_{\text{out}} \sim 5 \times 10^{-6}$, so a jump measured against the nominal radius is being asked the right question. Fitting the slope of Δp against $1/R$ through the origin over all four radii at once gives $\sigma = 9.350098 \times 10^{-3}$ against a prescribed 1.000000×10^{-2}, a relative error of 6.50×10^{-2}, while the spread of $\Delta p\, R$ across the sweep is 1.06×10^{-2}.

The law is asserted tightly, the constant is not. The recovered σ sits 6–7% low at every radius, and that is a discretisation error rather than a wrong coefficient. The two failures are distinguishable: a broken stress destroys the $1/R$ law, whereas a coarse lattice only shifts the constant in front of it, which is exactly what a 1% spread accompanying a 7% deficit says. Holding the droplet's shape fixed — $R/W = 5$ and R/N both constant, so that only the lattice spacing changes — and refining settles it:

N	R	W	σ	rel. err
96	20	4	9.316×10^{-3}	6.84×10^{-2}
144	30	6	9.689×10^{-3}	3.11×10^{-2}
192	40	8	9.837×10^{-3}	1.63×10^{-2}

with fitted orders 1.94 and 2.24, which is the order the scheme is. This ladder is a separate mode of the executable (`-refine`) and carries its own assertion, that the worst order exceed 1.7.

Refining W at fixed R would not measure an order, and tuning it would be worse than useless. At fixed R two errors of opposite sign compete: under-resolution of the profile, $O((h/W)^2)$ and negative, and the finite thickness of the interface against the curvature, $O(W/R)$ and positive. Measured at $R = 28$, the deficit runs -6.51×10^{-2} at $W = 4$, -2.68×10^{-2} at $W = 6$, -9.7×10^{-3} at $W = 8$ and $+1.00 \times 10^{-2}$ at $W = 12$. It passes through zero near $W = 10$, and a width chosen there would look excellent while measuring nothing but the cancellation.

Three operator forms, and the density ratio. Surface tension reaches the flow by two independent routes, and the case runs both. The stress form (`MultiphaseBGK`) puts the capillary tensor into the equilibrium's second moment and carries density in the zeroth, as every single-phase operator in this code base does; it is matched-density only. The potential form (`MultiphasePotentialBGK`) applies $F_s = \mu_\phi \nabla \phi$ as a body force on a distribution whose zeroth moment is a normalised *pressure*, and it is the one that reaches a ratio. The third form is that same potential physics collided in central moments (`MultiphaseCentralMoments`, selected by `-op cm`), which is what §12.2.2 runs by default. At a matched density the first two share no code beyond the phase field that feeds them, so their disagreement bounds both.

form	ρ_H/ρ_L	Δp	σ	rel. err	$\max \lvert \boldsymbol{u} \rvert$
stress	1	4.65468×10^{-4}	9.30935×10^{-3}	6.91×10^{-2}	5.34×10^{-5}
potential	1	4.63803×10^{-4}	9.27607×10^{-3}	7.24×10^{-2}	1.67×10^{-5}
potential	10	4.85739×10^{-4}	9.71478×10^{-3}	2.85×10^{-2}	3.21×10^{-6}
potential	100	4.91735×10^{-4}	9.83470×10^{-3}	1.65×10^{-2}	4.90×10^{-7}

The two forms differ by 3.58×10^{-3} at $R = 20$ with matched densities. Surface tension is a property of the interface, so the recovered value must not move with the density ratio; the total spread over a hundredfold ratio is 5.59×10^{-2}, and the table shows where it comes from.

The measurement gets *better* as the ratio rises, from 7.24×10^{-2} low at a matched density to 1.65×10^{-2} at a ratio of 100, so most of the spread is the matched-density row's ordinary resolution deficit rather than any dependence on the ratio itself. The spurious currents fall by more than an order of magnitude across the same sweep.

The pressure column is worth reading against §10.7. Under the stress form the jump rides on a background of ρc_s^2: $p_{\text{in}} = 0.334116$ against $p_{\text{out}} = 0.333333$ at $R = 12$. Under the potential form, seeded at $p = 0$, the far field sits at -6.8×10^{-5} and the interior at $+3.96 \times 10^{-4}$, so the whole of what is stored is the Laplace jump. Reconstructing that pressure is where a mistake was made and then measured: the potential form stores $\tilde{p}$, and $p = \rho(\phi)c_s^2\tilde{p}$ requires the *local* density under the default normalisation. Using a constant instead turned a 1.7% error at a ratio of 100 into 19.5% with the physics untouched — a diagnostic reading the wrong quantity, not a scheme going wrong. The spread above is likewise normalised by the *prescribed* σ rather than by the smallest measured one, because a measured value returning negative would otherwise flip the sign of the quotient and turn a gross failure into an apparent pass; it did, once.

The phase lattice. `-plat` selects a D2Q5 phase field with a first-order equilibrium or the D2Q9 default with the full second-order one. A static droplet cannot tell them apart in σ — 9.738×10^{-3} against 9.722×10^{-3} at $R = 20$, $W = 6$, since $\boldsymbol{u}$ is zero and the equilibrium's $\boldsymbol{u}$-truncation has nothing to act on — but the spurious currents halve between them, 3.63×10^{-5} to 1.64×10^{-5}. The advection accuracy the second-order form buys needs a moving interface to show, which this case does not have.

Acceptance. Eight criteria, all passing on the run tabulated above:

criterion	tolerance	measured
interface width maintained	1%	2.79×10^{-4}
flat interface drives no flow	10^{-6}	2.22×10^{-9}
$\Delta p\, R$ constant across R	2%	1.06×10^{-2}
σ at this resolution	10%	6.50×10^{-2}
spurious currents	10^{-3}	5.65×10^{-5}
ϕ conserved to round-off	10^{-11}	8.72×10^{-13}
stress and potential forms agree	2%	3.58×10^{-3}
σ independent of ρ_H/ρ_L	8%	5.59×10^{-2}

The tolerance on σ is deliberately loose because `-refine` asserts the order that the absolute value at any one resolution cannot.

12.2.2 Layered Poiseuille flow

`ViscousInterfaceForce` sat in the tree from the moment the pressure form landed, implemented and unexercised: it is identically zero in a static droplet, and the cases that do excite it — Rayleigh–Taylor, water entry — have no exact answer to check against. This is the case that does, and it is registered under `ctest` as `layered_poiseuille`.

Two immiscible layers lie between parallel plates, driven along the channel by a uniform force G per unit volume, fluid 1 (μ_1, ρ_1) below and fluid 2 above, no-slip at $y = \pm h$. The flow is fully developed, $\boldsymbol{u} = u(y)\hat{x}$ with $\partial_x u = 0$, so the nonlinear term vanishes *identically* rather than approximately and the steady profile solves the full Navier–Stokes equations for any pair of densities:

$$u_k(y) = -\frac{Gy^2}{2\mu_k} + a_k y + c, \qquad a_1 = \frac{Gh}{2}\,\frac{\mu_1 - \mu_2}{\mu_1(\mu_1 + \mu_2)}, \qquad a_2 = a_1\frac{\mu_1}{\mu_2}, \qquad c = \frac{Gh^2}{\mu_1 + \mu_2}, \tag{114}$$

from $u(\pm h) = 0$, continuity of u, and continuity of the *shear stress* $\mu\,\mathrm{d}u/\mathrm{d}y$ at the interface. The interface velocity c is the cleanest number in the problem: one closed form, no fitting, and dependent on both viscosities. That the density does not appear anywhere in (114) is the whole point of using it here.

What each ingredient is on the hook for. The interpolation $\mu(\phi)$ owns the kink in $\mathrm{d}u/\mathrm{d}y$: get it wrong and the two parabolas join at the wrong slope ratio. $F_\nu = \nu(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^\top)\cdot\nabla\rho$ is proportional to $\nabla\rho$, so it is active here and only here among the cases with exact answers — a viscosity ratio alone leaves it zero. And $\rho(\phi)$ enters through F_ν and through the forcing prefactor but not through the answer, so a density ratio changes what the code must do while leaving what it must produce fixed. That is the kind of contrast that catches a term rather than a constant.

The driving force is applied through the operator's external force slot as a uniform G, not through the body-force acceleration $\boldsymbol{b}$: $F_b = \rho\boldsymbol{b}$ would drive the two layers unequally and give a different, much less clean exact solution. G is chosen so that c hits a fixed target of 0.02, which holds the Mach number as the grid refines — otherwise a finer grid is also a faster flow and the fitted order picks up the Mach error along with the discretisation. The reference fluid is $\mu_2 = 0.1$, $\rho_2 = 1$; the contrasts are a viscosity ratio of 5 and a density ratio of 10, interface width 4, mobility 0.02, on a channel 16 cells long with solid rows top and bottom.

Two things measured before anything is scored. The field is seeded with (114) rather than with rest. Relaxation to steady state is diffusive and takes $O(H^2/\nu)$ steps, and the first version of this case duly reported an unconverged answer as a converged one; starting from the analytic profile leaves only the discretisation difference to relax. This does not beg the question, because the scheme relaxes to *its own* steady state and the error measured is the distance between that and the exact one — and the steadiness detector, a relative change below 10^{-10} over 500 steps, still has to fire. The single-phase control reaches it at 18 500 and 63 000 steps; every configuration with a contrast runs to the 120 000-step cap.

The interface also does not stay exactly where it is placed. Its $\phi = 1/2$ crossing settles a fraction of a cell off centre while the conserved integral of ϕ is unchanged, and that offset does not shrink with resolution. The exact solution is therefore generalised to an arbitrary interface height s, with $h_1 = h + s$ and $h_2 = h - s$,

$$a_1 = \frac{G\,(h_2^2\mu_1 - h_1^2\mu_2)}{2\mu_1(\mu_2 h_1 + \mu_1 h_2)}, \qquad c = \frac{G h_1^2}{2\mu_1} + a_1 h_1, \tag{115}$$

which reduces to (114) at $s = 0$, and every profile is scored against the interface position it actually has. Comparing against a centreline solution would measure where the interface ended up rather than whether the profile around it is right, and those are separate questions deserving separate answers.

The contrasts, one at a time.

configuration	H	$L_2(u)$	order	u_{iface} err	y_{iface}	$\max\lvert F_p\rvert/G$
single phase	32	2.6751×10^{-4}	—	1.13×10^{-3}	0.1025	0
	64	6.6877×10^{-5}	2.00	2.85×10^{-4}	0.0905	0
viscosity only	32	3.6187×10^{-2}	—	5.02×10^{-2}	0.1093	0
	64	2.3587×10^{-2}	0.62	2.60×10^{-2}	0.0893	0
density only	32	2.5078×10^{-2}	—	3.41×10^{-2}	0.1967	2.20×10^{2}
	64	2.8995×10^{-2}	−0.21	3.80×10^{-2}	0.2992	4.08×10^{2}
both	32	4.2967×10^{-2}	—	1.99×10^{-2}	0.1840	7.33×10^{1}
	64	4.2896×10^{-2}	0.00	1.56×10^{-2}	0.2683	1.36×10^{2}

The first pair is the control and says the base scheme is exact and second order on a problem it should solve exactly. The second is the diffuse interface smoothing a kink in $\mathrm{d}u/\mathrm{d}y$ over W cells, which is first order in W/h and converges accordingly; with `-fine` adding $H = 128$ the column reads 3.6×10^{-2}, 2.4×10^{-2}, 1.3×10^{-2} at orders 0.62 and 0.82, against 2.7×10^{-4}, 6.7×10^{-5}, 1.6×10^{-5} at 2.00 and 2.08 for the control. The third pair should not happen at all. The exact profile does not contain ρ, so a density ratio alone must leave the answer unchanged, and instead it leaves a few per cent that refinement does not remove — 2.5×10^{-2}, 2.9×10^{-2}, 2.4×10^{-2} at $H = 32$, 64, 128, with no order to fit. With both contrasts on, that residual dominates and the total stops converging as well.

Splitting the error by distance from the interface separates a term that lives *at* the interface from one that biases the whole profile, and the two need different explanations. For the control at $H = 64$ the split is 3.343×10^{-5} near against 5.792×10^{-5} far. For the density-only case it starts concentrated, 2.386×10^{-2} near against 7.706×10^{-3} far at $H = 32$, and spreads across the channel under refinement, 2.254×10^{-2} against 1.824×10^{-2} at $H = 64$. That is the fingerprint of a force that must cancel globally rather than of a local stencil error.

It is conditioning, and it is reported rather than asserted. The diagnosis — that F_p scales with the pressure *level*, must cancel against $\rho c_s^2 \nabla \tilde{p}$ to a few hundred times the driving force, and so becomes harder to cancel as the grid refines — is given in §10.7, along with the gauge experiment that confirms it. What this case contributes is the measurement. Under the default local-ρ normalisation the density-only residual runs at order -0.21 with $|F_p|/G$ reaching 2.20×10^2 at $H = 32$ and 4.08×10^2 at $H = 64$; under the constant-ρ_0 normalisation the same configuration converges at order 0.95:

normalisation	H	$L_2(u)$	order	$\max \lvert F_p \rvert / G$
local ρ (default)	32	2.5078×10^{-2}	—	2.20×10^2
	64	2.8995×10^{-2}	-0.21	4.08×10^2
constant ρ_0	32	1.8572×10^{-2}	—	3.79×10^2
	64	9.6381×10^{-3}	0.95	7.47×10^2

Note what does not improve: $|F_p|/G$ is *larger* under the constant reference, 3.79×10^2 against 2.20×10^2 at $H = 32$. The gain is not that the term shrinks but that it stops tracking the pressure level. Neither normalisation dominates and neither is asserted here; the default remains local-ρ, and §10.7 says where each belongs.

Does F_ν earn its place? Switching it off with both contrasts on is the measurement that answers this, and it is the reason the case was written.

H	L_2 with F_ν	L_2 without	ratio
32	4.2967×10^{-2}	4.2837×10^{-1}	9.97
64	4.2896×10^{-2}	5.9350×10^{-1}	13.84

An order of magnitude, growing with resolution, on top of a residual that the pressure force already keeps from converging. The form was checked as well as the presence: the LBE with this equilibrium recovers $\nabla \cdot (\nu S)$, not $(1/\rho)\nabla \cdot (\mu S)$, and the difference is exactly $\nu S \cdot \nabla \rho$, so the implemented expression is the right one. Substituting the plausible alternative $(\nabla \mu) \cdot S$ makes the error larger, 1.8×10^{-1} at $H = 32$; that reading was tried and reverted.

One limitation of the term is not tested here. Its stencil reads $\boldsymbol{u}$ at every neighbour including non-fluid ones, where the velocity field holds whatever the solver left there, so an interface touching a wall takes a one-sided derivative as though it were central. In this case the interface

sits at mid-channel and never meets the plates, so the defect cannot appear — which means the case establishes the term's form and its necessity, not its behaviour at a wall.

Acceptance. Three assertions, all passing: the single-phase control is second order, 2.00 against a threshold of 1.8; F_ν is necessary, the best ratio being 13.8 against a threshold of 3; and ϕ is conserved to 1.19×10^{-11} against a tolerance of 10^{-9}. The non-convergence at a density ratio is printed with the results and deliberately not asserted, because it is a known property of the formulation rather than a regression to be guarded.

12.2.3 A floating body: Archimedes and the metacentre

Water entry has no closed form to check against, which is why it is a demonstrator. Hydrostatics of a floating body has two, both sharp, and neither carries a tunable constant; `validation/floating_body` puts the volume penalisation of §10.8 against both.

The first is the draft. A box of height H and density ρ_b floating between fluids of density ρ_w and ρ_a sits at

$$d = H\,\frac{\rho_b - \rho_a}{\rho_w - \rho_a}, \tag{116}$$

which is Archimedes with the air's buoyancy *kept* rather than dropped. At the density ratio of 50 used here the air carries 2 per cent of the answer, so discarding it would be a larger error than anything measured below.

The second is the stability. A box floats upright only if its metacentric height is positive; in two dimensions, with waterplane second moment $B^3/12$ and displaced area Bd,

$$GM = BM - BG = \frac{B^2}{12\,d} - \frac{H - d}{2}, \tag{117}$$

and the *sign* of that number decides whether a heeled box returns to upright or lies down.

Why the second one is the test worth having. Two boxes are run: a raft with $B = 2H$ and a pillar with $B = H/2$. They have the same area, the same density, the same gravity, the same resolution and the same code path, and differ only in which side is longer — $GM = +13.61$ against -6.80 from (117). A computed quantity has to change sign between them, and no tolerance has to be chosen to notice. Almost any error in the coupling — a sign in the righting moment, the first moments $\boldsymbol{S}$ dropped, the inertia taken from the nominal rectangle rather than the smoothed one — would make both boxes do the same thing, and that is precisely what this pair cannot tolerate.

Both cases use $\rho_w/\rho_a = 50$, $\rho_b = 0.5\,\rho_w$, $g = 2 \times 10^{-4}$, $\nu = 0.02$, an interface width of 4 cells and $H = 32$, on 192×128 (raft, upright) and 128×128 (pillar), with the pressure-form central-moment operator and a D2Q9 phase field. Surface tension is set to exactly zero on purpose: both answers are hydrostatic, the Bond number at these proportions is in the thousands, and capillarity would contribute far less than the discretisation while adding its own spurious currents to the measurement. The interface stays sharp regardless — that is the conservative Allen–Cahn operator's work, and it does not go through κ. Run lengths are set from the linearised roll stiffness $(\rho_w - \rho_a)\,g\,Bd\,|GM|$ against the body's own inertia: 2538 steps to the period for the raft, 1796 for the pillar, where the same number is instead the e-folding time of the capsize. Averages are taken over the second half of each run only, because the tank is closed and periodic in x, so the body's own wake wraps round and comes back; a single instantaneous draft carries that wave and the mean does not.

The wall-sided correction, and a discrepancy that was nearly charged to the solver. The righting couple is read at step 0, where the fluid is still at rest and hydrostatic, so there is nothing in it but the integral of $\chi\rho\,\boldsymbol{r}$ and it can be put straight against metacentric theory. The raft is released at 15°, which is not a small angle: as it heels the emerging and immerging wedges stop being mirror images and the arm grows to

$$GZ = \left(GM + \tfrac{1}{2}BM\tan^2\theta\right)\sin\theta, \qquad T = -(\rho_w - \rho_a)\,g\,B\,d\,GZ. \tag{118}$$

For the raft the quadratic term is +5.7 per cent of the linear arm, and it holds while the deck stays dry and the bilge stays wet — out to 26° at these proportions. Against the textbook small-angle formula the measured couple was about 7 per cent out, of which (118) accounts for most; the correction was applied only after the discrepancy had been found, and without it the remainder would have been charged to the solver. With it the measured −37.070 stands against −36.628, 1.2 per cent. For the pillar at 5° the quadratic term is negligible and the comparison is 1.4205 against 1.4559, 2.4 per cent; the upright box at zero heel returns -6.65×10^{-16}, which is the couple being zero rather than small.

	measured	expected
draft $/H$ (Archimedes)	0.4991	0.4898
waterline slip on the hull $/d$	0.0175	0
righting couple, raft (wall-sided)	−37.070	−36.628
righting couple, pillar (wall-sided)	+1.4205	+1.4559
raft tilt, rms over second half (deg)	1.96	< 7.5
pillar tilt, final (deg)	64.51	> 45

The draft and slip are held to 6 per cent, the two couples to 5. The raft, released at 15°, settles to an rms tilt of 1.96°. The pillar, released at 5°, is at 64.5° after three e-folding times and still turning; its resting state is a raft with B and H exchanged, $GM = +6.8$, but reaching that far equilibrium is a much longer run than a regression test should be, so what is asserted is only that it has left upright for good — past 45°, where the rotated raft's righting arm has taken over — and the final angle is printed for whoever wants to look.

The draft is measured twice, and the pair is the interesting part. The geometric route takes the body's centre height against the free surface, the latter read as the water column height $\int \phi\,dy$ far from the body, which absorbs the level rise the body's own displacement causes in a closed tank. The mass route inverts (116) through $m_f = \int \chi\rho$, the fictitious fluid mass the rigid-body solve actually uses for buoyancy. The second is nearly circular — $m_f = m_b$ is the fixed point of the solve, so on its own its $0.4905\,H$, 0.14 per cent off Archimedes, says only that the solve converged. The *difference* is not circular: at $0.4991\,H$ against $0.4905\,H$ it is 1.8 per cent, and it *bounds* the amount by which the phase field inside the penalised region has drifted away from the free surface outside it. That is the one failure mode this coupling has that nothing else in the suite would catch, because there is no contact-line model: the waterline on the hull is advected with the body rather than pinned to the water, and if it slides then buoyancy is being computed for a body that is not the one on screen. It is a bound and not a measurement, because the same gap also contains the smoothed hull edge, the diffuse interface and the column integral of ϕ, and those do not separate here. What can be said is that the gap is small, and that it is the right quantity to watch if a case ever puts a body somewhere the interface has to move along it.

The unit test. `tests/test_body` runs no flow. It pins down the three things a simulation would hide rather than expose, in 29 checks. First, that the indicator turns: χ is evaluated in the

body frame, so a rotation applied to the wrong side of the transform gives a shape that turns the wrong way, which in a symmetric case looks like nothing at all. The shape used is deliberately not square — a square would rotate onto itself and every check would pass without the transform doing anything — and its area is rotation invariant to 6.1×10^{-6} at 30° and 2.2×10^{-16} at 90°, matching $2h_x \cdot 2h_y$ to 4.8×10^{-5}. Second, that the 3×3 sway–heave–roll system, which is solved in closed form through its Schur complement, actually satisfies its own equations: the solution is substituted back and the largest row residual, scaled by the size of the terms in the row, is zero to round-off at m_b/m_f of 5, 1 and 0.1. The middle one is the case the classical Uhlmann arrangement divides by zero on and the last is the one it gets the sign wrong on, and the check that this is nevertheless the *same* equation is that with the first moments zeroed and rotation locked the closed-form reaction still satisfies $(m_b - m_f)\, d\boldsymbol{U} = \boldsymbol{R} + (m_b - m_f)\boldsymbol{g}$ to 6.0×10^{-17} wherever that relation is defined. Third, that the reported reaction is the force that was written: $\boldsymbol{R}$ and the torque are returned in closed form rather than reduced from the force array, which is worth a full sweep a step and is exact only if the closed form is right, so the test sums the array that was actually populated — in a sheared velocity field over a stratified density, so that all seven integrals are non-zero and nothing cancels by symmetry — and finds R_x and R_y agreeing to 7.1×10^{-15} and 1.4×10^{-14} on a reaction of magnitude 17.469, and the torque to 1.4×10^{-14} on 58.544. Alongside these: a body alone with gravity falls at exactly g and does not start spinning, locking rotation leaves no roll, locking translation leaves the roll row standing alone rather than solving the 3×3 and discarding two rows, the out-of-plane force is identically zero, and no force is left behind on ground the body has vacated.

The second moment is larger than the rectangle's, and must be. The indicator's area matches the sharp rectangle to 5×10^{-5}, but its second moment exceeds it by 2.55 per cent at a smoothing width of 1.5 cells, and the test asserts that excess is positive and below 6 per cent rather than asserting it away. The tanh smoothing moves equal amounts of indicator across each face, and the r^2 weight values what went out more highly than what came in, so a larger second moment is the correct answer. This is the whole reason mass and inertia are taken as integrals of χ rather than from textbook formulae: a body given the nominal rectangle's inertia would roll a few per cent too fast.

FP32. Every acceptance row reproduces to the digits printed above except the waterline slip, 0.0176 against 0.0175, and the mass draft's departure from Archimedes, 0.15 per cent against 0.14. The unit test passes all 29 checks; the 3×3 residuals rise from zero to 7.0×10^{-9}, 1.5×10^{-8} and 5.8×10^{-8} at the three mass ratios, against a tolerance of 2×10^{-5}. Everything algebraic in the test is checked relative to the size of the terms in it — the 3×3 carries masses of order 10^5 against accelerations of order 10^{-4}, so an absolute tolerance would be meaningless in FP64 and unsatisfiable in FP32 — and the Schur complement is nowhere near cancelling for these numbers, $|\boldsymbol{S}|^2/A$ being a few per cent of B, so single precision loses only its own epsilon.

What is not tested. The roll *period*, which would need the added inertia of the entrained water and has no closed form; only the sign of the stiffness and the magnitude of the couple at release are asserted. The draft of a body small enough for surface tension to matter, at Bond number of order one, which this formulation cannot produce because it has no contact line and therefore no surface-tension contribution to the draft at all. Ballast, since the body's centre of mass is taken to be the centre of the rectangle — which is what lets its own weight drop out of the roll equation, and is exactly the standard way of making a tall body float upright that this cannot express. Three dimensions is no longer on that list: the 6×6 with a rotating inertia tensor and a quaternion pose is `RigidBody3D.hpp`, reached by `Box` and `Disc` (§10.8). What this case does not do in 3-D is its own *validation* — see the entry in §20.

12.2.4 A published multiphase campaign: De Rosis and Enan (2021)

De Rosis and Enan [16] is the source paper for this tree's phase-field formulation (§10), and its Sec. III is nine test cases with tabulated numbers. That makes it the same kind of target as §12.1.6: not one benchmark but a campaign, where the pattern across the cases matters more than any single row. Everything below runs the central-moment collision of §10 on both lattices, which is what their own drivers use.

The reference implementation is the authority, not the prose, and this section was rewritten once because of it. The paper states $Pe = U_0\xi/M$. Every one of their drivers computes

$$M = U_{\mathrm{ref}}\, d \,/\, Pe,$$

with d the *domain size*. On their Table III at $Pe = 80$ that is $M = 0.0497$ and $\omega_\phi = 1.540$; the interface-width reading gives $M = 7.5 \times 10^{-4}$ and $\omega_\phi = 1.991$, sixty-six times less mobility and hard against the stability edge. Read that way, BGK diverges on three of the four Zalesak rows and the central-moment operator does not — a difference written up here, in an earlier revision, as the reason that operator exists. It was an artefact of the mobility. **Across every case in the paper their ω_ϕ never exceeds 1.988.**

The same source settles several other points where the text and the code disagree, and the code wins each time: their Tables II, IV and V hardcode $M = 0.001$ rather than deriving it; the interface width is per-case rather than the 3 the text quotes ($\xi = 1.99$ for the Zalesak disk, 2 for Table V, 5 for *both* Rayleigh–Taylor cases); the Rayleigh–Taylor surface tension is $\sigma = \nu_H U/Ca$ with no ρ_H, though $Ca = \mu_H U/\sigma$ implies one; and — the one that mattered most — **the three-dimensional Rayleigh–Taylor reference time is $\sqrt{W/g}$ with no Atwood factor** where the two-dimensional one is $\sqrt{W/(g\,\mathrm{At})}$. Using the 2-D form in 3-D stretches the clock by $1/\sqrt{\mathrm{At}} = 1.41$, so a spike reported at $t/t_0 = 3$ has been run 41% further than theirs.

Two differences remain by choice rather than oversight. Their scheme is D3Q19; this one is D3Q27, because the shifted product basis the collision is written in needs a product lattice (§5.3). That distinction is real only in three dimensions — D3Q19 and D3Q27 have the same $z = 0$ slice, so the two-dimensional cases run on the same effective D2Q9 either way. And their Zalesak slot is a sharp cut where this one is tanh-smoothed.

What is compared, and to which column. The paper computes $\boldsymbol{\nabla}\phi$ by finite differences (its Eq. 20) and from moments (its Eq. 70) and tabulates both. This tree implements the finite-difference route only: Eq. (70) reads the phase field's own populations at the node, and under Esoteric Pull the slots a node reads are the slots it is about to write (§3.2), so the gradient would be taken from a mixture of pre- and post-collision states.

The results, and the one that is not there. Table 7 collects them. Their Table I is *not* run: it is three Peclet numbers at 100 000 steps each, and the campaign was stopped before it completed rather than reporting the earlier numbers, which were taken at a mobility sixty-seven times too small and answer a different question.

Four of the eight rows sit within 21% of the published column and one within 2%. The Zalesak sweep grows from +13% to +41% as ω_ϕ rises toward their ceiling of 1.988, which is a trend rather than scatter. Two rows need separating from the rest.

The translation case is inside the published spread, not outside it. Their Table II collects five methods at that operating point: 0.0074, 0.0134 (theirs, FD), 0.0458 (theirs, moments), 0.0874 and 0.1176 — a factor of sixteen. This scheme's 0.0397 lies between their own two entries, 13% below their moment-based one. An earlier revision of this section quoted

Table 7: Interface capture, their III.A–III.F, at their mobilities. The velocity is prescribed in all of them, so no fluid operator is exercised. Pe is their domain-based definition. Their Tables VI and VII give a D3Q7 column as well, at 0.0754 and 0.1726.

Case	Table	Pe	ω_ϕ	CM	Theirs
Diagonal translation, Geier setup	II	2000	1.988	0.0397	0.0134
Zalesak disk	III	80	1.540	0.0667	0.0593
		400	1.887	0.0707	0.0590
		800	1.942	0.0673	0.0558
		4000	1.988	0.0711	0.0505
Circular interface in shear	IV	3980	1.988	0.0292	0.0244
Circular interface, smoothed shear	V	10220	1.988	0.0195	0.0199
Spherical interface in shear	VI	200	1.887	0.0200	0.0490
Swirling deformation of a sphere	VII	200	1.887	0.1469	0.1133

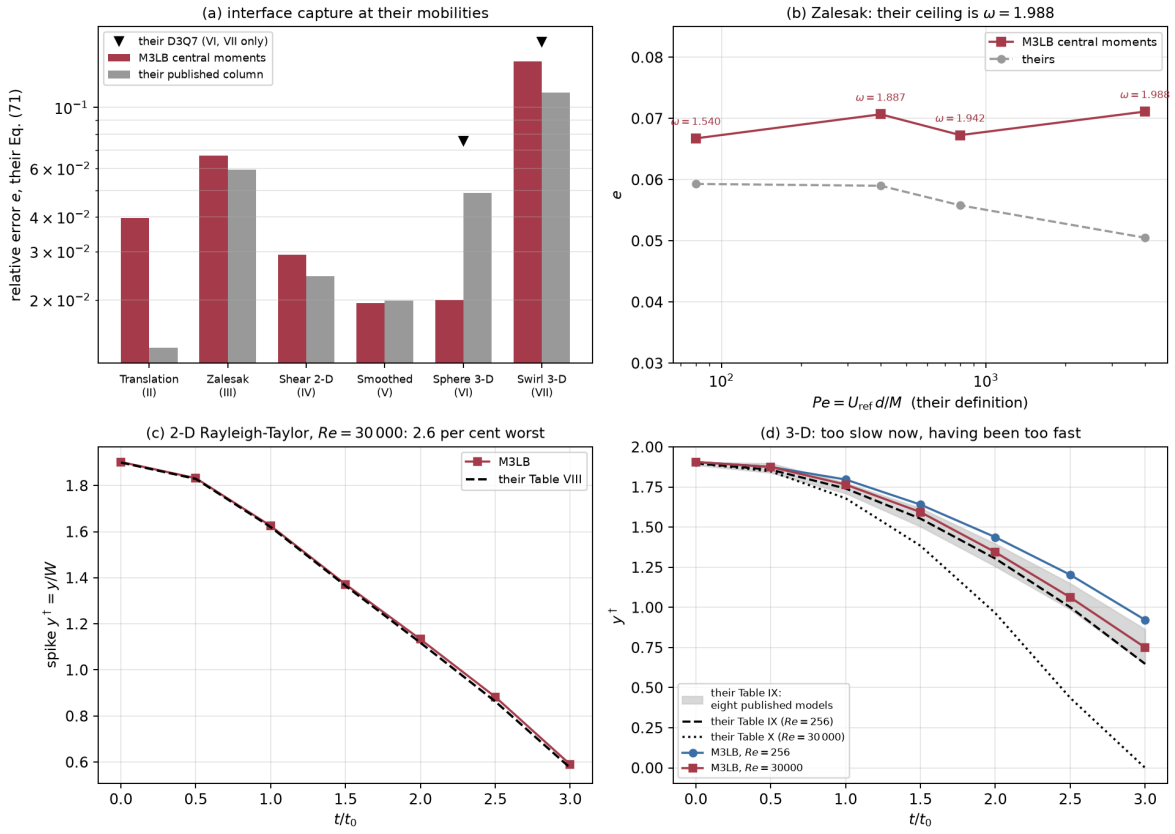


Figure 5: The campaign at their parameters, central moments throughout. (a) interface capture against the published column, with the D3Q7 entries their Tables VI and VII also give. (b) the Zalesak Peclet sweep with ω_ϕ marked: their domain-based mobility puts the ceiling at 1.988, and an earlier revision of this section read Pe as $U_0\xi/M$, ran at 1.9998, and drew conclusions from a regime the paper never visits. (c) the 2-D Rayleigh–Taylor spike against their Table VIII — the pass. (d) the two 3-D cases against Tables IX and X, with the band their Table IX collects from eight published models; both now sit above it, having sat far below it before the reference time was corrected.

it as "+196% against their FD column", which is picking one end of a wide range and calling it a failure.

The two three-dimensional cases disagree with each other. The spherical interface comes in 59% *below* their D3Q19 result and the swirling deformation 30% above it. A richer velocity set explains the first — their own D3Q7 column is worse than their D3Q19 by 35%, and D3Q27 continues that trend — and then fails to explain the second. The swirling case is also the one insensitive to the mobility correction, moving only from 0.1488 to 0.1469, so whatever limits it is not the operating point. It is the most severe deformation in the group, the one that draws the interface into filaments a few cells across, and that is as far as this goes.

Rayleigh–Taylor instability. Their III.H and III.I are the first cases here where the fluid is solved: the phase field is coupled to `MultiphaseCentralMoments` through the interfacial force, and both lattices run central moments. The measured quantity is the vertical position of the spike normalised by the domain width.

Table 8: Rayleigh–Taylor spike position against their Tables VIII, IX and X, at their $\xi = 5$, their $\sigma = \nu_H U/Ca$, their domain-based mobility, and the 3-D reference time $\sqrt{W/g}$. Their values in italics. Table X ends at exactly zero, so that instant is excluded from the worst relative deviation.

Case	Table	$t/t_0 = 0$	$t/t_0 = 0.5$	$t/t_0 = 1$	$t/t_0 = 1.5$	$t/t_0 = 2$	$t/t_0 = 2.5$	$t/t_0 = 3$	V
2-D, $Re = 30000$	VIII	1.902	1.832	1.625	1.371	1.133	0.883	0.590	
theirs		*1.900*	*1.829*	*1.620*	*1.365*	*1.118*	*0.863*	*0.575*	
3-D, $Re = 256$	IX	1.906	1.875	1.797	1.641	1.438	1.203	0.922	4
theirs		*1.898*	*1.858*	*1.741*	*1.553*	*1.304*	*1.001*	*0.648*	
3-D, $Re = 30000$	X	1.906	1.875	1.766	1.594	1.344	1.062	0.750	14
theirs		*1.898*	*1.848*	*1.680*	*1.384*	*0.964*	*0.436*	*0.000*	

Two dimensions: a pass, measured without F_ν. Table 8 and the numbers in this paragraph were produced before 2026-09-19, when `enan_rt` was found to have bound a `ViscousInterfaceForce` into the collision and never refreshed it — so the viscous interface force was identically zero throughout, on a case running At = 0.5, a ratio of three, where §10.4 argues the term is not one to drop. They are reported as they stand and are SUPERSEDED; the fix moved the 2-D spike from 1.3125 to 1.2969 at $t/t_0 = 2$ and $|u|_{\max}$ by 3.1%, so the conclusion survives and the digits do not. The 2-D case agrees with their Table VIII to 2.6% at worst and to under 0.5% through the first three quarters of the run. That is the whole of III.H, and it is the strongest evidence in this section that the coupled multiphase solver is right, because unlike the interface-capture group it exercises the fluid operator, the interfacial force and the coupling order together. It is also insensitive to the corrections above: at the wrong ξ and the wrong mobility it gave 2.7%.

Three dimensions: still wrong, but wrong differently. Both 3-D cases now fall too *slowly* — +42% at $t/t_0 = 3$ on their Table IX, +144% at $t/t_0 = 2.5$ on their Table X — where before the reference-time correction they fell through the floor, reaching $y^\dagger = 0.047$ against a published 0.648. That the sign flipped is itself the evidence that $t_0 = \sqrt{W/g}$ was most of the discrepancy: rescaling the old trajectory onto the corrected clock predicted ≈ 0.88 against their 0.648, and the rerun gave 0.922.

It is still outside the band their Table IX collects from eight published models, from $t/t_0 = 1$ onward.

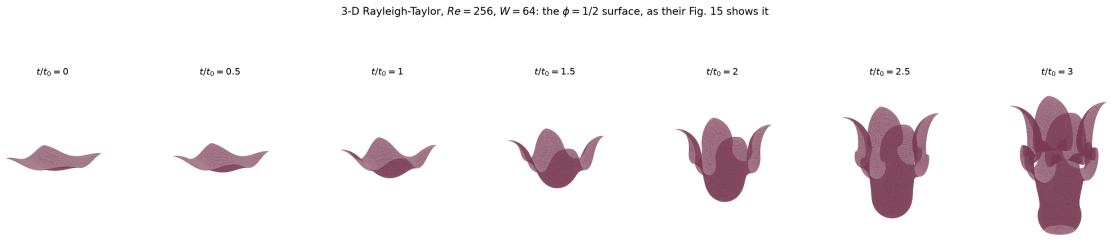


Figure 6: The $\phi = 1/2$ surface of the 3-D case at $Re = 256$, the counterpart of their Fig. 15, extracted by marching cubes from the volumes `validation/enan_rt -field -vol` writes. The morphology is what their text describes — a mushroom-shaped spike, rising bubbles, and four saddle points at the middle of the four sides — which is why the disagreement in Table 8 is read as a rate rather than as a broken force.

A second, independent model says the phase field is not the cause. `validation/enan_rt_cg` runs the identical case with the colour gradient of §10: same D3Q27, same Esoteric Pull, same domain and walls, same perturbation from their Eq. (83), same reference time $\sqrt{W/g}$, same three spike measures. What it shares with the phase field is everything *except* the interface treatment — there is no Allen–Cahn equation, no mobility, no prescribed width, and the interface thickness is an outcome of the recolouring. Their $Pe = 1024$ has no counterpart and is not matched; Re, At and Ca are.

t/t_0	Colour gradient	Phase field	Their Table IX
1.0	1.781	1.797	1.741
1.5	1.641	1.641	1.553
2.0	1.422	1.438	1.304
2.5	1.172	1.203	1.001
3.0	0.891	0.922	0.648
worst	+37.4%	+42.3%	

The two agree with each other to within a cell or two at every instant and miss the published column together. That is a stronger constraint than a null result: it eliminates the entire class of explanations specific to the conservative Allen–Cahn formulation — its source, its mobility, its prescribed width — and points instead at what the two models share. Their Table IX is the right place for this comparison, since three of the eight models it collects are colour-gradient methods.

The isosurface sequence of Fig. 6 says the same thing from a different direction: the morphology is right. The light fluid rises into bubbles, the heavy falls into a mushroom-shaped spike, and the four saddle points at the middle of the four sides that their text describes are all present from $t/t_0 = 1$. A broken interfacial force or coupling order would distort that; instead the instability develops in the correct shape and simply lags.

So the remaining candidates are narrower than they were, and none is settled:

- **The Reynolds number is ambiguous in their own sources.** The paper says $Re = 256$ for Table IX; one of their 3-D drivers sets `Reynolds` = 128 and another 3000. This runs 256. Since the spike falls too slowly and a lower Re is more viscous, their 128 would make it worse and 3000 better, so the ambiguity is not incidental to the disagreement.
- $\xi = 5$ on $W = 64$ is an interface 8% of the domain width.

- D3Q27 against their D3Q19, which is a real difference in three dimensions — though the two interface-capture cases above show it cutting both ways.

Their 3-D initial condition is a sharp step where this one is a tanh of width ξ; both relax to the same equilibrium profile, so that is expected to be a transient rather than a cause, and it is not offered as one.

One observation belongs with the Re ambiguity rather than with the explanations, because it is a coincidence until something accounts for it. The $Re = 30\,000$ run here tracks *their Table IX*, which is $Re = 256$, far better than the nominal $Re = 256$ run does — within 3.1% through $t/t_0 = 2$ and 15.7% at $t/t_0 = 3$, against +42% for the matching label. Before the reference-time correction the same pattern held with the pairing reversed: the $Re = 256$ run then resembled their Table X at $Re = 30\,000$. Two viscosities and two clocks admit more than one way to line up by accident, so this is recorded as a lead and not as a finding.

Vertical penetration of a wedge. Their III.G drives a symmetric wedge of deadrise angle φ into quiescent water at constant speed v and compares the vertical force with Wagner's potential-flow slamming result,

$$F_W(t) = \rho_H \frac{\pi^2 v^2 a(t)}{4\tan\varphi}, \qquad a(t) = \frac{\pi v t}{2\tan\varphi},$$

$a(t)$ being the wetted half-width. Those combine to $F_W = \rho_H \pi^3 v^3 t/(8\tan^2\varphi)$, which is *linear in time* — and that linearity is most of the test, being a statement the simulation can fail in a way no fitted constant hides. Normalising,

$$F^* = \frac{F}{\rho_H v^2 b} = \frac{\pi^3}{8\tan^2\varphi}\,\tau, \qquad \tau = \frac{vt}{b},$$

so the comparison is a dimensionless slope depending on nothing but the deadrise angle. The case is stated in metres and seconds, and *their* discretisation is stated too, so the conversion below follows theirs rather than being chosen here.

Penalisation in place of an immersed boundary. The paper uses a Peskin IBM; this tree has volume penalisation, and `PenalisedBody.hpp` argues the substitution: for a body whose interior moves rigidly it enforces the same condition, keeps the pose continuous rather than lattice-quantised, and needs no refill scheme for uncovered nodes. The wedge is a new indicator on that existing machinery — a signed distance to the two faces with the same tanh smoothing the rectangle uses — and the body was already able to be driven rather than free, which is what Wagner's constant-velocity entry needs.

The one thing that costs is surface accuracy, and on this case it is the leading suspect for what does not reproduce. The product of two tanh indicators rounds a convex corner, and a wedge apex is sharper than a rectangle's; measured as the integral of χ over the nominal area, the penalised wedge is larger than nominal by 0.03% at 45°, 0.23% at 15° and 0.50% at 10° for $b = 100$ and a one-cell smoothing — and by 5.6% at 10° if the body is only $b = 30$. The excess sits at the apex, which for water entry is exactly where the impact starts, so a shallow-deadrise run wants a large body rather than a small one with the smoothing turned down.

Their discretisation, which is stated and is worth following. The semi-wedge is 500 grid points, the lattice is 2000×1000, $v = 1$ m/s and $D = 1$ m, so $\mathrm{d}x = 2$ mm and $\mathrm{d}t = u_{\mathrm{lat}}\,\mathrm{d}x/v = 2 \times 10^{-5}$ s at their $u_{\mathrm{lat}} = 0.01$. The whole run to 6.1 ms is then *305 steps*.

That last number is what makes the rest of it possible. Water at $\nu = 10^{-6}$ m^2/s gives $\nu_{\mathrm{lat}} = 5 \times 10^{-6}$ and $\tau = 0.500015$, which is $Re = 10^6$ — a Reynolds number this document

records elsewhere as out of reach, because the relaxation time sits on the stability floor. It is reachable here for one reason: in 305 steps nothing has time to go unstable. The temptation is to stretch the run instead by lowering u_{lat}, which forces ν_{lat} up and the effective Reynolds number down by orders of magnitude; that is the wrong trade for this case and it was the first one tried here.

Two further details are taken from the same source rather than from the paper's prose, because they differ. The density ratio is **100**, not the 816 of water over air that the text quotes; and the light phase is 15 times more viscous in kinematic terms, $\nu_{\text{air}}/\nu_{\text{water}}$, rather than equally viscous.

The measurement is the surface pressure, not the body reaction. This is the substantive difference between the two ways of doing this case, and getting it wrong costs more than a factor. Their driver sums the nodal pressure at the surface markers over half the wedge; the immersed-boundary reaction is never used for the Wagner comparison at all.

A reaction force carries the added-mass impulse of starting the body. Released with its apex on the undisturbed surface, that impulse put $F^* = 35.8$ at the instant of contact against a Wagner value of zero there — five times the signal, with the force then *decaying* through the window instead of growing. A surface-pressure sum does not see it, which is why their wedge can start on the surface and fall with no release height at all, while a reaction-based version of this case needs one and needs it tuned against the domain's acoustic time.

So the force here is $\int p \, \mathrm{d}s$ over the wetted semi-wedge, with $p = \tilde{p}\, \rho\, c_s^2$ and markers at their spacing of half a cell.

Table 9: Wedge entry against Wagner at their two deadrise angles and their discretisation. The slope is of F^* against $\tau = vt/D$; Wagner's is $\pi^3/(8 \tan^2 \varphi)$. MEASURED BEFORE 2026-09-19, when `enan_wedge` was found to have two couplings wired and never driven: it refreshed ϕ AFTER the fluid stepped, so the fluid collided against $\nabla\phi(t-1)$, and it never refreshed $\boldsymbol{F}_\nu$, so the viscous interface force was identically zero on a case running water over air at a 15× kinematic viscosity contrast. Both are fixed; re-measured, the 2° slope moves 2518.6 → 2517.0 and R^2 improves 0.9904 → 0.9907. Neither touches the deviation from Wagner, which is therefore still unexplained — see below.

φ	Slope	Wagner	Dev.	R^2
2°	2518.6	3178.3	−20.8%	0.9904
3°	1727.1	1411.1	+22.4%	0.9906

Results. Both angles sit within about 22% of Wagner, and the force is linear in time to $R^2 = 0.99$ — which is the half of Wagner's result that a simulation can fail outright, and it does not.

What does not reproduce. The ratio between the two angles is 1.46 where Wagner's $1/\tan^2\varphi$ gives 2.25, so the scaling *with* deadrise is under-reproduced even though both magnitudes are close. The 2° case comes in low and the 3° high.

The candidate is resolution at the apex, and it is a candidate rather than a conclusion. At $b = 500$ the 2° wedge is 17.5 cells tall against 26.2 at 3°, so the shallower angle — the one that reads low — is the less resolved of the two, and volume penalisation represents a surface by a diffuse indicator a cell or so thick where a Peskin immersed boundary puts markers on it exactly. That difference is at its worst on a shallow apex, which is precisely where a slamming pressure lives. Testing it needs a resolution ladder at fixed deadrise, which has not been run.

12.3 Thermal flows

12.3.1 Thermal transport

Two exact solutions of the advection–diffusion equation, with amplitude and Peclet number both scaled diffusively ($U \sim 1/N$) so that the comparison is a convergence study rather than a measurement of how far the wave travelled:

case	order (FP64)	order (FP32)	D_{eff} rel. err	phase err
D2Q5 diffusion	2.00	2.03	1.5×10^{-3}	4.5×10^{-14}
D3Q7 diffusion	2.00	2.00	1.3×10^{-3}	2.8×10^{-14}
D2Q5 advection–diffusion	2.00	2.02	1.1×10^{-3}	2.5×10^{-4}
D3Q7 advection–diffusion	1.99	2.00	7.2×10^{-4}	2.3×10^{-4}

The phase is measured as well as the amplitude: an error in the coupling term puts the wave in the wrong *place*, which an amplitude check alone would not detect.

12.3.2 Open boundaries: the Gaussian plume

The open boundary of §8.8 is validated against the classical plume solution. A continuous point source of strength Q at the origin in a uniform wind U along x, with isotropic diffusivity D, satisfies $U\,\partial_x C = D(\partial_{yy}C + \partial_{zz}C)$ once streamwise diffusion is neglected. That is the two-dimensional diffusion equation with x/U in place of time, so

$$C(x,y,z) = \frac{Q}{4\pi D x}\,\exp\!\left(-\frac{U(y^2+z^2)}{4Dx}\right), \qquad \sigma(x) = \sqrt{\frac{2Dx}{U}}, \tag{119}$$

and integrating across the plume gives $U\iint C\,\mathrm{d}y\,\mathrm{d}z = Q$ at *every* station — which is the property the boundary condition has to preserve.

Two tests are run, and they are deliberately of different kinds.

Uniform-state invariance. A constant field under a constant wind is an exact fixed point of equilibrium, collision, streaming and (69) alike, so this needs no analytic solution and no judgement about what "close" means. On 24^3 with every face open, after 6×10^4 steps:

	$\max\lvert C - \bar{C}\rvert$	$\max\lvert\Delta C\rvert$ over 2000 steps	$\lvert\bar{C} - 1\rvert$
$u = 0$	0	0	0
$u = 0.05$	3.1×10^{-13}	1.4×10^{-15}	6.8×10^{-3}

The first two columns are the criterion and are round-off. The third is a diagnostic, not a failure: with every face zero-gradient the problem is pure Neumann, so no constant is preferred and the level is not determined. The offset is D3Q7's first-order equilibrium, the $O(u^2)$ defect of §5.5, and it scales accordingly — 5.0×10^{-4}, 1.9×10^{-3}, 6.8×10^{-3}, 2.2×10^{-2} at $u = 0.0125$, 0.025, 0.05, 0.1, i.e. as $u^{1.9}$. It is *not* the initialiser laying down equilibrium at zero velocity, which was the obvious first suspicion: attaching the wind only after the field is laid down and settled reproduces the offset to six significant figures. Any problem with a Dirichlet cell in it, the plume below included, does not see this at all.

The plume. On $128 \times 64 \times 64$ with $U = 0.05$, $D = 0.01$ ($\omega = 1.852$, $\mathrm{Pe} = 640$), a source at $x = 8$ and $Q = 1$ per step, $x = 0$ held at $C = 0$ and the five remaining faces open:

x	σ	$\sqrt{2Dx/U}$	err	$C_{\max}$	$Q/4\pi Dx$	flux$/Q$
20	2.370	2.191	+8.2%	5.386×10^{-1}	6.632×10^{-1}	0.99962
44	3.901	3.795	+2.8%	2.048×10^{-1}	2.211×10^{-1}	0.99962
68	4.982	4.899	+1.7%	1.266×10^{-1}	1.326×10^{-1}	0.99962
92	5.867	5.797	+1.2%	9.158×10^{-2}	9.474×10^{-2}	0.99962
122	6.813	6.753	+0.9%	6.807×10^{-2}	6.981×10^{-2}	0.99963

The flux column is the boundary-condition test, and it is flat to five decimal places from $x = 20$ to the last interior cell: worst deviation 0.038%. A reflecting exit would pile concentration up and make it climb; an over-draining one would make it fall. Neither happens, and nothing propagates back upstream. This is the same measurement as the aorta's inlet/outlet balance of §13.2, for the same reason.

The width column tests something the uniform-state test cannot see at all, namely $D = c_s^2(1/\omega - 1/2)$ with $c_s^2 = 1/4$: a wrong speed of sound appears immediately as a slope error. The agreement improves monotonically downstream, from +8.2% at $x = 20$ to +0.9% at $x = 122$, which is the signature of the finite source: (119) is a *point* source and the code releases from a 3^3 box, so the two can only agree once the plume is much wider than the box. The residual near 1% is consistent with the $O(1/\mathrm{Pe}) \approx 0.2\%$ error of neglecting streamwise diffusion in (119) together with D3Q7's $O(u^2)$ equilibrium defect.

12.3.3 Natural convection

Differentially heated square cavity, hot and cold vertical walls, adiabatic horizontal walls, no-slip throughout, against the benchmark of de Vahl Davis [7] at $\mathrm{Pr} = 0.71$. Fixing the characteristic velocity $u_c = \sqrt{g\beta\,\Delta T\,H}$ sets the Mach number, and then $\nu = H u_c \sqrt{\mathrm{Pr}/\mathrm{Ra}}$ follows.

Ra	N	op	Nu	benchmark	dev.	$u_{\max}$	reference
10^3	64	BGK	1.1176	1.118	−0.04%	3.65	3.649
		TRT	1.1179	1.118	−0.01%	3.65	
10^4	64	BGK	2.2431	2.243	+0.00%	16.14	16.178
		TRT	2.2479	2.243	+0.22%	16.14	
10^5	96	BGK	4.5169	4.519	−0.05%	34.47	34.730
		TRT	4.5323	4.519	+0.29%	34.52	

Within 0.3% everywhere. Note that BGK slightly edges TRT here, contrary to the expectation that the magic parameter would help: the wall placement it fixes is a *momentum* boundary, whereas what limits this case is the thermal boundary layer, resolved by the scalar lattice.

12.3.4 Rayleigh–Bénard convection

A layer of depth H heated from below and cooled from above, periodic horizontally, no-slip and isothermal on both plates. This exercises the scalar Dirichlet condition in the configuration it was designed for, and couples it to the buoyancy force of §7.3.

Wall pairing. The scalar uses anti-bounce-back and the fluid halfway bounce-back, so *both* wall planes sit at $y = 0.5$ and $y = H + 0.5$ and the layer is exactly H lattice units deep. Pairing an on-node condition for one field with a midway condition for the other would place the thermal and viscous boundaries half a spacing apart, which corrupts Ra itself rather than merely adding an error. The box is one critical wavelength wide, $2\pi/k_c = 2.0158\,H$.

Three references, not one. Linear stability of a rigid–rigid layer gives $\mathrm{Ra}_c = 1707.762$ at $k_c H = 3.117$, and near onset a growth rate proportional to the supercriticality. All three are now measured (`-conv`, `-marginal`, `-rate`).

Convergence in H. Ra_c deviates by +3.29, +1.33, +0.65 and +0.25 % at $H = 16, 24, 32, 48$ for the midway pair — observed order 2.36 ($R^2 = 0.9996$). The on-node pair (regularised velocity + moment scalar) gives +3.61, +1.73, +1.05, +0.53 %, i.e. order 1.75 ($R^2 = 0.9997$) and twice the error at $H = 48$. That is the curvature-induced wall slip of Eq. (143) showing up as a *convergence* signature rather than only as an offset: a slip proportional to $\partial^2 u_\parallel / \partial n^2$ drags second order toward first. The bisection bracket resolves Ra to 0.68, which is 16% of the $H = 48$ deviation, so that finest point carries that uncertainty.

The preferred wavelength, and why it is the sharper test. Sweeping the box width samples the neutral curve $\mathrm{Ra}(k)$, whose minimum locates k_c independently of any box chosen by hand. Measured: $k_c H = 3.1107$ (−0.20%) at $H = 16$ and 3.1196 (+0.08%) at $H = 32$, against Ra_c errors of +3.27% and +0.65% from the *same* runs — a sixteenfold difference in accuracy at the coarse resolution. The reason is structural: the $O(1/H^2)$ error shifts the whole curve vertically in Ra, and a vertical shift does not move a vertex sideways. So the wavelength is the more discriminating of the two references at modest resolution, which inverts how this section previously used them.

A trap in locating that minimum. The neutral curve is strongly asymmetric — measured $\mathrm{d}^2\mathrm{Ra}/\mathrm{d}k^2$ of 839 on the low-k side against 393 on the high-k side, a ratio of 2.13, reproduced as 2.19 at $H = 16$ — so a parabola fitted across the whole sampled range is the wrong model and places the vertex toward the shallower side. It returned $k_c H = 3.2045$ (+2.81%) where a three-point fit about the minimum gives 3.1196 (+0.08%) and a quartic through all seven gives 3.1085 (−0.27%). The residual uncertainty is therefore the estimator rather than the simulation, and the driver prints the biased global fit beside the local one so the size of the bias stays on the record.

The growth rate. Near onset $\sigma \propto (\mathrm{Ra}/\mathrm{Ra}_c - 1)$, which gives two parameter-free predictions: σ linear in Ra, and a zero crossing at Ra_c. Measured at $H = 32$ over $1650 \le \mathrm{Ra} \le 1800$: $R^2 = 0.999961$, zero crossing 1719.6 against the independently bisected 1718.9 — agreement to 0.04% between two unrelated routes to the same constant. Extending to $\mathrm{Ra} = 2200$ the local slope drifts monotonically from 0.335 to 0.300 per 50 in Ra, the expected departure from the leading-order result. The slope itself, $\mathrm{d}\sigma/\mathrm{d}(\mathrm{Ra}/\mathrm{Ra}_c) = 11.28$ per diffusive time near onset, is reported as a measurement and has *not* been checked against a source. Bracketing only the sign of σ, which the onset mode does, uses one bit of each run by comparison.

Onset. Linear stability of a rigid–rigid layer gives $\mathrm{Ra}_c = 1707.762$ at $k_c H = 3.117$. This is the sharpest single number this configuration can be asked for: the buoyancy coupling, both boundary conditions, the diffusivity calibration and the wall placement all feed into it, and an error in any one moves it. It is measured here by bisecting on the sign of the growth rate of the critical mode.

	bounce-back + anti-bounce-back		regularised + moment	
H	Ra_c	deviation	Ra_c	deviation
16	1764.0	+3.29%	1769.5	+3.61%
24	1730.5	+1.33%	1737.4	+1.73%
32	1718.9	+0.65%	1725.7	+1.05%
48	1712.1	**+0.25**%	1716.8	+0.53%

Both families converge on 1707.762, the midway pair at $O(h^{2.3})$ and the on-node pair at $O(h^{1.6})$.

Measuring the growth rate. The obvious estimator — the total kinetic energy — does not work. Seeded at 10^{-6} it sits near round-off, and the resulting "growth rate" is not even monotonic in Ra, which is impossible for a linear instability; it put Ra_c at 1749 rather than 1719 at $H = 32$. Projecting the vertical velocity onto the critical mode $\sin(kx)\sin(\pi y)$ instead rejects acoustic transients, round-off and the spurious currents of the initial state, and recovers a growth rate that varies smoothly and crosses zero cleanly.

Nusselt number. At $H = 48$, $\mathrm{Pr} = 0.71$. Two independent estimators are reported — the wall gradient and the volume-averaged convective flux $1 + \langle vT\rangle H/(D\,\Delta T)$. They measure different things, so their agreement is a convergence check and not a tautology.

Ra	$\mathrm{Ra}/\mathrm{Ra}_c$	Nu (wall)	Nu (volume)	spread	$\lvert u\rvert_{\max}$
1 500	0.88	1.0000	1.0000	6.0×10^{-13}	0.0000
2 500	1.46	1.4692	1.4690	1.3×10^{-4}	0.0038
5 000	2.93	2.1079	2.1073	3.0×10^{-4}	0.0059
10^4	5.86	2.6490	2.6475	5.7×10^{-4}	0.0073
3×10^4	17.6	3.6144	3.6089	1.5×10^{-3}	0.0090
5×10^4	29.3	4.1523	4.1424	2.4×10^{-3}	0.0096
10^5	58.6	4.9869	4.9659	4.2×10^{-3}	0.0103

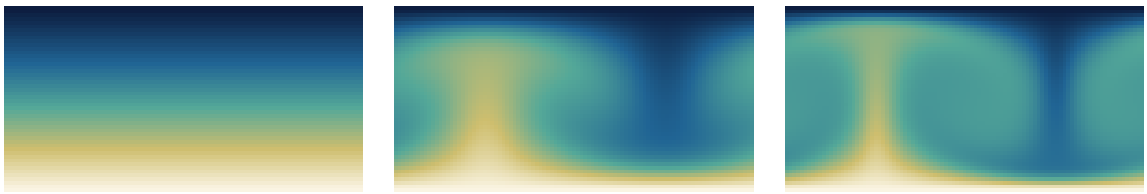
Below onset the result is exact: $Nu = 1.0000$ with the two estimators agreeing to 6×10^{-13} and $\lvert u\rvert_{\max}$ identically zero, so the conductive state is a fixed point of the coupled system to round-off. Above onset the estimators agree to better than 0.1% up to $\mathrm{Ra} = 10^4$ and to 0.42% at 10^5; the growth of that spread is the honest indicator that the top of the table is the least settled part. A power-law fit gives $Nu = 0.132\,\mathrm{Ra}^{0.319}$ overall and an exponent of 0.267 for $\mathrm{Ra} \geq 3 \times 10^4$.

What is not claimed. These Nu have *not* been compared against published tables. Values for two-dimensional rolls depend on aspect ratio and Prandtl number, and no reference set has been verified here, so no deviation column is given. What is verified is internal: the subcritical exactness, the agreement of two independent estimators, and Ra_c itself. Two further caveats apply to the highest rows: the box is one critical wavelength wide, which is the right choice for onset but restricts the roll structure at large Ra, and above $\mathrm{Ra} \sim 10^5$ two-dimensional convection may become time-dependent, in which case a single final-state Nu should be replaced by a time average.

12.3.5 A published thermal suite: Zhou, De Rosis and Revell (2026)

The campaign of §12.1.6 is fluid-only. The tests that follow work through a second published suite, this time a thermal one: Sec. 3 of Zhou, De Rosis and Revell [17], five of its seven cases. Sections 3.3 (a side-open cavity packed with conductive blocks) and 3.7 (Rayleigh–Bénard with a melting boundary) are out of scope — the first needs a porous-media drag model and the second an enthalpy formulation for phase change, and neither exists here.

What can and cannot be reproduced. That paper is about a *coupled* FVM–LBM framework. Its domain is split into an LBM subregion, an FVM subregion and an overlap, and most of its tables measure the coupling: the width of the overlap, the number of LBM sub-iterations per FVM step, the continuity of mass flux across an interface. M3LB has no FVM side, so none of that is testable and none of it is attempted. What is testable is the physics each case was posed to check, against the same external references the paper itself is measured against.



(a) Ra = 1500
$Nu = 1.0000$

(b) Ra = 10^4
$Nu = 2.6490$

(c) Ra = 10^5
$Nu = 4.9869$

Figure 7: Rayleigh–Bénard convection, temperature, $H = 48$, one critical wavelength wide, D2Q9 fluid with a D2Q5 scalar (§12.3.4). The three panels are the same run at three Rayleigh numbers and show what the Nusselt column of that section measures. (a) is *below* the critical $\mathrm{Ra}_c = 1707.762$: the perturbation decays and the field relaxes to pure conduction — a linear vertical gradient, no horizontal structure, and $Nu = 1$ to 6×10^{-13}. (b) is supercritical, and the same perturbation grows into a steady convection cell with a hot plume rising against cold fluid descending. By (c), at $58.6\,\mathrm{Ra}_c$, the thermal boundary layers at the two plates have thinned and the core is nearly isothermal — which is the mechanism behind Nu rising to 4.99, since Nu is precisely the wall gradient normalised by the conductive one. All three share a colour scale.

The comparison is therefore three-cornered, and the three corners are not equivalent. Where a table below carries a column for the paper's own coupled result, it is there for context and not as a target: a monolithic solver pays no coupling error, so being closer to the reference than the coupled framework is the expected outcome and says nothing on its own. The reference column is the one that matters.

Discretisation. The fluid is D3Q27 with the central-moments operator, shifted storage, Esoteric Pull and Guo forcing in the Boussinesq form; the temperature is D3Q7 with the BGK scalar operator. The two-dimensional cases are three-dimensional runs one cell deep with periodic z. For D3Q27 that reduction is the familiar one (§12.4.2). For D3Q7 it is worth spelling out, because it is less obvious: with $n_z = 1$ the $\pm z$ populations stream onto their own node, so they never transport, and since $c_z \cdot u = 0$ they relax to $w_z\,\delta T$ exactly as the rest population does. The seven-speed set collapses to a five-speed one with $w_0 = \frac{1}{4} + 2 \cdot \frac{1}{8} = \frac{1}{2}$ and $w_{x,y} = \frac{1}{8}$, whose second moment is still $2 \cdot \frac{1}{8} = \frac{1}{4} = c_s^2$. A legitimate D2Q5, just not the usual one, and with the same diffusivity calibration — which case 3.4 below confirms rather than assumes.

The five cases are taken out of the paper's order and in two groups: 3.1 and 3.4 first, because both have exact solutions and so measure the scheme rather than a comparison, then 3.2, 3.5 and 3.6, where the answer is somebody else's number. Their headings below carry the paper's own case number in parentheses.

Wall conventions, which are the trap in every coupled case. There are two consistent families, and mixing them puts the viscous and the thermal boundary half a spacing apart and corrupts Ra or Gr without any symptom — the argument of §12.3.4. *Midway*: halfway bounce-back for the fluid and anti-bounce-back for the scalar, both planes at 0.5 and $N - 1.5$, cavity $N - 2$ wide. *On-node*: the regularised velocity condition for the fluid and Dellar's moment condition for the scalar, both planes on nodes 0 and $N - 1$, cavity $N - 1$ wide. Cases 3.2 and 3.6 use the midway family throughout. Case 3.5 has a moving lid, which requires the on-node family for the fluid, so it uses the on-node family for the isothermal walls too; its two adiabatic flanks are the one unavoidable exception, since the scalar module has no on-node zero-flux condition.

12.3.6 A convergence criterion that was wrong by two orders of magnitude

Every case here declares steady state on the whole-field relative change over a probe interval,

$$\mathrm{res} = \frac{\|q(t+\Delta) - q(t)\|_2}{\|q(t+\Delta)\|_2}, \tag{120}$$

taken over u and T together. A per-step residual measures the step size as much as the drift and can be made small by shrinking the step.

It is not enough, and the shortfall was found rather than anticipated. The residual is not the distance to the fixed point. Near steady state the interval change decays geometrically, $\mathrm{res}_{k+1} \approx q\,\mathrm{res}_k$, so the change still to come is

$$\mathrm{drift} = \mathrm{res}\,q + \mathrm{res}\,q^2 + \cdots = \frac{\mathrm{res}\,q}{1-q}, \tag{121}$$

and for a diffusive relaxation with a probe interval far shorter than the diffusive time, q sits within a percent of unity and drift is a hundred times res. Case 3.1 makes the consequence concrete because its exact answer is a straight line: stopping at $\mathrm{res} < 10^{-6}$ left the $\kappa = 1$ profile wrong by $1.9 \times 10^{-4}\%$ and called it converged. Every case below thresholds drift; the same case then returns $1.1 \times 10^{-9}\%$, which is round-off. Both numbers are printed, and $q \geq 1$ falls back to drift = res so that a run still growing can never be declared converged.

12.3.7 Conjugate heat conduction (their 3.1): exact

A layer of length $L = 1$, solid over $0 \leq x \leq 0.2$ and fluid beyond it, held at $\theta = 1$ and $\theta = 0$ at the two ends and periodic elsewhere. With ρc_p uniform — which is what the paper's Eq. (36) assumes, writing the balance in terms of α — the conductivity ratio $\kappa = \lambda_s/\lambda_f$ is also the diffusivity ratio and the steady solution is piecewise linear,

$$\theta = \begin{cases} a_s x + 1, & 0 \leq x \leq L_s, \\ \kappa a_s x + b_f, & L_s < x \leq 1, \end{cases} \qquad a_s = \frac{-1}{L_s + (1-L_s)\kappa}. \tag{122}$$

This is the one case that needed a new capability. `ScalarBGK` carried a single relaxation rate; it now carries an optional field of them (`omega_of`), and `ScalarSolver` asks for the rate per node through `Collision::omega_at(n)`. Nothing imposes the interface condition: continuity of θ and of $\alpha\,\partial_n\theta$ emerges from the streaming, with the effective interface midway between the last node of one material and the first of the other. Whether that is exact or merely second-order was the question.

It is exact. Table 10 reports the relative error of Eq. (38), the two branch slopes fitted by least squares, and the abscissa at which the two fitted lines cross. The slopes reproduce a_s and κa_s to six figures and the crossing lands on 0.2000000 at every ratio from $\kappa = 0.1$ to $\kappa = 100$; driving the tolerance from 10^{-11} to 10^{-13} takes the error from $10^{-9}\%$ to $10^{-10}\%$, so what is being measured is the residual tolerance and not a discretisation floor.

One choice in the table deserves naming because it is a choice and not a property of the problem. The ratio is imposed as $D_s = D_0\sqrt{\kappa}$, $D_f = D_0/\sqrt{\kappa}$, a geometric split about $D_0 = 0.05$. Putting the whole ratio on one side would either drive one τ to the 1/2 floor — and with it the run length to $L^2/D_{\min}$, a hundred times longer at $\kappa = 100$ — or to tens, where the closures whose error grows as $(1/\omega - 1/2)^2$ stop being accurate.

12.3.8 Normal plate velocity (their 3.4): second order, and a hypothesis that died

A uniform stream u_y crosses an isothermal layer, injected at the cold bottom and withdrawn at the hot top, with periodic flanks. The steady solution is the paper's Eq. (44),

$$\theta(Y) = \frac{e^{\mathrm{Pe}\,Y} - 1}{e^{\mathrm{Pe}} - 1}, \qquad \mathrm{Pe} = \mathrm{Re}\,\mathrm{Pr} = \frac{u_y H}{\alpha}, \tag{123}$$

κ	τ_s	τ_f	ϵ [%]	slope$_s$	slope$_f$	x_{int}	Ref. [17] [%]
0.1	0.5632	1.1325	1.5×10^{-9}	-3.571429	-0.357143	0.2000000	1.23
1	0.7000	0.7000	1.1×10^{-9}	-1.000000	-1.000000	—	0.48
10	1.1325	0.5632	2.4×10^{-9}	-0.121951	-1.219512	0.2000000	0.37
100	2.5000	0.5200	1.5×10^{-8}	-0.012469	-1.246883	0.2000000	0.35

Table 10: Conjugate conduction at $L = 200$ cells, drift tolerance 10^{-11}. Exact slopes are $a_s = -1/(0.2 + 0.8\kappa)$ and κa_s; the crossing point is undefined at $\kappa = 1$, where the branches are parallel. The last column is the coupled framework's own error at a finer resolution (1/500).

so the three cases the paper labels $\text{Pr} = 0.1, 1, 10$ at $\text{Re} = 10$ are $\text{Pe} = 1, 10, 100$ here: nothing in the scalar module can see Re and Pr apart from their product. Both wall families are run, because anti-bounce-back is derived assuming no normal velocity through the wall and this case has exactly that, whereas the moment condition makes no such assumption.

Pr	Pe	u_y	ϵ anti-b.b. [%]	ϵ moment [%]	Ref. [17] [%]
0.1	1	3.75×10^{-4}	0.00060	0.000036	0.41
1	10	3.75×10^{-3}	0.0233	0.0151	2.00
10	100	3.75×10^{-2}	2.331	1.971	3.65

Table 11: Normal plate velocity at $H = 200$, the paper's resolution, $\tau = 0.8$ fixed so that u_y follows Pe.

The $\text{Pe} = 100$ entry is two per cent, and the obvious suspect was wrong. This equilibrium is only first order in u (§5.5), so a defect growing as u^2 — and therefore as Pe — looked like the explanation. Re-running $\text{Pe} = 100$ at $\tau = 0.8$, 0.65 and 0.575, a factor of four in u_y and in D at fixed u_y/D, moves the error in no printed digit: 2.3313% three times over. At fixed Pe the steady discrete solution is a function of u_y/D alone, so the whole two per cent is grid resolution of the boundary layer — which at $H = 200$ and $\text{Pe} = 100$ is two cells thick. The refinement ladder confirms it:

H	100	200	400	800	1600	order
ϵ anti-b.b. [%]	9.353	2.331	0.5824	0.1456	0.0364	2.00
ϵ moment [%]	11.267	1.971	0.4218	0.0983	0.0238	2.12

Table 12: Grid convergence at $\text{Pe} = 100$. The order is fitted over $H \geq 200$; at $H = 100$ the boundary layer is one cell and neither family is in its asymptotic range.

12.3.9 Natural convection in a square cavity (their 3.2)

The differentially heated square cavity again, but this time on D3Q27 and D3Q7 rather than the D2Q9 and D2Q5 of §12.3.3, and at the two Rayleigh numbers the paper runs. The reference is de Vahl Davis [7].

Two remarks on the diagnostics before the numbers, because the first version of this case got one of them wrong.

The hot/cold Nusselt difference is not a convergence check. The configuration is invariant under $(x, y) \mapsto (L - x, L - y)$ with $T \mapsto -T$ and $u \mapsto -u$; the boundary conditions, both lattices, bounce-back and anti-bounce-back all respect that map, and the initial state $T = 0$, $u = 0$ is a fixed point of it. The discrete solution is therefore centro-symmetric at every

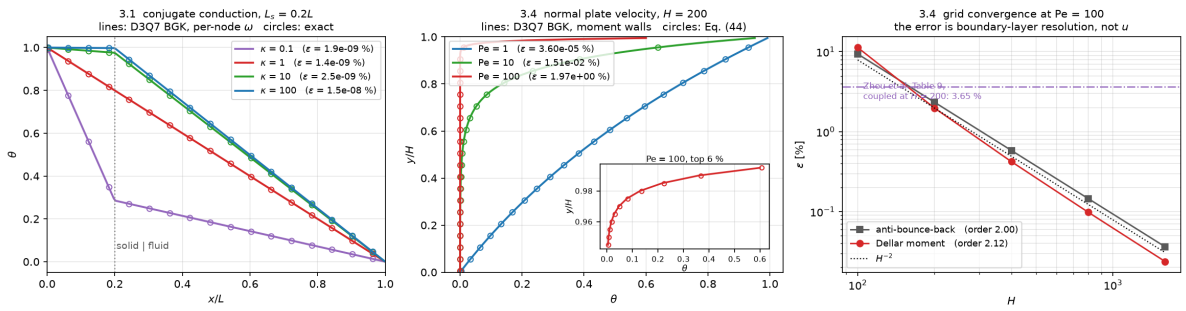


Figure 8: The two cases with exact solutions. Left: conjugate conduction, four conductivity ratios, against the piecewise linear steady state; the kink is the material interface at $x/L = 0.2$. Centre: the normal-plate profile at $\mathrm{Pe} = 1, 10, 100$ against Eq. (44), with the top six per cent inset because at $\mathrm{Pe} = 100$ that is where the entire profile lives. Right: the refinement ladder, second order in both wall families, with the coupled framework's 3.65% at $H = 200$ marked for scale. Rendered by `doc/fig/zhou_analytic.py`.

step, the two wall Nusselt numbers agree identically — the measured imbalance is $10^{-13}\%$, on a run and on a run that has barely begun alike — and their difference says nothing whatever about convergence. It is still worth printing, as a check that the symmetry is intact, but it was originally presented as a conservation test and it is not one.

The independent estimator is the volume one. At steady state the horizontal heat flux through every vertical plane is equal, and averaging that statement over x removes the wall gradient entirely:

$$\mathrm{Nu}_{\mathrm{vol}} = 1 + \frac{\langle u_x \theta \rangle H}{\alpha}, \tag{124}$$

the average taken over the fluid volume, with $\langle u_x \rangle = 0$ in a closed cavity so that the temperature gauge drops out. It shares no arithmetic with the wall stencil. The runs report $\langle u_x \rangle H/\alpha$ alongside it, at 10^{-15} or below, because the identity is worthless if the cavity is not closed to round-off.

Peak velocities are the plain maximum, not the largest magnitude. The two extrema on a centreline are nearly equal and opposite, so an absolute-value search flips sign between resolutions — it did, between $N = 66$ and $N = 98$ — and the reported location then stops matching de Vahl Davis, who tabulates the positive one.

N	H	τ_f	$\mathrm{Nu}_{\mathrm{wall}}$	$\mathrm{Nu}_{\mathrm{vol}}$	spread	dev. from 4.519
50	48	0.5384	4.5147	4.4999	0.330%	−0.10%
66	64	0.5512	4.5177	4.5107	0.156%	−0.03%
98	96	0.5767	4.5196	4.5173	0.051%	+0.01%
130	128	0.6020	4.5203	4.5192	0.023%	+0.03%

Table 13: Natural convection at $\mathrm{Ra} = 10^5$, $\mathrm{Pr} = 0.71$, $u_c = 0.1$. The spread between the two estimators falls by roughly N^{-2} and is the honest discretisation error; the deviation column is measured against de Vahl Davis's Richardson-extrapolated 4.519.

The collision operator, and the paper's Figs. 13–14. That paper's central algorithmic claim is that a central-moments collision keeps driving the residual down where BGK stalls — at 10^{-9} for $\mathrm{Ra} = 10^5$ and 10^{-8} for 10^6 in its Fig. 13. Re-measuring it here needs one thing

quantity	this work	dVD83 [7]	dev.	Luan [18]	Ref. [17]
$\mathrm{Ra} = 10^5$, $N = 130$					
$\overline{\mathrm{Nu}}$	4.5203	4.519	+0.03%	4.507	4.505
$\mathrm{Nu}_{\max}$	7.7019	7.717	−0.20%	7.738	7.742
$(y/L)_{\max}$	0.0820	0.081	—	0.083	0.082
$\mathrm{Nu}_{\min}$	0.7319	0.729	+0.4%	0.746	0.700
$u^*_{\max}$	34.733	34.73	+0.01%	—	—
$v^*_{\max}$	68.654	68.59	+0.09%	—	—
$\mathrm{Ra} = 10^6$, $N = 98$					
$\overline{\mathrm{Nu}}$	8.8151	8.800	+0.17%	8.807	8.803
$u^*_{\max}$	64.846	64.63	+0.33%	—	—
$v^*_{\max}$	219.65	219.36	+0.13%	—	—

Table 14: Natural convection against de Vahl Davis. The extremum locations are reported as $(y-0.5)/H$, which cannot reach 0 or 1: the nearest sample to a corner is half a cell away, which is why both this work and Luan et al. place $(y/L)_{\min}$ near 0.997 rather than at 1.000.

changed and nothing else, and since the temperature field is D3Q7 BGK in both runs, what is being compared is the fluid operator alone.

Ra	op	steps	final residual	$\overline{\mathrm{Nu}}$	dev. from dVD83
10^5	BGK	108 000	1.2×10^{-13}	4.5129	−0.14%
	CM	104 000	1.8×10^{-13}	4.5177	−0.03%
10^6	BGK	228 000	2.1×10^{-13}	8.7894	−0.12%
	CM	228 000	2.8×10^{-13}	8.8022	+0.03%

Table 15: $N = 66$, $u_c = 0.1$, drift tolerance 10^{-13}.

The accuracy half of the claim holds and is not marginal: central moments are four to five times closer to the benchmark at both Rayleigh numbers. The convergence half does not reproduce. Both operators drive the residual smoothly to round-off with no plateau anywhere, in step counts that agree to four per cent, and the two histories are indistinguishable on a log scale. The floor in the paper's Fig. 13 is therefore unlikely to be a property of BGK; the natural candidate is the FVM–LBM exchange, which injects a fixed perturbation every FVM step. One caveat on the comparison: that paper's residual is per step and this one is per 2000-step interval, so the two are not the same number — but a per-step residual of 10^{-9} is four to seven orders above round-off however it is normalised, and nothing here sits above round-off at all.

12.3.10 Thermal lid-driven cavity (their 3.5), and two conclusions drawn too early

A square cavity with the lid sliding right, the other three walls no-slip, the *bottom* hot and the top — the moving lid — cold, and adiabatic flanks. Three groups are independent: $\mathrm{Pr} = 0.71$, $\mathrm{Gr} = 10^6$ and $\mathrm{Ri} = \mathrm{Gr}/\mathrm{Re}^2$, swept over 10, 1 and 0.1, so $\mathrm{Re} = 316.2$, 1000 and 3162.3. Setting $u_c = \sqrt{g\beta\Delta T L}$ fixes $\nu = u_c L/\sqrt{\mathrm{Gr}}$, and then the lid speed is *not* free: $U = u_c/\sqrt{\mathrm{Ri}}$. That is the whole numerical difficulty of the case. At $\mathrm{Ri} = 0.1$ the lid must run 3.16× the buoyancy scale, so u_c has to come down to keep the lid subsonic, and $\tau = \frac{1}{2} + 3u_c L/1000$ comes down with it: $u_c = 0.03$ caps the Mach number at 0.16 and then $L \geq 128$ is needed to clear $\tau = 0.51$. The two constraints pull opposite ways, and the largest u_c each case tolerates is also the cheapest, because $\langle\theta\rangle$ relaxes on $L^2/(\alpha\,\mathrm{Nu})$ and α grows with u_c — so u_c is chosen per Ri (0.15, 0.08, 0.03) rather than held fixed, with a separate control on that choice reported below.

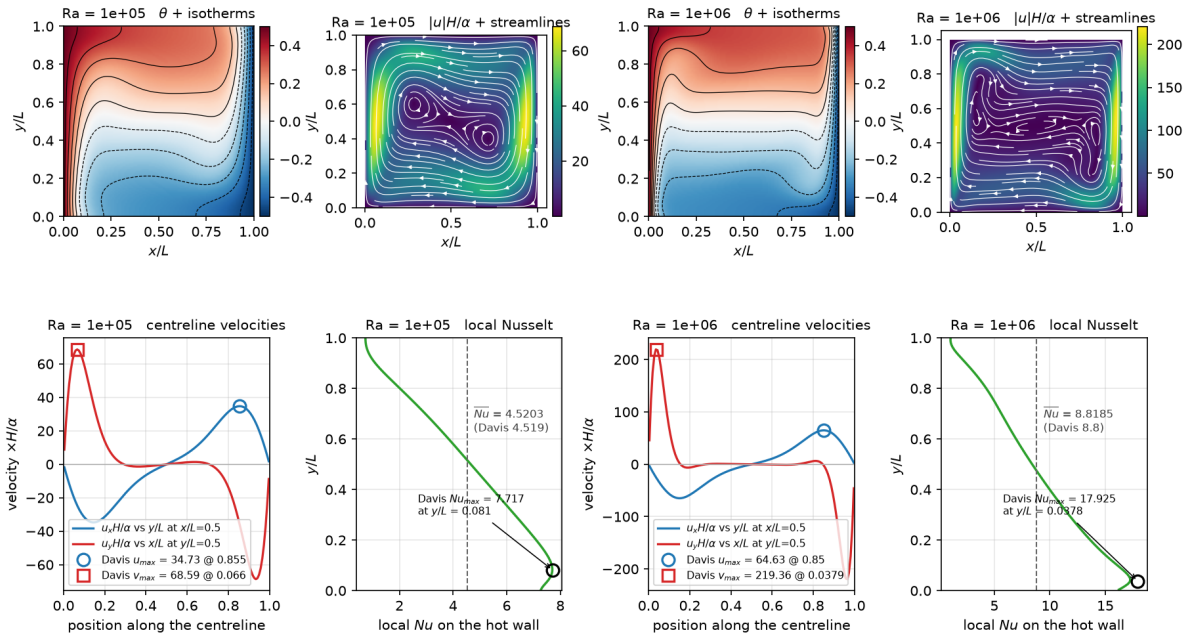


Figure 9: Natural convection in a square cavity, Ra $= 10^5$ and 10^6. Top: the temperature field with isotherms, and the speed with streamlines. Bottom: the two centreline velocity profiles and the local Nusselt number on the hot wall, with de Vahl Davis's tabulated peaks marked — they land on the curves. Rendered by `doc/fig/zhou_natural_convection.py`.

Three things had to be fixed before this case produced a number worth quoting, and they are worth recording because two of them were invisible in the residual.

A body force along both side walls. The moving lid forces the fluid side into the on-node family, so the isothermal walls use the moment condition. The obvious partner for the adiabatic flanks was `ScalarAdiabatic`, and it is wrong here: bounce-back puts the insulated plane at 0.5, which makes the boundary node a ghost outside the fluid, and `ScalarSolver` says so by writing $T = 0$ there. In cases 3.2 and 3.6 that costs nothing, because those nodes are `Solid` for the fluid and never collide. Here they are `RegWall` nodes — fluid nodes that do collide and do get forced — so `BoussinesqGuo` read $T = 0$ against $T_0 = 0.5$ and applied $-\frac{1}{2}g\beta$ down two whole columns. The gauge is now symmetric about zero, which makes a node reporting $T = 0$ neutrally buoyant instead of maximally cold, and `-flank` selects between bounce-back and `ScalarOutflow`, which is on-node and zero-gradient and reports the real temperature. This is the concrete cost of mixing the two wall families, and it is now in `CLAUDE.md`.

The hot/cold Nusselt imbalance is real here, and it was the mean temperature. Unlike case 3.2, the moving lid breaks the centro-symmetry, so the two wall Nusselt numbers are independent and their difference is an honest check. It read -1.5% to -4% and would not go away. It was not a leak at either flank: the domain-mean temperature was still falling, and

$$\frac{\mathrm{d}\langle\theta\rangle}{\mathrm{d}t} = (\mathrm{Nu}_{\mathrm{hot}} - \mathrm{Nu}_{\mathrm{cold}})\frac{\alpha}{L^2} \tag{125}$$

predicted -0.128 against a measured imbalance of -0.111 — fifteen per cent, so the imbalance *is* the drift and nothing is being destroyed at a wall.

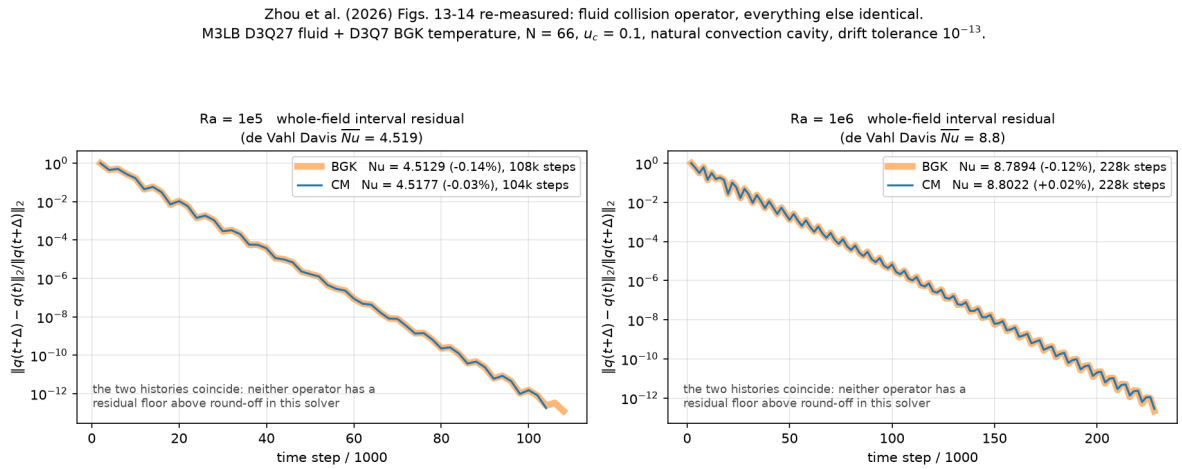


Figure 10: The paper's Figs. 13–14, re-measured. BGK is drawn wide and pale underneath, central moments thin and dark on top, because the two histories coincide — a same-width dashed line simply vanishes under the solid one. Neither operator has a residual floor above round-off in this solver. Rendered by `doc/fig/zhou_operator_residuals.py`.

Which means the convergence criterion had to change. The mean temperature relaxes on $L^2/(\alpha\,\mathrm{Nu}) \sim t_{\mathrm{diff}}/\mathrm{Nu}$, about 10^5 steps at $N = 65$ and growing with N, because α itself grows with N here. Several hundred lid turnovers is nowhere near it: after 190 000 steps $\langle\theta\rangle$ was 0.386 and still moving. And the whole-field residual of §12.3.6 cannot see the problem: it is dominated by the velocity field, which has already settled, so it floors while $\langle\theta\rangle$ is still walking. So this case declares steady state on the heat budget itself: $|\mathrm{Nu}_{\mathrm{hot}} - \mathrm{Nu}_{\mathrm{cold}}|/\mathrm{Nu}$ below a tolerance for three consecutive probes, and not before twenty probes have elapsed — a state that is still *symmetric* balances the budget trivially, which is how a seed with no asymmetry satisfied the test within a few thousand steps before the flow existed. By Eq. (125) the criterion is exactly the statement that $\langle\theta\rangle$ has stopped moving.

A seed that cost 130 000 steps, and two wrong conclusions. The relaxation above is the expensive part, so `-tseed` replaces the conduction profile with a well-mixed core at a chosen mean. It cannot change the answer — the fixed point is unique — but the first pass chose that mean badly: 0.38, taken from a coarse $N = 65$ run that was itself still moving. With the core seeded some 0.08 too cold the hot-wall gradient is inflated, and $\mathrm{Ri} = 10$ read $\mathrm{Nu} \approx 7$ for 130 000 steps, looking quasi-steady, before falling back to 4.7 as $\langle\theta\rangle$ climbed to its real value of 0.463.

Two conclusions were drafted from that state and both were wrong: that the case has no steady state, and that the buoyancy-dominated cases disagree with Cheng & Liu by some 44%. Nothing in the field residual flagged either. The imbalance did, at the 2–4% level, which is the whole reason it is the criterion. The runs below are seeded at the measured mean, use a four-fold tighter tolerance, and average over sixty intervals.

The agreement is close at all three Richardson numbers, and the flow structures are the paper's. Its Fig. 23 describes, at $\mathrm{Ri} = 10$, two counter-rotating vortices stacked vertically with the lower one larger and a secondary vortex in the lower-left corner; at $\mathrm{Ri} = 1$ two of comparable strength with the buoyancy-driven one distorted; at $\mathrm{Ri} = 0.1$ a single primary vortex with three small corner vortices. Fig. 11 has all of it, feature by feature.

The $\mathrm{Ri} = 0.1$ error bar is the one exception worth naming. Its two wall Nusselt numbers differ by 12% (13.05 against 11.60) and that gap is *not* a heat-budget violation: Eq. (125) turns it into a mean-temperature drift a hundred times larger than the 5×10^{-4} actually measured over the window. It is a difference between the two wall-gradient stencils in the thinnest thermal layer of the set — $\mathrm{Nu} = 12$ over $L = 128$ puts it at five cells — one of which sits under a sheared

Ri	Re	u_c	U	τ_f	Ma	$\langle\theta\rangle$	$\overline{\mathrm{Nu}}$	[19]	[20]	[17]	dev.
10	316.2	0.15	0.0474	0.5576	0.098	0.463	4.689 ± 0.021	4.860	4.848	4.588	−3.5%
1	1000.0	0.08	0.0800	0.5307	0.139	0.373	5.749 ± 0.008	5.750	5.739	5.402	−0.0%
0.1	3162.3	0.03	0.0949	0.5115	0.164	0.248	12.323 ± 0.723	12.161	12.138	12.610	+1.3%

Table 16: Thermal lid-driven cavity, $N = 129$, on-node isothermal walls, bounce-back flanks. $\overline{\mathrm{Nu}}$ is the mean of the two wall values over the last sixty probe intervals and the quoted spread is half their gap; the fluctuation within the window is smaller still, 0.05% to 0.5% of the mean, so these are steady states rather than time averages of a wandering one. Deviations are measured against Cheng & Liu.

lid. The mean is quoted with the gap as the uncertainty.

The u_c control did not do its job, and what it found instead. Re-running Ri = 10 at $u_c = 0.03$ rather than 0.15 was meant to show that the answer does not depend on the compressibility knob. It came back at $\overline{\mathrm{Nu}} = 7.439$ with $\langle\theta\rangle = 0.394$, against 4.689 at 0.463 — a different state, with its own balanced heat budget, reached because that run was seeded near 0.394 and stayed there. So the control is inconclusive on u_c; what it demonstrates instead is that this configuration admits more than one steady state and that the seed selects among them. A second reading of the same thing: with the lid stopped altogether, a seed carrying one half-wavelength across the box settles at $\overline{\mathrm{Nu}} = 5.71$ while one carrying two settles at 3.94, both with a balanced budget and $\langle\theta\rangle$ back at 0.500 as the symmetry of the lid-off problem requires. The Ri = 10 row above is the branch the physical evolution actually reached, having drifted there on its own from a colder seed. Anyone using this case should expect branch sensitivity and say which state they are reporting.

12.3.11 Natural convection in a cubic cavity (their 3.6)

The same problem in three dimensions: hot at $x = 0$, cold at $x = L$, the other four walls adiabatic and no-slip. Cases 3.2 and 3.6 share one routine here, because they are the same problem one cell deep and N cells deep, and running them through the same code is what makes the 2-D and 3-D numbers comparable.

The paper gives no table for this case — it compares centreline profiles on the $z = L/2$ plane against Fusegi et al. [22], plotted as symbols in its Figs. 27 and 29. So the reference numbers below are the peaks *read off those plots*, which makes them good to about five per cent and no better. They are used anyway, for two reasons: they come from the source rather than from a remembered table, and the paper's own normalisation $u_{\mathrm{ref}} = \sqrt{g\beta h\Delta T}$ is exactly the u_c these runs are pinned with, so the comparison needs no conversion.

Ra	N (H)	$\mathrm{Nu}_{\mathrm{wall}}$	$\mathrm{Nu}_{\mathrm{vol}}$	u_x/u_{ref} at y/L	u_y/u_{ref} at x/L
10^5	50 (48)	4.3296	4.3154	0.1409 at 0.844	0.2429 at 0.073
	Fusegi via Figs. 27, 29:			∼0.14 at ∼0.85	∼0.22 at ∼0.075
10^6	50 (48)	8.5716	8.4499	0.0809 at 0.865	0.2450 at 0.031
	Fusegi via Figs. 27, 29:			∼0.11 at ∼0.88	∼0.26 at ∼0.055

Table 17: Cubic cavity, Pr = 0.71, $u_c = 0.2$. The hot/cold Nusselt difference is 10^{-14}% here as well — the centro-symmetry of §12.3.9 survives the third dimension — so $\mathrm{Nu}_{\mathrm{vol}}$ is again the independent number.

At $\mathrm{Ra} = 10^5$ the agreement is as good as reading a plot allows, and the two estimators sit 0.33% apart — the same figure the 2-D case reaches at the same H. Note also that the 3-D Nusselt

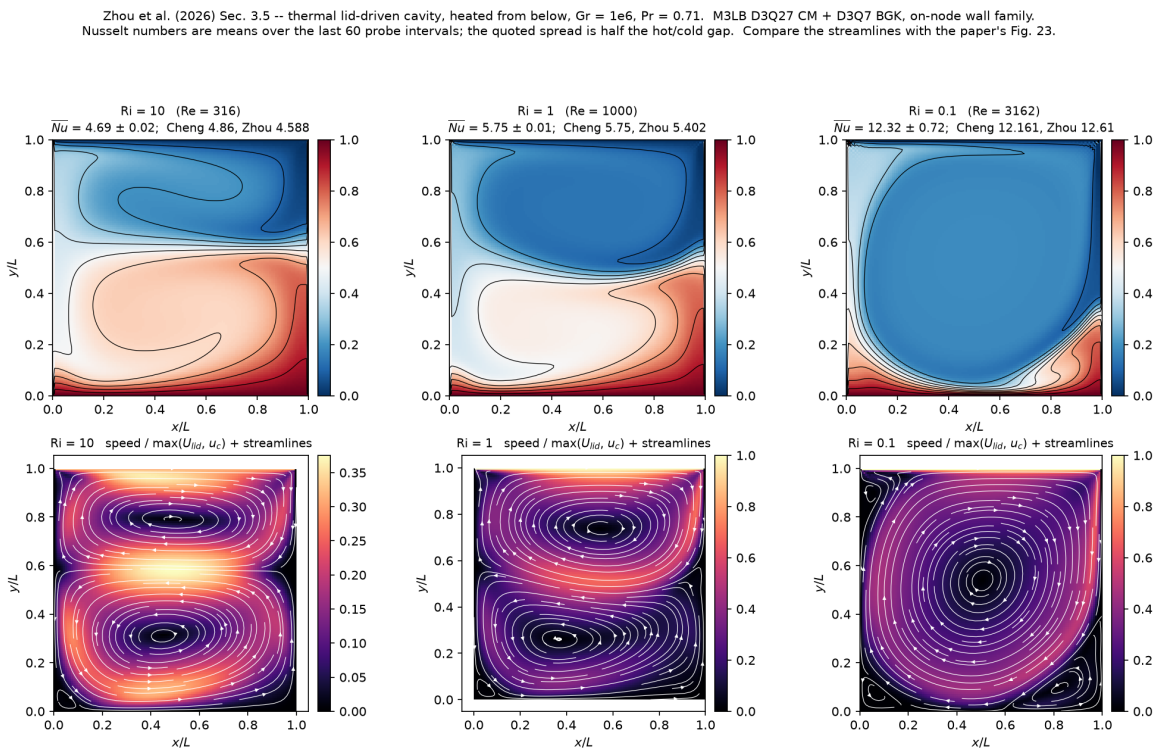


Figure 11: Thermal lid-driven cavity, $\mathrm{Gr} = 10^6$, $\mathrm{Pr} = 0.71$. Top: the temperature field with isotherms at $\Delta\theta = 0.1$. Bottom: the speed with streamlines. Compare with the paper's Fig. 23 — two counter-rotating vortices stacked vertically with the lower one larger and a secondary vortex in the lower-left corner at $\mathrm{Ri} = 10$; two of comparable strength at $\mathrm{Ri} = 1$; a single primary vortex with three corner vortices at $\mathrm{Ri} = 0.1$. Rendered by `doc/fig/zhou_lid_cavity.py`.

number, 4.330, is four per cent *below* the 2-D value 4.520 at the same Rayleigh number, which is the expected effect of the two extra no-slip walls.

At $\mathrm{Ra} = 10^6$ the same resolution is not enough and the diagnostics say so without needing the reference: the two estimators separate to 1.44%, and the u_x peak — which sits at $y/L = 0.865$, inside a boundary layer roughly $H/2\mathrm{Nu} \approx 2.8$ cells thick — comes in 26% low. This is a resolution shortfall, not a model failure, and it is stated rather than presented as a result: $H = 48$ at $\mathrm{Ra} = 10^6$ in three dimensions is under-resolved, and the resolution that would fix it is beyond what this machine can reach in a session.

12.4 Magnetohydrodynamics

The magnetic walls themselves are validated in the boundary-condition survey (§12.1.5), where Hartmann flow appears as the test that exercises Dellar's moment condition against an exact solution; what follows is the physics rather than the boundary.

12.4.1 Exact solutions: resistive decay and the shear Alfvén wave

case	order	recovered	rel. error
resistive decay, L_2	2.00	$\eta = 0.050110$	2.2×10^{-3}
shear Alfvén wave, phase speed	2.00	$v_A = 0.050017$	3.4×10^{-4}
shear Alfvén wave, damping	—	$(\nu + \eta)/2$	5.3×10^{-3}

The shear Alfvén wave is an exact solution of the *full nonlinear* equations, not merely the linearised ones: with $\boldsymbol{u} \perp \boldsymbol{b}_0$ and everything depending only on x, $(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}$ vanishes identically

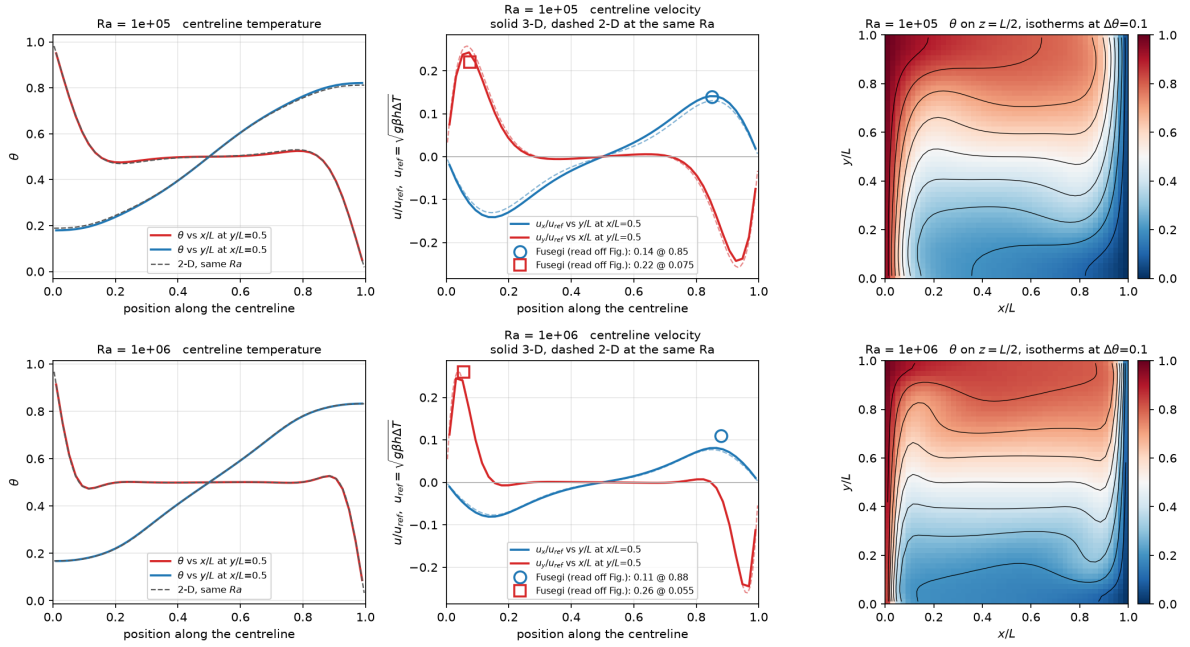


Figure 12: Cubic cavity, the same quantities the paper's Figs. 26–29 plot. Velocities are on the paper's own axis, u/u_{ref} with $u_{\mathrm{ref}} = \sqrt{g\beta h \Delta T}$, which is exactly the u_c these runs are pinned with; the circles and squares are the Fusegi peaks read off those figures and are good to about five per cent. The dashed curves are the 2-D result at the same Rayleigh number. At $\mathrm{Ra} = 10^6$ the u_x peak sitting well below its marker is the resolution shortfall discussed in the text, visible without needing the reference. Rendered by `doc/fig/zhou_natural3d.py`.

while $(\boldsymbol{b}\cdot\nabla)\boldsymbol{b}$ does not, so the Lorentz coupling and the induction equation are both driven and both must be correct. Phase is measured as well as amplitude, because an error in the coupling produces the wrong wave *speed* rather than the wrong amplitude.

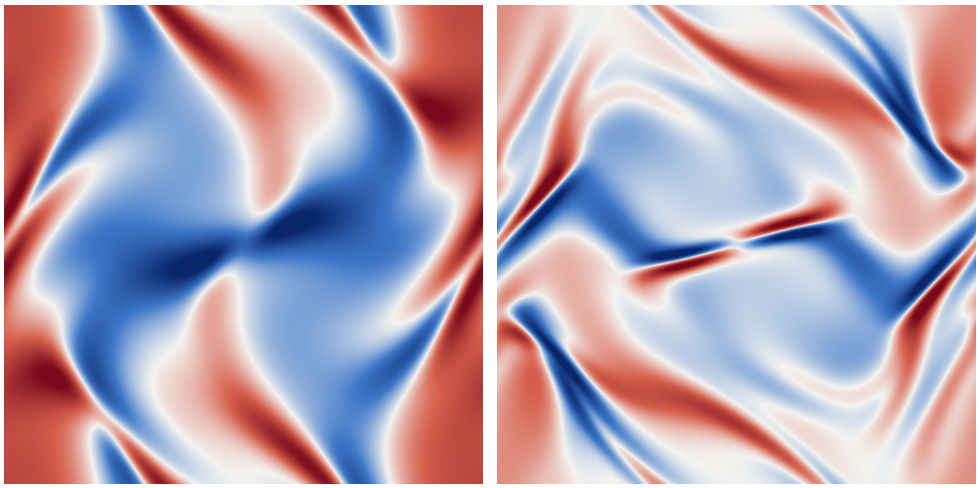
On $\nabla\cdot\boldsymbol{b}$. The two wave cases report $|\nabla\cdot\boldsymbol{b}|$ at round-off, and *this is meaningless*: in both, b_x depends only on y and b_y only on x, so the divergence is structurally zero whatever the scheme does. Orszag–Tang is the only case in the suite that exercises the property, and there $\nabla\cdot\boldsymbol{b}$ is *not* preserved to machine precision: measured against the field's own gradient scale $k|\boldsymbol{b}|$ it sits at truncation level and converges away at order 1.65. That is the honest claim, and it is stated here in place of the stronger one.

12.4.2 Orszag–Tang vortex

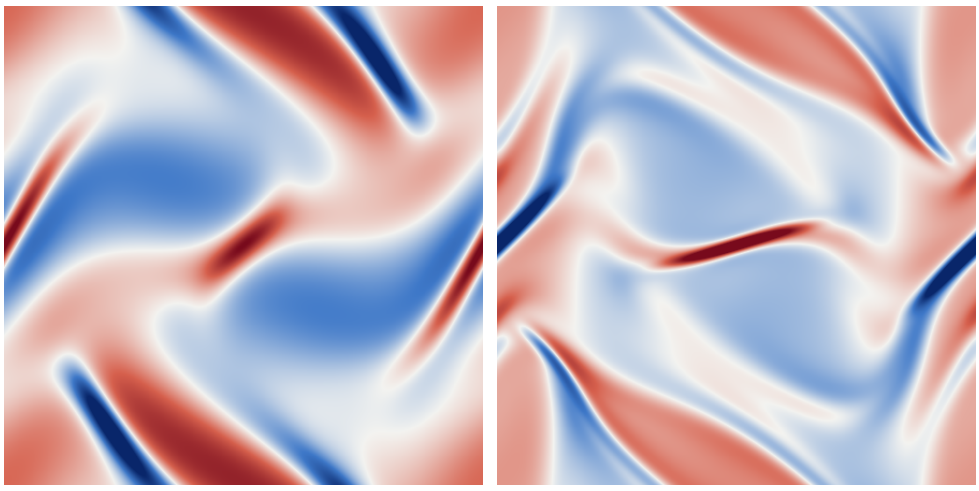
The Orszag–Tang vortex [8] has no closed-form solution, so it is not validated against a formula. It is validated against published reference data and against structural properties. Initial conditions on a periodic box of side L:

$$\boldsymbol{u}(\boldsymbol{x},0) = u_0\big[-\sin y,\ \sin x\big], \qquad \boldsymbol{b}(\boldsymbol{x},0) = b_0\big[-\sin y,\ \sin 2x\big]. \tag{126}$$

Comparison with published data. Reproducing the setup of [6]: $L = 2\pi\,\mathrm{m}$, $u_0 = b_0 = 2$, $\rho = 1\,\mathrm{kg\,m^{-3}}$, $N = 1024$, $\Delta t = 5\times10^{-5}\,\mathrm{s}$. The derived lattice parameters reproduce the published values exactly ($u_{0,\mathrm{lat}} = 1.6297\times10^{-2}$, $\mathrm{Ma} = 0.0282$), with $\mathrm{Re} = u_0 N/\nu$ in lattice



(a) vorticity ζ, $t = 0.5\,\mathrm{s}$ (b) vorticity ζ, $t = 1\,\mathrm{s}$



(c) current j, $t = 0.5\,\mathrm{s}$ (d) current j, $t = 1\,\mathrm{s}$

Figure 13: Orszag–Tang vortex at $\mathrm{Re} \approx 628$, $\mathrm{Pr}_m = 1$, on a 512^2 grid with the hybrid central-moment operator. Diverging colour scale, blue negative, red positive. The scale *saturates at the 99th percentile of* $|v|$ — 6.16 and 11.13 for ζ, 12.88 and 22.67 for j — because the current sheets are strongly localised: at $t = 1$ the peak $|j|$ is 46.4 against a 99th percentile of 22.7, so scaling by the maximum would leave the whole field pale. Between $t = 0.5$ and $t = 1$ the smooth initial vortex steepens into the thin diagonal current sheet visible in panel (d), which is the feature this benchmark exists to resolve and the reason $j_{\max}$ is the more grid-sensitive of the two reported peaks.

units and $\mathrm{Pr}_m = 1$. The reference column is the high-resolution pseudo-spectral data quoted there.

	t (s)	spectral	published LB	here, BGK	here, hybrid CM
$j_{\max}$	0.5	18.24	18.24	18.23	18.23
$j_{\max}$	1.0	46.59	46.65	46.55	46.55
$\zeta_{\max}$	0.5	6.758	6.756	6.756	**6.756**
$\zeta_{\max}$	1.0	14.20	14.18	14.182	**14.180**

Errors against the spectral reference are 0.07/0.09/0.03/0.14 %, where the published LB errors are 0/0.13/0.03/0.14 %: the same accuracy from an independent implementation, with the hybrid scheme landing on the published vorticity values to every quoted digit. Convergence toward the reference is second order, $j_{\max}$ at $t = 1$ going -1.92% ($N = 256$), -0.46% ($N = 512$), -0.09% ($N = 1024$).

The one residual is $j_{\max}$ at $t = 1$, 0.22% below the published value. That is the quantity in which to expect it: the reference computes the current locally from the distributions whereas this code uses second-order central differences, and j is the more gradient-sensitive of the two peaks. Vorticity agreeing exactly while the current is 0.2% low points at the derivative operator rather than at the scheme.

Stability at high Reynolds number. At $\mathrm{Re} = 5000$ the lattice viscosity gives $\tau = 0.510$ on the 1024^2 grid, close enough to 1/2 that the BGK ghost modes go unstable. Runs are terminated at the first non-finite field or at $\mathrm{Ma} > 0.5$.

grid	τ	BGK	hybrid CM	published BGK
512^2	0.505	blow-up $t = 0.38$ s	stable to $t = 1.00$ s	—
1024^2	0.510	blow-up $\boldsymbol{t = 0.54}$ s	stable to $\boldsymbol{t = 1.20}$ s	$t \approx 0.52$ s

The BGK failure reproduces the published behaviour quantitatively — within one probe interval (0.02 s) — and shows the correct resolution dependence. Its signature is instructive: $j_{\max}$ grows smoothly to 24.0 at $t = 0.50$, triples to 73.7 at $t = 0.52$, then overflows, while the Mach number is 0.055 at the last clean sample. This is a collision instability, not a gradual compressibility breakdown.

The hybrid central-moment scheme runs 2.2× longer on the same grid and passes the $t \approx 0.99$ s mark that defeated BGK even on a finer grid in [6]. Its $j_{\max}$ peaks at 166.2 near $t = 1.00$ and then *decays* smoothly to 155.4 by $t = 1.18$, with the Mach number bounded at 0.069 throughout. That turnover is the clearest evidence the run is genuinely resolved rather than merely not-yet-diverged: an unstable run grows monotonically, it does not turn over.

Quasi-two-dimensional reduction: 3D lattices with $n_z = 1$. The Orszag–Tang vortex is a two-dimensional problem, so running it on a 3D lattice with a *single* cell in z is a reduction test: with $n_z = 1$ and periodic z the neighbour wrap sends the out-of-plane neighbour back to the node itself, the corresponding populations stream in place, and the answer must agree with the native 2D lattice. No code change was required — the wrap of §3.3 and the pair swap of Table 6 handle it.

The magnetic field follows the fluid's dimensionality by default (D2Q5 beside D2Q9, D3Q7 beside D3Q27), but since $b_z \equiv 0$ for this flow a 2D magnetic lattice is a legitimate pairing and isolates its contribution. Error against the spectral reference, per cent:

fluid	magnetic	$j_{\max}(0.5)$	$j_{\max}(1)$	$\zeta_{\max}(0.5)$	$\zeta_{\max}(1)$
$N = 256$					
D2Q9	D2Q5	-0.89	-1.92	-0.25	-1.57
D3Q27	**D2Q5**	$\mathbf{-0.89}$	$\mathbf{-1.92}$	$\mathbf{-0.25}$	$\mathbf{-1.57}$
D3Q27	D3Q7	-0.77	-1.66	-0.24	-1.51
$N = 512$					
D2Q9	D2Q5	-0.25	-0.46	-0.07	-0.48
D3Q27	D3Q7	-0.22	-0.39	-0.07	-0.46

Two results, one exact and one asymptotic.

D3Q27 reduces to D2Q9 exactly. Not approximately: the agreement in the highlighted row is structural, and both conditions for it can be checked in exact arithmetic. First, summing the D3Q27 weights over z reproduces the D2Q9 weights term for term,

$$\tfrac{8}{27} + 2 \cdot \tfrac{2}{27} = \tfrac{4}{9}, \qquad \tfrac{2}{27} + 2 \cdot \tfrac{1}{54} = \tfrac{1}{9}, \qquad \tfrac{1}{54} + 2 \cdot \tfrac{1}{216} = \tfrac{1}{36}, \tag{127}$$

so for a z-invariant field the projected equilibrium and the projected BGK update are exactly D2Q9's. Second — and this is the part that is not automatic — the out-of-plane Maxwell-stress contribution cancels within every in-plane group:

$$\sum_{c_z} w_i \left(c_z^2 - c_s^2\right) = 0 \qquad \text{for each } (c_x, c_y) \text{ on D3Q27.} \tag{128}$$

The spread converges away. Between $N = 256$ and $N = 512$ the lattice-to-lattice spread falls from 0.12, 0.26 and 0.06 percentage points (on $j_{\max}(0.5)$, $j_{\max}(1)$, $\zeta_{\max}(1)$) to 0.03, 0.07 and 0.02 — factors of 4.0, 3.7 and 3.0, i.e. second order. The lattice pairings are therefore converging to the same answer, and what separates them at finite resolution is truncation error rather than different physics. Of the two choices, the magnetic lattice matters more than the fluid one: swapping D2Q5 for D3Q7 moves $j_{\max}$ by about 0.25 percentage points at $N = 256$, which is unsurprising given $c_s^2 = 1/3$ against $1/4$ and correspondingly different truncation in the induction equation.

12.4.3 Orszag–Tang vortex in three dimensions

Section III.D and Fig. 6 of [13], with the initial condition from Mininni, Pouquet and Montgomery [11], that paper's Ref. [38]. Cubic periodic box of side 2π on M^3 points, $\delta = 2\pi/M$:

$$\boldsymbol{u}(\boldsymbol{x}, 0) = v_0 \left[-2\sin(\delta y),\ 2\sin(\delta x),\ 0\right], \tag{129}$$

$$\boldsymbol{b}(\boldsymbol{x}, 0) = 0.8\, b_0 \left[-2\sin(2\delta y) + \sin(\delta z),\ 2\sin(\delta x) + \sin(\delta z),\ \sin(\delta x) + \sin(\delta y)\right]. \tag{130}$$

Both are solenoidal: each component depends only on coordinates it is not differentiated by, so every term of the divergence vanishes identically. $\mathrm{Re} = 100$ on the lattice definition $\mathrm{Re} = v_0 M/\nu$, $\mathrm{Pr}_m = 1$, D3Q27 for the fluid and D3Q7 for the field, central moments throughout.

Three things the paper leaves open. Stated here because they change the numbers, not buried in the code.

- *A sign.* Equation (28) as printed has $+2\sin(\delta y)$ in the first velocity component; Ref. [11] has $-2\sin y$, which is what is used here. Unlike the two-dimensional Taylor–Green case in the same paper, the divergence does *not* discriminate — both forms are solenoidal — so this rests on the cited source rather than on a consistency check. It is a genuine ambiguity, not a demonstrated typo.

- v_0. The paper says only that "the values of v_0 and b_0 lead to a Mach number, Ma ~ 0.034", and that depends on which speed Ma is built on. Taken here on the peak initial speed $2\sqrt{2}\,v_0$, giving $v_0 = 6.94 \times 10^{-3}$.
- b_0. Not given; set to v_0, the usual equipartition choice.

What can be checked. Figure 6 is a plot against a high-resolution pseudospectral run and those values are not available numerically, so no per-point comparison is possible. What the figure *does* supply is its axes — panel (a) plots the volume-averaged kinetic energy Γ normalised from 1, panel (b) runs from 5 to 25 in $J_{\max}$ — and the text states that "the current shows a maximum at about $t = 1$".

M	$J_{\max}$ peak	at t	$\Gamma(4)$
32	15.71	1.194	0.1240
64	19.19	1.198	0.1356
128	20.43	1.200	0.1374

Every check the figure permits comes out right. The peak time converges to $t = 1.200$ against the stated $t \approx 1$; $J_{\max}$ reaches 20.4 on the finest grid, inside the 5–25 axis; Γ falls from 1 to 0.137, inside the 0–1 axis; and the curves converge monotonically upward with M, the finer grids resolving the current sheet better, which is the trend the figure reports.

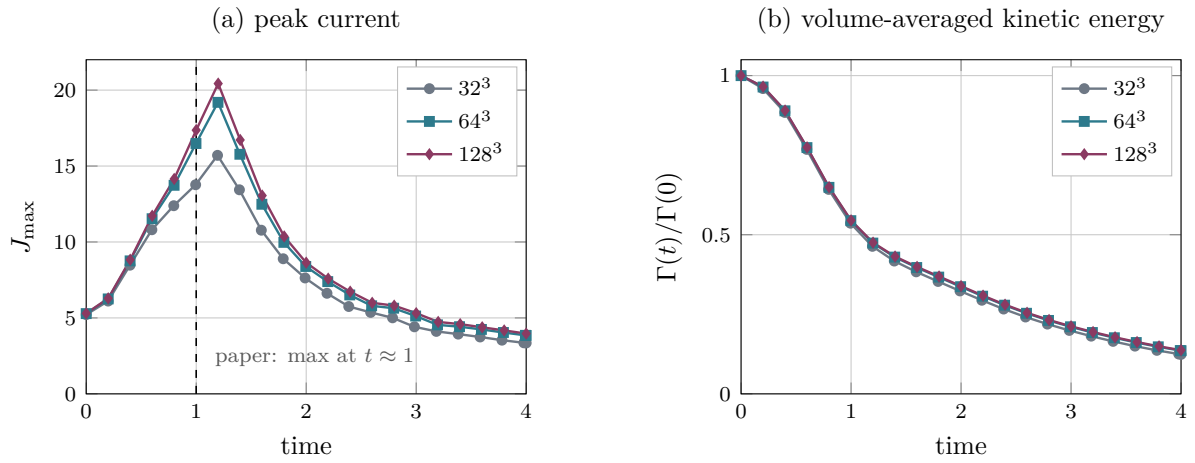


Figure 14: Three-dimensional Orszag–Tang vortex, D3Q27 with central moments, to be read against Fig. 6 of [13]. (a) The peak current magnitude rises with resolution as the current sheet is resolved, and its maximum converges to $t = 1.200$ against the paper's stated $t \approx 1$; the value on the finest grid, 20.4, sits inside that figure's 5–25 axis. (b) The energy, normalised as the paper normalises it, falls from 1 to 0.137. Both panels use a linear time axis; the published figure uses a logarithmic one.

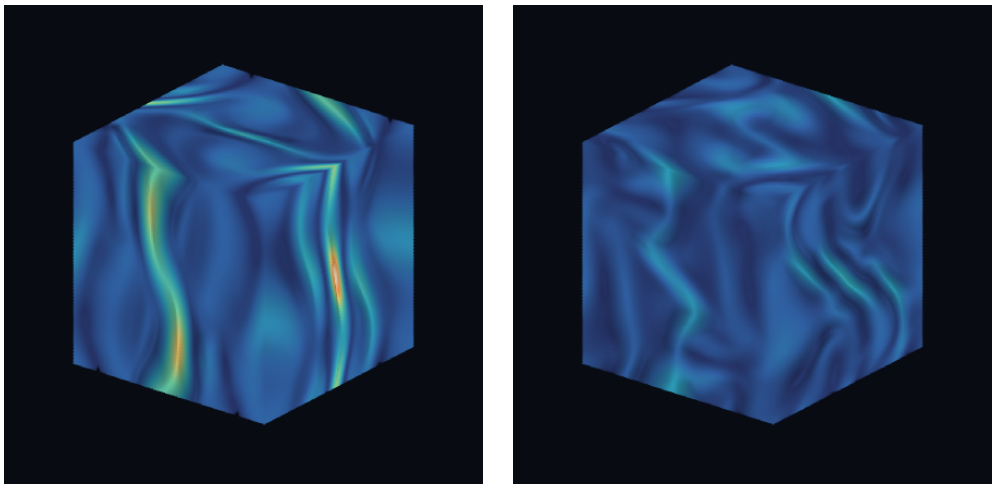
Convergence, and what it does and does not establish. Equation (30) of [13] defines χ as the L_∞ norm of the relative discrepancy against the spectral reference, quoted there as of order 10^{-2} for the finest grid. With no reference values to hand, the same measure is computed against $M = 128$ as a stand-in:

against $M = 128$	Γ	$J_{\max}$
$M = 32$	1.56×10^{-2}	2.31×10^{-1}
$M = 64$	1.9×10^{-3}	6.05×10^{-2}

The successive differences in the $J_{\max}$ peak are 0.171 and 0.061, a ratio of 2.83 — close to second order on a quantity that is a derivative of the field and therefore the hardest thing here to converge. The $M = 64$ figures, 6.1×10^{-2} on $J_{\max}$ and 1.9×10^{-3} on Γ, bracket the 10^{-2} the paper quotes.

This is self-convergence, not agreement. Substituting our own finest grid for the spectral reference flatters systematically, because it cannot see any error common to all three grids. It establishes that the scheme converges and roughly how fast; it does not independently establish that it converges to the right answer. Read the table above as a consistency check, not as a score.

One incidental result worth recording: at $M = 32$ this runs at $\tau = 0.507$, close enough to $1/2$ that BGK ghost modes are normally a problem, and it completes the full $t \in [0, 4]$ without difficulty — consistent with the stability advantage [13] claims for central moments.



(a) $t = 1$, peak current (b) $t = 2$

Figure 15: Three-dimensional Orszag–Tang vortex, current magnitude $|\nabla \times \boldsymbol{b}|$ through the whole 128^3 box, D3Q27 with D3Q7 induction (§12.4.3). Volume rendered rather than sliced: a cut plane shows the sheets only where they happen to intersect it and says nothing about how they are oriented, which for this flow is the point. The current organises into curved sheets wrapping the box, and (a) is the instant the peak is reached — the same $t \approx 1$ the $J_{\max}$ curve of Fig. 14 locates. *Both panels share one colour scale*, set by the $t = 1$ peak, so the visible dimming from (a) to (b) is the real decay: $J_{\max}$ falls from 19.1 to 8.32, less than half, in the same units as Fig. 14. Normalising each panel to its own maximum — the default in most rendering tools — would have made the later, weaker field the brighter of the two. The reference is recomputed at this resolution rather than carried over from the 64^3 volumes: the raw lattice values differ between the two by the factor $\Delta t \propto 1/M$ alone, so reusing the coarse-grid number would have rescaled the pair by about two. Orthographic view, azimuth 38°, elevation 24°; emission–absorption compositing with the weak field left transparent, the local gradient used as a surface normal for a diffuse term so the sheets read as surfaces rather than fog, and 2× supersampling. Rendered by `doc/fig/vol3d.py`, which is self-contained.

12.5 Electrohydrodynamics

Two cases from Patnaik, Skillen and De Rosis [21]: the hydrostatic solution, which has a closed form, and electroconvection, which has a published table and a described flow structure. Throughout, $C = 10$, $M = 10$, $\alpha = 10^{-4}$ and $\beta = 0.3$, the paper's own values.

12.5.1 Hydrostatic injection: an analytic profile

With $\boldsymbol{u} \equiv 0$ by construction only Poisson and charge transport are live, and the steady charge profile is exact [21]: $q(y) = q_0\, a/2\sqrt{y+b}$, with (a, b) tabulated per C. The l_2-norm of the percentage relative error, their Eq. (79):

C	this code, $H = 80$	[21], model C, $H = 320$	ratio
0.1	0.0272 %	0.0203 %	1.34
1	0.2607 %	0.1415 %	1.84
10	1.4889 %	1.1135 %	1.34

Within a factor of two of the published errors at a *quarter* of the resolution.

The half cell is harmless until you differentiate it. This tree's default wall family puts the plate at the halfway plane, and §12.3.4 measures that as costing "1% in H, 3% in Ra" — negligible. It is *not* negligible when the quantity of interest is a derivative of the field carrying the boundary value. Here $\boldsymbol{E} = -\nabla\varphi$, and with halfway plates the one-sided stencil works purely from interior nodes and never sees the imposed φ: measured as a uniform +3.3% bias in the bulk charge at $H = 40$, halving to +1.3% at $H = 80$, i.e. $O(1/H)$, which dragged the whole convergence rate to 1.0. On-node plates (`ScalarMoment`, Dellar's condition) fixed it — $C = 10$ at $H = 80$ went from 11.19% to 1.49%. Ask which family a case needs before defaulting to halfway.

Two defects the case found in its own setup. A *uniform* initial charge makes $C = 10$ diverge: for uniform q the Poisson solution carries a parabolic bump of height $qH^2/8$, which at $C = 10$, $H = 40$ is 1.25 — larger than the applied $\Delta\varphi$. Measured, φ reached 1.19 by $t = 400$, $\boldsymbol{E}$ reversed in the lower layer and q went negative by $t = 800$. $C = 0.1$ and $C = 1$ survived the same initial condition only because the same bump is 100× and 10× smaller. And the error norm must *exclude* the injector node, where q is imposed and its magnitude is the largest in the profile: including it inflated the denominator and moved $C = 10$ from 11.2% to 1.2%, which looked like an improvement in the scheme and was an improvement in the bookkeeping.

What is not reproduced. The published convergence rate is 1.6. This implementation gives 1.57 at $C = 0.1$ but 0.89 and 1.00 at $C = 1$ and $C = 10$, all verified at steady state, and excluding the injector layer entirely cuts the magnitude fivefold while leaving the rate near 1.1 — so it is not a boundary-layer effect. Two candidates, neither yet separated: the potential's BGK source, whose magnitude scales with C and whose $\frac{1}{2}\partial_t G$ correction is what makes it second order, and the charge operator at $\omega_q \to 2$. The profile and the error *magnitudes* are reproduced; the rate is not.

12.5.2 Electroconvection: the coupled case

Both equations of (74) now feed the fluid and vice versa. Their Sec. 3.2.1: aspect ratio $A = 0.614$, the least unstable half-wavelength, at $T = 190$ and $T = 420$.

The side walls, and a measurement that reversed the reasoning. Their Eqs. (14)–(16) put zero-gradient φ and q and $\boldsymbol{u}\cdot\boldsymbol{n} = 0$ on the lateral sides: free-slip, neither periodic nor no-slip. The default here is a *doubled periodic box* — a free-slip box of width L_x is the mirror-symmetric half of a periodic box of width $2L_x$, on whose mirror planes u_x is odd and φ and q are even, which is Eqs. (14)–(16) exactly, and whose smallest admissible wavenumber π/L_x is precisely the most unstable mode.

An earlier version of this section justified that choice by saying the tree had no free-slip wall and could not have one. Both halves were wrong. It can (§8.2), it now does, and `-freeslip` runs the reference's literal boundary on one half-box — at half the cells. What the measurement then showed is that the doubled box is not a substitute but the *better* discretisation, for a reason worth stating precisely: it has no lateral boundary at all, so it never has to discretise one.

n_y	`-freeslip`	doubled	gap	
41	3.7419	3.5995	+3.96%	both converged
81	3.8759	3.7522	+3.30%	both still creeping
163	3.8209	3.7597	+1.63%	both still creeping

The two are exactly equivalent in the continuum, so a gap that large is a discretisation difference and had to be chased. It is not the seed's phase: the doubled solution is mirror-symmetric to 0.000×10^{0} about $x = 0$ and $x = n_{xh}$ and to 3.5×10^{-2} about the half-cell planes, and shifting the seed by half a cell moves the answer 0.09%. It is the lateral scalar wall. 85.7% **of the squared difference lies within three cells of a lateral wall**, in columns that are 24% of the domain, and the free-slip charge saw-tooths there — $0.248, 0.142, 0.205, 0.142, 0.153$ against the doubled box's smooth $0.190, 0.187, 0.176, 0.161, 0.142$. The cell-to-cell mode amplitude is 0.4808 at the wall against 0.0167 for the doubled box: twenty-nine times larger, and ten times its own mid-box level.

The mechanism is $\omega_q \to 2$, arriving through the boundary. $\omega_q = 1.99952$ at these parameters. `ChargeCentralMoments` annihilates the ghost moments in the *bulk* — that is the whole point of its relaxation schedule (§9.3) — but the lateral wall is `ScalarAdiabatic`, i.e. bounce-back, applied outside the collision, and at $\omega \to 2$ bounce-back re-injects an odd–even mode that never damps. It is the same mechanism `ScalarRegularised` exists to fix for the collision at $\mathrm{Ra} = 10^{14}$, reaching the solution through the boundary instead. A fixed-width boundary layer is why the gap falls at roughly *first* order rather than second. Substituting `ScalarOutflow` does not fix it and is kept only to record that: it is an *open* boundary, not a mirror, and it bleeds charge — q/q_0 reaches -0.1652 against -0.0211, an eightfold worse violation of the maximum principle. What this case wants is an on-node zero-flux scalar wall, and a fluid mirror on the same plane to pair it with.

So `-freeslip` is kept as an *independent lateral discretisation*: the gap between the two geometries measures the lateral boundary error the way the D2Q9/D3Q27 gap below measures the interior, and neither geometry alone can show it. The doubled box remains the default because removing a boundary beats discretising it badly.

12.5.3 `-fsnode`: the same wall, on-node

Both halves of what the paragraph above asks for now exist — `ScalarSpecular` (§8.3) and `SpecNode` (§8.4) — and `-fsnode` is the pairing, with every plane on a node so that the lateral family matches the on-node plates instead of sitting half a cell from them. It reduces the gap by about three times and does not close it:

n_y	doubled	`-freeslip`	gap	`-fsnode`	gap
41	3.5995	3.7419	+3.96%	$m = 2$; see below	
81	3.7472	3.8658	+3.17%	3.7270	−0.54%
163	3.7727	3.8311	+1.55%	3.7517	−0.56%

All six runs above are one matched set, taken together at $t_f = 40\,t_0$; the halfway column therefore differs by up to 0.3% from the older table, whose runs were stopped by a different

clock. Both are explicitly still creeping at those two resolutions, so that is the size of the disagreement two unconverged runs of the same case are entitled to. Verified against the pre-`--fsnode` binary: 3.8658 either way at $n_y = 81$, to four figures.

Read $n_y = 163$: there the on-node and doubled runs both *converged* ($t/t_0 = 37.4$ and 36.3) while the halfway run was still creeping. The halfway gap falls at roughly first order; the on-node one is flat at about -0.55% over the single refinement available, and two points are not an order. What is claimable is the factor 2.8 at the finest grid.

At $n_y = 41$ it lands on the wrong branch, and that is not the seed. The on-node box converges to $m = 2$ — two cells per half-box — at $u_{\max}/u_0 = 1.34$, where both other geometries give $m = 1$ at 3.6 to 3.7, and it does so for seeds of 10^{-2}, 10^{-3} and 10^{-4} alike. So it is not the seed kick described above; it is a coarse-grid trap that $n_y = 81$ escapes. A lateral boundary can select the wrong branch of a subcritical bifurcation without failing, without diverging, and while converging cleanly.

`-frozen`, and why the better answer is not a better scalar wall. `-frozen` drops the fluid and the seed, leaving the charge and the potential alone. Their solution is then exactly one-dimensional, so any x-dependence *is* the lateral wall — the doubled box is laterally flat to 0.00% there. Measured at $T = 140$, $C = 10$, as the lateral deviation of q at mid-depth and sampled at matched distance from each geometry's own mirror plane (the halfway grid's first fluid node sits at 0.5, the on-node grid's at 0):

n_y	halfway, peak / rms	on-node, peak / rms
41	14.41% / 4.260%	16.68% / 5.542%
81	4.49% / 1.003%	7.06% / 1.802%
163	1.14% / 0.224%	2.23% / 0.493%
order	1.97 / 2.16	1.66 / 1.87

The on-node *scalar* wall is about twice as large at every resolution. So the better coupled answer above is not coming from it: what `-fsnode` changes that helps is the *fluid* mirror, which finally sits on the plates' plane. Two effects of opposite sign, and only the frozen run can tell them apart.

What `ScalarSpecular` did fix is the ringing, which is not the same thing. The halfway profile saw-tooths into the wall — $0.1200, 0.1034, 0.1012, 0.1027$ — and the on-node one is monotone — $0.1220, 0.1155, 0.1057, 0.1024$. The odd–even mode is gone; a larger *smooth* wall layer took its place, and that layer is itself an $\omega \to 2$ effect. At weak injection, where the field is nearly uniform and a correct zero-flux wall would be nearly exact, sweeping α at $C = 0.5$:

ω_q	1.99952	1.99521	1.95312	1.61290
halfway	0.87%	0.21%	0.10%	0.01%
on-node	9.76%	6.60%	1.97%	0.04%

Ten times worse, and collapsing as ω_q leaves 2. It is the charge's own wall and not the potential's: at $C = 0.05$ the charge bends φ by 0.006% and its wall error is still 12.60%. **The mechanism is not diagnosed.** Two pairings are untested and are the obvious suspects: `validation/scalar_specular.cpp` exercises the wall with `ScalarBGK` only, with a *solenoidal* velocity and no source, at $\omega = 1.99$, whereas the charge here runs `ChargeCentralMoments` with a drift whose divergence is $Kq/\varepsilon > 0$ and the potential carries a source. §12.5.4's specular-vs-periodic check is consistent with all of this and could not have seen it: it is a *volume integral*,

measured at 0.42% (0.05% interior-only) at $N = 41$, $T = 100$, which is what a 10% excess in two columns of forty-one contributes to a volume average. A wall layer hides inside an integral diagnostic.

What the doubled box *admits* that free-slip does not is laterally antisymmetric modes and a continuous family of translations of the symmetric one. So the wavelength is measured rather than assumed: every run reports the dominant lateral wavenumber of u_y at mid-depth and its share of the lateral kinetic energy.

Peak velocity, and the structure behind it.

case	lattice	$u_{\max}/u_0$	mode	vs. [21]	vs. FVM
$T = 190$, $n_y = 81$	D3Q27	3.7413	$m = 1$	+1.12%	+0.03%
$T = 190$, $n_y = 81$	D2Q9	3.7522	$m = 1$	+1.41%	+0.33%
$T = 190$, $n_y = 163$	D2Q9	3.7597	$m = 1$	+1.61%	+0.53%
$T = 420$, $n_y = 81$	D3Q27	4.2804	$m = 2$	−3.60%	−3.16%
$T = 420$, $n_y = 81$	D2Q9	4.4752	$m = 2$	+0.79%	+1.25%
$T = 420$, $n_y = 163$	D2Q9	**4.4322**	$m = 2$	**−0.18**%	+0.28%

Their Table 4 gives, at the converged grid, $T = 190 \to$ FVM 3.74, BGK 3.70, present 3.70; $T = 420 \to$ FVM 4.42, BGK 4.44, present 4.44. Note that the three published methods span 1.1% at $T = 190$, so they do not pin a fourth to better than that.

The mode column is a separate assertion from the velocity, and it is the one that says the box found the right *flow*. At $T = 190$ the dominant wavenumber is $m = 1$: one convection cell per free-slip half-box. At $T = 420$ it is $m = 2$ — four vortex cores in the doubled box, at $x = 11, 39, 61, 89$ of 100, mirror symmetric about both $x = 0$ and $x = 50$, hence two per half-box. Against the paper's own words: "a single, weakly defined vortex-like structure" at $T = 190$ and "a strong, symmetric pair of counter-rotating vortices" at $T = 420$. A peak velocity that matches with the wrong pattern behind it is a coincidence, not a reproduction.

The gap between two lattices is a resolution meter. Running both D2Q9+D2Q5 and D3Q27+D3Q7 costs a factor 3.7 in time and buys something no single lattice provides. At $T = 190$ they agree to 0.29% and the grid barely matters (3.5995 → 3.7522 → 3.7597 at $n_y = 41, 81, 163$, i.e. +4.24% then +0.33%) — with two caveats found while chasing the free-slip gap above: the two finer runs both stopped on the clock still creeping, so that 0.33% separates two values that were each still rising, and a refinement that holds the lateral *alignment* fixed is blind by construction to an error that depends on it. At $T = 420$ the pattern doubles its wavenumber, so each cell spans ~ 25 lattice units instead of ~ 50, and the two lattices diverge to 4.55% — while each converged cleanly and, alone, looked perfectly healthy. That gap is what identified $n_y = 81$ as under-resolved at $T = 420$ *before* any refinement was run; refining then confirmed it, moving D2Q9 from 4.4752 to 4.4322. Two independent discretisations agreeing is evidence; the size of their disagreement is an instrument.

The seed, and why it had to be ruled out. Electroconvection above threshold is an instability, and a laterally uniform layer has nothing to break its symmetry but round-off — which, on a symmetric grid in a deterministic code, may be exactly symmetric. So the charge carries a small non-negative perturbation in the injection layer, non-negative because q must stay in $[0, q_0]$ and an initial condition outside the bound it is judged by is not a starting point. This is a *subcritical* bifurcation, so a finite-amplitude perturbation can in principle select the branch: measured across three decades of seed amplitude at $n_y = 81$, $T = 190$, the converged $u_{\max}/u_0$ is 3.7578, 3.7559, 3.7522, 3.7528 — a spread of 0.149%, non-monotonic, with identical mode content. The seed sets the transient.

It can still throw the run off the branch on the way. At $n_y = 163$ the seed drove $u_{\max}/u_0$ to 5.0 by $t/t_0 = 0.55$, and the flow then collapsed *all the way* to the hydrostatic solution — $q_{\min}/q_0 = 0.0732$ against the analytic 0.0748, so it really was the base state and not a quiet patch. It regrew from there with a pure $m = 1$ mode and converged to 3.7597. The physics found its own way back.

The maximum principle, which is not the trivial one. The charge is not passively advected: $\nabla\cdot(\boldsymbol{u} + K\boldsymbol{E}) = Kq/\varepsilon > 0$, so in non-conservative form the equation carries a reaction term $-Kq^2/\varepsilon$. That term is negative but *vanishes* at $q = 0$, so $q = 0$ remains an invariant lower bound and $q = q_0$ an upper one, exactly as for a passive scalar. Negative charge is therefore the scheme failing and nothing else — the same instrument §12.3.4 uses on T, and for the same reason: it fails long before a norm does. Worst excursion $-0.046\,q_0$ during the injection front at $T = 420$, $n_y = 81$; the $n_y = 163$ runs stay in bounds at both temperatures.

What is not reproduced, and its shape. Refinement moves $T = 420$ *onto* the published value (-0.96% from $n_y = 81$ to 163) and moves $T = 190$ *away* from it ($+0.20\%$), leaving $T = 190$ grid-converged and seed-independent at $+1.6\%$. A single resolution-*independent* offset in the effective T produces exactly that asymmetry: near a bifurcation $u_{\max} \sim (T - T_f)^{1/2}$, so $\delta u/u = \frac{1}{2}\,\delta T/(T - T_f)$ and the same offset is amplified three to nine times more at $T = 190$ than at $T = 420$. The leading candidate is u_0: these runs use 10^{-2} against the paper's 10^{-3}, $\mathrm{Ma}^2 = 0.42\%$, and Mach error does not refine away (§20). Ruled out already is the aspect ratio — $n_x = \mathrm{round}(A\,n_y)$ realises $A = 0.625$ at $n_y = 81$ and 0.617 at $n_y = 163$ against 0.614, but $A = 0.614$ is the *minimum* of the neutral curve, so $\mathrm{d}(\text{growth})/\mathrm{d}A = 0$ there and the effect is second order, 0.03% and 0.003%; it also shrinks under refinement while $u_{\max}$ rises, which is the wrong sign.

Their Table 5 reports 3.33 at 50×81, ten per cent below its own converged value. This case gives 3.75 there, because it is already within 0.4% of converged at that grid — it has no coarse-grid error to match. The plate-placement result above is a plausible cause and an inference about an implementation not visible from here, so it is stated as a candidate. The case scores against Table 4 for that reason, and prints the Table 5 disagreement on every run rather than letting a passing test bury it.

12.5.4 The closed square cavity, and the electric Nusselt number

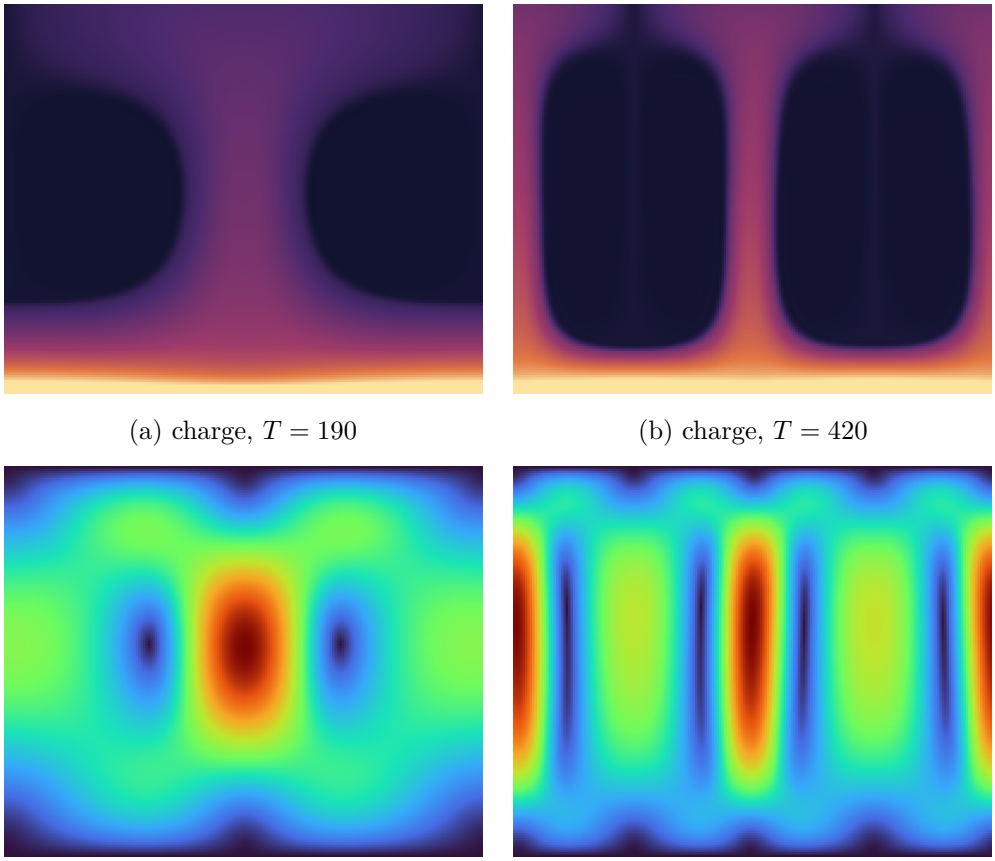
Their Sec. 3.2.3, after Zhang et al.: a square $A = 1$, $L_x \equiv L_y$, with *every* wall closed,

$$\boldsymbol{u} = \boldsymbol{0}, \qquad \partial_x q = 0, \qquad \partial_x \varphi = 0 \qquad \text{at } x = 0,\, L_x. \tag{131}$$

Three things change from §12.5.2, and each changes what the case can be trusted to say.

The side walls are real. §12.5.2 has free-slip sides and is run in a doubled periodic box precisely so that there is no lateral boundary to discretise. Equation (131) is not a mirror — $\boldsymbol{u} = \boldsymbol{0}$ is not $\boldsymbol{u}\cdot\boldsymbol{n} = 0$ — so that escape is closed and the lateral scalar wall must be built. It is the wall of §8.3, and this case is why that section exists: measured here with the *fluid frozen*, so that nothing but the charge/potential pair is in play, the two conditions the solver already had both fail.

lateral scalar wall	$N = 41$	$N = 81$	
none (periodic, not the paper's problem)	$-0.026\,q_0$	$-0.022\,q_0$	reference
`ScalarAdiabatic` (bounce-back)	non-finite	non-finite	inside $0.05\,t_0$
`ScalarOutflow` (open, on-node)	$-0.083\,q_0$	$-0.279\,q_0$	I_0 158 % wrong
`ScalarSpecular` (mirror, on-node)	reproduces the periodic interior		

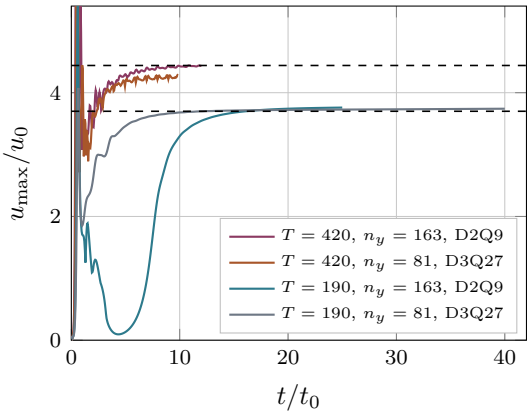


(a) charge, $T = 190$ (b) charge, $T = 420$

(c) speed, $T = 190$ (d) speed, $T = 420$

Figure 16: Electroconvection at $C = 10$, $M = 10$, $A = 0.614$, $n_y = 163$, D2Q9 fluid and charge with D2Q5 potential (§12.5.2). The box is periodic and twice the reference's width; a free-slip wall is the mirror plane of a doubled periodic box, so each panel shows *two* of the reference's cells. Charge is scaled to $[0, 0.5\, q_0]$ and saturates: the injector row is $q = q_0$ by construction, and the dark lobes are the charge-depleted void regions the reference reads as a hallmark of the subcritical bifurcation. Speed is scaled to its own peak, $3.76\, u_0$ in (c) and $4.43\, u_0$ in (d); its dark points are the vortex cores, two in (c) and four in (d). The change of lateral wavenumber between the columns — $m = 1$ to $m = 2$ — is measured, not imposed, and reproduces the reference's description of a single cell at $T = 190$ becoming a counter-rotating pair at $T = 420$.

(a) approach to the steady state



(b) hydrostatic profile against Eq. (78)

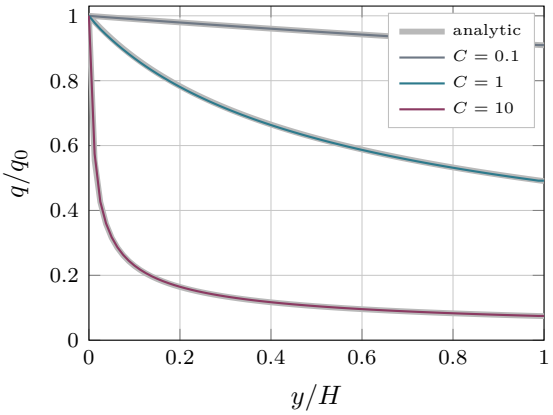


Figure 17: Electrohydrodynamics. (a) Peak velocity against time for the four production runs (§12.5.2); dashed lines are the reference's converged values, 3.70 and 4.44. The $T = 420$ pair separates: D3Q27 converges 4.55% below D2Q9 at $n_y = 81$, and refining brings D2Q9 down onto the published value. All four overshoot near $t/t_0 \approx 1$ before settling. (b) Hydrostatic charge profiles at $H = 80$ against the analytic solution (§12.5.1), thick grey.

The entries are the worst excursion of q outside its bound $[0, q_0]$. Two things are worth reading off. The open boundary gets *worse* under refinement, which is the signature of a boundary condition that is wrong rather than merely inaccurate. And $\omega_q \to 2$ is unavoidable here rather than incidental: α is tied to T (below), so the interesting end of the sweep is the low-diffusion end.

Every wall in this case is therefore on-node — `RegWall` for momentum, `ScalarMoment` for the plates, `ScalarSpecular` for the sides — so it is one wall family throughout and not two mixed.

The diffusion number is tied to T. Their Sec. 3.2 quotes $\alpha = 10^{-4}$ and Sec. 3.2.3 additionally fixes $Sc = 10^3$. Those are the same statement only at $T = 1000$, and the definitions say which is the rule:

$$\alpha = \frac{D}{K\Delta\varphi}, \quad Sc = \frac{\nu}{D}, \quad \nu = \frac{\varepsilon\Delta\varphi}{KT}, \quad \varepsilon = M^2K^2 \implies \alpha = \frac{M^2}{T\,Sc}. \tag{132}$$

$M = 10$, $Sc = 10^3$ gives $\alpha = 0.1/T$, i.e. 10^{-4} at $T = 1000$ exactly. Their Eq. (81) writes the diffusive flux with the coefficient $M^2/(T\,Sc)$ rather than with α, which settles it: Sc is the invariant across the sweep. So α follows T, and $\alpha = 10^{-5}$ at $T = 10^4$ puts ω_q at 1.9994.

The answer is an integral, and its divisor is measured. Their Eqs. (81)–(82),

$$I_e = \frac{1}{V}\int_V \Big[q\,(u_y + KE_y) - \frac{M^2}{T\,Sc}\,\partial_y q\Big]\,\mathrm{d}V, \qquad N_e = \frac{I_e}{I_0}, \tag{133}$$

with I_0 the same integral in the hydrostatic state at the same forcing. Every run here computes its own I_0 by solving the same problem with the Coulomb force switched off and the seed removed, on the same grid, with the same stencils and the same walls. That is deliberate: N_e is a ratio, so discretisation error common to the two states divides out, and what survives is the enhancement due to motion rather than the sum of that and a grid error. Taking I_0 from a formula would put the grid error into N_e undivided.

The formula is still printed, as a *check* on the numerical I_0 rather than as its source. At $D = 0$ the hydrostatic current is uniform, $j = KqE$ with $\mathrm{d}E/\mathrm{d}y = q/\varepsilon$, so

$$E(y)^2 = E_0^2 + \frac{2jy}{K\varepsilon}, \qquad E_0 = \frac{j}{Kq_0}, \qquad \int_0^H E\,\mathrm{d}y = \Delta\varphi, \tag{134}$$

and one bisection in j closes it. The strongest evidence that the pair is consistent is not that agreement, though: it is that below the onset of convection the code returns $N_e = 1.0000 \pm 0.0000$ with the fluid at rest, to four figures, because numerator and denominator are then literally the same computation.

What it measures. $N = 129$, $u_0 = 5 \times 10^{-3}$, forty drift times with the last fifteen averaged, from the reference's own initial condition (zero charge, zero velocity, so the only seed is round-off). Fig. 8's values are digitised from its log–log axes by eye and carry roughly ± 0.05 on the plateau, so nothing here is scored against them.

T	N_e	Fig. 8	diff	Ma	$\mathrm{Re}_{\mathrm{cell}}$
150	1.0000	—	—	0.000	0.0
250	1.0003	1.03	−2.9%	0.001	0.0
500	1.5950	1.75	−8.9%	0.030	0.1
1000	1.6963	1.88	−9.8%	0.048	0.4
1500	1.7505	1.93	−9.3%	0.054	0.7
3000	1.9785	2.35	−15.8%	0.087	2.3
5000	2.5974	2.80	−7.2%	*0.138*	*6.2*
10000	3.3810	3.20	+5.7%	*0.163*	*14.7*

The shape is reproduced: hydrostatic below onset, a steep rise, a shallow plateau, then renewed growth, with $\kappa = +0.35$ over $1500 \to 10^4$ against Fig. 8's own points at 0.27 and its drawn guide at 0.50. The level sits 9–13 % low wherever the grid is trustworthy.

Read the last two rows against the last two columns first. $T = 5000$ and $T = 10^4$ are the only entries that look like agreement, and they are the only two that break this solver's own rules — Ma against a 0.087 guideline, and a cell Reynolds number where the reference's $H = 499$ gives 1.6 and 4.2. At $T = 5000$, N_e is 2.8558 at $N = 81$, 2.8017 at $N = 129$ and 2.4903 at $N = 201$ ($\mathrm{Re}_{\mathrm{cell}} = 8.5$, 6.1, 3.6); unseeded at $N = 201$ it is 2.4366. So the coarse grid was *passing through* the published value rather than reproducing it, which is why the apparent agreement there means nothing. Halving u_0 at $N = 129$ moves 2.8017 to 2.7104, so compressibility is a 3 % term and not the explanation.

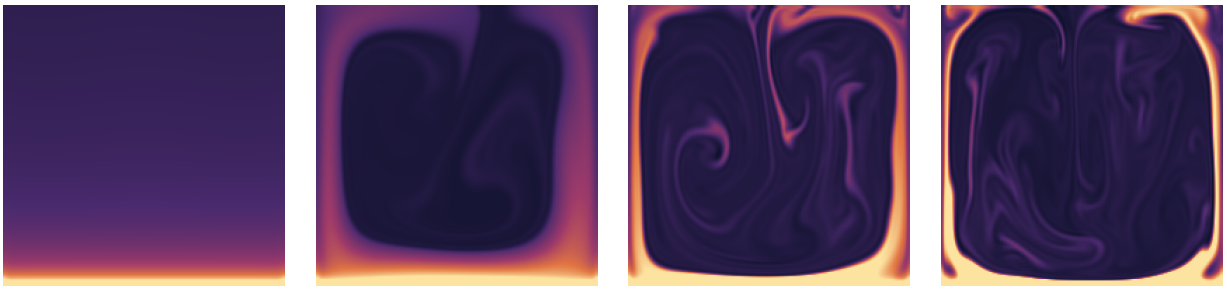
But that ladder must not be extrapolated, and this document did. An earlier version concluded from those three grids that refinement moves *away* from the reference and that the converged deficit is ∼13 % low. All three are under-resolved. §16.13 has since run the reference's own 500^2 on a T4 at $\mathrm{Re}_{\mathrm{cell}} = 1.5$ — the first properly resolved point in either implementation — and it gives $N_e = 3.1010 \pm 0.4139$, i.e. 10.8 % *high*. The sequence is not monotonic through to the resolved grid, and three points that never reach a regime cannot be extrapolated into it. At that grid the reference's 2.80 lies *inside* the run's own r.m.s., which makes it consistent with the reference rather than converged to it: the r.m.s. is 13 % of the mean over three drift times. What survives is the narrower statement — when the runs that agree are the only ones breaking your own Mach and cell-Reynolds rules, distrust the agreement.

The seed chooses a branch, and the usual check cannot see it. This solver's standing defence of a seed is that it sets the transient and not the answer, tested by changing its amplitude and watching the converged value hold. In a *subcritical* bifurcation that test passes and still misleads:

$T = 250$	amp 10^{-2}	amp 10^{-4}	amp 0	Fig. 8
N_e	1.3922	1.2400	1.0003	1.03
$u_{\max}/u_0$	2.000	1.990	0.078	—

Two decades of seed give the same saturated amplitude — exactly what seed-independence looks like — and no seed at all gives no convection. The seed was not setting the amplitude, it was deciding which branch the flow reached, and the reference starts from exactly zero. At $T = 500$ the unseeded run is also *steady* (r.m.s. 0.0007 against 0.107) and closer to the reference, 1.6583 against 1.75: there the seed was selecting a different and worse attractor rather than merely triggering the instability. The same bistability is what the reference's own §3.2.1 hysteresis loop is about.

The fields. Animations of q/q_0 at $N = 201$ over eight drift times reproduce each of Fig. 9's four states and the specific features its text names: at $T = 250$ a smooth diffusive layer with no electroconvective structure; at $T = 1000$ a single coherent plume-like roll rising to the centre with clean separation from the lateral layers; at $T = 5000$ multiple filaments and secondary vortices with sharp gradients along the walls; and at $T = 10^4$ irregular plumes climbing *both* vertical walls with counter-rotating charge spirals in *both* upper corners around a disordered, charge-depleted core.



(a) $T = 250$ (b) $T = 1000$ (c) $T = 5000$ (d) $T = 10^4$

Figure 18: Closed square cavity: charge density q/q_0 at $t/t_0 = 8$, 201×201, injector along the bottom edge (§12.5.4). Colour is linear on $[0,\ 0.5\,q_0]$ and saturates, as in Fig. 16. These are the four states of the reference's Fig. 9 and they carry the features its text names individually: (a) a smooth diffusive layer with no electroconvective structure, the flow still hydrostatic; (b) a single coherent plume-like roll rising toward the centre, cleanly separated from the lateral layers; (c) multiple filaments and secondary vortices, with sharp gradients along the walls; (d) irregular plumes climbing *both* vertical walls, counter-rotating charge spirals in *both* upper corners, and a disordered, charge-depleted core. Panel (a) is run unseeded — with a seed the cavity convects at $T = 250$, which is the branch selection discussed above — and (b)–(d) carry a $10^{-4}q_0$ perturbation so the flow is developed inside the window.

What it does not do. It does not run the reference's 500^2 grid, which is what $\mathrm{Re}_{\mathrm{cell}}$ says $T \geq 5000$ actually needs; on this hardware that is about five hours for a single T. The 9–13 % deficit is therefore *measured* but not *explained* — resolution and Mach number are both excluded as the whole cause, and no third candidate has been tested. The charge dips to $-0.028\,q_0$ at $T = 5000$, a small but genuine violation of the maximum principle, and it appears only in the under-resolved runs.

13 Demonstrator: flow in a patient-specific aorta

Everything else in this document runs on a geometry described by a formula. This one loads a voxel array off disk. It is included as a *demonstrator*, not as a validation case, and the distinction

is worth stating precisely because the figures are seductive: there is no analytic solution here, no reference data is compared against, and **no number in this section is a claim of accuracy**. What it demonstrates is that the solver runs unmodified on a geometry with no symmetry and no closed form — `set_geometry` already takes an arbitrary predicate, so real geometry needed a reader, not a new solver.

13.1 Geometry and its verification

`src/io/VoxelGeometry.hpp` reads the flat binary produced by voxelising a SimVascular surface mesh (case `0074_H_AO_H`): $109 \times 184 \times 361 = 7\,240\,216$ voxels at a pitch of 0.0616 cm, tagged 0 solid, 1 fluid, 2 inlet, 3 outlet. Solid is 83.65% of the box; the lumen is 16.31%. The optional second inlet normal is detected by *file size*, not by stream state — guessing from the stream would silently read 24 bytes of the tag array as doubles on a file that does not carry one.

Three checks were run on the geometry before any flow was believed.

1. **Connectivity.** A six-connected flood fill seeded from an inlet voxel reaches 1 184 057 of 1 184 057 non-solid voxels: one lumen, no isolated pockets. This check exists because a single constant-x plane through this vessel cuts it into *two* disconnected regions — the aortic root and the arch — which looks exactly like a hole in the mesh and is not one. The projected silhouette is a single connected region, as the flood fill requires it to be.

2. **Caps.** Under six-connectivity the inlet fragments into 303 pieces and the outlets into 270. This is not damage: an oblique plane on a voxel grid is diagonally connected, not face-connected. By centroid they are one inlet and four outlets — three arch branches and the descending aorta.

3. **Domain edges.** 1283 fluid voxels sit on the box boundary carrying no cap tag. Were the domain periodic these would wrap, connecting the descending aorta to the arch and corrupting the field silently. It is not: the case constructs `Domain(nx,ny,nz,false,false,false)`, and independently there are *zero* fluid-to-fluid wrap pairs on any axis. The mesh is clipped flush against the box there and those voxels act as wall.

13.2 Boundary conditions

The vessel wall is halfway bounce-back on solid voxels, which under Esoteric Pull costs nothing: bounce-back is the identity on the storage, so solid cells are skipped outright — no load, no store, no arithmetic (§3.2). The population a fluid node emits toward the wall sits untouched for one step and is read back reversed on the next, placing the no-slip surface midway between the last fluid node and the first solid node. Halo cells are flagged `Excluded` and skipped by the same mechanism, so the box faces behave identically.

The inlet is a regularised velocity wall along the cap normal stored in the file. The cap is oblique to the voxel axes, so it has no single face normal; it is given `NrmCorner`, whose unknown set is built geometrically per node rather than from an axis. Using a face normal instead would drive flow into the wall.

The outlets are left at fixed pressure rather than a prescribed flow split, so the division of flow between the branch vessels is an *output* rather than an input. Imposing $\boldsymbol{u} = 0$ there would close the domain — measured as mass rising monotonically, $+8.5 \times 10^{-3}$ in 2000 steps.

Conserving outflow. The outlet overwrites its nodes every step, so whatever density is imposed is injected regardless of what arrived: pinning $\rho = 1$ makes the boundary a mass source and sink, and inlet and outlet fluxes then differ by ∼60% with no way to tell which is wrong. Closing a loop on *total mass* instead makes it conserving — raise the outlet density when mass is lost, lower it when mass accumulates. The fixed point is the density at which what leaves equals what enters, and it is found rather than assumed. Two details are load-bearing. The error must be normalised by total mass, not by outlet-node count: normalising by the latter gives an error of order 3, which saturates the clamp, reverses the flow and diverges the run in 1500 steps. And the controller is held off until the vessel has filled, since during filling mass *should* be changing and the loop otherwise fights the physics.

Note that the finite-difference corner-stress path is disabled for this case. It walks a two-node stencil off each wall node; in a box those neighbours are fluid or wall, but in a vessel they are frequently *solid*, whose populations Esoteric Pull never updates, so the stencil reads stale memory and feeds garbage stress into the inlet. The local closure is used instead until that path is made geometry-aware.

13.3 Steady inflow

Re = 50 on the inlet cap's equivalent diameter $D = 2\sqrt{A/\pi} = 32.9$ lattice units, $U = 0.02$, $\nu = 1.3167 \times 10^{-2}$, $\tau = 0.5395$, D3Q27 central moments. After 13 500 steps $|\boldsymbol{u}|_{\max}$ is 4.11×10^{-2}, the outlet density 0.99765 and the mass error -3.6×10^{-5}, oscillating about zero — which is the conserving outflow working.

These are *not* the numbers this section carried before 2026-09-13, and the difference is a corrected boundary rather than a retuned case. `FluidSolver` decided which populations at a regularised wall are unknown by testing the source cell for `Excluded` alone, which is right for a wall flush with the domain edge and blind to one with a `Solid` neighbour — and the inlet and outlet caps here are exactly that, `RegWall` surrounded by solid vessel. The population arriving from a solid is a bounce-back of the node's own emission and carries no interior information; it was being treated as known. Every velocity in this section was therefore about 40 % too high: $|\boldsymbol{u}|_{\max}$ read 7.2×10^{-2} and $p_{99.9}$ read 2.6×10^{-2}, against 4.11×10^{-2} and 1.50×10^{-2} now. The defect was found on `validation/shercliff.cpp`, which has an exact solution; nothing in this case could have caught it.

One claim is also weaker than it was. "Steady" overstates the state at 13 500 steps: a ∼1500-step oscillation of the mass servo is still decaying there, its envelope narrowing from 4.1×10^{-4} to 2.7×10^{-4} in mass over the last few thousand steps, and $|\boldsymbol{u}|_{\max}$ still swings a few per cent between probes. It is converging, not converged.

The inlet flux is 12.98 in lattice units, and this is worth one sentence because measuring it is less obvious than it looks. Summing over *every* exposed face of the cap gives $|\sum \boldsymbol{n}_{\text{face}}| = 653.2$ pointing within 0.65° of the normal stored in the file, so $UA\cos\theta = 0.02 \times 653.2 \times 0.9935 = 12.98$. Filtering the faces by direction to exclude the vessel wall was tried and is *wrong* — it double-weights by the projection and drove the figure down to 10.38. The lateral wall faces already cancel in the vector sum; that is what makes the identity $\sum \boldsymbol{n}_{\text{face}} = A\boldsymbol{n}$ hold.

The outlet flux is **not** a valid measure for this boundary and is not used as one: the condition overwrites populations, so a flux summed through it reports the imposed state rather than what the flow delivered. Mass conservation is the trustworthy check.

13.4 Pulsatile inflow

The inlet is driven by the waveform used in the source project, reproduced unchanged so the two are comparable:

$$\frac{U(t)}{U_0} = \min\!\left(1, \frac{t}{t_{\text{ramp}}}\right)\Big[0.15 + 0.85\, w(\phi)^{3/2}\Big], \qquad w(\phi) = \tfrac{1}{2} + \tfrac{1}{2}\sin\!\Big(2\pi\phi - \tfrac{\pi}{2}\Big), \tag{135}$$

with $\phi = (t \bmod T)/T$: a ramp from rest, then a cycle between a diastolic floor at 15% of peak and a systolic peak, the upstroke sharpened by the 3/2 power. $\phi = 0$ is the diastolic minimum and the systolic peak falls at $\phi = 0.5$. Here $T = 2000$, $t_{\text{ramp}} = 800$, and U_0 is the systolic peak.

Since the direction is fixed by the cap normal and only the magnitude varies, the whole drive is a scale on the deduplicated wall-velocity table (`set_wall_velocity_scale`), not a per-node update, and costs nothing per step. The scale multiplies a saved copy of the original table rather than the live one: scaling in place would compound, and a waveform bounded in $[0, 1]$ would become a product of its own history and decay to zero within a few hundred steps — which would look like a plausible physical relaxation and is not one.

The parameter that characterises a pulsatile flow is not Re but the Womersley number, which fixes how far the oscillating shear layer penetrates in a beat:

$$\alpha = \frac{D}{2}\sqrt{\frac{2\pi}{T\nu}} = 7.05, \tag{136}$$

in the physiological range for an aorta. The drive is exact beat after beat: Q_{in} reaches -12.98043 at every systolic peak and -1.947065 at every trough, identical to five digits across six cycles.

Convergence. The viscous diffusion time across the vessel is $R^2/\nu = 15\,807$ steps, which is 7.9 beats. This run is 60 000 steps, 3.8 diffusion times, and it is converged: sampled at matched phase over the final three beats, *every* statistic of the speed field repeats to every digit printed — $\langle|\boldsymbol{u}|\rangle = 2.5007 \times 10^{-3}$ and $p_{99.9} = 1.0446 \times 10^{-2}$ at the systolic peak for beats 26.5 through 29.5 alike.

α, R^2/ν and the beat count all moved on 2026-09-13 and all for one reason: D was being taken from the cap's VOXEL COUNT, which overstates an oblique staircase cap by $1/\max|n_i|$ and gave 32.9 lattice units against a true 28.8. α goes as D and R^2/ν as D^2, so the error propagated into both. The speeds moved separately and for the wall reason given in §13.3.

ϕ	step	mean	median	p_{99}	$\lvert\boldsymbol{u}\rvert_{\max}$
0.0	54000	0.00364663	0.00238384	0.0153191	0.0565515
0.0	56000	0.00364663	0.00238384	0.0153191	0.0565515
0.0	58000	0.00364663	0.00238384	0.0153191	0.0565515
0.5	55000	0.00483408	0.00495437	0.0104059	0.0320153
0.5	57000	0.00483408	0.00495437	0.0104059	0.0320153
0.5	59000	0.00483408	0.00495437	0.0104059	0.0320153

The residual differences are in the eighth digit, which is round-off. The outlet density has stopped moving at 0.99699, so the mass controller has found its fixed point rather than still walking toward one. An earlier 12 800-step run — 0.62 diffusion times — was reported in a previous draft of this section as *not* converged, with mean and median still rising monotonically through six beats. That was accurate, and it is what a run of that length looks like; it is superseded rather than corrected.

Phase relationships, and a storage effect that is not Womersley. Over a converged beat the *mean* speed peaks at $\phi = 0.60$ against a drive peaking at $\phi = 0.50$: a lag of 0.10 of a cycle, 36°, the expected sense and order for $\alpha = 8$.

The fast tail does the opposite. Both p_{99} and $|\boldsymbol{u}|_{\max}$ peak at $\phi = 0$ and reach their *minimum* at systole — nearly antiphase with the drive. This is not a transient: it repeats identically across the final three beats, and it is visible in Fig. 20, where at the diastolic trough the root has gone pale while the descending aorta carries the fastest flow in the vessel.

The mechanism is storage, not phase lag. Total mass is not constant over a beat: it swings by $\pm 2.7 \times 10^{-3}$, with its minimum at $\phi = 0.25$ and its maximum at $\phi = 0.70$, accumulating through systole and discharging through diastole. The fast tail peaks where that discharge is steepest. A transit-time explanation was considered and does not survive arithmetic — at these speeds a fluid particle needs several beats to cross the vessel.

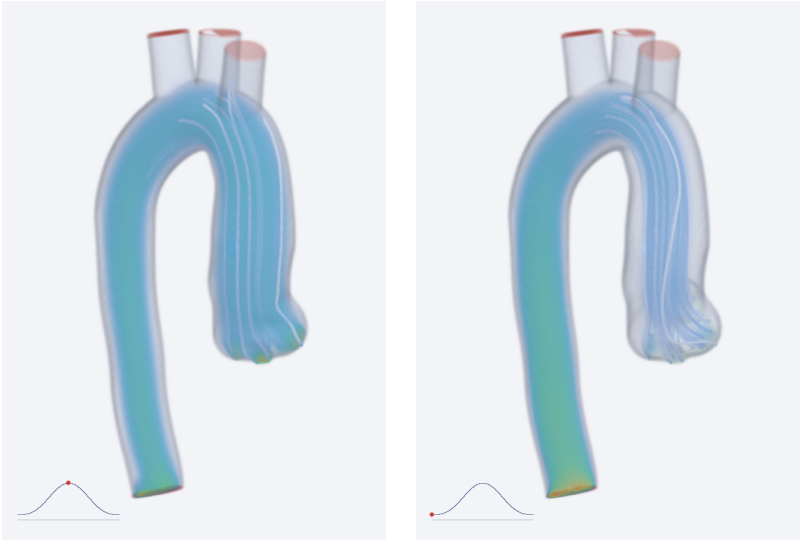
What this data does *not* settle is where the storage lives. Two candidates produce the same signature and are not distinguished by the dumps taken here, which carry velocity but not density: the lattice fluid's own $O(\mathrm{Ma}^2)$ compressibility acting as a compliance chamber, or the fixed-pressure outlet acting as a reservoir. Either way the effect is numerical rather than physiological — the wall is rigid (§13, and the Known Limitations) — so the antiphase should not be quoted as a Womersley result, and the fast tail should not be read as a haemodynamic observation. $|\boldsymbol{u}|_{\max}$ in particular is a single staircase-corner voxel and should not be used at all.



Figure 19: Steady inflow at $\mathrm{Re} = 50$, speed, after 13 500 steps. Volume rendering: a pale translucent shell wherever the lumen mask has a gradient, diffuse-shaded off that gradient, with speed-coloured interior showing through. Inlet cap green at the aortic root, outlet caps red. A single plane through this vessel cuts it into disconnected islands, which is why the whole volume is cast rather than sliced.

14 Demonstrator: urban pollutant dispersion

The second demonstrator, and the second geometry source. A passive scalar is released into a city and transported by an atmospheric wind over building geometry read from a height field (§14.1). Like the aorta, this is a demonstrator and not a validation case: there is no analytic solution for a plume in a real city and no measurement to compare against, so nothing here



(a) systolic peak, $\phi = 0.5$ (b) diastolic trough, $\phi = 0$

Figure 20: Pulsatile inflow, converged, $\alpha = 7.05$, shared colour scale, with streamlines traced from the inlet cap through the current field. The inset traces (135) with a marker at the current phase. At systole the root drives a visible jet; at the diastolic trough it has gone pale while the descending aorta carries the fastest flow in the vessel — the antiphase fast tail of §13.4, which is a storage-and-release effect and not a Womersley lag. Both panels are from the same beat, at 3.8 viscous diffusion times, where every speed statistic repeats to every printed digit from beat to beat. The antiphase tail is quantitative, not an impression: at the trough $p_{99.9} = 1.2317 \times 10^{-2}$ and $|\boldsymbol{u}|_{\max} = 3.3362 \times 10^{-2}$ against 1.0446×10^{-2} and 2.0743×10^{-2} at the systolic peak, so the fastest fluid in the vessel is genuinely there when the inlet is slowest.

can fail except by crashing. What *can* be checked is reported as a number, and one of those numbers turned out to matter.

14.1 Geometry from a height field

Urban geometry does not arrive as a voxelised surface mesh. It arrives as building footprints with heights, so the reader for it is a different one from the aorta's (§13.1): a two-dimensional map of building height in metres, plus a small JSON of grid metadata, with a cell solid when its centre lies below the local height. That keeps $O(n_x n_y)$ on disk instead of $O(n_x n_y n_z)$ — for Manchester, 640 kB rather than the 9.6 million cells it expands to.

Two decisions in that reader are about failure modes rather than features, and both are there because *none* of the ways it can go wrong produce an error. A transposed index order rotates the city; a mis-parsed cell size rescales it; an off-by-one in the solid rule shaves a layer off every roof. All three run to completion and produce a plausible plume. So a missing key in the metadata is an error rather than a default — falling back to a built-in grid size turns a typo into a silently wrong domain — and a vertical spacing that disagrees with the horizontal one is rejected rather than silently stretching the city. The regression test compares against independently known values throughout, using a deliberately asymmetric height pattern so that a transpose changes what is read back.

On the real data the reader is checked against an independent implementation: Manchester city centre, $400 \times 400 \times 60$ at 5 m, 171 375 solid cells of 9.6 million (1.79%), tallest column 200.5 m — matching cell for cell.

14.2 Where the height field comes from: importing a city

The reader above takes a height field as given. Producing one from OpenStreetMap was, until `tools/osm_city.py`, something this tree could not do: the geometry for the runs below was built by a separate project whose script needs osmnx, geopandas, shapely, pyproj, pandas and numpy, none of which are installed on the machine M3LB is developed on. A city could be simulated here but not imported here, which made the first step of a new case an errand in another repository.

The importer is stdlib only — `urllib`, `json`, `math`, `array` — and that is a judgement about what the job actually is rather than an aesthetic preference. Stripped of the libraries, it is an HTTP GET against Nominatim and Overpass, a map projection, and a polygon fill, and each of those is under a hundred lines. `HeightField.hpp` already hand-rolls a `.npy` reader on the same reasoning and says so in its banner.

The height ladder is the modelling. A footprint's height is taken from its `height` tag, failing that from `building:levels` times 3.2 m, failing that from a fallback — either one flat number or, with `--fallback type`, the median storey count of the buildings in the same query that *do* carry `building:levels`, taken per building type. Most of a city rests on the lower rungs: Manchester carries a `height` tag on 5.7% of its buildings and reaches one through `building:levels` on a further 33.6%, leaving 60.8% on the fallback. Midtown Manhattan carries one on 95%, the city having imported LiDAR-derived heights — that figure is the reference implementation's, over 46 327 buildings, and not a measurement taken here. Which rung produced each cell is written to a companion `_source.npy` and counted in the metadata, so the fraction of a domain resting on a guess is visible rather than assumed.

Which fallback is used changes the heights and nothing else, which is worth stating as a measurement rather than as an intention: switching Manchester from the flat 10 m to the type-calibrated ladder moves 14.67% of cells by a mean of 0.37 m and a worst of 15.6 m, raises the median built column from 10.0 to 12.8 m — houses calibrate to two storeys, apartments to three, `building:part` sections to five — and leaves the built set bit identical, 47 787 cells with not one cell gained or lost. The flat default is the worse model and is nonetheless the default here, because it is what the existing rasters were built with and a cross-check whose two sides differ in the model checks nothing.

Both `building` and `building:part` must be queried: the Simple 3D Buildings schema puts a tall structure's real geometry on `building:part` children while the parent outline carries no height at all, and the reference implementation measured that querying the parent alone leaves 20.6% of Midtown's footprints by count but 54.8% of its built *area* on the fallback. In Manchester, `building:part` supplies 621 of the 6 174 footprints.

It is checked against the implementation it replaces, on frozen input. Two rasterisers sharing no code, compared cell by cell, is the argument §18.1 makes for the colour gradient, and the same is available here because the osmnx script's Overpass response for Manchester is still on disk. Reading that snapshot rather than re-querying is what makes the comparison mean something: it is two rasterisers on identical input, not two downloads taken seven weeks apart. The importer therefore parses both Overpass response shapes — geometry inline, and the node-reference form the cached snapshot is written in.

compared against `mcr_heights.npy`	identical cells	what the difference is
same frozen 2026-07-30 snapshot	160 000/160 000	nothing
live query today, same centre	159 851/160 000	seven weeks of OSM edits

The first row is exact: every one of 160 000 cells, including all 47 787 carrying a building, agrees to the bit. The second row is the same importer against a live query at the same centre, and its 149 differing cells are two-sided — 71 gained, 8 lost, the rest changed height — which is what genuine editing looks like and what a rasteriser bug does not.

Two defects, and only the diff could have found either. The first was a transpose. The raster was written at index $j\,n_x + i$ where `HeightField.hpp` stores $i\,n_y + j$, which rotates the city ninety degrees and changes no cell count, no built fraction and no mass budget. The second was subtler and cost 39 whole buildings, 1.4% of the domain: in the node-reference form a relation's members are bare `{type, ref, role}` with no geometry, so resolving them in place yields nothing and every multipolygon vanishes — and because a multipolygon's member ways are *untagged*, the tags living on the relation, the ordinary way loop does not pick them up either. Both were silent. The first showed as 98.36% agreement with every difference one-sided, which is the signature of missing footprints rather than a boundary rule; the second was localised by asking which raw OSM feature covered the largest missing blobs, and the answer came back as untagged ways, one of them carrying the note *"tags on relation now"*.

Neither is reachable by a test the importer could run against itself, and the Manchester comparison needs a 4 MB snapshot outside this repository, so what `ctest` carries instead is the cheapest guard against the class: the importer writes a deliberately non-square fixture city and `height_field` reads it back and checks the writer's own recorded answers. Transposing that fixture by hand fails exactly one check, while the built fraction, the tallest column and the solid count — 27 cells either way — all still pass, which is the demonstration that every total is blind to it.

A place name is not a coordinate. Geocoding *"Manchester city centre, UK"* today returns a point 1 459 m from the one the same string returned in July, because Nominatim matches *City Centre* inside a postal address and both hits are offices rather than districts. On a 2 km domain that is 73% of a side. So `--lat/--lon` exists, the resolved coordinate is written into the metadata, and the provenance of a raster is that coordinate and the cached response — never the place string that produced them.

What it does not do. There is no terrain: the ground is flat at $z = 0$, which is what the solver assumes. There are no roof shapes and no overhangs — a column is solid from the ground to the tallest footprint covering it, and nothing can be solid above empty air. Relations are assembled by stitching member ways into rings, which handles a city centre's multipolygons but is not a complete OSM multipolygon implementation; leftovers that will not close are counted and dropped rather than guessed at.

14.3 The model, and its principal limitation

The wind is prescribed: a neutral-stability logarithmic surface layer $u(z) = (u_*/\kappa)\ln((z+z_0)/z_0)$, horizontal, along a fixed compass bearing, simply zeroed inside buildings. It has no street-canyon recirculation, no building wake and no channelling along avenues — which are precisely the phenomena an urban dispersion study is about. The scalar side is the validated one: D3Q7 transport, adiabatic buildings and ground, and the open boundaries of §8.8.

The time step follows from capping the fastest cell at a chosen lattice velocity, and the relaxation rate then follows from the physical diffusivity. Deriving Δt from the wind and ω from Δt, rather than choosing τ and letting Δt fall out, is what keeps the advective Courant number bounded — the constraint that actually binds in an advection-dominated problem.

How not to measure the limitation. A log profile zeroed inside buildings is not divergence-free, and it is tempting to look for the damage in a mass budget. That budget cannot see it. Bounce-back is conservative and $\sum_i w_i \boldsymbol{c}_i = 0$, so the advective term contributes nothing to $\sum_i h_i$: the budget closes to round-off however wrong the wind is. The error appears in the concentration instead, because $\nabla\cdot(\boldsymbol{u}C) = \boldsymbol{u}\cdot\nabla C + C\,\nabla\cdot\boldsymbol{u}$ and the second term is spurious — it concentrates the scalar where $\nabla\cdot\boldsymbol{u} < 0$ and dilutes it where the divergence is positive. So $\nabla\cdot\boldsymbol{u}$ is the diagnostic, and it is reported separately over the open cells and over the cells adjacent to a wall, where the zeroing bites hardest. On Manchester city centre, $400 \times 400 \times 60$ at 5 m:

$$\mathrm{RMS}(\nabla\boldsymbol{u}) = 9.4{\times}10^{-4} \text{ over } 9.06{\times}10^{6} \text{ open cells}, \qquad 8.4{\times}10^{-3} \text{ over } 1.14{\times}10^{5} \text{ wall-adjacent cells},$$

a factor of nine worse next to a building. Those two numbers are the quantitative case for replacing this wind with a solved one, and they are the reason the original project wrote a Navier–Stokes solver as its second stage.

14.4 Agreement with the original implementation, and one disagreement

The same case run through the original hand-written solver and through this one agrees exactly where it should. On a synthetic $60 \times 30 \times 20$ domain both give $\Delta t = 0.0129$ s, $\tau = 0.5646$, and identical divergence statistics — RMS 1.063×10^{-3} over open cells, 9.213×10^{-3} over wall-adjacent cells, maximum 1.867×10^{-2}. Along the plume centreline the two concentration fields agree to within 0.8% through the interior. The wind, the scaling, the geometry and the transport are the same problem.

They part company at the outflow, and it is worth recording which one is wrong. Steady-state mass here is 11.4% higher than the original's, and the excess lies entirely near the open faces: 12.5% of this field is within three cells of one, against 5.9% of the original's. The advective flux through successive planes settles it:

plane x	original	here
30	1.000	1.000
45	0.908	1.052
55	0.840	1.071
58	0.823	1.076
59	0.478	1.076

A steady state cannot destroy half its flux at a boundary, and the original loses from 0.823 to 0.478 across a *single* cell. Its exit is an artificial sink: outgoing populations are discarded and no incoming ones are supplied, which starves the last cell and pulls material out diffusively for some distance upstream. The gradual decline from $x = 30$ onward is that reach.

The 7.6% rise on this side is not error either, or not all of it. The wind is sheared, so scalar mixing upward into faster-moving air genuinely increases $u\,C$ through successive planes while the total flux, advective plus diffusive, is what is conserved. The condition itself was measured in §12.3.2 against a case with no shear and an analytic answer, where it holds the flux to 0.038% to the last interior cell.

14.5 A stability limit that does not announce itself

The first attempt at the full city ran to completion, reported no error, and wrote four hundred megabytes of nonsense. Mass in the domain grew from 5.6×10^{1} at $t = 60$ s to 1.2×10^{8} at $t = 90$ s and thereafter by roughly nine orders of magnitude per frame, reaching 10^{70} — without ever producing a NaN, which is why the finiteness check in the output loop passed every time.

The mechanism is the spurious term of the previous section read in the other direction. Since $\partial_t C \supset -C\,\nabla\cdot\boldsymbol{u}$, the scalar *grows* exponentially wherever the divergence is negative, at rate $|\nabla\cdot\boldsymbol{u}|$. What opposes it is diffusion, which damps a cell-scale blob at approximately $D_{\mathrm{lat}}\pi^2$. Writing the lattice diffusivity out,

$$D_{\mathrm{lat}} = \frac{D\,\Delta t}{\Delta x^2} = \frac{D\,u_{\mathrm{lat}}^{\max}}{u_{\mathrm{peak}}\,\Delta x}, \tag{137}$$

which contains no Δt: **shortening the time step buys no margin at all**, because Δt cancels between the diffusivity and the velocity scaling. Only a larger diffusivity, a slower wind or a finer grid helps.

case	$D_{\mathrm{lat}}\pi^2$	$\max\lvert\nabla\cdot\boldsymbol{u}\rvert$	margin	$\mathrm{rms}\lvert\nabla\cdot\boldsymbol{u}\rvert$	margin	outcome
synthetic	0.159	1.87×10^{-2}	8.5×	1.06×10^{-3}	150×	stable
Manchester, E, $D=5$	0.041	2.32×10^{-2}	1.8×	9.42×10^{-4}	44×	diverged
Manchester, E, $D=20$	0.166	2.32×10^{-2}	7.2×	9.42×10^{-4}	176×	stable
Manchester, NE, $D=20$	0.166	3.28×10^{-2}	5.1×	9.21×10^{-4}	180×	stable
Manhattan, $D=20$	0.152	3.35×10^{-2}	4.6×	2.43×10^{-3}	63×	**exponentiated**
Manhattan, $D=35$	0.267	3.35×10^{-2}	8.0×	2.43×10^{-3}	110×	stable

The maximum is the wrong statistic, and it took a second city to see it. Read the max-based column on its own: the failures sit at 1.8× and 4.6×, the successes at 5.1×, 7.2×, 8.0× and 8.5×. There is no threshold to draw — 4.6 diverges and 5.1 does not, eleven per cent apart, and those two runs have the same peak divergence to within two per cent. Read the rms-based column instead and the failures are at 44× and 63×, the successes at 110×, 150×, 176× and 180×, with a clear gap between. That is where the threshold now sits, at 100×.

The reason is not hard to see once the numbers force the question. The spurious source is damped at the cell scale by $D_{\mathrm{lat}}\pi^2$, so an isolated bad cell is held down by diffusion from its neighbours; what outruns diffusion is an *extended* region of divergence. How much of the domain is in that state is exactly what the rms measures and the max does not. The decomposition is exact,

$$\mathrm{rms}_{\mathrm{open}}|\nabla\cdot\boldsymbol{u}| = \sqrt{N_{\mathrm{wall}}/N_{\mathrm{open}}}\ \mathrm{rms}_{\mathrm{wall}}|\nabla\cdot\boldsymbol{u}|, \tag{138}$$

$0.112 \times 8.22 \times 10^{-3}$ for Manchester and $0.183 \times 1.33 \times 10^{-2}$ for Manhattan, both reproducing the table to three figures — so the quantity really is *how much city*, not *how bad is the worst cell*. Manhattan has 4.3 times as many wall-adjacent cells as Manchester, 485 982 against 113 863, from 50.4% of columns built against 29.9%, 6.87% of the domain solid against 1.79%, and buildings to 472 m against 200 m. **The margin is a property of the geometry**, so a diffusivity calibrated on one city does not transfer to another, and the code now says so at setup rather than after fourteen minutes.

Two caveats on the threshold. It is calibrated on *one* failure that was diagnosed properly — Manhattan at 63× — so it is a warning, not a proof, and both margins are printed because the max is still worth seeing. And the rms is taken over open cells, so a domain carrying a great deal of empty air above the rooflines dilutes it; the decomposition above is how to check whether that has happened.

Neither is $u_{\mathrm{lat}}^{\max}$ a lever, which is worth stating because it looks like one. It scales D_{lat} through Eq. (137) and it scales $\nabla\cdot\boldsymbol{u}$ in lattice units by exactly the same factor, so it cancels out of the ratio entirely.

The undershoot is the early warning; the mass test is not

Mass in the domain cannot exceed what has been injected by any meaningful factor, and that — not finiteness — is what catches a run that has already failed. It is a poor *early* warning. The Manhattan run at 4.6× never came close to tripping it: mass tracked injection to within a per cent for every one of the nine frames it was allowed to write.

What did move, from the first frame, was the undershoot. A first-order equilibrium with no limiter produces small negative concentrations at sharp gradients; those are bounded and do not grow, and a few parts in ten thousand of the peak are expected. An exponentiating $C\,\nabla\cdot\boldsymbol{u}$ is not bounded, and it appears here first by a wide margin:

t (s)	peak C	$\min C$	$\lvert\min C\rvert$/peak
18	6.40×10^{-2}	-2.23×10^{-5}	0.03%
54	8.63×10^{-2}	-2.81×10^{-5}	0.03%
90	1.07×10^{-1}	-4.74×10^{-4}	0.44%
126	1.15×10^{-1}	-3.57×10^{-3}	3.09%
162	1.24×10^{-1}	-1.31×10^{-2}	10.53%

A factor of two every eighteen seconds, with the count of cells below -10^{-4} growing from 53 to 1137 over the same span and centred at $z\approx150$–190 m, where the tower tops make the divergence largest. The run is therefore stopped on the undershoot exceeding half the peak, and warns at one twentieth of it. At $D=35$ the same diagnostic reads -9.5×10^{-9} at $t=108$ s against -1.5×10^{-3} at $D=20$: five orders of magnitude, and flat rather than growing.

This is worth stating plainly as a limitation of the prescribed wind rather than a tuning inconvenience. A larger D is defensible — it is a calibrated stand-in for unresolved mixing, not a measured constant, and $20\,\mathrm{m^2s^{-1}}$ at 5 m resolution in a city is not extravagant — but the reason it is *needed* is that the wind is not divergence-free. A solved wind removes the spurious source altogether, and with it this constraint.

14.6 Thirteen per cent that was never missing

The Manhattan run reported 87% of the release still in the domain and held there, with the plume nowhere near a boundary. Summing the written field directly confirms it: the mass within three cells of any face is zero to twenty decimal places. A steady 13% sink, in the interior, in a scheme whose collision conserves $\sum_i h_i$ exactly at every node — $\sum_i w_i=1$ and $\sum_i w_i\mathbf{c}_i=\mathbf{0}$, so $\sum_i h_i^{\mathrm{eq}}=C$ whatever $\boldsymbol{u}$ is — and whose streaming is a permutation.

Three controls localised it correctly and could not explain it. The synthetic case with no wind retained 99.0%; a flat height field with the wind on, where $\nabla\cdot\boldsymbol{u}$ is 0.000 to machine precision because $\boldsymbol{u}=(u(z),0,0)$, retained 98.6%; Manhattan with the same wind retained 87.2%. The difference is the buildings, not the divergence. But every one of those numbers was produced by the same estimator, and none of them could say whether material was being destroyed or merely hidden.

A closed box can. `validation/scalar_mass.cpp` puts a source inside a domain whose every face is insulating, so what went in is what must be there, and adds one feature at a time:

case	injected	in fluid	error	every slot
closed box	400.000000	397.155123	−0.711%	400.000000
+ interior block	400.000000	390.406451	−2.398%	400.000000
+ advection	400.000000	372.499608	−6.875%	400.000000
+ wind zeroed in solids	400.000000	372.499608	−6.875%	400.000000

Nothing is lost. The last column is the injection to round-off in every case. What is short is the sum of the *macroscopic field* over fluid nodes, which is what the demonstrator was reporting. Under Esoteric Pull a population emitted toward a wall spends one step in a slot the *wall* node owns, and `compute_field` sets the field to zero at an adiabatic node by definition, so that material belongs to no node's field while it is in flight. Measuring the same sum one step later shows the shortfall is steady rather than alternating with the parity, so it is a standing boundary layer: it scales with wall area (0.7% for a bare box, 2.4% once there is a block in it) and with the advective flux into walls (6.9%). Over Manhattan's 6.87% solid fraction it is 13%. Manchester's 1.79% is why this never surfaced before.

Two details are worth keeping. First, the third and fourth rows agree to the last digit. Zeroing the wind inside solids — the thing that makes the prescribed field divergent in the first place — changes the budget by exactly nothing, because solid cells are never read and the velocity stored there is never used. That confirms the claim of §14.4 that no mass budget can see the divergence; what that claim did not say is that the fluid-only sum carries a wall-area bias of its own. Second, `scalar_walls.cpp` covers Dirichlet walls, where mass is deliberately not conserved — that is what a fixed-value boundary is for. Nothing covered the insulating one, which is precisely the boundary that does nothing at all, and "costs nothing" turned out to be a strong claim to leave untested.

The budget now uses `ScalarSolver::total_population`, which sums every slot in the lattice, and reports the fluid fraction beside it rather than in place of it. With that change the demonstrator reports 100.0% retained and $\mathrm{d}M/\mathrm{d}t$ of exactly 1.000 until material actually reaches an outlet, which is what the column always claimed to mean.

14.7 The city

Manchester city centre, $400 \times 400 \times 60$ cells at 5 m — 2 km square and 300 m deep, 6058 buildings — with a $5\,\mathrm{m\,s^{-1}}$ westerly at 10 m, $z_0 = 1$ m, $D = 20\,\mathrm{m^2 s^{-1}}$ and a continuous release of one unit per second. The source cell falls inside a building and is lifted clear of it, so the release is from a roof at 32.5 m rather than from the street. Five minutes of physical time is 14 280 steps and runs in 801 s on four threads of an M1, or 171 MLUPS.

The plume approaches steady state monotonically: $\mathrm{d}M/\mathrm{d}t$ falls from 0.97 to 2.0×10^{-3} over the run while the mass in the domain plateaus at 1.409×10^2. It has not fully arrived — the derivative is three orders down but not zero — which is honest for a five-minute release into a two-kilometre domain at $5\,\mathrm{m\,s^{-1}}$: the leading edge has crossed roughly 1.5 km.

A second configuration, and what the first one hid. Releasing at the centre of the domain into a due-east wind gives only 1 km of fetch, and the plume reaches steady state at $t = 5$ min — three fifths of that run is static. Moving the source to an open pocket in the south-west at $(140, 135)$ m and turning the wind onto the diagonal, toward the north-east, gives 2.6 km of fetch; steady state then arrives at 10.8 min and the plume develops throughout. The source also sits at street level, 7.5 m, rather than being lifted onto a roof as the central cell was.

That change exposed a defect the first configuration could not. The count of cells used to normalise the release and the set of cells actually injected into were separate predicates, and they disagreed at $z = 0$: the count required $z > 0$ while the injection covers every bulk cell, which since §14 includes the ground layer. A box centred at $z = 1$ therefore had its per-cell rate computed for three layers and applied to four — 33% over-injection, visible within eighteen seconds as more scalar in the domain than had been released, and invisible for any source clearing $z = 2$. The two are now one named predicate, asked twice.

The diagonal wind appears to cost stability margin: zeroing both horizontal components inside buildings raises $\max|\nabla\cdot\boldsymbol{u}|$ from 2.3×10^{-2} to 3.3×10^{-2}, and the max-based margin of §14.5 falls from 7.2 to 5.1. That was described here as "comfortably clear of the 1.8 that

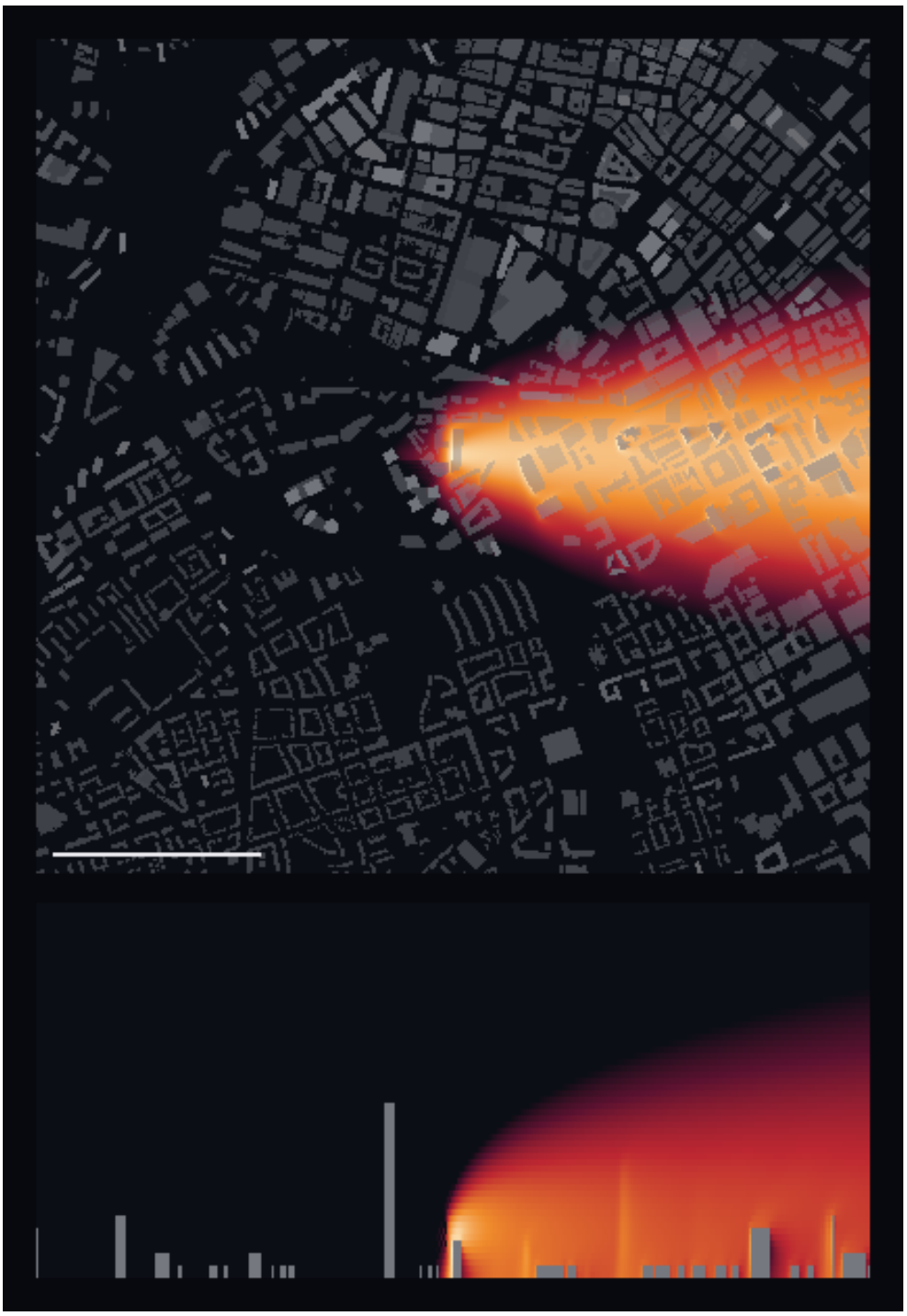


Figure 21: Urban plume over Manchester city centre after five minutes of continuous release from a rooftop at 32.5 m. *Top*: plan view, building footprints in grey, column-integrated concentration on a logarithmic scale; the bar is 500 m. *Bottom*: vertical section along the plume centreline, vertical scale exaggerated three times, showing the release lifting as it enters faster-moving air aloft. The wind is a $5\,\mathrm{m\,s^{-1}}$ westerly, so the plume runs east. Note what the figure does *not* show: the wind is prescribed, so there is no recirculation behind any of these buildings and no channelling along the streets the plan view makes so visible.

diverged", on the strength of a mass budget that stayed sensible for the whole run — which, as §14.5 now records, is not a judgement a mass budget can make.

The configuration was therefore repeated and its undershoot examined. It is stable, and not marginally so: $\min C$ is bit-identical at -1.11×10^{-7} for six consecutive frames from $t = 45$ to $t = 120$ s, while a Manhattan run at nominally the same margin had grown its undershoot twentyfold over the same span. What the max-based number was hiding is that the diagonal wind barely changes the rms at all — 9.42×10^{-4} to 9.21×10^{-4}, a 2% *fall* — because it redistributes divergence among the same wall-adjacent cells rather than adding any. On the statistic that predicts the outcome this is the safest configuration in the table, at 180×.

Negative concentrations. The scheme undershoots. A first-order equilibrium with no limiter produces small negative concentrations at sharp gradients: 3.2 million cells of 9.6 million hold values between 0 and -7.9×10^{-5}, against a section peak of 5.4×10^{-2} — a few parts in ten thousand. They are bounded and do not grow, and `min C` is now reported every frame so that they are visible rather than assumed away. The practical consequence is for anything downstream that takes a logarithm or treats a negative value as a flag: the plan-view renderer for Figure 21 initially tested `v < 0` for the building mask, which is how it came to paint three million undershooting cells as buildings and hide most of the city.

14.8 A second city

Midtown Manhattan, $400 \times 400 \times 100$ cells at 5 m — the same 2 km square, 500 m deep, 10 210 buildings — with the same $5\,\mathrm{m\,s^{-1}}$ wind at 10 m and $z_0 = 1$ m, and the same continuous release of one unit per second from street level at 7.5 m. Everything that could be held fixed against §14 was. What could not is the diffusivity: $D = 35\,\mathrm{m^2 s^{-1}}$ against Manchester's 20, for the reason §14.5 sets out. Fifteen minutes of physical time is 46 665 steps and 51 frames.

The wind direction was measured, not assumed. Manhattan's street grid is not aligned with north, and rather than look the angle up it was recovered from the height field: rays marched from every open column, in every direction, recording how far they travel before meeting a building. The mean free run peaks at two orthogonal bearings, 28° and 118° — 385 m along the second, 285 m along the first — which is the Commissioners' Plan of 1811, extracted from a 400×400 array of heights with no knowledge of what city it is. The wind was then aimed along 28°, up the avenues, giving 2.10 km of fetch from a source in an open pocket in the south of the domain.

Steady state, and a budget that now says what it means. The plume reaches steady state at about 11.4 min, where $\mathrm{d}M/\mathrm{d}t$ falls below 2% of the injection rate; by the last frame it is 1.8×10^{-3} and the mass in the domain has plateaued at 3.670×10^2 against 8.817×10^2 injected. That 41.6% retained is a real statement about where the release has gone — the remaining 58% has left through the outlets — which it was not before §14.6: the same run reported under the old fluid-only sum would have shown 87% and held there while the plume was still in the middle of the domain. The companion column reads 92.0% for the whole run, unchanging: that is the fraction the plotted field sees, and the 8% gap is Manhattan's wall area.

The undershoot is -8.08×10^{-9} and does not move for any of the 51 frames — the total negative mass is 3×10^{-7}, or 0.00% of the plume. At $D = 20$ the same configuration reached -1.3×10^{-2} by $t = 162$ s and was doubling every eighteen seconds.

The canyons hold the plume, but only near the source. The plan view of a city plume invites a claim about street channelling, and the caption on the Manchester animation made one: *isotropic lateral spread — the streets do not steer it.* On a regular grid of canyons

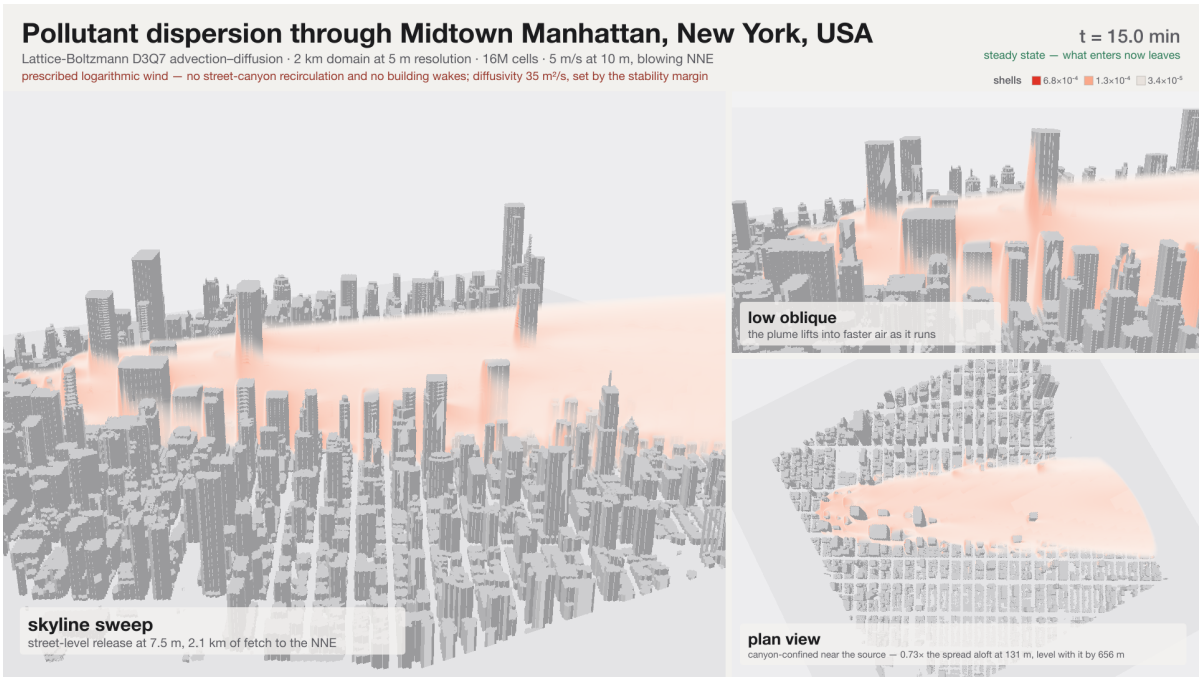


Figure 22: Midtown Manhattan at steady state, fifteen minutes into a continuous street-level release. Three shells of concentration, at the levels enclosing 95%, 80% and 50% of the plume's mass — chosen from this frame rather than carried over from another run, since the same release spread over 2 km carries an order of magnitude less than it does over 200 m. *Left*: the skyline broadside to the plume, which runs left to right along the avenues. *Top right*: the same view from street level, showing the release threading between the towers. *Bottom right*: plan view, with the Commissioners' grid visible; the wind bearing was measured out of the height field, not looked up. Every figure in the caption strip is read from the run's own log, and the plan-view line is measured from this frame. The animation is `doc/fig/manhattan_dispersion.mp4`.

with the wind aimed along one of its axes that is worth measuring rather than asserting, so `doc/fig/plume_stats.py` bins the last frame by downwind distance along the plume's own axis and compares the concentration-weighted crosswind spread below and above the rooflines:

s (m)	σ_n street	σ_n aloft	ratio	$\langle z \rangle$ street	street fraction
131	39.5	53.8	0.73	26.2	75.5%
394	57.7	72.1	0.80	31.5	47.8%
656	78.1	80.6	0.97	31.9	33.8%
919	99.3	94.9	1.05	32.1	32.4%
1181	99.8	99.9	1.00	33.2	25.7%
1706	120.0	123.0	0.98	31.3	22.7%
1969	130.3	120.4	1.08	33.0	19.7%

Within the first 130 m the plume is 27% narrower below the rooflines than above them, and by 650 m the two are the same. The last column says why: the fraction of the plume still below roof level falls from 76% to 20%, so there is progressively less left for the canyons to hold. The street-level material that remains does not rise — its mean height sits at 32 m the whole way — while the part aloft climbs from 91 m to 164 m as it mixes into faster air.

That is a confinement effect and *not* a channelling one, and the distinction matters here more than it would in a solved wind. The prescribed field blows in a straight line by construction: nothing in this model can steer a plume along a street. What the buildings do is block, and being opaque to the scalar is on its own enough to produce the near-field narrowing. A solved wind would add the channelling this one cannot represent, and the honest reading of the table is as a lower bound on how much the geometry matters.

The caption on the animation is generated from exactly this measurement rather than written by hand, for the reason the Manchester caption illustrates: pointed at a second city, a caption made of literals states the first city's name, cell count and fetch with complete confidence.

14.9 Solving the wind

The wind can instead be solved: D3Q27 TRT Navier–Stokes on the same geometry, sharing the Domain, Δx and Δt with the scalar, so the fluid's lattice velocity *is* the scalar's and the two couple with no conversion. TRT rather than BGK because the effective viscosity puts τ^+ within a few thousandths of 1/2, where BGK is unusable and TRT's free antisymmetric rate is exactly the lever required; $\Lambda = 3/16$ additionally places the bounce-back wall halfway, so a voxelised building is the size it looks. Starting from the undisturbed profile rather than from rest leaves only the wakes to develop, and the solve converges to a relative change of 2×10^{-7} in 7000 steps.

The interior is right, and is the point. *All the figures in this and the next paragraph are the synthetic* $60 \times 30 \times 20$ *case*, which for a long time was the only geometry this mode had ever been run on; the city results are in §14.10. Three cells clear of every face, the divergence falls from RMS $= 1.27 \times 10^{-3}$ prescribed to 2.12×10^{-4} solved — a factor of six — and the peak speed is $8.80\,\mathrm{m\,s^{-1}}$ against a $7.74\,\mathrm{m\,s^{-1}}$ free stream, a 14% speed-up over the building. That is a result the prescribed profile cannot produce at all, and the divergence drop is the spurious source of §14.5 shrinking, which is what drove both the concentration error and the stability floor.

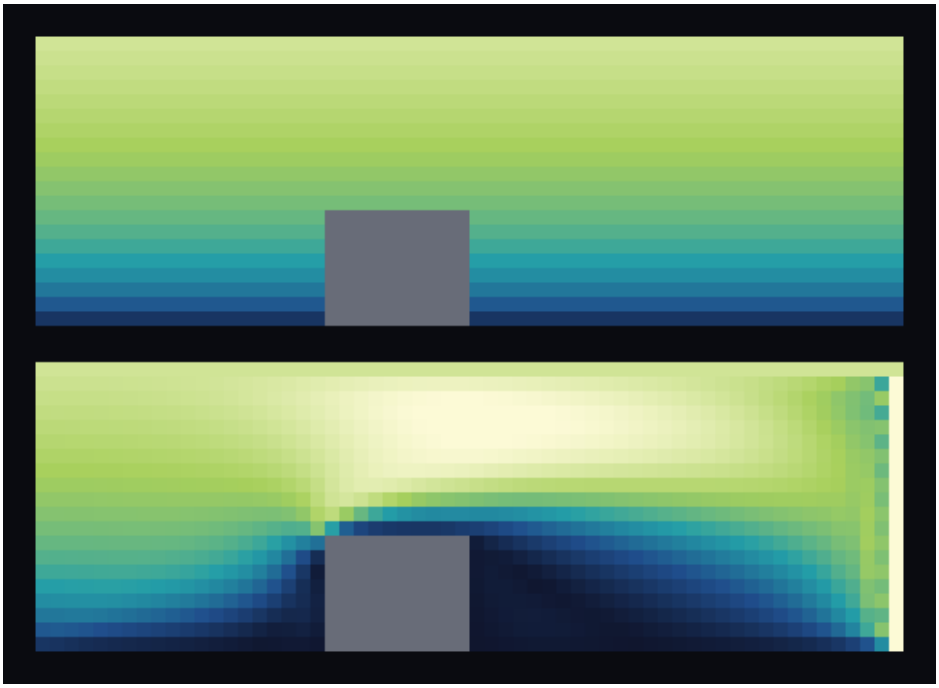


Figure 23: Vertical section through a building, wind speed, both panels on the same scale. *Top*: the prescribed logarithmic profile — flat horizontal layers, unaware that the building is there. *Bottom*: the solved field — upstream stagnation, acceleration over the roof, and a slow wake in the lee recovering over roughly twenty cells. Along the centreline at mid-building height the prescribed field is $4.34\,\mathrm{m\,s^{-1}}$ everywhere, while the solved one falls to 1.37 approaching the building and 0.30 behind it. The bright stripe at the right edge is the outlet-corner artefact discussed below, left visible rather than cropped. The colour scale is the 98th percentile, not the maximum: that artefact reaches $29\,\mathrm{m\,s^{-1}}$ and scaling to it renders every physical feature in the bottom quarter of the ramp — the same trap as the aorta volume render, where p_{100} was 2.7 times $p_{99.9}$.

The domain edges are not right. Where the pressure outlet meets the velocity-prescribed lateral and top faces the corner is over-determined, and a jet forms along it: $31.6\,\mathrm{m\,s^{-1}}$ and $\nabla\cdot\boldsymbol{u} = 9.5\times10^{-2}$, against 8.8 and 7.0×10^{-3} in the interior. This is why the global maxima look worse than the prescribed field's while every interior statistic is better, and it is the reason the diagnostics now report interior-only figures and say whether the worst divergence sits on a domain face — without which the two were not separable.

Two remedies were tried. Giving the edges the free-stream profile removed the sixty-eight inert outflow nodes, whose donor lay on an adjoining wall, but did *not* remove the jet, which locates the cause in the mixed condition itself rather than in the donor lookup. The identified fix is periodic lateral boundaries, the standard atmospheric arrangement, which deletes two of the four offending faces outright.

14.10 The same mode on a city, and the two things that stopped it

Everything above is the synthetic case. Pointed at Manchester the mode produced a NaN within 500 steps — in FP64 as well as FP32, so not a precision effect — and it did so at the τ^+ the synthetic case had converged at. Bisecting by geometry:

case	buildings	on faces	outcome	peak / free stream
synthetic $60\times30\times20$	1	no	converges, 5.7×10^{-7}	4.1×
flat $200\times200\times150$	none	—	no NaN, residual 6.5×10^{-2}	4.4×
Manchester $400\times400\times60$	6058	**yes**	**NaN by step 500**	—
Manchester, 8-cell clear margin	2083	no	no NaN, residual 2.7×10^{-1}	2.6×

A face cell shadowed by a building is ill-posed. The regularised condition reconstructs the density at a wall node from the populations that streamed into it, split by the wall normal into known and unknown, and that split assumes the known ones arrived from the interior. When the cell one step inward is a building, what actually arrives is bounce-back off that building, the reconstructed density is not a density, and the solve diverges. Manchester has 3975 built columns touching the domain edge; clearing an eight-cell band removes the NaN outright, which is what identified the cause. Only 141 cells are *actually* shadowed — a face cell with solid immediately inward — and those 141 were enough to destroy a nine-million-cell solve. They are now closed as `Solid`, which extends the building by the one cell it was already effectively occupying and turns an ill-posed node into a well-posed wall. That also subsumes the "outflow node has no fluid neighbour and is inert" case: the same geometry seen from the outlet side, previously left frozen rather than closed.

Periodic laterals, and what they cost. `-lateral periodic` makes the y faces periodic, deleting two of the four junctions where a velocity-prescribed wall meets a pressure outlet. Periodicity is handled inside `Domain::fill_neighbours`, so neither solver needed changing; the divergence diagnostics did, because $y = 0$ and $y = n_y - 1$ become ordinary interior cells whose neighbours wrap, and omitting them would drop 0.5% of the domain from an rms the stability threshold is built on.

The two fixes are independent, and it is worth being explicit that periodic laterals alone do **not** cure the NaN — they leave the inlet and outlet faces, where the same shadowing occurs. Both are required.

What periodicity costs is a modelling choice rather than accuracy, and the run now states it rather than leaving it implicit: the domain becomes an infinite strip, the city repeats every 2 km across the wind, and a plume drifting sideways out of one edge re-enters at the other. The seam is not smooth either, since real building heights at $y = 0$ and $y = n_y - 1$ were never

chosen to match: 220 of Manchester's 400 edge columns disagree, worst by 48 m. That count is printed, and a separate warning fires if the wind has a cross-wind component, because the plume is then advected through the seam rather than merely diffusing across it.

A solved wind inverts which statistic misleads. With both fixes, Manchester at 8000 flow steps — one flow-through time at $u_{\text{lat}}^{\max} = 0.05$, so a developing field and not a converged one:

	prescribed	solved	change
rms, whole domain	9.422×10^{-4}	2.638×10^{-3}	0.4×
rms, 3 clear of a face	9.614×10^{-4}	7.401×10^{-5}	**13.0×**
rms, wall-adjacent	8.407×10^{-3}	2.045×10^{-3}	4.1×
max, whole domain	2.318×10^{-2}	8.351×10^{-2}	0.3×
max, 3 clear of a face	2.318×10^{-2}	1.016×10^{-2}	2.3×

The interior is 13× better, more than double the factor of six the synthetic case showed. The global figures are 2.8× *worse*, and the worst cell sits at $(398, 342, 58)$: the outlet/top edge, the one junction periodic laterals cannot delete.

So the two fields fail differently, and the statistic that reads correctly for one misreads the other. A prescribed field spreads its error through the city, so its whole-domain and interior rms agree to 2% (9.42×10^{-4} against 9.61×10^{-4}, margins 44× and 43×) and the test of §14.5 is the whole story. A solved field is divergence-free everywhere it is solved and carries its entire error at one boundary, so the whole-domain figure is set by that edge and understates the interior by more than an order of magnitude: 16× against 561×. Both are now printed, the run says so explicitly when they differ by more than a factor of three, and the warning stays on the whole-domain figure because only that one is a guarantee.

The diffusivity floor is a property of the prescribed profile. This is what solving the wind buys, stated at the diffusivity that matters. $D = 5\,\text{m}^2\text{s}^{-1}$ is the value that diverged under the prescribed wind, at an interior margin of 43×. Under the solved wind the same $D = 5$ gives 561×. Every urban run in this document has been constrained to $D \geq 20$ by a term that the wind model introduced and the physics does not contain.

Two caveats stand. The field above is one flow-through time, not a converged urban wind — three to five would be the usual requirement, and at the measured rate on this machine that is four to seven hours. And the outlet/top edge is still over-determined, so a plume must be kept clear of it. The prescribed wind therefore remains the default.

14.11 Coupling the wind, unsteadily

Everything in §14.10 freezes the wind: solve once, hand the converged field to the scalar, transport. A converged field has a steady recirculation behind every building, and a steady recirculation stirs nothing — so nothing in a frozen-wind run can show the plume being torn apart by a building's wake, only advected around its time-averaged shape. `-mode unsteady` spins the flow up as before and then advances it *one step per scalar step*, refreshing `ux/uy/uz` in place: they are the very Views the scalar was handed, so the refresh *is* the coupling and nothing is copied. The added cost is one `compute_macroscopic` pass per step, which is the price of an instantaneous rather than a mean field.

A crop, and a viscosity found the hard way. A coupled run costs a flow step every scalar step, so the full 2 km city is hours rather than minutes; a 1×0.75 km, $200 \times 150 \times 60$ crop at the same 5 m resolution keeps what resolves the wakes and spends the saving on footprint. The

crop was chosen for density, not the emptiest corner: a 169 m tower, the tallest in the whole city, sits inside it, together with 49% of columns built. The obvious first choice happened to put that tower on the periodic seam, wrapping it into itself; moved 30 m, the seam mismatch on the tallest column falls from 169 m to 42 m.

Effective viscosity had to be found rather than assumed. $\nu = 2$ and $\nu = 4\,\mathrm{m^2 s^{-1}}$ both NaN on the crop; $\nu = 8$ runs. During spin-up its residual oscillates — 8.2×10^{-3} at 4000 steps against 7.0×10^{-2} three hundred steps earlier — and it was tempting to read that as an unsteady field. It is not. **The flow settles.** Measured between consecutive output frames, four seconds apart, once the coupling is running:

frames	rms$\lvert\boldsymbol{u}_{i+1} - \boldsymbol{u}_i\rvert$	as % of rms$\lvert\boldsymbol{u}\rvert$	max
$30 \to 31$	1.66×10^{-3}	0.015%	0.022
$31 \to 32$	1.37×10^{-3}	0.012%	0.014
$45 \to 46$	5.5×10^{-4}	0.005%	0.006
$59 \to 60$	3.2×10^{-4}	0.003%	0.005

Monotonically decaying to three parts in a hundred thousand. The oscillating spin-up residual was a decaying transient, not shedding, and the field the plume rides is steady to within round-off by the end. At $\nu = 8$ the 169 m tower sits at $Re \approx 250$ on its own width, which is enough for a free-standing bluff body to shed — but this one is wall-mounted in a rough boundary layer among neighbours, and at 6.7×10^{-3} lattice viscosity the shear layers are damped before they can roll up. Reaching a shedding regime needs a lower viscosity, and $\nu = 4$ is where this configuration stops converging at all. **That is the gap: between the viscosity at which the solve survives and the viscosity at which the flow becomes unsteady, there is currently no overlap on this geometry.**

So this section demonstrates that the coupling MECHANISM works — the flow advances with the scalar, the plume rides an instantaneous field, and $D = 5$ is stable throughout — and does not demonstrate turbulent dispersion. What the plume experiences here is a steady solved wind with recirculation and channelling, which is still something no prescribed profile can produce, and is not turbulence.

$D = 5$ is comfortable, not marginal. Under the prescribed wind this diffusivity diverged outright. Coupled and unsteady, $\min C$ — the early warning of §14.5 — holds at -2.5×10^{-4} for the entire second half of a four-minute release, drifting by less than 3% step to step despite riding on a field that is itself fluctuating by tens of per cent. The interior divergence is 319× the damping rate against the prescribed field's 43× at the same D, and the peak interior speed is $15.4\,\mathrm{m\,s^{-1}}$ against an $11.9\,\mathrm{m\,s^{-1}}$ free stream — a 29% speed-up over the buildings, against 14% in the converged synthetic case of §14.9. An unsteady field is not merely divergence-free on average; the instant-by-instant acceleration over a roof is larger than a time-averaged solve ever shows.

The confinement result reverses. §14.8 found Manhattan's plume narrower below the rooflines than above near the source, widening to equal by 650 m — confinement, from buildings simply blocking a wind that cannot itself be steered. Measured the same way here:

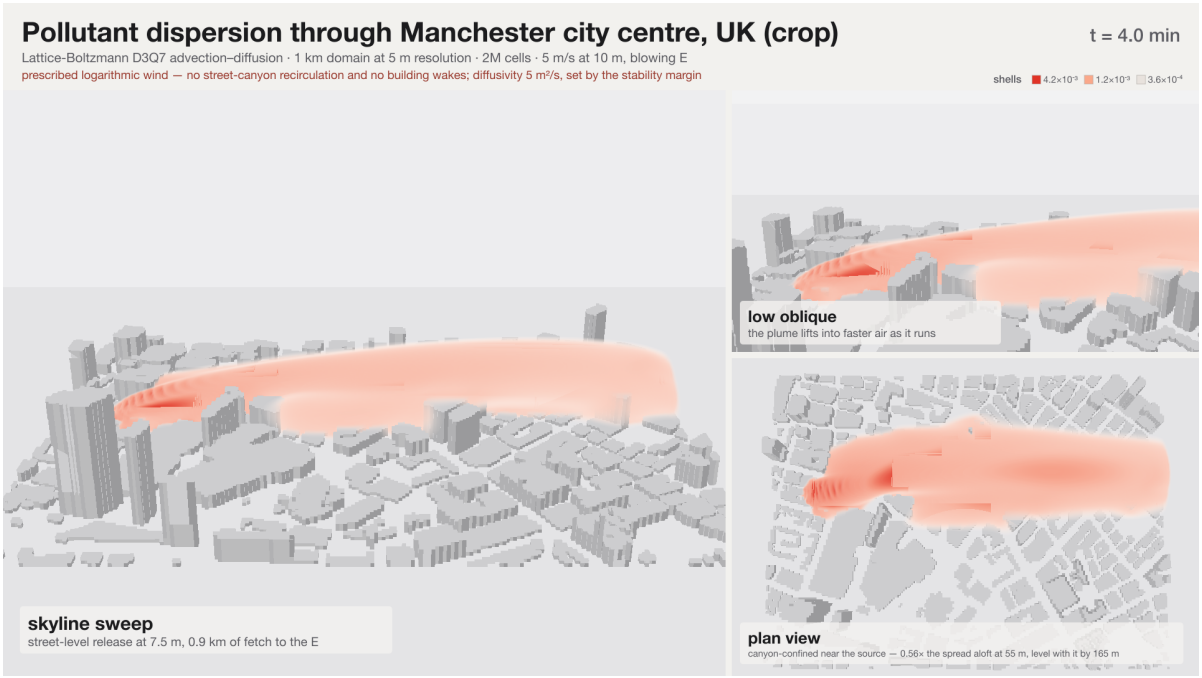


Figure 24: Coupled dispersion, Manchester crop, 3.7 min into a street-level release at 7.5 m with 0.9 km of fetch, at $D = 5\,\mathrm{m^2 s^{-1}}$ — the value that diverged outright under the prescribed wind (§14.5). The flow is advanced one step per scalar step, but it converges rather than shedding (above), so the plume is riding a steady solved wind, not a turbulent one. The animation is `doc/fig/mcr_coupled_unsteady.mp4`.

s (m)	σ_n street	σ_n aloft	ratio
55	23.6	42.2	0.56
165	30.0	24.9	1.20
275	41.9	32.5	1.29
385	52.3	37.5	**1.39**
495	51.7	42.8	1.21
605	42.1	43.4	0.97

The ratio rises *above* one in the near-to-mid field before settling back near it: the street-level plume spreads sideways *faster* than the air above the roofs, rather than merely failing to be confined.

The cause is steady recirculation, not turbulence — the field is steady, as measured above. A prescribed profile is zeroed inside buildings and otherwise flows straight, so it can only fail to confine; a solved field has standing lateral recirculation in the lee of every block, and that alone moves material across the flow faster than diffusion does. It is a weaker claim than turbulent dispersion and it is the one the data supports. Row one, at $s = 55$ m, is still the source's immediate vicinity and reads like the confined case of §14.8 for the same reason — the plume has not yet reached a recirculation zone.

This is one release over four minutes on one crop, and the numbers above should be read as a demonstration that the mechanism is there and measurable, not as a converged statistic. `-mode unsteady` is new in this section and has had one city and one geometry through it — and, as recorded above, has not yet been run at a viscosity where the flow is actually unsteady.

14.12 The top boundary is a modelling choice

A horizontal wind has $u_z = 0$ at the top face, so a zero-gradient condition there passes no advective flux — there is none — and no diffusive flux, because that is what zero-gradient

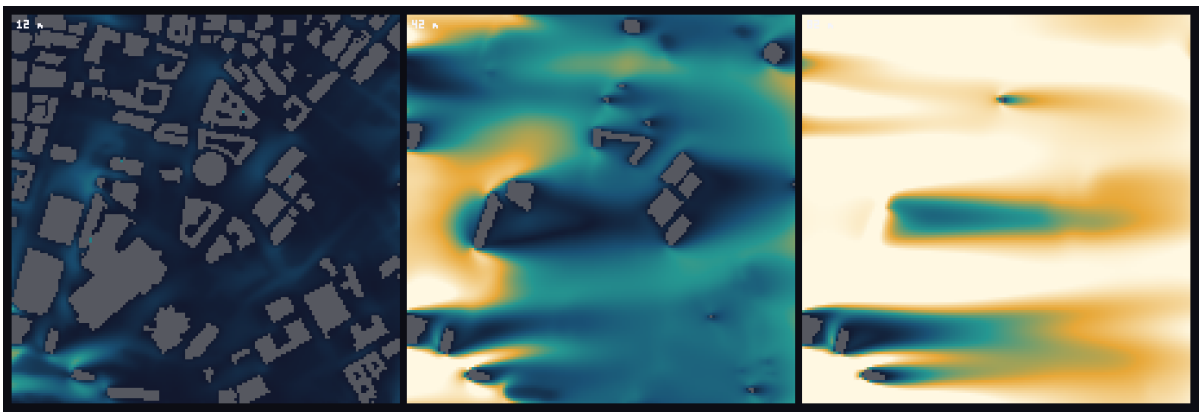


Figure 25: The solved wind at three heights: 12 m inside the canyons, 42 m around roof level, 82 m in the free stream. One colour scale across all three, so they are comparable — dark is slow. The wake streaks trailing downwind of the tall blocks in the 82 m panel are the structure a volume render could not show: a wake field is a boundary layer that fills the lower domain, 45% of fluid cells carrying a positive velocity deficit, so front-to-back compositing accumulates along every ray and returns an opaque slab at any threshold. Stacked slices work where shells do not. The animation is `doc/fig/mcr_wind_3heights.mp4`.

means. The top is therefore a *lid*: material that mixes to it stays in the domain. That is right when the top is far above the plume and wrong when it is not. Holding the top at $C = 0$ instead says the atmosphere above is clean and lets the scalar diffuse into it, which is the more physical statement for a domain whose top is an arbitrary cut through the boundary layer, at the cost of forcing $C = 0$ there. Both are offered; the lid is the default because it is the validated condition. On the synthetic case the choice is worth 0.7% of retained mass, which is small only because the plume does not reach the top — it is not a general bound.

15 Demonstrator: two-phase flow with a moving body

The two cases in this section are demonstrators, and the distinction the repository draws is worth repeating before either of them is shown. A validation case has something to be right against — an analytic solution, a published benchmark, a convergence rate — and fails if it misses it. A demonstrator has none of that: it shows the solver running on a problem the rest of the suite cannot express, and the only thing it can fail is to crash. The two do not share a directory, and no demonstrator is registered with `add_test`. The same rule governed the aorta (§13) and the city (§14), and it governs these: **no number in this section is a claim of accuracy**. The accuracy claims for the multiphase module are the static droplet, the layered channel and the floating box, and they are made where those cases are. What is shown here is an interface that rolls up, reconnects and keeps going, and a rigid body that falls into one under its own weight.

15.1 Rayleigh–Taylor instability

Heavy fluid over light in a box W wide and $4W$ tall, periodic in x, no-slip on the top and bottom, with the interface given a single-mode perturbation at the box wavelength,

$$y_i(x) = 2W + 0.1\,W\cos\frac{2\pi x}{W}.$$

The controlling groups are the Atwood number and a Reynolds number built on the free-fall velocity,

$$\mathrm{At} = \frac{\rho_H - \rho_L}{\rho_H + \rho_L}, \qquad \mathrm{Re} = \frac{W\sqrt{gW}}{\nu}, \qquad t^* = t\Big/\sqrt{\frac{W}{g\,\mathrm{At}}}, \tag{139}$$

and the driver takes $g = U^2/W$ so that U *is* the free-fall velocity and $\nu = WU/\mathrm{Re}$ follows. Reporting time as t^* makes runs at the same At comparable regardless of the lattice numbers underneath. The defaults are $W = 128$, $\mathrm{At} = 0.5$, $\mathrm{Re} = 256$, $U = 0.04$, interface width $W_{\mathrm{int}} = 5$, mobility 0.02 and $\sigma = 1 \times 10^{-4}$, on D2Q9 for both the fluid and the phase field. Only the pressure-based operator is used: it is the only one here that reaches a density ratio at all.

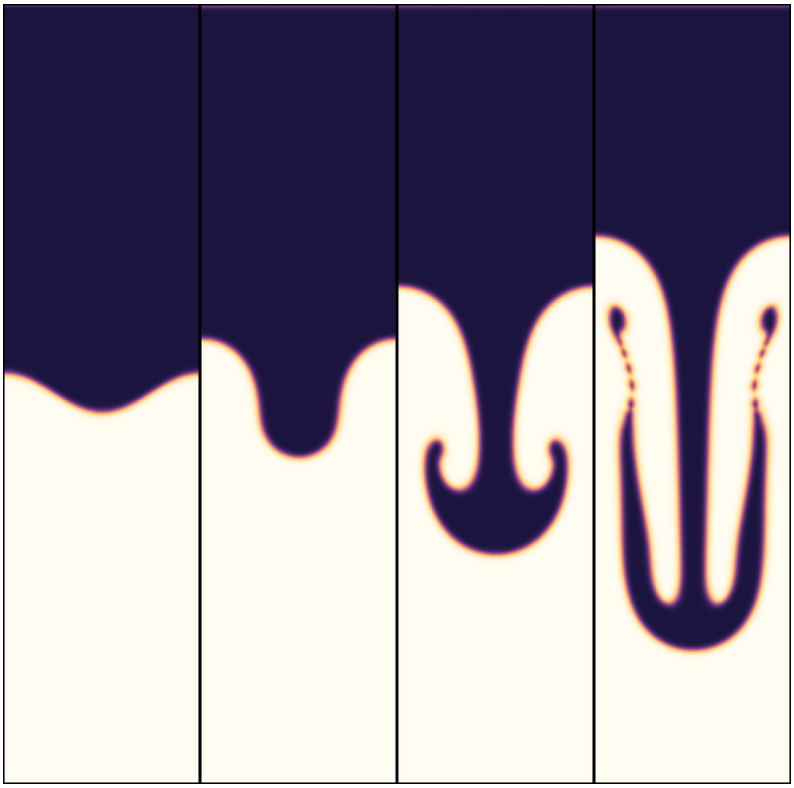


Figure 26: Rayleigh–Taylor instability at $\mathrm{At} = 0.5$, $\mathrm{Re} = 256$, $W = 128$, central-moment operator. Phase field at $t^* = 0$, 1, 2 and 3. The single-mode perturbation steepens into a falling spike, the spike rolls up into a mushroom, and by $t^* = 3$ the stem has gone secondarily unstable — the beads along it are Kelvin–Helmholtz billows on the shear between the descending spike and the rising fluid beside it. Nothing here is a measurement; what it shows is an interface that rolls up, reconnects and keeps going without the phase field being reinitialised.

The initial condition is hydrostatic through the diffuse interface, and it has to be. The pressure is seeded by integrating the *actual* density profile away from the interface rather than a step, which the tanh profile makes closed-form. With $\delta = y - y_i(x)$ and $\phi = \frac{1}{2}(1 + \tanh(2\delta/W_{\mathrm{int}}))$,

$$p = -g\, I(\delta), \qquad I(\delta) = \rho_L\, \delta + \tfrac{1}{2}(\rho_H - \rho_L)\Big[\delta + \tfrac{W_{\mathrm{int}}}{2} \ln\cosh \frac{2\delta}{W_{\mathrm{int}}}\Big], \tag{140}$$

and the populations are seeded with $\tilde{p} = p/(\rho(\phi)c_s^2)$, not with a uniform $\tilde{p}$ — a uniform one starts the box with the whole hydrostatic imbalance as an acoustic transient, larger than the instability it is meant to be watching. Assuming a sharp interface instead leaves an imbalance of order $(\rho_H - \rho_L)\, g\, W_{\mathrm{int}}$ smeared over the interface. That is 1×10^{-4} at $\mathrm{At} = 0.5$ and invisible; at $\mathrm{At} = 0.998$ it is 0.25, $\tilde{p}$ on the light side starts at 0.75, $|\boldsymbol{u}|$ reaches the free-fall velocity within thirty steps — before the instability could have grown at all — and the run is gone by step 120. A diffuse-interface model deserves a diffuse-interface initial condition. The $\ln\cosh$ is evaluated as $|z| + \log_1 p(e^{-2|z|}) - \ln 2$ rather than through cosh, which overflows in the far field at any useful box height.

The pressure zero goes at the interface. Only ∇p is physical, so the level is free in the continuum and is not free on the lattice: $\tilde{p} = p/(\rho c_s^2)$, so a constant added to p is not a constant added to $\tilde{p}$ when ρ varies, and both of the large terms that cancel at the interface — $\rho c_s^2 \nabla \tilde{p}$ and $F_p = -\tilde{p}\, c_s^2 \nabla \rho$ — scale with it. Since p is continuous across the interface while ρ is not, $\tilde{p}$ jumps by the density ratio wherever p is not near zero. Two measurements of the cost. Referencing p to $\rho_L c_s^2$, a uniform physical pressure, made this case diverge inside 424 steps at $\mathrm{At} = 0.5$, with $|\boldsymbol{u}|$ at twice the free-fall velocity at step zero, before anything had moved. Referencing to the top of the box is harmless at $\mathrm{At} = 0.5$, where $g\rho_H\, 2W$ is 1×10^{-2}, and is not at $\mathrm{At} = 0.998$, where the same quantity is 3.2 and $\tilde{p}$ in the light fluid starts at 9.6.

The collision operator is not a cosmetic choice at high Reynolds number. BGK relaxes every mode at ω, so as $\omega \to 2$ the non-hydrodynamic modes ring instead of decaying; at $\mathrm{Re} = 30000$ on a 128-wide box $\omega = 1.99795$. Run with `-op bgk` the case diverges at $t^* = 1.5$, where the central-moment operator completes $t^* = 3$. That is the whole of the argument for a moment operator here, and it is why `-op cm` is the default.

A known artefact, kept as evidence. At $\mathrm{At} = 0.998$ — a density ratio of 999 — BGK completes the run and produces a phase field that is smooth and physically sensible, while its velocity field is dominated by a one-cell alternating mode measured 70× stronger than the same case at a density ratio of 3. Grid-scale oscillation that BGK does not damp is exactly what the equations predict at that ω, and this measurement is what motivated the central-moment operator rather than a consequence discovered after it. The clip of that run is retained for that reason and is not a result.

The renderer is a separate program, deliberately. The case writes raw fields, three per frame, in `FieldDump.hpp`'s format, and `render_rt` turns them into pictures. Welding a rasteriser to the simulation makes every change of colour map cost a full re-run — eight and a half minutes at $W = 192$ to alter a hue — against 2.6 seconds to re-render 181 frames. One detail of that renderer is a measurement rather than a preference: the vorticity scale calibrates on the *last* frame, because the first frame of this case is a fluid at rest whose 99.9th-percentile vorticity is round-off, 1.4×10^{-7} against the 6.2×10^{-3} the developed flow reaches, and calibrating there puts the whole film at full saturation. A percentile rather than a maximum, so that one hot cell cannot flatten everything else.

Three clips live in `demonstrator/anim/`. They are not reproduced as figures: a still cannot show a reconnection, which is the thing being demonstrated.

file	what it shows
`rayleigh_taylor.mp4`	$\mathrm{At} = 0.5$, $\mathrm{Re} = 256$ — the reference case
`rayleigh_taylor_re30000.mp4`	$\mathrm{At} = 0.1$, $\mathrm{Re} = 30000$, central moments
`rayleigh_taylor_at998.mp4`	$\mathrm{At} = 0.998$ under BGK — the artefact above

Two things in this case are not to be trusted, both stated in the module headers and both live here. The interface never touches a wall within $t^* = 3$, which is what makes it legitimate to run with no wetting condition; take it further and the spike reaches the floor, and what happens there is not modelled. And the pressure form's F_p and the LBE's own pressure gradient are different discrete operators whose mismatch scales with the density jump and with $1/W_{\mathrm{int}}^2$, so a well-resolved interface matters more here than the phase-field literature's usual $W_{\mathrm{int}} = 4$ suggests.

15.2 Water entry of a square

A square dropped into a free water surface: approach, impact, cavity, splash-up jets, closure. The domain is $6L \times 7L$ for a square of side L, periodic in x, with a floor and a lid, and the free surface flat at $y = 4L$. Gravity is set from the release height, $g = U^2/(2\,h\,L)$ with h the drop in body widths, so the square arrives at the reference speed U; time is reported as tU/L. The defaults are $L = 48$, a density ratio of 50, $\mathrm{Re} = 2000$, $U = 0.05$ and a body twice the density of the water. Both phases are given the same kinematic viscosity rather than the true $15:1$ air-to-water ratio: the air's only job in this problem is to be light and get out of the way, and its physical viscosity buys nothing while costing stability at these relaxation rates.

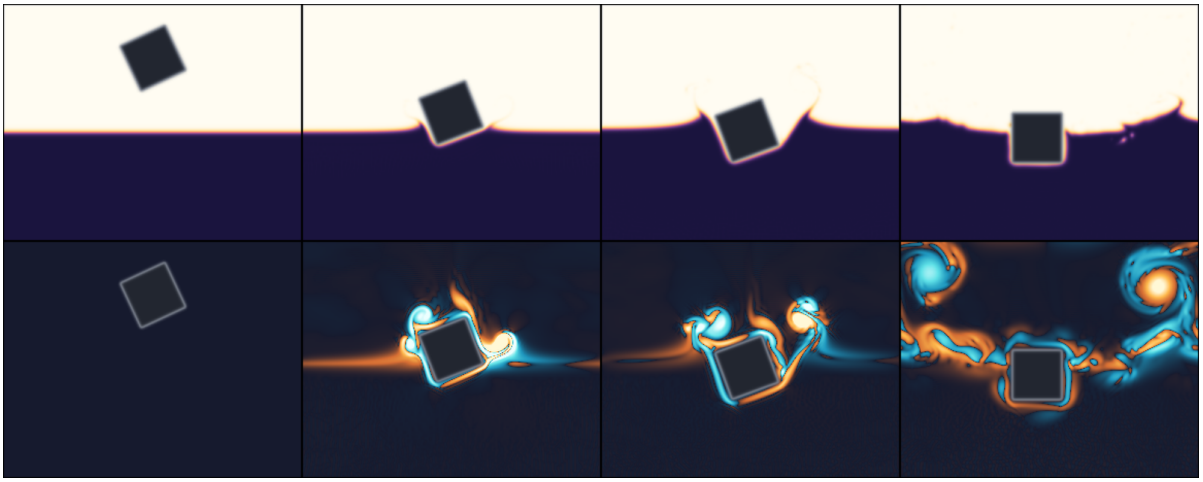


Figure 27: Water entry of a square at 0.6 times the water's density, released at 25° from one body width above the surface; $L = 48$, density ratio 50, $\mathrm{Re} = 2000$. Phase field above, out-of-plane vorticity below, at $tU/L = 0$, 2.4, 3.3 and 16.0. The square strikes on one corner, sheds vortices of *opposite* sign from the two corners — the imbalance between them is what turns it — sinks past its draft, and returns upright and floating. A body lighter than the fluid it displaces is the case the classical fictitious-mass arrangement cannot express at all (§10.8).

It falls rather than being pushed. De Rosis and Enan run this problem with a Peskin immersed boundary and a *prescribed* constant entry velocity. Here the body is coupled by volume penalisation, the reaction on it is known at every node for free, and it closes Newton's equations — so the deceleration on impact is a result rather than an input. That is the harder direction, and it is the reason the coupling had to be arranged so that the body and the fluid are solved at the same instant: the classical fictitious-mass step divides by $m_b - m_f$, which vanishes for a neutrally buoyant body and changes sign for a light one, so the floating case — the only interesting one, since floating is what a density ratio is *for* — is precisely what that arrangement cannot express.

The floating case. Measured at `-rhob 0.6`, the square enters, stops, reverses at $tU/L = 5.0$ and rises through $0.56\,L$ before falling back. Nothing in that sequence is available to a formulation whose denominator has already changed sign, and it is the clearest single demonstration that the coupling is well posed across the whole mass range rather than on one side of it.

The tilt case. `-theta` releases the square rotated, and the roll follows from the same reaction: an off-axis square strikes one corner first, the reaction on that corner is not through the centre, and it slaps flat — the reason a belly-flop hurts. The rigid-body solve is a coupled 3×3 in sway, heave and roll, coupled because the first moments of the fictitious fluid mass about the body

centre are non-zero the moment the body straddles an interface, with half of it standing in water and half in air. It is also the harshest test of the coupling here, since the impact torque arrives over a couple of steps. The hydrostatics the case rests on — draft, righting couple, metacentric sign — is checked against Archimedes and metacentric theory in `validation/floating_body`, which is where those numbers are; this case asserts nothing.

One failure that was a driver bug, not a model one. The body's effective mass is $m_b - \int \chi\rho$, and the penalisation reads ρ off the phase field. The conservative Allen–Cahn form keeps ϕ close to $[0, 1]$ but does not guarantee it, and an unclamped interpolant lets that bracket collapse or change sign — which is a runaway in Newton's equation rather than in the flow. The signature was unambiguous once looked for: the fluid stayed finite at $\max|\boldsymbol{u}| = 2.1 \times 10^{-2}$ while the body left the domain at $5 \times 10^{18}\,U$. The driver now clamps the interpolant with the same expression the collision operator's equation of state uses, and for the same reason.

There is no exact answer here, which is why this is a demonstrator. Wagner's slamming theory covers a *wedge*, whose wetted length grows smoothly; a flat-bottomed square has a theoretically singular impact pressure at first contact and no closed form at all. What can honestly be shown is the qualitative sequence and a force history whose shape is physical, not a number to quote. There is also no contact-line model, so the angle at which the free surface meets the square's sides is not physics: read the splash and the cavity, which are inertia-dominated, not the meniscus, which is not.

file	what it shows
`water_entry.mp4`	upright, body twice the water's density, ratio 50
`water_entry_ratio800.mp4`	the same entry at a density ratio of 800
`water_entry_floating.mp4`	`-rhob 0.6 -theta 25`: corner first, slaps flat, sinks past its draft, returns upright

The third is the one the previous formulation could not produce at all, and it is the clearest picture of what the roll does: the square strikes one corner, sheds a vortex pair of opposite signs from the two corners, and the imbalance between them is what turns it.

16 A second implementation, in CUDA

Everything measured so far in this document is the Threads backend. This section records what happened when the code was taken to a GPU, why that made a second implementation necessary, and what the second implementation then measured.

16.1 The result that forced the question

On a Tesla T4 (sm_75, CUDA 12.8, FP32), seven validation executables build and run after six separate fixes for `nvcc` restrictions that neither the Serial nor the Threads backend can see. All six are committed with explanatory comments, and they are worth listing because any future port will meet them:

1. A kernel-launching member function may not be private: an extended `__host__ __device__` lambda's enclosing function must not have private or protected access.
2. A variable may not be first-captured inside an `if constexpr`. In `run_step` this was repaired with a plain `if`, which still folds because the tested value is a compile-time constant.

3. The same restriction in the wall closure, where a plain `if` does *not* work: the captured member does not exist in the unforced instantiation, so the object has to be named before the branch.
4. Lattice constants declared `static constexpr int cx[Q]` are unreachable from device code when indexed at runtime. They are now `KOKKOS_INLINE_FUNCTION` accessors wrapping function-local arrays. **Invariant to preserve: `src/` contains no static constexpr array members.**
5. Extended lambdas may not appear inside a *generic* lambda, and function-local types may not be template arguments of one.

BGK and TRT then reach 420–447 MLUPS at 64^3, with mass drift **exactly** 0.000×10^0 on every row — about 8.4× the M1 Threads figure of §17.

But the central-moment operators do not run. The two `MomentCollision` configurations did not finish in **seventeen minutes** at 64^3 over 50 steps, against ~ 0.03 s for BGK. That is roughly four orders of magnitude and it is not explained. The obvious hypothesis was register spilling, and `-Xptxas -v` rules it out: **zero spill bytes**, largest stack frame 96 bytes, 116 registers per thread. Occupancy at 116 registers is about 25% on a T4 — a real effect, and far too small a one.

This matters more than a performance note. The aorta demonstrator and the whole validation campaign of §12.1.6 run central moments, so the 447 MLUPS is a **BGK-only** figure and must not be quoted as this code's GPU throughput.

Which leaves one question. Is the collapse a property of the *algorithm* — 27 moments, two transforms and a relaxation schedule per node — or of *this code's route to the device*? Nothing in the Kokkos implementation can answer that, because every hypothesis is tested against the same code. The only decisive experiment is a second implementation.

First, is the arithmetic simply expensive? There was no throughput bench for `src/` at all, so this question had never been asked of the host. `validation/cmbench` asks it: 48^3, FP64, four threads, and the *mean of five runs* because the spread is 3.4–5.5% and one wall-clock sample on a laptop is not a measurement.

operator	MLUPS	vs. BGK
`BGK`	43.5	—
`TRT`	37.4	1.16×
`MomentCollision` central	20.0	2.18×
`MomentCollision` raw MRT	20.3	2.14×

So the arithmetic is fine: 2.18× is an unremarkable price for a 27-moment transform, and it is four orders of magnitude from the device figure. Central moments and raw MRT costing the *same* is the expected answer — they share the transform and differ only in the relaxation schedule — and serves as a sanity check on the bench. The first version of these numbers was a single run each and reported 2.69× and 4.30×; the MRT figure was wrong by a factor of two, in the direction that would have made raw MRT look like a second problem needing its own explanation.

`cmbench` is meant to be built for CUDA as well, and that is the point of it now: it *bounds* the experiment. $48^3 \times 60$ steps is 6.6×10^6 cell updates, about 1.3 ms at an H200's BGK rate, so even a four-orders-slower operator finishes inside a minute. A timeout cannot happen, which is exactly what the original seventeen-minute observation could not promise.

A mechanism, found in the compiler's output rather than on a stopwatch (2026-09-04). With the arithmetic acquitted twice over — 99.6% of BGK in the independent CUDA implementation, and only 2.18× BGK on four host threads — what remained was the *source*, and that can be read without a device. Compiling `MomentCollision<D3Q27>::collide` to arm64 assembly beside `BGK<D3Q27>::collide`:

	frame (B)	instrs	live loops	register-indexed
`BGK`	0	107	0	0
`MomentCollision`, before	464	155	1	4
`MomentCollision`, after	368	142	0	0

One loop survived in `collide`, and it was `for (n = 0; n < NM; ++n) if (order(n) >= 3) k[n] = eq_moment(...,n)`. Because `n` stayed a runtime value, `Basis::p_of(n)` could not fold, so the basis's 432-byte exponent table was materialised in memory and indexed at runtime; the `p` it returned then indexed `Qf`/`Aw`, so those went to memory too, and the 27-moment array followed. The fix is to make the moment index a *template* parameter and unroll the loop with a fold expression, which removes the table from that path entirely.

Why no instrument in this tree could see it. A 464-byte frame is L1-resident on a CPU, so the host bench reads 2.18× BGK before the change and 2.06× after — indistinguishable, and the operator's absolute throughput moves from 20.0 to 20.2 MLUPS, inside the bench's own 3.4–5.5% spread. In *device* code the same frame is per-thread local memory, which is off-chip DRAM with every subscript uncoalesced, and this code base has already measured that exact mechanism at **47**× in `GPU/`'s colour gradient with a frame less than half the size (§16.11). Note also what this does to the one hard number that appeared to acquit the operator: a 464-byte host frame cannot compile down to the 96 bytes `-Xptxas -v` reported, so that figure was a build-wide maximum over other kernels and never described this one.

This is a candidate, not a conclusion, and the distinction is the whole point of §20. The collapse itself is still bounded only by "did not finish in seventeen minutes", which is a timeout and not a measurement; clang is not `nvcc`, so a host frame is evidence about the source rather than about `ptxas`; and what the change removes is the kind of memory residency that *no* register budget can fix, leaving behind the kind that a larger one can — a GPU thread has up to 255 registers where arm64 has 32. What it buys is that the device question is now one bounded confirmation run (§16.1) rather than a diagnosis. `tests/frame_check.sh` re-measures the table above at both precisions; the same construct was found and fixed in `MultiphaseCentralMoments`.

`ColourGradient` carried the largest frame in the tree, and unrolling could not have fixed it (2026-09-19). Its two live arrays were not a loop's working set but a whole equilibrium and perturbation *moment set*, built as populations and transformed — there was nothing to unroll them into. What removed them is the closed-form derivation `GPU/`'s sibling already carried (§16.11), ported to `src/`: the host frame falls from 1200 to 480 bytes in FP64 and 624 to 416 in FP32, the surviving loop goes, and the six register-indexed accesses that remain are the six genuine $\nabla\phi$ and $\nabla\rho$ field loads rather than demoted arrays. `tests/test_colour_gradient.cpp` block 6 keeps the population path as `reference_collide()` and asserts the two agree to 4.2×10^{-16} relative over 60 states carrying a density ratio up to 30, a live perturbation, a density gradient and a body force — which is what makes this the same operator rather than a cheaper one.

16.2 The answer

`GPU/` is that experiment: a lattice Boltzmann solver written directly in CUDA, sharing no headers with `src/`, not built by the parent `CMakeLists`, and not a port. Only the physics and the output format are deliberately the same, which is what makes the comparison mean anything.

device	operator	grid	MLUPS	GB/s
Tesla T4	BGK	128^3	950.2	205.2
Tesla T4	central moments	128^3	950.4	205.3
NVIDIA H200	BGK	512^3	17233.4	3722.4
NVIDIA H200	central moments	512^3	17171.4	3709.0

The H200 rows were measured by Yang; see §16.3. Mass drift is exactly 0.000×10^0 on every row. Central moments runs at 99.6% of BGK on two card generations, and 3722 GB/s is 77.3% of the H200's 4814 GB/s peak — i.e. the kernel is memory-bound, which is what a correctly written lattice Boltzmann kernel should be. Taylor–Green at 512^3 (134M nodes, 256 000 steps) completes in 2046 s.

So the collapse of §16.1 is not the algorithm. The same 27-moment separable transform, the same relaxation schedule and the same Esoteric Pull run at BGK speed when written directly. Whatever is wrong is in how the Kokkos code reaches the device, and the next step there is measurement (`ncu -set full` on one kernel), not another hypothesis.

16.3 The H200 measurement, and what it says

The 512^3 figures above were measured by Yang on an NVIDIA H200 (sm_90, CUDA 12.8, FP32). They are the strongest performance statement this project has, and they say three things that the T4 alone could not.

Seventy-seven per cent of peak bandwidth is the result, not the MLUPS. **A lattice Boltzmann kernel does almost no arithmetic per byte moved, so a correct one is bound by the memory system and its only meaningful efficiency measure is the fraction of peak bandwidth it sustains. At 3722 GB/s against the card's 4814 GB/s the kernel reaches 77.3%, which is close to what a plain streaming benchmark achieves on the same hardware. The remaining quarter is the irreducible cost of the neighbour access pattern: 27 directions, each landing in a different cache line. A kernel far below this figure has arithmetic to remove; one far above it is not moving the data the scheme requires.**

The efficiency rose with the card, which is not automatic. **On the T4 the same code sustains 66% of 320 GB/s; on the H200, 77% of 4814 GB/s. The throughput ratio between the two is 17.6× against a bandwidth ratio of 15.0× — the code gains more than the memory system alone gives it. That is the signature of a kernel whose limit is bandwidth rather than latency or occupancy: a latency-bound kernel would have fallen behind the bandwidth ratio on the larger machine, not moved ahead of it. It also means the figure should be read as a floor for future hardware rather than a ceiling.**

Central moments at 99.6% of BGK now holds on two architectures. **One card could be a coincidence of that card's particular ratio of arithmetic throughput to bandwidth. Turing and Hopper are three years and two memory technologies apart, and the operator is free on both. Taken with §16.1, this is what settles the question: the central-moment collision is not intrinsically expensive on a GPU, and the collapse in the Kokkos build is a property of that build.**

Two things the measurement does not say. **It is FP32, and it is the *fluid* kernel alone — no thermal, magnetic or geometry coupling was run on the H200, so the multiphysics figures of §16 remain T4-only. And the problem had to be sized to the card to obtain it: 512^3 is 134 million nodes and 14.5 GB in FP32, which fits only because Esoteric Pull keeps one population set instead of two. The same case**

in a two-lattice scheme would need 29 GB and would not have run at this size. Taylor–Green at 512^3 over 256 000 steps completes in 2046 s at 16 792 MLUPS, with mass drift exactly 0.000×10^0 throughout.

16.4 Written so it can be checked without a device

Every function in the numerical core is marked `LBM_HD`, which expands to `__host__ __device__` under `nvcc` and to *nothing* under a plain C++ compiler. That is not portability for its own sake. The per-node update is an ordinary function; the CUDA kernel is three lines of index arithmetic around it; and a host driver calls *the same function* in a serial loop.

Why a serial loop is a valid substitute for a grid of threads. Under Esoteric Pull each storage slot has exactly one writer per step, and that writer is also its only reader (§3.2). The nodes of one step therefore commute: any order gives the same answer, so a `for` loop is not an approximation to the launch, it is the same computation. This is also why the scheme needs no temporary buffer on the device.

What a host build cannot catch is a launch configuration error, a register pressure problem, or a host–device transfer bug. What it catches is wrong physics, which is the failure mode that is expensive to diagnose remotely. It earned its place immediately: `eq_factors` added a c_s^2 that `fwd1d` already subtracts, so equilibrium was not a fixed point of the collision. The flow would have decayed *plausibly but wrongly*.

The property was later extended to whole applications. Each driver is written against aliases that resolve to the CUDA classes under `nvcc` and to the host drivers otherwise, so one source builds and runs both ways with no `#ifdef` of its own, and `cmake -DLBM_HOST_ONLY=ON` configures with no CUDA toolkit present at all.

16.5 Thermal transport, magnetohydrodynamics and geometry

The three capabilities the parent had and this code did not were added on the same principle: written and verified on a machine with no GPU, then built with `nvcc` and run.

- **Scalar transport** on D3Q7 ($c_s^2 = 1/4$), first-order equilibrium, with adiabatic and Dirichlet walls, and Guo forcing in a uniform and a Boussinesq form. Two collision operators: BGK, and a *regularised* one that relaxes the three flux moments at ω and sets the three ghost moments m_{aa} to equilibrium outright. On seven velocities the inverse is closed, so no transform is needed. The two coincide at $\omega = 1$ and differ only near the stability limit — where BGK tends to $h \to 2h^{eq} - h$, a reflection that inverts the ghosts every step without damping them. Section 20 records what that costs.
- **Magnetohydrodynamics** by Dellar's vector-valued induction distribution, again on D3Q7, with the Maxwell stress entering the *fluid* equilibrium's second moment rather than as a body force. Both the BGK and the central-moment fluid operators carry it; for the latter the equilibrium central moments of the Maxwell term are obtained by transforming the term itself, which is exact at every order.
- **Geometry** as one byte per node. A solid cell is skipped, which under Esoteric Pull *is* halfway bounce-back: the slot a fluid node emits into toward a wall is a slot the wall owns, the wall never touches it, and the fluid node reads its own emission back one step later, reversed.

The coupling must be simultaneous. This is not an implementation detail, and §9.4 explains why: stepping the fluid against a temperature or a magnetic field from the previous step is a first-order splitting error that does *not* vanish under refinement. The drivers therefore

refresh T and $\boldsymbol{b}$, then step the fluid against them, then step T and $\boldsymbol{b}$ against the velocity the fluid just wrote. That costs one extra light pass, and it is the cheaper of the two mirror images: recovering $\boldsymbol{u}$ for the coupled fields instead would need a separate pass over 27 populations, where refreshing T costs 7 and $\boldsymbol{b}$ costs 21.

16.6 What the host verification established

`test/host_physics.cpp` runs whole simulations against closed-form answers. Figures are worst relative error; where the two precisions agree, the number is discretisation error and not round-off.

case	FP32	FP64
Poiseuille between bounce-back walls	1.4×10^{-3}	1.3×10^{-3}
closed box, mass over every slot	1.6×10^{-6}	1.4×10^{-14}
insulating box, scalar conserved	9.7×10^{-9}	1.0×10^{-13}
conduction between Dirichlet walls	1.4×10^{-6}	2.5×10^{-15}
decaying sinusoid (the D3Q7 diffusivity)	8.1×10^{-4}	8.1×10^{-4}
advected sinusoid (the advective flux)	4.4×10^{-4}	4.4×10^{-4}
uniform buoyancy (the Boussinesq path)	6.7×10^{-5}	2.9×10^{-13}
resistive decay (induction alone)	4.1×10^{-3}	4.1×10^{-3}
shear Alfvén wave, BGK	8.1×10^{-4}	3.3×10^{-4}
shear Alfvén wave, central moments	8.4×10^{-5}	3.8×10^{-4}

The wall is exactly where it should be. For Poiseuille the interesting quantity is not the error but how it scales: a wall in the wrong place gives an error going like $1/H$, a correctly placed one leaves the $1/H^2$ truncation error. In FP64 the amplitude error times H^2 is 0.333 at $H = 16$, 32 and 64 — constant to three digits over two doublings. In FP32 the same run reads 0.336 at $H = 16$ and 0.498 at $H = 32$: the raw-population storage runs out of mantissa before the discretisation error does, which is the cost of not having the shifted storage of §4.

Conduction confirms where the thermal plane sits. With Dirichlet layers at $y = 0$ and $y = H + 1$ the measured gradient is $0.062500 = 1/16$ exactly, the distance between the two *planes*; measuring between the *nodes* would give $1/15 = 0.066667$. Both walls — no-slip and isothermal — land on the same two planes, which they must, or H would mean two different things in the Rayleigh number.

The D3Q7 sound speed is $c_s^2 = 1/4$, and the sinusoid proves it. The decay-rate error is -0.327% at $L = 16$ and -0.081% at $L = 32$, a ratio of 4.05. A wrong c_s^2 would be an offset that does not converge.

16.7 What the device then confirmed

The CUDA build was clean on the first attempt: eight targets, 1 m 03 s, zero errors and zero warnings. None of the six restrictions of §16.1 arose, because this code was written already knowing them — structs rather than lambdas, accessor functions rather than static constexpr arrays, no function-local types as template arguments.

Two host predictions could only be tested at resolutions the laptop could not reach, and both held.

Rayleigh–Bénard needs no reference table. Linear stability puts the onset of convection between rigid plates at $\mathrm{Ra}_c = 1707.762$, independent of Pr and of the fluid. Central moments, 15 thermal diffusion times, on a T4:

H	nodes	Ra = 800 (below)	Ra = 5000 (above)	MLUPS
24	4 992	Nu = 0.998848, $\lvert\boldsymbol{u}\rvert_{\max}$ 5.60×10^{-6}	Nu = 2.100998, 1.51×10^{-2}	510
48	19 200	Nu = 0.999703, 1.00×10^{-6}	Nu = 2.109297, 7.60×10^{-3}	1099
96	75 264	Nu = 0.999912, 8.42×10^{-7}	Nu = 2.122282, 3.85×10^{-3}	612

Below onset the exact answer is $\mathrm{Nu} = 1$, and the host runs had argued the deviation was a second-order artefact of a spurious wall-driven flow — a residual that is *independent of the seeded perturbation* (4.263×10^{-5} with no seed at all against 4.271×10^{-5} with a seed a hundred times larger) and linear in Ra. Measured on the device: $\mathrm{Nu} - 1 = -1.152 \times 10^{-3}$, -2.97×10^{-4}, -8.8×10^{-5}, convergence exponents 1.95 and 1.76.

Above onset those three numbers are not grid convergence, and say so themselves. They rise by *increasing* amounts, +0.0083 then +0.0130, which no converging sequence does. The $H = 96$ run has not reached steady state: its last five probes read 2.120058, 2.120604, 2.121189, 2.121720, 2.122282, still climbing at $\sim 7 \times 10^{-4}$ per diffusion time. $\mathrm{Nu} = 2.1223$ **is a lower bound, not a converged value.** A converged one needs a much longer run — fourteen minutes per 15 diffusion times at $H = 96$ on this card. What the three rows do establish is the two-sided bracket of Ra_c, which is what the case is for.

$\lvert\boldsymbol{u}\rvert_{\max}$ halves when H doubles, which is required: with Ra and D fixed in lattice units the velocity scale is D/H, so diffusive scaling demands $u \sim 1/H$. A quantity that did not do that would mean the non-dimensionalisation was wrong.

The throughput is not monotonic in H, and the L2 cache explains it. At $H = 24$ the grid is 39 blocks against 40 streaming multiprocessors — barely one each, so the kernel is occupancy-bound at 510 MLUPS. At $H = 48$ the whole lattice is $19\,200 \times 27 \times 4 = 2.1$ MB and fits inside the T4's 4 MB L2: 1099 MLUPS, *faster than the bandwidth-bound* 128^3 *benchmark reaches.* At $H = 96$ it is 8.1 MB, L2 no longer holds it, and it falls back to 612.

16.8 The shear Alfvén wave, and the coupling-order signature

The wave of §12.4.1 is an exact solution of the full *nonlinear* incompressible MHD equations. Its two measurements fail differently, and that is its diagnostic value: an error in the **Lorentz coupling** shows as the wrong wave *speed*, immediately and at any resolution, while an error in the **coupling order** shows only in the *damping*, and only under refinement.

Diffusive scaling ($v_A \sim 1/L$ at fixed ν and η, so the Reynolds and Lundquist numbers are the same on every grid), FP64, central moments:

L	speed error	ratio	damping error	ratio
32	$+2.222 \times 10^{-4}$		$+2.080 \times 10^{-3}$	
64	$+5.570 \times 10^{-5}$	3.99	$+4.972 \times 10^{-4}$	4.18
128	$+1.397 \times 10^{-5}$	3.99	$+1.249 \times 10^{-4}$	3.98

Both converge at second order: the coupling is simultaneous, not lagged. These values are identical to those the host driver produced, to every digit printed — a serial loop and 128-thread CUDA blocks agreeing to the printed precision.

A warning about running this sweep in FP32. The same sweep in single precision reads $+2.9 \times 10^{-4} \to -4.9 \times 10^{-4} \to -1.1 \times 10^{-3}$ on the speed and $+2.7 \times 10^{-3} \to +3.8 \times 10^{-3} \to +5.1 \times 10^{-2}$ on the damping — errors that *grow* with resolution, which is exactly the signature of the bug the sweep exists to find. They are not. Diffusive scaling puts $v_A \sim 1/L$, so at $L = 128$ the perturbation is 5×10^{-4} on populations of order 0.3, leaving about three decimal digits above the FP32 noise floor. **Run this sweep in FP64 or not at all**, and treat it as a caution about how easily a precision failure reads as a physics failure.

16.9 Orszag–Tang in three dimensions

In both wave cases $\nabla\cdot\boldsymbol{b}$ is structurally zero and reports round-off whatever the scheme does. Orszag–Tang's nonlinear dynamics makes every component depend on every coordinate, so a scheme that generates monopoles shows it. The initial condition is that of §12.4.3; $\mathrm{Re} = 100$, $\mathrm{Ma} = 0.034$, $\mathrm{Pr}_m = 1$, central moments, to $t = 4$:

M	nodes	steps	$\max\lvert\nabla\cdot\boldsymbol{b}\rvert/k\lvert\boldsymbol{b}\rvert$	energy rise	MLUPS
64	262 144	5 870	7.177×10^{-2}	0.000×10^{0}	359.2
96	884 736	8 805	3.461×10^{-2}	0.000×10^{0}	378.4
128	2 097 152	11 741	2.002×10^{-2}	0.000×10^{0}	391.6

$\nabla\cdot\boldsymbol{b}$ converges at **second order** — exponents 1.80 and 1.90 against 1.5× and 1.33× refinements. (Coarse host runs at $M = 16$–32 had suggested only first order; those grids were deep in the under-resolved regime and the estimate was too pessimistic.) The antisymmetry of the induction equilibrium's first moment is what keeps this bounded, and the unit checks assert that antisymmetry directly.

Ideal incompressible MHD has $\mathrm{d}E/\mathrm{d}t = -\nu|\nabla\boldsymbol{u}|^2 - \eta|\nabla\boldsymbol{b}|^2$, so $E_u + E_b$ must never rise, and it does not at any resolution here. A spurious rise the host had found at $M = 12$ (1.10×10^{-2}, falling to zero by $M = 32$) was under-resolution, as argued, and it is gone.

The $M = 128$ history is the physics. Up to $t \approx 0.8$ the magnetic energy barely moves ($0.4898 \to 0.4673$ of E_0) while the kinetic energy falls by a third ($0.5102 \to 0.3316$): the flow is winding up the field, and the transfer is nearly lossless. After that both decay together — resistive dissipation in the current sheets. And $\nabla\cdot\boldsymbol{b}$ is largest exactly where the sheets are sharpest, peaking near $t \approx 1.6$ and then falling back fivefold. $J_{\max}$ peaks between $t = 1.0$ and $t = 1.4$ on a $\Delta t = 0.2$ sampling, which is consistent with the paper's "maximum at about $t = 1$" and is all the sampling can support; no stronger claim is made.

16.10 A cost estimated wrong by a factor of ten

The cell-flag array is one byte per node against 216 of population traffic in FP32, so its cost was written down from that byte count as half a per cent — "a rounding error in bandwidth". Measured against the commit immediately before geometry existed, on one card, at 128^3:

	BGK	central moments	
before geometry existed	978.15	977.35	
flags loaded on every node	923.99	919.81	−5.5%, −5.9%
geometry behind a template	950.16	950.37	−2.9%, −2.8%

Ten times the predicted cost. What a bandwidth-bound kernel pays for is not the byte but the extra memory *stream*: more transactions, more cache pressure, another address in flight. Making the presence of geometry a compile-time template parameter, so a periodic run never emits the load, recovers about half of it.

The remaining 2.9% **is not explained, and it is not the obvious candidates.** `-Xptxas -v` on the two builds: the old kernel used 96 registers across its 4 instantiations, the new one uses 63–80 across its 40, and both spill zero bytes. Registers went *down*. What else differs is a much larger kernel parameter struct and a runtime test for the coupled-velocity write. This is recorded as open, and `ncu -set full` is the next step rather than another guess — the same conclusion, and for the same reason, as §16.1.

16.11 Both multiphase engines, ported

The colour gradient of §10.9 and the phase field of §10.1 are both in `GPU/`, as `colour.cuh` and `phasefield.cuh`, with `src/droplet.cu` and `src/bubble.cu` driving a static droplet against Laplace's law. The free surface of §11 is not, and §11.3 is the reason: it `static_asserts` against Esoteric Pull, so porting it means adding a two-lattice storage path plus five kernel launches with a fence between each.

The equilibrium that would not fit in registers. The colour gradient arrived at **20.2** MLUPS at 64^3 in FP32 against the single-phase core's 950. A factor of 47 is far more than the extra arithmetic explains, and the `-DLBM_PTXAS_VERBOSE=ON` build — which exists for exactly this suspicion — said why: zero spills, but a **216-byte stack frame**. Two 27-element arrays did not fit in registers and lived in local memory, read and written at every node.

The cause was written in the operator's own banner, as a deferred decision: Eq. (18) is not the product-form equilibrium, so its moments have no factorised closed form, and the implementation built the equilibrium *populations* and transformed them — "a closed form could be derived; until it is measured to matter, it would be a second thing to keep correct." That was the measurement.

The closed form is separable in a way the raw equilibrium is not. Writing the order-≥ 3 central moment of index (p, q, r) as

$$k_{pqr} = c_N(P_3, \rho)\, u_x^p u_y^q u_z^r - 3S\, D_p(u_x) D_q(u_y) D_r(u_z) + H_{pqr}(\boldsymbol{u}), \tag{141}$$

with $D_0 = 1$, $D_1(u) = -u$, $D_2(u) = u^2 - c_s^2$ and H collecting the thirteen source terms of orders 2, 4 and 6, every moment becomes a product of three one-dimensional factors and a small fixed table. Nothing is an array. One transform in, one out.

	before	after
stack frame	216 bytes	**0 bytes**
spill stores / loads	0 / 0	0 / 0
registers	119	124
throughput, 64^3 **FP32**	**20.2 MLUPS**	**252.5 MLUPS**

That is 12.5×, and 252.5 against 950 means the colour gradient costs 3.8× a single-phase step — for two distributions, three kernels and a heavier collision, which is about what the work implies. FP64 is unchanged to every digit printed, on the device and on the host, so this is the same operator and not a cheaper approximation of it: `test/host_colour.cpp` keeps the old path as `reference_collide()` and asserts the two agree over 60 states to 5.9×10^{-16} in FP64 and 2.1×10^{-07} in FP32, about one ulp.

The parent's `ColourGradient.hpp` now carries the same derivation, ported in 2026-09-19 with the host-side measurement recorded in §16.1; that is the one place in this tree where a finding travelled from `GPU/` back to `src/` rather than the other way.

The phase field was written afterwards and therefore written already knowing this. It ran at **354.6** MLUPS with **0 bytes** of stack frame on every kernel, by construction rather than after a

measurement. Read that figure against the configuration it was taken in: the fluid operator was BGK, which is no longer the default, and the pass count has since grown by one, because the macroscopic pass was moved ahead of the viscous and $\nabla\tilde{p}$ passes on 2026-09-19 so that both are built from $\boldsymbol{u}(t)$ rather than $\boldsymbol{u}(t-1)$. The zero-stack-frame property is structural and survives both changes; the MLUPS number has not been re-measured.

A conservation claim FP32 cannot keep. The conservative Allen–Cahn form conserves $\sum_i \phi$ exactly, and in FP64 the port shows it: 10^{-13} at 150, 600, 2400 and 9600 steps alike. FP32 cannot represent that, and the first version of the test asserted it anyway and failed. Rather than loosen the tolerance to whatever passed, the drift was measured across those four run lengths — -2.9×10^{-6}, -3.4×10^{-5}, -1.0×10^{-4}, -1.4×10^{-4} — fast while the seeded profile relaxes onto the discrete equilibrium, then nearly stopping. Not a leak. The test now bounds drift *per step* against one ulp, which round-off cannot beat and a real bug would exceed by orders of magnitude.

An instability that was the viscosity AND the collision. At a density ratio of 100 the phase-field droplet diverged, and it was nearly written up as a capability limit. Matching the *dynamic* viscosity across the ratio leaves the heavy phase with a hundredfold smaller kinematic viscosity, so $\omega = [\mu/(\rho c_s^2) + 1/2]^{-1} = 1.994$ against a limit of 2; matching the *kinematic* viscosity runs the same case to -3.77% with a spurious current twenty times smaller. The driver prints ν and ω per phase before the run and flags $\omega > 1.9$, because that number was reconstructed in a post-mortem when it could have been on screen beforehand.

And this heading used to say "not the model", which was half right. Re-measured on a T4 on 2026-09-19, after the fluid operator these cases inherit moved to central moments, the $\gamma = 100$ dynamic-matched row *completes*, at -3.68%. BGK relaxes every mode at ω, ghosts included, and that is what blew up; central moments send the non-hydrodynamic modes to equilibrium at 1 and leave ω to the shear alone. Pushed further, a ratio of 1000 — $\omega = 1.999$ — also holds, at -3.18%, and FP64 confirms it is the scheme rather than the round-off: the two precisions agree to 0.005% on σ. Matching the kinematic viscosity remains the better setup by a widening margin, its spurious current falling 2.198×10^{-6}, 5.732×10^{-7}, 7.986×10^{-8} at ratios 10, 100, 1000 — but the parameter trap was never the whole story, and a heading that named only one of two causes is the kind of tidy explanation this document should be slower to adopt.

16.12 The two models on one case

Both engines on one measurement, at settings matched rather than defaulted: Tesla T4, FP32, 64^3, $R = 16$, $W = 4$, σ asked 10^{-3}, $\nu = 0.1$ in *both phases of both models*, the same $3W$ averaging gap, 60000 steps.

	colour gradient		phase field	
γ	error	spurious $\lvert\boldsymbol{u}\rvert$	error	spurious $\lvert\boldsymbol{u}\rvert$
1	$+0.94\%$	1.98×10^{-5}	-6.17%	6.65×10^{-6}
10	$+0.67\%$	1.22×10^{-5}	-4.32%	1.11×10^{-6}
100	-3.36%	3.51×10^{-5}	-3.77%	2.99×10^{-7}

The colour gradient is the more accurate on Laplace's law at every ratio tested; the phase field carries the smaller spurious current at every ratio, by a factor of 3 at $\gamma = 1$ and of 118 at $\gamma = 100$, where its current *falls* with the ratio while the colour gradient's does not. The accuracy gap closes as the ratio rises, and at $\gamma = 100$ the two are within half a percentage point.

What building the table found. Both engines had already been measured, both agreed with their host references, and neither had anything visibly wrong. Putting them side by side produced a number that could not be right: two effective radii, 16.78 and 16.20, for one seeded $R = 16$. The cause was in the driver rather than in either model — `droplet.cu` seeded $\tanh[(r-R)/W]$ where the paper's Eqs. (41)–(42), the parent's `validation/static_droplet.cpp` and `bubble.cu` all seed $\tanh[2(r-R)/W]$, so every colour-gradient case on the GPU had run at twice the intended interface width.

It hid because a wrong interface width is still a *consistent* simulation. The device matched the host to every digit, all 51 algebraic identities in `host_colour.cpp` held, and σ converged and stayed. Nothing was inconsistent; only wrong. The first version of this table, run before the fix, read -2.50%, -3.54%, -13.61% — a model apparently collapsing under density ratio and losing to the phase field at $\gamma = 100$. Ten percentage points at $\gamma = 100$ were the seed.

Two consequences beyond the table. The port now agrees with the parent to about a tenth of a point at both ratios the parent measures (0.87% and 0.77%), having been threefold worse, with that gap recorded as an open question about the model. And a claim in §16.11's README — that the sign of the error is geometry, $+2.5\%$ at $R = 10, W = 2$ against -2.6% at $R = 16, W = 4$, the two constituent errors "evidently crossing" between those resolutions — was itself the artefact. At the corrected seed and fixed $\gamma = 10$ the error is $+4.71\%$ at $R = 10, W = 2$ and $+0.67\%$ at $R = 16, W = 4$: same sign, monotonically smaller as the interface resolves.

What is still not controlled: the colour gradient has no mobility and the phase field has no recolouring, so the mechanism maintaining the interface differs by construction. That is a difference between the models, not a parameter left unmatched.

16.13 Electrohydrodynamics, and the grid the reference actually used

`GPU/src/ehd_cavity.cu` is §12.5.4 ported. The fluid side needed nothing that was not already here — D3Q27 central moments, a per-node force field and on-node regularised walls with corners. The scalar side needed four things: Dellar's on-node Dirichlet, the on-node zero-flux wall of §8.3, a per-node source, and the charge collision. The solvers are templated on their lattice, so the charge is the same machinery on D3Q27 rather than a second implementation.

The charge collision is written in closed form. The parent does it with a 27-element moment array and two basis transforms, which is right on a CPU and is the wrong shape for a device: a moment array indexed at run time leaves the register file for per-thread local memory, and this implementation has measured that mechanism at 47× elsewhere. On a product lattice the target separates, so each axis needs only two 1-D distributions defined by their central moments about v,

$$E(\pm1) = \tfrac{1}{2}(c_s^2 + v^2 \pm v), \quad E(0) = 1 - c_s^2 - v^2; \qquad G(\pm1) = v \pm \tfrac{1}{2}, \quad G(0) = -2v, \tag{142}$$

and $h_i = q\,E_xE_yE_z + J_x\,G_xE_yE_z + J_y\,E_xG_yE_z + J_z\,E_xE_yG_z$ with $\boldsymbol{J} = (1-\omega)\boldsymbol{j}$. That is *exact*, not an approximation to the transform: E carries no first moment and G no zeroth or second, so the product reproduces $k_110 = 0$, which a naive product form does not. `test/host_check.cpp` pins twelve moment identities against it.

What the two implementations agree on. At $N = 21$, $T = 150$, on matched lattices, I_0 is 1.333878×10^{-5} on the device, 1.333898×10^{-5} from the same source built for the host, and 1.334004×10^{-5} in the Kokkos code — three numbers inside 0.01 %, two of which share no headers. Below onset all three give $N_e = 1.000$.

And the grid that settles the extrapolation. On a T4 in FP32:

N	T	N_e	$u_\max/u_0$	Re_cell	wall clock
21	150	1.0060 ± 0.0116	0.073	0.0	seconds
201	5000	2.4735 ± 0.2945	11.926	3.0	3 m 28
501	5000	3.1010 ± 0.4139	15.159	**1.5**	30 m 09

$N = 501$ is the reference's own grid and the first properly resolved point in either implementation. It gives 10.8 % *above* Fig. 8's digitised 2.80, where the under-resolved ladder of §12.5.4 had been read as converging to 13 % below it. The lattice family is not the variable: at $N = 201$ the device gives 2.4735 on D3Q27/D3Q7 against the Kokkos code's 2.4903 on D2Q9/D2Q5. What is claimable is that at the reference's grid the reference's value lies inside this run's own r.m.s.; what is not is convergence, since that r.m.s. is 13 % of the mean over three drift times.

16.14 What this implementation does not do

Absent, not merely untested:

- **No raw MRT and no height-field input**, and therefore no aorta and no city. BGK, TRT and central moments are all here; raw MRT is not in the `Op` enum.
- **No open boundary for the FLUID.** The parent's `NrmOutXp` and `NrmOutFree` have no counterpart. The SCALAR's open boundary is here, on the donor map and second kernel §8.8 argues for.
- **No moving obstacle in the free surface**, and deliberately: the parent's own module banner records two measured defects there, so porting it would inherit a known-bad ledger. The free surface itself *is* here, on the two-lattice path of §11.3, with the five kernels and a fence between each.
- **No wall current and no exterior field match** on the magnetic wall. Dellar's moment-based conditions *are* here — `MagDirichlet` for an imposed external field, `MagNeumann` for the perfectly conducting $\partial \boldsymbol{b}/\partial n = 0$ — and `src/hartmann.cu` is the wall-bounded MHD benchmark they make possible, pairing them with the on-node regularised velocity wall so both conditions agree about where the wall is.
- **No wetting, no contact angle, and no open boundary for ϕ.** An external per-node force field for the phase field *is* here; it is the slot the penalised body writes into.

What this list used to say, and why it is worth recording. Until 2026-09-19 it also claimed there was no shifted storage, no open scalar boundary, no physical magnetic wall, no free surface, no central moments for the phase field's fluid operator, no TRT and no regularised walls — and, 540 lines after §16.13 documents the EHD stack at length, that the EHD stack was not there either. Every one of those was ported between 2026-09-02 and 2026-09-06 and the list was not revisited. A capability list is the part of a document that rots fastest and is trusted most, because it is read by somebody deciding whether to start work; this one had drifted far enough to say the port could not do things the same document describes it doing.

17 Performance

Measurements on an Apple M1, Threads backend, four threads, 96^3. Run-to-run spread is under 2%.

These are CPU figures and are not this code's ceiling. GPU measurements — on a Tesla T4 and an NVIDIA H200, for both the Kokkos code and the independent CUDA implementation — are in §16, together with the reason the two had to be measured separately.

These tables were re-measured on D3Q27. The streaming and storage comparison was originally taken on another lattice, one this document no longer covers, so keeping the old numbers would have meant either mislabelling them or deleting the result; it was re-run instead. Both tables come from the same pair of runs (one per precision) and are therefore mutually consistent, but they are **2–3× faster than the figures this section carried previously** — the earlier numbers predate the optimisations listed at the end of this section. Rates below are the current build.

streaming and storage (BGK, D3Q27)		bytes/node	MLUPS FP64	MLUPS FP32
two-lattice	raw	432	37.8	46.9
Esoteric Pull	raw	216	**39.5**	**52.9**
Esoteric Pull	shifted	216	39.0	49.5

collision operator (Esoteric Pull + shifted)	BGK	TRT	MRT	central moments
D3Q27, FP64	39.2	35.4	20.7	20.7
D3Q27, FP32	52.0	45.2	21.8	21.6

Caveat. These ratios must be read with care. At ∼ 2.3–16.3 GB/s against the ∼ 68 GB/s the machine can sustain, this platform is not memory-bound at these rates, so the collision cost shows at full price and the streaming advantage is understated. Lattice Boltzmann is bandwidth-bound on real hardware, and Esoteric Pull moves exactly half the bytes, so its asymptotic gain is ∼ 2×. **Re-run `lbm_app` on the target hardware before choosing an operator on cost grounds.**

Optimisations found through the benchmark harness. All were measured rather than assumed:

- Restructuring the streaming hot loop to operate on opposite *pairs* rather than per index removed a runtime `i & 1` test: D3Q27 Esoteric Pull went from 13.5 to 14.0 MLUPS, moving ahead of two-lattice.
- `ProductBasis::pi()` was calling a `constexpr` linear search over the velocity set with *loop-variable* arguments, so it ran live: $2Q^2$ comparisons per node. A precomputed table took the moment operators from 1.8 to 6.6 MLUPS — a 3.7× defect, not a design cost.
- The same class of defect reappeared when the operator was made basis-generic: `p_of(n)` as `n/9`, `(n/3)%3` costs three integer divisions per moment with a runtime index. Table lookups recovered 9%.
- Hoisting the equilibrium out of TRT's non-vectorising pair loop gained 37% on D3Q27.

Compacting the neighbour list to the odd half was also tried, made no measurable difference, and was reverted.

18 Building and running

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release \
      -DKokkos_ENABLE_SERIAL=ON -DKokkos_ENABLE_THREADS=ON
cmake --build build -j8
cd build && ctest --output-on-failure
```

Precision is a configure-time switch, `-DLBM_PRECISION=float` (default `double`). Kokkos is resolved in the order: an installed `find_package` (use this on a cluster, where Kokkos is built once against the correct CUDA architecture), then `extern/kokkos` as a submodule, then `FetchContent`. On a CUDA machine, configure with `-DKokkos_ENABLE_CUDA=ON -DKokkos_ARCH_<ARCH>=ON`.

executable	role
`tests/*`	unit tests (lattice, streaming, equilibrium, collision, moments)
`validation/poiseuille`	walls, magic parameters, resolution behaviour
`validation/decaying_flows`	convergence, isotropy, cross-lattice consistency
`validation/galilean`	frame invariance across operators
`validation/thermal`	scalar transport + one coupled convection case
`validation/mhd`	resistive decay, Alfvén wave, Orszag–Tang divergence
`validation/laplace`	$\Delta p = \sigma/R$, spurious currents, density-ratio sweep
`validation/layered_poiseuille`	two-fluid channel: $\mu(\phi)$, $\boldsymbol{F}_\nu$, the pressure force
`validation/floating_body`	Archimedes' draft and the sign of the metacentric height
`validation/grid_convergence`	full lattice × operator sweep, CSV
`validation/natural_convection`	Rayleigh sweep against the benchmark
`validation/orszag_tang`	published comparison and stability runs
`lbm_app`	throughput benchmark

The four in the lower block are analysis runs, not regression tests: they take minutes and are excluded from `ctest`.

18.1 One overlap between the two code bases is enforced, not checked by hand

§16 states that `src/` and `GPU/` agree where they overlap and that a disagreement is a bug in one of them. Until `tests/cross_colour` nothing enforced it: the rule was checked by running a case on each and comparing printed diagnostics, which is slow, noisy, and exercises only whatever states that case happens to visit.

The test drives *both* colour-gradient operators directly — no simulation, milliseconds — over 320 identical randomised states at $\gamma = 1, 10, 100, 1000$ under both rest-term readings, and diffs all 27 populations. It reaches the branches a droplet never visits: $|\nabla\phi| = 0$, $A = 0$, ρ_{ref} set and unset, ω_{bulk} away from ω, a body force, and an input deliberately off equilibrium. The worst difference is **0 exactly, at both precisions** — sharper than it sounds, since two independently written implementations also *round* identically.

Four things about it are deliberate. Both trees put their lattice and their operator in `namespace lbm`, so one translation unit including both headers collides on `D3Q27` and half a dozen other names; the two objects are therefore joined by `extern "C"`. It lives in the *parent* and reaches into `GPU/include` rather than the reverse, because the parent already links Kokkos — which `ColourGradient.hpp` needs for its node-field Views — and `GPU/` deliberately does not: making `GPU/`'s CMake acquire Kokkos so that it could be cross-checked would spend exactly the independence that makes the comparison mean anything. The two precision defaults are *opposite* (`GPU/`'s `Real` is `float` unless `LBM_DOUBLE` is defined, the parent's is `double` unless `LBM_SINGLE_PRECISION` is), and mismatching them still compiles and reports a disagreement of about 10^{-7} — FP32 epsilon, indistinguishable at a glance from a physics bug, and it cost a real detour while this was being written; the CMake sets `LBM_DOUBLE` on the `GPU` object exactly when the parent is FP64. And the velocity orderings are *checked rather than assumed*: both are D3Q27 and both number their velocities identically today, but the harness matches slot to slot by integer velocity first and reports what it found, because comparing two differently

ordered sets would produce a large, structured and entirely spurious disagreement — the kind that reads as physics.

It has been shown to fail. A test that passes and has never been demonstrated to fail is a test of nothing, and this document records that lesson elsewhere under `GPU/`'s gather/scatter bug. Perturbing one coefficient in one of `cg_source_high`'s seventeen machine-generated slots by 10^{-7} *relative* — 1/3 written as 1/3.0000001, far smaller than any transcription error a human makes — turns seven of the nine checks red at a worst population difference of 4.1×10^{-11}. The tolerance therefore sits about four orders below the smallest plausible real defect.

The target is skipped, with a message, when `GPU/` is absent; that this is one repository today is an assumption confined to that one `CMakeLists`. The pattern generalises to any other operator the two trees share.

19 Reproducing Neffaa, Bos & Schneider (2008)

Decaying two-dimensional MHD turbulence in a circular domain, *Phys. Plasmas* **15**, 092304. The reference asks whether the four Ting–Matthaeus–Montgomery decay regimes survive confinement, and imposes the wall by volume penalisation: $u_0 = 0$ but $B_0 = B_\parallel$, so only $B \cdot n$ is driven to zero. `demonstrator/mhd_decay -regime 1..4` is that case. Numbers, logs and films are in `results/N_neffaa`.

Run at $N = 321$, $r = (19/20)\pi$ in a 2π box, $\mathrm{Re} = 1000$, $\mathrm{Pm} = 1$, fifty domain crossings — their $t = 450$ on the advective clock.

19.1 The initial condition is now exact in both columns

A regime is the *pair* $(E_u/E_B, H_c)$. The energy ratio was always imposed; the cross helicity used to be whatever the random draw gave, scattering by ± 0.28 between seeds, so regime III — which *is* its cross helicity — could not be run at all. Eq. (11) is now implemented, with β solved by bisection against $\cos\theta = H_c/(0.5\sqrt{E_u/E_B})$. Measured at $t = 0$: 0.043818, 0.000001, 0.473611, 0.090000 against Table I's 0.043818, 0.000001, 0.473611, 0.090000, and the same for seeds 12345, 99, 1 and 2026. The seed now picks the realisation, not the case.

Two departures were forced. Taken literally $u_\perp$ is u rotated pointwise by $\pi/2$, which for $u = \nabla_\perp \psi$ is $-\nabla\psi$, whose divergence is ω — not solenoidal, and this code carries B directly rather than as a curl. The independent solenoidal field the initial condition already draws is used instead; $\nabla \cdot b$ starts at 4×10^{-16}. And that field is not independent enough — ψ and a share a mode set, an amplitude and the taper, giving $\langle u \cdot w \rangle = +0.24$, a *floor* on the achievable alignment — so one Gram–Schmidt step precedes the mixing.

19.2 $A = \frac{1}{2}\int a^2$, and what it says

A is Table I's first column, the centre panel of Fig. 2, and regime II's only published signature; it was never computed. The vector potential is now carried through the initial condition rather than recovered by inverting a curl ($\max|\nabla \times a - b| / \max|b| = 1.9 \times 10^{-15}$) and then advected as its own D2Q5 scalar at the resistivity, which is the reference's Eq. (15).

A is gauge dependent and the paper's own note added in proof says so. The gauge used is zero mean over the fluid disc, the one a spectral inversion gives and the one that minimises A; naming it is not optional.

regime	E/A ours	E/A paper	ratio
I	5.49	16	0.34
II	7.96×10^4	3.4×10^5	0.23
III	10.6	31	0.34
IV	8.51	16	0.53

Low by 1.9× to 4.3× at $t = 0$, before any dynamics — and that shortfall is *not ours*. E/A follows from the spectrum analytically: since $\mathbf{B} = \nabla \times a$ the shell spectrum of a is $E_b(k)/k^2$, so

$$E/A \;=\; (1 + \mathrm{alf})\,\frac{\sum_k E_b(k)}{\sum_k E_b(k)/k^2},$$

a weighted mean square wavenumber, and the paper's 2π box makes the mode index the wavenumber itself so no conversion enters. Eq. (6) then predicts $E/A = 4.76$ for regime I against Table I's 16, converged in the cut (4.58 at $k_{\max} = 24$, 4.76 at 170 and at 400). Measured against that prediction this tree is faithful — 1.20, 1.19, 1.31, 1.21 across the four regimes, which is finite-mode realisation scatter — while Eq. (6) itself sits at 0.20–0.44 of Table I. **The disagreement is between Table I and Eq. (6), not between this tree and the paper.**

Three other readings of Eq. (6) were tested and none reaches Table I. Taking it as the enstrophy and current spectrum makes it *worse* (0.07–0.16), which is immediate once E/A is recognised as a mean square wavenumber: moving the spectrum onto ω and j removes two powers of k from $\mathbf{B}$ and smooths the field, where raising E/A needs more small-scale magnetic power. Taking it as the spectrum of the potentials overshoots by 8–18×. The gauge cannot close it either, for the same directional reason: a non-zero mean in a only ever increases A, and zero mean already minimises it. Solving instead for the spectrum Table I would need gives $E_b(k) = k^{0.86}$ Eq. (6), a large-k slope near $k^{-2.1}$ against the k^{-3} stated in the sentence following Eq. (6), with the four regimes agreeing on that exponent independently. `results/N_neffaa/spectrum_check.py` carries the arithmetic.

19.3 The clock, and a correction that changed the result

E_u is the TOTAL kinetic energy, $\frac{1}{2}\int |u|^2\, d^2x$ over V_f, as Eq. (7) writes it. The velocity scale is therefore $u_{\mathrm{rms}} = \sqrt{2E_u/A}$ with $A = \pi r^2 = 27.983$, not $\sqrt{2E_u}$. One eddy turnover is $2r/u_{\mathrm{rms}}$ in the paper's units and $2R/u_0$ in ours — the same dimensionless turnover — so crossings map one-for-one and only the scale differs: 57.65, 0.229, 27.69 and 31.58 paper time units per crossing for regimes I to IV.

An earlier version of this section used 9.0 for all four, taken from reading Table I's Re column as the velocity scale. That column is the printed formula $2r\sqrt{2E_u}/\nu$ applied literally to a total, which carries units of LENGTH and equals the physical Reynolds number times $\sqrt{A} = 5.29$. The consequence was that the runs were read 6.4×, 3.1× and 3.5× too late for regimes I, III and IV and 39× too early for II. Every time-series comparison moved when it was fixed, and mostly toward the reference.

The same correction removes the Reynolds gap. The reference's PHYSICAL Reynolds numbers are 618, 155536, 1287 and 1128 against this tree's 1000: the campaign sits within a factor 1.3 of regimes III and IV and 1.6× ABOVE regime I. The earlier claim of "Re = 1000 against their 3868–7920" compared our physical Reynolds number against their length-dimensioned column.

Table I therefore fails to close against the paper's own equations in three places: E/A against Eq. (6), the Re column against the ratio column (Eq. (10) fixes $E_B = 1/2$, but the implied E_B is 0.70, 4.6×10^{-5}, 0.29, 0.46), and the Re column against its own formula. None of it touches the campaign, which uses only the ratio columns and Figs. 2 and 4.

19.4 What reproduces

The structural progression of Figs. 7–8 for all four regimes, including two regime-specific claims: case II is the only one forming pronounced circular vortices that roll up the current sheets, and case III shows almost identical velocity and magnetic fields. Both are plainly visible in the films.

Selective decay in regime I is reproduced in rate if not in magnitude: E/A falls by a factor 11.6 against the reference's 12.4, agreeing to 6 % while the absolute values differ threefold. The disagreement is in the initial length scale, not in the decay.

Regime II's late-time decay rate gives $\alpha = 1.76$ against the paper's 2 and the circle's largest Stokes eigenvalue 1.64 — the best quantitative agreement in the comparison.

19.5 What does not, and why

On the corrected clock most of what looked like disagreement was a reading error. Regime III's dynamic alignment IS reproduced through the window the paper plots: $\cos\theta$ is 18 % high at $t = 100$ (0.866 against 0.733) and 6.6 % low at $t = 200$ (0.691 against 0.740), falling away only at $t = 450$, some 28× further into the decay than $t = 0$. Regime IV's negative alignment is reproduced to within 33 % at every time, sign included, and its decay exponent is -0.70 against the stated -0.4 rather than the -1.33 first reported.

Two genuine disagreements remain. Regime I's alignment climbs to 0.906 in the reference while ours goes negative and stays there — and since the campaign ran 1.6× ABOVE that regime's Reynolds number, insufficient Re cannot be the explanation. And regime III loses the equipartition its regime is defined by, E_u/E_B falling to 0.035 where the reference holds 0.72.

Regime II is not a disagreement but out of range: its velocity scale puts $t = 450$ at 1964 crossings where the campaign ran 50, reaching $t = 11$. Extending it needs a 39× longer run.

The largest remaining caveat is therefore duration and realisation rather than Reynolds number. Every regime here is a single draw, the paper does not state how many realisations its curves average, and a single two-dimensional decay at these Reynolds numbers is not self-averaging — which is the most likely reason regime I's alignment goes the other way.

Not attempted: Fig. 9's final-state scatter plots, which need ψ and hence a Poisson solve; Fig. 5's PDF of the local $\cos\theta$; and Fig. 3's periodic control. The E/A shortfall is not among the open items: it is a property of the reference, as above.

20 Known limitations

Stated plainly, because each has been measured.

1. **An on-node regularised fluid wall under a non-uniform body force diverged over $\sim 10^5$ steps, and the mechanism is not diagnosed** (§7.6). With the density-anomaly force the first version of `ice_equilibrium` went non-finite at 1.7×10^5 and 4.2×10^5 steps, the growth beside the water wall where no drag acts; the halfway pair ran 2.7×10^6 steps clean. The on-node Rayleigh–Bénard numbers are growth rates over 0.75 diffusive times and cannot see it.

2. **An isothermal enthalpy front is first order at equilibrium.** Its steady position is any link in a window about a cell wide, and which one it reaches depends on the direction it came from (§7.6). A mushy band removes the ambiguity at the cost of a model error set by its width.

3. **The closed cavity's electric Nusselt number is 9–16 % below the reference in the mid-range, and that is measured rather than explained** (§12.5.4). The *shape* of Fig. 8 is reproduced — $N_e = 1$ below onset, a steep rise, a plateau, renewed growth at $\kappa = +0.35$ — but at $T = 500$–3000, where the $N = 129$ grid *is* resolved ($\mathrm{Re}_{\mathrm{cell}} \leq 2.3$), the

level sits low. Mach number is excluded (3 %), the seed is excluded by the unseeded sweep, and the wall columns contribute under 1 %. Resolution is *not* excluded there and cannot be from this implementation, since the reference's 500^2 costs about five hours per T on one machine. At the top of the range it has now been run on a device (§16.13) and the deficit does not survive: $N_e = 3.10 \pm 0.41$ against 2.80. A matched-resolution comparison across the whole sweep has still not been made.

4. **The perfectly conducting magnetic wall cannot close a domain, and the failure is not what any of three plausible explanations predicted.** `MagNeumann` ($\partial \boldsymbol{B}/\partial n = 0$) is validated on Hunt's duct, where it sits on *one* opposing pair of walls and the other pair is Dirichlet. Applied to every face of a closed domain — a confined decaying MHD field, `demonstrator/mhd_decay.cpp` — it goes non-finite inside sixty steps at the *grid scale*: the magnetic energy rises by a factor 263 over twenty steps, $\max|\boldsymbol{b}|$ goes $0.14 \to 19.2$, and the magnetic integral scale collapses to one cell. Three explanations were tested and falsified. It is not the staircase: a square domain, where 0.9 % of Neumann nodes read another wall node one step inward against a disc's 14.6 %, fails identically. It is not $\omega \to 2$: $\omega_{\mathrm{mag}} = 1.949$, 1.901, 1.769 and 1.586 all fail identically. It is not a slow drift of the unpinned field level, which cannot produce a factor 263 in twenty steps at the grid scale. What holds is the control — Neumann on one face pair with Dirichlet on the other decays cleanly and tracks the insulating run to two figures. So the condition requires a Dirichlet boundary somewhere on the domain, and the drift it leaves behind is visible even then: in that control the domain mean $|\langle \boldsymbol{b} \rangle|$ climbs two decades while $\max|\boldsymbol{b}|$ falls one. The mechanism of the fast growth is *not* diagnosed. What has changed (2026-09-25) is that a closed domain no longer needs it: the on-node parity wall, `MagneticSolver::set_parity_walls`, imposes the conducting condition as B_n odd and B_t even about the wall — $\boldsymbol{B}\cdot\boldsymbol{n} = 0$ *and* $\partial_n \boldsymbol{B}_t = 0$, which is what $\boldsymbol{n} \times \boldsymbol{E} = 0$ gives at a wall with $\boldsymbol{u} = 0$ — where `MagNeumann` sets ∂_n of *every* component to zero and so leaves B_n free. It closes a box on all six faces, reproduces the periodic box it mirrors to 10^{-14} (`validation/mhd_parity_wall.cpp`), and keeps B_n at 10^{-32} on the wall nodes. That the unpinned normal component is what made the all-Neumann box grow is consistent with this and has not been isolated.

5. **The ghost-cell free-slip wall `SpecWall` is exact only for a steady flow.** Its 5.5×10^{-14} is a steady, wall-invariant channel. Against the periodic box whose symmetry planes lie halfway between nodes — the planes it claims — a decaying shear wave differs by 2.4×10^{-2}, 1.2×10^{-2} and 6.1×10^{-3} of the field at $M = 16$, 32, 64 (first order), and a mode varying along the wall by 13 % at $M = 16$ (`validation/specwall_unsteady.cpp`). The ghost permutes at its own update, so a population returns two steps after it was sent and a diagonal one lands a cell further along the wall. The on-node `SpecNode` is an unsteady identity and is the one to use where the transient is measured.

6. **A regularised wall on the edge of a `GPU/` array has no unknown directions.** That array wraps, and `build_reg_walls` marks a direction unknown only when its source is Solid or Excluded, so a box flush against the array edge built its wall stress from the opposite face, with no message: the no-slip Taylor–Green twin's wall vorticity came out at half the parent's after forty steps. Every earlier `GPU/` wall case had solid rows beyond its walls. The builder now warns, and `src/tg_mhd.cu` surrounds its box with a one-cell Solid shell, after which the two codes agree to every printed digit.

7. **Volume penalisation cannot reach the permeability its source method uses, and the boundary condition is therefore imposed to a few per cent rather than to zero.** An explicit penalisation is stable only for $\epsilon \gtrsim \Delta t$, and $\Delta t = 1$ on a lattice, so ϵ is $O(1)$ where Neffaa, Bos & Schneider use $\epsilon = 10^{-3}$ with $\Delta t = 5 \times 10^{-4}$ — three orders of magnitude, and a property of the discretisation rather than of a parameter choice. Measured on the magnetic

condition in `demonstrator/mhd_decay.cpp`, where the penalised wall claims $\boldsymbol{B}\cdot\boldsymbol{n} = 0$: the ratio $\max|\boldsymbol{B}\cdot\boldsymbol{n}|/\max|\boldsymbol{B}|$ on the boundary ring starts at 5.7 % for all settings, grows to 26 % with the term removed, and falls to and holds 2.7 % at $\epsilon_m = 2$, with $\epsilon_m = 20$ in between. So the condition is genuinely imposed and monotonic in strength, and its floor is a few per cent. The same arithmetic bounds the velocity side: `validation/stokes_disc.cpp`'s exact azimuthal decay mode puts the penalised wall at $R-0.431$ cells against a staircase's $R-0.096$, both constant to 1 % over a 4× refinement, so penalisation is not a different convergence class but the same fixed offset 4.5× larger. What penalisation buys in exchange is that $\boldsymbol{u}$ and $\boldsymbol{B}$ are penalised through one indicator and are therefore co-located by construction, with no half-cell mismatch to price.

8. **The penalised perfectly conducting wall has a resolution ceiling between $N = 96$ and $N = 128$, and the two parameters that postpone the failure do not cure it.** Distinct from the sharp `MagNeumann` entry above: this is the *volume-penalised* $\boldsymbol{S} = -\chi(\boldsymbol{B}\cdot\hat{\boldsymbol{n}})\hat{\boldsymbol{n}}/\epsilon_m$ on a confined sphere, which does close a domain — but only up to a resolution. Measured in `GPU/src/mhd_sphere.cu` at $Re = 500$, $Pm = 1$, $k_{\max} = 6$, $k_0 = 4$, matched seed, two turnovers ($32N$ steps), one flag apart from an insulating run that survives every N:

N	ω	outcome	blow-up
96	1.912	survives $2\,T_e$	—
128	1.884	non-finite	step 448 ($t/T_e = 0.219$)
160	1.857	non-finite	step 460 ($t/T_e = 0.180$)
192	1.831	non-finite	step 432 ($t/T_e = 0.141$)
224	1.806	non-finite	step 420 ($t/T_e = 0.117$)

Three things make this a measurement rather than an anecdote. The blow-up step is 440 ± 20 *independent of* N while t/T_e falls monotonically, so the timescale is fixed in lattice units and not convective. It is not $\omega \to 2$, and the ladder runs the wrong way for that reading: $\nu = u_0 2R/Re$ grows with N, so ω moves *away* from 2 as the failure sets in. And it is not a port defect — at $N = 96$ the CUDA run agrees with its Kokkos parent (`demonstrator/mhd_sphere.cpp`, no shared headers) to -0.008 % on E_u and -1.6 % on E_b at $t/T_e = 2$, with Bn/B identical to four decimals at every sampled point.

The stiffness bound of the entry above is visible directly, and it moves with N. At $N = 128$, sweeping the magnetic penalisation time gives non-finite at step 208, 288 and 448 for $\epsilon_m = 0.5$, 1, 2 and survival at $\epsilon_m = 4$ and 8 — monotonic, and exactly the $\epsilon \gtrsim \Delta t$ rule with the threshold no longer $O(1)$. The layer width is *not* a clean control: `smooth` = 1, 1.333, 2, 2.667 gives fail, fail, survive, fail. Neither knob transfers. At $N = 224$, `smooth`= 2, `smooth`= 2.5 and $\epsilon_m = 4$ all still go non-finite, at steps 448, 544 and 672: they buy delay proportional to the softening and nothing else, and $\epsilon_m = 4$ has by then degraded Bn/B from 0.0005 to 0.0141, i.e. it is no longer the boundary condition asked for. Matching the layer as a *fraction* of R fails too — `smooth`= 1.333 at $N = 128$ is the same 2.6 % of R as `smooth`= 1 at $N = 96$ and dies at step 480 — so what matters is an absolute number of cells, which is the signature of a grid-scale mechanism rather than an under-resolved wall layer.

The obvious objection — that ϵ, ϵ_m and `smooth` are frozen in *lattice* units, so the ladder silently stiffens the wall as it refines — is real and was checked. They are a time in steps and a length in cells, and $T_e = 16N$ steps, so $\epsilon_m/T_e = 1/(8N)$: at $\epsilon_m = 2$ the wall is physically 2.33× stiffer at $N = 224$ than at $N = 96$, in the one direction the sweep says is fatal. Repeating the ladder with all three scaled by $N/96$ — the same physical problem refined — still goes non-finite at every N, at steps 432/440/408/420 for $N = 128/160/192/224$ against 448/460/432/420 unscaled: at most 4 % apart, and a blow-up step is not a reproducible

metric at that precision. So the ceiling is not an artefact of the parameter freezing. It also sharpens the statement: at $N = 128$ the physical match $\epsilon_m = 2.667$ fails while $\epsilon_m = 4$ survives, and at $N = 224$ the match 4.667 fails and so does 4, so the ϵ_m required for stability grows *faster than linearly* in N — no fixed physical wall stiffness survives refinement, and stability is only ever bought by making the wall softer than physical, which gives the boundary condition up rather than imposing it.

Two things this is not. It is not a convergence study: the ladder is *acoustic*, $u_0 = 0.05$ at every N, so $Ma = 0.0866$ throughout and the compressibility error does not refine away — a diffusive ladder ($u_0 \sim 1/N$, ν fixed, steps $\sim N^2$) has not been run. And the mechanism is *not* diagnosed. Series, both ladders and the full sweep in `results/N_mhd_sphere/cond_ladder/`.

9. **The on-node zero-flux scalar wall is about twice the halfway one's error, and the mechanism is not diagnosed.** With the fluid frozen, where the charge/potential solution is exactly one-dimensional and any lateral structure is the wall, `ScalarSpecular` gives peak lateral deviations of 16.68/7.06/2.23 % at $ny = 41/81/163$ against the halfway wall's 14.41/4.49/1.14 %, both near second order (§12.5.2). What it did remove is the saw-tooth; a larger smooth wall layer took its place, and that layer is itself an $\omega \to 2$ effect — ten times the halfway wall's at weak injection, collapsing as ω_q leaves 2. It is the charge's own wall, not the potential's. `validation/scalar_specular.cpp` cannot see it: that identity uses `ScalarBGK`, a solenoidal velocity and no source, at $\omega = 1.99$. Two untested pairings — the central-moment charge operator, and a drift with $\nabla\cdot\boldsymbol{v} \neq 0$ — are the obvious suspects and neither has been checked.

10. **Two specular fluid walls, on different planes, and picking the wrong one is silent.** §8.2's is a ghost cell, half a cell outside the last fluid node; §8.4's is on-node and pairs with the on-node scalar family. Each is exact against the box it mirrors — 5.5×10^{-14}, 1.9×10^{-15} and 5.3×10^{-15} — but exact about different planes, and the same box closed by the wrong one is out by 16 % of the field. Axis-aligned faces only in both, and the ghost version additionally refuses edges. The on-node pairing reduces §12.5.2's lateral gap by 2.8× at the finest grid but does not close it, and at $ny = 41$ it lands on the wrong branch of the subcritical bifurcation for every seed tried.

11. **The central-moment operators are unusable on a GPU in this code.** Two `MomentCollision` configurations did not finish in seventeen minutes at 64^3 over 50 steps, where BGK took ~ 0.03 s — roughly four orders of magnitude (§16.1). Since the campaign of §12.1.6 and the aorta both run central moments, the 447 MLUPS BGK figure is *not* this code's GPU throughput and must not be quoted as such. The cause is not the algorithm: the same scheme runs at 99.6% of BGK in the independent CUDA implementation, and at only 2.18× BGK on four host threads. A *mechanism* was identified on 2026-09-04 and removed — a moment loop that did not unroll, which forced the basis's exponent table and all three moment arrays into memory — but the collapse is not thereby closed. The original figure is a *timeout*, not a measurement; the diagnosis was made with clang rather than `nvcc`; and no device has been available to confirm either. Treat this as narrowed, not solved, until `cmbench` has printed a number on a GPU.

12. **The CUDA implementation of §16 no longer covers only a fraction of this one.** What it still lacks is raw MRT, height-field input and an open boundary for the *fluid*, so it cannot run the aorta or a city; and no moving obstacle in the free surface, deliberately. What it has is shifted storage, TRT, regularised velocity walls, the open SCALAR boundary, Dellar's moment-based magnetic wall and with it the wall-bounded `src/hartmann.cu`, the free surface on a two-lattice path, the EHD stack of §16.13, and both immiscible models of §10 (§16.11). Its lattices are D3Q27 for the fluid and D3Q7 — or D3Q27 where the

central-moment phase operator is wanted — for the scalar, the induction distribution and the phase field. It exists to answer one question and to carry the multiphysics forward on a device, not to replace this code.

13. **No MPI.** The code is single-rank. `Domain` carries an `mpi_halo` hook and the halo machinery exists, but domain decomposition is not implemented. Note that Esoteric Pull will require a two-way, parity-dependent halo exchange (§3.3).

14. **The scalar module has no on-node zero-flux wall, and that now costs a measured 4%.** `ScalarAdiabatic` is halfway bounce-back and `ScalarOutflow` is an open boundary; neither is an on-node mirror. With a free-slip lateral wall the electroconvection case therefore rings: the cell-to-cell mode reaches 0.4808 in the first columns against 0.0167 for the doubled box, because $\omega_q = 1.99952$ and bounce-back at $\omega \to 2$ re-injects an odd–even mode the collision would have annihilated (§12.5.2). The peak velocity moves +3.96% at $n_y = 41$, falling at roughly first order. The doubled periodic box avoids this by having no lateral boundary, and is the default for that reason — but a geometry that genuinely needs one free-slip wall *without* a mirror partner, and carries a scalar, will meet this. The free-slip wall itself is exact (§8.2); it is the scalar's companion condition that is missing.

15. **Electroconvection at $T = 190$ sits 1.6% above the reference, and refinement does not close it.** Grid-converged (+0.20% from $n_y = 81$ to 163) and seed-independent (0.149% over three decades of seed amplitude), so neither resolution nor the initial perturbation explains it. $T = 420$ agrees to −0.18% on the same grid. The asymmetry is itself the clue — near a bifurcation the same effective-T offset is amplified far more at the lower T — and points at u_0, which is 10^{-2} here against the reference's 10^{-3}. *Untested:* the u_0 sweep that would settle it has not been run. Their Table 5's coarse-grid values are likewise not reproduced (§12.5.2), for the opposite reason: this case has no coarse-grid error to match.

16. **The hydrostatic EHD convergence rate is not reproduced away from weak injection.** 1.57 at $C = 0.1$ against the published 1.6, but 0.89 and 1.00 at $C = 1$ and $C = 10$ (§12.5.1). The profile and the error magnitudes are reproduced. Two candidates remain unseparated: the potential's BGK source, whose magnitude scales with C, and the charge operator at $\omega_q \to 2$.

17. **The EHD stack *is* in `GPU/`**, as of 2026-09-06 — `src/ehd_cavity.cu`, `include/lbm/ehd.cuh` and `include/lbm/specular.cuh`, described in §16.13. This entry used to say the opposite while that section sat 540 lines earlier in the same document. What the port cost is recorded there: Dellar's on-node Dirichlet, the on-node zero-flux wall, a per-node source and the central-moment charge collision, the last as a closed form rather than the parent's moment transform. Cross-checked against the Kokkos twin at matched lattices, I_0 agrees to 0.008% with no shared headers.

18. **Convergence studies must watch the Mach number.** The inlet Poiseuille of §12.1.12 reaches a velocity-dependent error term that does not refine away, and at $U_0 = 0.005$ it dominates by $L_y = 641$ and pulls the fitted rate from 2.067 down to 1.964. This is the cubic aliasing defect $c_{i\zeta}^3 = c_{i\zeta}$ discussed in [12], not a boundary-condition problem: it survives moving the outlet, changing its order, and refining. Any new convergence test should sweep U_0 before a shortfall in the fitted rate is blamed on the scheme.

19. **The scalar module has no shifted-storage default.** T_{ref} must be set to the mean temperature by the user for FP32 accuracy; leaving it at 0 reproduces unshifted storage.

20. **The scalar module has no on-node zero-flux condition.** `ScalarAdiabatic` is bounce-back, so an insulated wall always sits half a cell inside the boundary node. That is consistent

with halfway bounce-back for the fluid and inconsistent with the regularised condition, which is on-node — so a case whose *velocity* boundary must be on-node, because a wall moves, cannot have an on-node adiabatic wall to go with it. Case 3.5 of §12.3.5 is exactly that case: its isothermal walls sit on nodes and its two adiabatic flanks half a cell in, making the cavity $N-1$ tall and $N-2$ wide. The aspect-ratio error is $1/(N-1)$ and refines away, but it is a gap in the boundary set rather than a property of the problem.

21. **Conjugate heat transfer varies the diffusivity, not the heat capacity.** `ScalarBGK::omega_of` gives every node its own relaxation rate, which reproduces a conductivity ratio only where ρc_p is uniform: the scheme transports $\sum_i g_i$, and that sum is the temperature rather than the enthalpy. A jump in volumetric heat capacity needs a different transported variable and is not implemented. Where ρc_p *is* uniform the interface is exact, not merely second order — §12.3.5 measures $10^{-9}\%$ at ratios from 0.1 to 100.

22. $\nabla\cdot\boldsymbol{b}$ **is not preserved to machine precision** (order 1.65 convergence). No divergence cleaning is implemented. **On a long unforced run it reaches parity with the field's own curl.** Orszag–Tang, the only other case exercising the property, is short and leaves the divergence at truncation level; the confined decay of `demonstrator/mhd_decay.cpp` runs 29 turnovers and measures $\mathrm{rms}|\nabla\cdot\boldsymbol{b}|/\mathrm{rms}|\nabla\times\boldsymbol{b}|$ growing from 8×10^{-16} to 1.04: 0.097 at 1.7 turnovers, 0.354 at 4.9, 0.976 at 19.4. The consequence is specific rather than general. A spurious gradient field contributes to E_{mag} and nothing to $\langle j^2\rangle$, so the magnetic integral scale $L_b=\sqrt{2E_{\mathrm{mag}}/\langle j^2\rangle}$ is inflated once the divergence is appreciable and its 5.5× growth over the full run is an upper bound — inside the window where the ratio stays under 10 % the growth is 1.26×. What is unaffected is anything built from $\nabla\times\boldsymbol{b}$, which is blind to a gradient field by construction. The *energy* fraction carried by the spurious part is *not* bounded by this measurement, since that ratio also carries $(L_{\mathrm{spurious}}/L_b)^2$ and the spurious field's length scale was not measured.

23. **Magnetic walls now carry the wall's conductivity, and this entry used to say they did not.** Both conditions exist: `MagDirichlet` sets $\boldsymbol{b}$ at the wall, which is the insulating case (the induced field has nowhere to go and vanishes), and `MagNeumann` sets $\partial\boldsymbol{b}/\partial n=0$, which is the perfectly conducting one (the tangential electric field vanishes instead). They give measurably different flows — Shercliff's duct against Hunt's — so the second is the counterpart of the first and not a refinement of it. What remains a limitation is the *closure*, which is the separate entry near the head of this list: the Neumann condition needs a Dirichlet boundary somewhere on the domain and cannot be applied to every face of a closed one.

24. **FP32 cannot resolve the finest convergence tests.** At $H=64$ the discretisation error falls below the FP32 round-off floor; the affected assertions are FP64-only by design.

25. **Stability results are bounded by the run length.** "Stable through $t=1.20\,\mathrm{s}$" means no divergence was detected before that cap, not that the run is unconditionally stable.

26. **The regularised condition has a curvature-induced wall slip.** Whenever the tangential velocity has wall-normal curvature the effective wall is displaced by

$$\text{offset}\times W=\frac{4}{3}\,\frac{\omega-1}{\omega^2}\qquad\Longleftrightarrow\qquad \delta u_{\mathrm{slip}}=-\frac{2}{3}\,\frac{\omega-1}{\omega^2}\,\frac{\partial^2 u_{\parallel}}{\partial n^2}. \tag{143}$$

It is zero for a linear profile — Couette is exact to 4×10^{-15} at every τ from 0.6 to 2.0 — and zero at $\omega=1$, but it is the whole of the offset column in §12.1.5 and it limits Ra_c for the on-node wall pair in §12.3.4. It is intrinsic to truncating $f^{(1)}$ at second Hermite order, which is what "regularised" means: a property of the method, not a defect of this

implementation. No correction is applied, but since (60) is exact a user needing better than $O(h)$ wall placement in a curved profile can run at $\omega = 1$ or subtract it.

Derivation. The source is in $f^{(2)}$ — curvature cannot appear in $f^{(1)}$, which carries only first derivatives:

$$f^{(2)} \supset \frac{1-\omega/2}{\omega^2}(\boldsymbol{c}_i\cdot\nabla)^2 f^{\mathrm{eq}} \supset \frac{1-\omega/2}{\omega^2}\frac{w_i}{c_s^2}c_{iy}^2 c_{ix}\,\rho\,u'', \tag{144}$$

odd in $\boldsymbol{c}_i$, hence exactly what the bounce-back scaffold reverses. Its half-space contraction is an exact lattice sum,

$$\sum_{c_{iy}>0} w_i c_{ix}^2 c_{iy}^3 = \frac{1}{18} \quad\Longrightarrow\quad \delta\Pi_{xy} = -\frac{1}{3}\frac{1-\omega/2}{\omega^2}\rho\,u''. \tag{145}$$

Decomposing f^{neq} at an interior node of a converged channel reproduces this source with ratio 1.000 at $\tau = 0.6, 0.8, 1.0, 1.5$, the probe first validated against $\sum_i f^{\mathrm{neq}} = 0$ (to 10^{-17}) and $\sum_i \boldsymbol{c}_i f^{\mathrm{neq}} = -\boldsymbol{F}/2$.

For the wall itself, group the populations by c_{iy} and take the x-odd combination $g_{c_y} = f_{(1,c_y)} - f_{(-1,c_y)}$. The steady recurrences give $(1-\frac{\omega}{2})r^2 - (2-\omega)r + (1-\frac{\omega}{2}) = 0$, i.e. $(r-1)^2 = 0$: a double hydrodynamic root and *no* kinetic boundary layer, so the discrete solution is exactly the parabola and the whole effect sits in the closure. With $u = 3(g_+ + g_-) + C$, $C = F[(2-\omega)/\omega + 3/2]$, the reconstruction giving $G_{c_y} = -w_{c_y}F/c_s^2 + 2w_{c_y}c_y\Pi_{xy}/c_s^4$ and the corrected scaffold giving $\Pi_{xy} = -2\hat{g}_- - F/6$, the wall closes to $g_+(0) + g_-(0) = -F/6$.

The step a one-way argument cannot see is that node 0 collides against the *imposed* $\boldsymbol{u}_w = 0$ rather than the bulk-extrapolated velocity, so continuity with node 1 requires $g_+(0)|_{\mathrm{BC}} = p(0) + \omega U/[6(1-\omega)]$. Eliminating,

$$U\frac{2-\omega}{6(1-\omega)} = \frac{C}{3} - \frac{F}{6}, \qquad 2C - F = \frac{4F}{\omega} \quad\Longrightarrow\quad U = \frac{4F(1-\omega)}{\omega(2-\omega)}, \tag{146}$$

and dividing by $du/dy|_0 = AW$ with $A = 3F\omega/(2-\omega)$ cancels both F and $(2-\omega)$, leaving (60). The coefficient $\frac{4}{3} = 2 \times \frac{2}{3}$ is rational because both factors are exact lattice quantities. Measured: exact at $\tau = 0.6, 0.7, 0.9, 1.0$, within 0.05% at $\tau = 0.8$, 1% at $\tau = 2$ where $\nu = 0.5$ is far outside the hydrodynamic regime assumed.

Note that the force corrections of §8.1 *raise* the reported Poiseuille error at $\tau = 0.8$, from 9.07×10^{-3} to 1.71×10^{-2}, because the force defect and the curvature slip have opposite signs there and were partially cancelling. That cancellation was accidental and τ-specific; at $\tau = 1$ the corrected condition is exact.

27. ω_3 **(bulk) in the MHD central-moment scheme** is not specified in [6]; the value 1 was used here and is exposed as a flag. The stability comparison should be read as close rather than exact reproduction.

28. **No wall compliance.** The arbitrary-geometry path (§13) treats the wall as rigid and static: halfway bounce-back on solid voxels, with no fluid–structure coupling. For the aorta this is the largest single departure from the physics being depicted. A real aorta distends in systole and stores volume elastically, and the pulse wave propagates along the compliant wall at ∼5 m/s; that Windkessel behaviour is the dominant feature of aortic haemodynamics. With a rigid wall the vessel responds essentially instantaneously and there is no pulse wave at all — the pulsatile case of §13.4 produces a pulsatile *flow rate*, not a pulse *wave*, and should not be read as the latter. Nothing in the solver approximates compliance, and none is faked.

29. **Voxelised walls are a staircase, and wall shear stress from them is not trustworthy.** The surface is quantised to half-integer lattice planes. Halfway bounce-back is second order

for a wall that genuinely sits at the halfway position, but an oblique vessel wall does not, so the geometric error is $O(h)$ and the effective radius is uncertain by about half a voxel. No interpolated curved-boundary treatment (Bouzidi, Filippova–Hänel) is implemented. Two consequences are measured rather than asserted: the staircase corners produce the speed outlier of §13.4 ($p_{99.9} = 0.026$ against $p_{100} = 0.070$), and because the central-moment operator relaxes all moments of order ≥ 3 to equilibrium there is no free rate left to set the magic parameter — the equivalent $\Lambda = (\tau - \frac{1}{2})/2 = 0.0197$ sits nearly ten times below the 3/16 at which the bounce-back wall is τ-independent (§12.1.1), so the effective no-slip plane is offset and moves with viscosity. Voxel-scale gradients at the wall should therefore not be used to report WSS.

30. **Performance figures are from a compute-bound machine** and do not predict GPU behaviour (§17).

31. **There is no contact-line or wetting model.** `PhaseWall` is zero-flux on the phase populations, which is correct for the transport but sets no contact angle, and the gradient stencil reads whatever ϕ the wall node holds; a penalised body is invisible to the phase field, which is advected by the fluid velocity alone. The consequence is measurable rather than notional: in `validation/floating_body` the draft taken from the fictitious mass and the draft taken from the geometry differ by 1.8%, and that gap *bounds* how far the waterline slides on the hull without separating it from the discretisation of either measurement. For a small enough body there is also no surface-tension contribution to the draft, so a Bond number of order one is out of scope. An interface should not be put against a wall, and the Rayleigh–Taylor demonstrator is legitimate only because its spike does not reach the floor within $t^* = 3$.

32. **The phase field has no open boundary.** The scalar's donor machinery (§8.8) would port, with the same second pass and the same fence, but it is not written. So an interface that must leave the domain is not available, and every WALL-BOUNDED multiphase case here is periodic in one direction and closed by no-slip walls in the other. Not every case is wall-bounded — `laplace` and `static_droplet` are triply periodic, as §12.2.1 says — and an earlier version of this entry drew the consequence too widely.

33. **There are two pressure normalisations and neither dominates.** With $p = \rho(\phi)c_s^2\tilde{p}$ the recovered mismatch is $F_p = -\tilde{p}\,c_s^2\nabla\rho$, which scales with the pressure *level* and is amplified by the density ratio; with a constant $p = \rho_0 c_s^2\tilde{p}$ it becomes $c_s^2(\rho - \rho_0)\nabla\tilde{p}$ and scales with the gradient. On the layered channel the first does not converge at all — fitted order -0.21, with $\max|F_p|/G$ at 220 and 408 for $H = 32$ and 64 — while the second converges at 0.95; a gauge shift in $\tilde{p}$ that changes no physics moves the error seventeenfold under the first and about twofold under the second. It is not a free win, and what does *not* improve is the size of the term: F_p/G is *larger* under ρ_0, 379 against 220 at $H = 32$. The recovered pressure term becomes $-(\rho_0/\rho)c_s^2\nabla\tilde{p}$, so one phase always pays: $\rho_0 = \rho_L$ diverges at a Laplace jump at a ratio of 10 and above, on a case the local-ρ form handles at a ratio of 100, and $\rho_0 = \rho_H$ gives the light phase an effective sound speed of 1.83 lattice units per step at a ratio of 10 and diverged inside 500 steps. The default stays local-ρ because that is what every validated result in this tree was obtained with.

34. **The multiphase central-moment operator requires a product lattice.** It is built on `ProductBasis`, and the `static_assert` says so: D2Q9 or D3Q27. Both *Navier–Stokes* lattices in this tree satisfy it; D3Q7 and D2Q5 do not and never did, because `is_product_lattice` demands all 27 or all 9 sign combinations and a seven- or five-velocity set does not have them. An earlier version of this entry said "both three-dimensional lattices", which is the sentence a reader would use to conclude that a D3Q7 scalar or phase field could take a

central-moment operator. It cannot: that is what the assertion catches, and it is why a 3-D phase field wanting central moments must transport on D3Q27. What D3Q19's removal on 2026-09-18 changed is narrower — it was the one Navier–Stokes lattice that was not a product set.

35. **Shifted storage is unavailable to the two PRESSURE-FORM multiphase operators.** `ShiftedPopulations` centres the stored variable on $\tilde{p} = 1$, which is precisely the pressure gauge that must be avoided at a density ratio, and it assumes a reference density of exactly 1 — at a ratio of 10 the stored populations are no longer small and the shift stops buying anything. `MultiphaseCentralMoments` therefore declares `RawPopulations` outright, and the FP32 accuracy argument that shifted storage exists for does not apply to those cases. It does *not* extend to `MultiphaseBGK`, the MATCHED-DENSITY capillary-stress operator, which takes its storage as a template parameter and is instantiated with `ShiftedPopulations` in `validation/laplace.cpp` — exactly the case the condition above excludes, since at a matched density there is no ratio to spoil the gauge.

36. **The free surface has no surface tension** (§11). ρ_G is uniform, so the gas pressure carries no curvature term, and every case at a Bond number where surface tension competes with gravity is out of scope. That excludes most droplet problems; it excludes none of the dam-break family, which is what the module is for.

37. **The free surface's moving obstacle is not reliable**, and the cause is located rather than suspected. `mass_exchange()` conserves mass by antisymmetry across a face, which assumes both cells agree on the mask for the whole pass; `move_obstacle()` changes the mask between passes, so every cell the body sweeps breaks the assumption. `transfer_covered_mass()` keeps the total right without restoring the per-face antisymmetry. Bounce-back, momentum exchange and refill are all in place and are not the problem. The fix is to make the mass exchange itself aware of the moving boundary, and it has not been attempted.

38. **The free surface cannot use Esoteric Pull**, and therefore cannot be ported to the CUDA implementation without a two-lattice storage path and five kernel launches with a fence between each (§11.3). It `static_asserts` against it rather than failing at run time.

39. **Neither immiscible model dominates the other** (§16.12). On one matched static-droplet case the colour gradient is more accurate on Laplace's law at every density ratio measured and the phase field carries the smaller spurious current at every ratio, by up to two orders of magnitude. The choice is a trade, and the measurement covers one case — a static droplet — not a flow.

40. **The rigid body's two paths are separate, and the 6-DOF one is demonstrated rather than validated.** The planar path carries one angle and one angular velocity about z; the six-degree-of-freedom path (`src/solver/RigidBody3D.hpp`) carries a quaternion and an inertia tensor held in the body frame and rotated out each step. They are chosen by the shape, and there are THREE groups rather than two. `Rect` and `Wedge` are planar prisms and use the validated 3×3. `Box` and `Disc` declare `six_dof = true` and take the 6×6. `Sphere` is neither: three-dimensional in geometry but `six_dof = false`, carrying three translations and no rotation at all, which is exact for an axial entry and for nothing else. Mixing a shape with the wrong solve is a compile error rather than a wrong answer. `tests/test_rigid3d` pins the 6×6 against that 3×3 on a planar problem to 1.7×10^{-18}, so the generalisation carries the older code's validation — but only for the planar limb. What the new three degrees of freedom do in a flow has *no* closed-form check in this tree: `demonstrator/cube_entry` is a demonstrator. A cube released exactly corner-down has a three-fold symmetry axis, which cannot be aligned with D3Q27's four-fold axis about y, so the discrete torque is not zero — and it does not refine away either: at matched $t/t_0 = 2.7$ the untilted cube's diagonal has

moved 0.027° at $D = 12$ and 0.042° at $D = 20$. What is reproducible is its ratio to the response of a deliberate 5° tilt, which is 10.3 at $D = 12$ and 10.2 at $D = 20$ over a matched window and rises to 13 in both by $t/t_0 = 2.7$. A ratio holding to 1 % across a resolution change while both of its terms move by 60 % says the pose is an unstable equilibrium seeded by the lattice, so a corner-down cube entering water tips further rather than righting itself. The drag is the larger error: the cube has fallen $1.492D$ at $D = 12$ against $1.765D$ at $D = 20$ by $t/t_0 = 2.75$, an 18 % difference, matching the sphere's own appetite for resolution.

41. **The body is of uniform density.** Its centre of mass is taken to be the centre of the rectangle, which is what lets its own weight drop out of the roll equation and leaves $-(\boldsymbol{S} \times \boldsymbol{g})$ as the whole righting couple. So the metacentric height follows from the shape and the draft alone — +13.61 for the raft and −6.80 for the pillar in `validation/floating_body`, the two boxes differing only in which side is longer — and ballast low in a hull, the standard way to make a tall body float upright, is exactly what cannot be expressed. It would need a second centre, a metacentric height that no longer follows from the draft, and a weight torque restored to the roll row.

42. **There is no collision model.** Two bodies, or a body meeting a wall, will interpenetrate: nothing computes a contact force. The floor stop in the water-entry demonstrator is a clamp in the driver on the square's diagonal reach, and it is a clamp, not physics.

43. **The only body shape is a rectangle.** `Rect::chi` is the sole indicator provided. Anything with a signed distance function drops straight in, and nothing else does.

44. **Penalisation has no sub-cell surface.** The body is resolved to the smoothing width of its indicator, so a pressure read at the face of the square is a smoothed pressure over a cell or two, and the draft is resolved to the interface width. Integrated force is far more trustworthy than local pressure here, which is the usual situation with this family of methods. The smoothing is not free of bias either: at a width of 1.5 cells the indicator's second moment exceeds the sharp rectangle's by 2.55%, which is why mass and inertia are taken from χ rather than from the nominal dimensions.

45. **Neither scalar operator diffuses at $\tau \to 1/2$, and the regularised one only removes the ringing.** Rayleigh–Bénard at $\mathrm{Ra} = 10^{14}$ on $H = 50$ puts $\alpha = 5.93 \times 10^{-8}$, so the conductive layer needs 8.4×10^6 steps to reach a single cell: the diffusion length is sub-lattice and no operator can represent it. What the operator choice does change is the character of the error. Under BGK ($\omega_T = 1.99999905$, i.e. $h \to 2h^{eq} - h$) the near-wall temperature oscillates without damping, $\mathrm{Nu}_{\mathrm{bot}}$ swinging between 34.9 and 93.6 over sixty free-fall times against an analytic $\simeq 100$, and $\mathrm{Nu}_{\mathrm{vol}}$ between 0.36 and 1.40 where conduction alone requires exactly 1. Annihilating the ghosts removes that entirely — $\mathrm{Nu}_{\mathrm{vol}}$ then reads 1.0000 and $\mathrm{Nu}_{\mathrm{top}}$ exactly 0 — but leaves $\mathrm{Nu}_{\mathrm{bot}}$ at 66 against the same 100, because the sub-lattice transport is ballistic rather than diffusive at this ω. Both operators are exact where the layer is resolved: steady conduction below Ra_c gives $\mathrm{Nu} = 1.0000$ on all three estimators with $|u| \to 3 \times 10^{-16}$, and the transient matches erfc to 0.045% falling to 0.004%. The consequence for `rb_high_ra` is that a Nusselt number from an unresolved run is a property of the discretisation, and the driver prints the volume and both plate estimators so the disagreement between them is visible rather than inferred. At the reference point itself ($H = 50$, $\mathrm{Ra} = 10^{14}$, FP64, central moments) the run does not survive: conduction is clean for some five hundred free-fall times and then it diverges as soon as buoyancy wins, $\mathrm{Nu}_{\mathrm{vol}}$ running $1.04 \to 6.50 \to 504 \to \mathrm{NaN}$ while $\max|u|$ climbs $2.8 \times 10^{-6} \to 5.8 \times 10^{-5} \to 1.0 \times 10^{-3}$. The step at which that happens is not quoted as a measurement — §20 already records that a blow-up step moves by 2× with the backend alone — but the order of events is reproducible, and the lever is H: since $\nu = UH\sqrt{\mathrm{Pr}/\mathrm{Ra}}$, going to $H = 1000$ moves $\tau_f - 1/2$ from 1.26×10^{-7} to 2.53×10^{-6} and

lifts the Nusselt ceiling from 25 to 500 at the same time.

Both codebases now carry that operator, and the two drivers were aligned on 2026-09-04. `src/collision/ScalarRegularised.hpp` is the parent tree's form of it: the same equilibrium, the same diffusivity, identical to `ScalarBGK` population-by-population at $\omega = 1$ (checked exactly, to 0.000000, on D2Q5 and D3Q7 in `tests/test_collision.cpp`), and the same scalar and flux moments at every ω — differing only in that the three ghosts land on equilibrium instead of being relaxed. Its inverse is closed-form rather than a transform, because on a rest-plus-axial-pairs lattice the axes decouple, and it compiles to 36 instructions against BGK's 78 with no stack frame at all.

The alignment mattered beyond that: the Kokkos driver had been running `ScalarBGK`, `omega_bulk` $= -1$ (so the trace relaxed at 1.9984 where both the reference and the CUDA twin use 1), and a default initial condition of `cold` against the twin's `cond`. Two independent implementations of one case are only evidence if they are the same setup, and those three made them three different setups. The temperature gauge is left deliberately different — symmetric about zero in the Kokkos driver, $[0, 1]$ in the CUDA one — since a gauge is not physics; the consequence is that $\mathrm{Nu}_{\mathrm{bot}}$, $\mathrm{Nu}_{\mathrm{top}}$ and $\mathrm{Nu}_{\mathrm{vol}}$ are comparable between them while $T_{\min}$, $T_{\max}$ and $\mathrm{Nu}_{\mathrm{ref}}$ are not.

$\mathrm{Ra}_{\max}$ **is a resolution rule, not a stability rule** (2026-09-05). It says nothing about whether a run survives, and two measurements at $\mathrm{Ra} = 10^{11}$ contradict it in opposite directions. At $H = 498$ on 1000×500, $\mathrm{Ra}/\mathrm{Ra}_{\max} = 0.62$ — the rule says comfortable — and the run *halted* at $t/t_{ff} = 25$ with $T = 1.19$ against a physical maximum of 0.5 and $\mathrm{Nu}_{\mathrm{bot}} = 80.4$ against $\mathrm{Nu}_{\mathrm{top}} = 57.4$, the overshoot running $0.0046, 0.0089, 0.0170$ and then 0.690 in successive free-fall times. The CUDA driver reproduced that failure from a *cold* start on an A100, so $H = 498$ fails from both initial conditions in both codebases. At $H = 98$ on 200×100, $\mathrm{Ra}/\mathrm{Ra}_{\max} = 169$ — the rule says hopeless — and it did *not* halt: $T_{\min}$ reached -0.92 at $t/t_{ff} = 20$ and stayed out of bounds to $t/t_{ff} = 64$, with $\mathrm{Nu}_{\mathrm{bot}} \simeq 95$ at *thirty times* $\mathrm{Nu}_{\mathrm{top}} \simeq 3.1$ and $\langle T \rangle$ at mid-depth still -0.5000 at $t/t_{ff} = 19$ — a layer filling, not a steady state. Neither is a measurement of $\mathrm{Ra} = 10^{11}$; what is worth noticing is that the grid 25× coarser survived where the finer one halted. The hypothesis, with two points behind it and stated as one: far above the $H/2$ ceiling the scheme clips and acts as an implicit LES, while *near* the ceiling it attempts a layer it cannot represent and fails. It predicts $H = 998$ (Nu/ceiling = 0.43) should behave better than $H = 498$ (0.87), which is what the CSF3 array in `GPU/csf3/` is for.

A cold start's seed has to go where the gradient is. Both drivers seeded a density mode at mid-depth, which is where the critical mode peaks for a *conductive* start and is useless for a cold one: a cold start has no gradient except in the diffusing layer at the bottom plate, and $\delta = \sqrt{\alpha t}$ is 2.55 cells at $t/t_{ff} = 7$ for $H = 498$, so a perturbation at $y = 250$ sits in neutrally stratified fluid. Measured there: the mode decayed monotonically, $9.69 \times 10^{-5} \to 6.26 \times 10^{-5}$ over seven free-fall times with its peak pinned to the seed row, while the conductive start at the same parameters had it growing at 1.51× per free-fall time. Since 2026-09-05 the cold start seeds the *temperature* inside the boundary layer instead, non-negatively so the initial field itself stays inside the maximum principle, and broadband because the layer selects its own wavelength from δ rather than from the box. Measured at 200×100: the fluctuation r.m.s. grows at 1.47× per free-fall time ($\sigma = 0.38/t_{ff}$) from $t/t_{ff} = 2$ with its peak at $y = 1$–2, against $\sigma = 0.45/t_{ff}$ for the conductive start — the same instability, now reachable.

The Kokkos twin (`demonstrator/rb_high_ra`) puts numbers on where the configuration is usable. A 200×100 layer at $U_f = 0.05$ with the conductive initial condition, out to a hundred free-fall times: $\mathrm{Ra} = 10^6$ gives $\mathrm{Nu}_{\mathrm{bot}} = 7.40$ and $\mathrm{Nu}_{\mathrm{top}} = 7.48$, agreeing with each other to 1% and with $0.14\,\mathrm{Ra}^{0.29} = 8.0$ to 7% — the positive control that makes the rest of the sweep

readable. $\mathrm{Ra} = 10^{10}$ gives 46.57 and 46.79, agreeing to 0.5% and landing on the CEILING $H/2 = 49$ rather than on the physical 111, which measures the ceiling rather than assuming it. $\mathrm{Ra} = 10^{14}$ diverges by $t/t_f \simeq 30$ with EITHER initial condition: the conductive profile removes the immediate undershoot the reference's cold start causes ($T_{\min} = -0.81$ against a floor of -0.5), but the run still dies once convection develops, so the seed and the τ floor are separate failures acting at separate times. The ceiling therefore sets the usable Rayleigh number, $\mathrm{Ra}_{\max} \sim (H/0.28)^{1/0.29}$: about 6×10^8 at $H = 98$ and 2×10^{12} at $H = 998$, so $\mathrm{Ra} = 10^{14}$ wants $H \gtrsim 3200$ for a single cell in the layer.

Two further cautions from that sweep. The volume estimator $\mathrm{Nu}_{\mathrm{vol}}$ carries a factor H/D — 1.7×10^8 at $\mathrm{Ra} = 10^{14}$ — so an accidental velocity–temperature correlation is amplified enormously; in a state that is provably not convecting it read 180, 606, 1216, 1719 while both plate estimators sat correctly at 1.0, and at $\mathrm{Ra} = 10^{10}$ it alternated sign. Its uncorrelated-noise floor $H u'_{\mathrm{rms}} T'_{\mathrm{rms}}/(D\,\Delta T\sqrt{N})$ is exceeded by a factor 28–36, consistently across a hundredfold change in D, because the spurious velocity is correlated over some 900 cells; the floor is a lower bound and the reliable signal is agreement between the two plate estimators. And the volume relation must be written on the FLUCTUATIONS, $\langle vT\rangle - \langle v\rangle\langle T\rangle$: Guo's half shift puts a uniform $F/(2\rho)$ into the reported velocity, so with a mean temperature away from T_0 the textbook $\langle vT\rangle$ form returned $\mathrm{Nu} = 53.77$ for a fluid at rest, where the answer is exactly 1.

46. **The aorta's outlet condition drives the vessel whenever the inlet waveform returns to zero, and the branch flow split is soft.** Measured 2026-09-13 with the physiological drive (`-wave physio`), whose inlet is at rest for 57 % of the cycle. `NrmOutFree` rescales a copied distribution to an *imposed density*, so with no through-flow left to copy that density becomes the only thing driving the domain: the descending aorta runs faster in diastole than at peak systole (p_{99} of $|\boldsymbol{u}|$ in the exit band 0.0543 against 0.0142), which momentum coasting cannot do. All of the top 0.1 % of speeds lie within two voxels of an outlet cap against none at systole, and 80.5 % belong to one cap. Two explanations were tested and killed — an acoustic echo, whose arrival time agrees to 0.7 % but which a space-time map shows never propagates; and a donor-selection fault, which is a real blind spot in the rule (outward is detected only as *Solid or off-domain*, so a neighbouring outlet node falls through) but does not bite here, the cap resolving its outward direction for 419 of 449 nodes. Demoting the 25 staircase fragment caps (`-fragmin`, 105 voxels, about 19 % of the outlet area at that end) makes it *worse* by a near-constant 11.8 % at every diastolic phase and 1.0 % at systole — a flux-driven boundary, so that flag is a diagnostic and not a repair. Systolic velocities in the ascending aorta and arch survive: those bands move by 0.6–2.9 % under the same perturbation. The flow split does not — the branch end moves by 16–23 % from a change made 350 voxels away, so the division of flow between branch vessels, which §13 calls an output of the simulation, must not be quoted without that caveat.

47. **The constant-pressure outlet cannot pass a vortex, and the symptom reads as a wake instability.** Measured 2026-09-13 on `validation/square_cylinder.cpp`, the confined square cylinder of Breuer et al. (2000). At $\mathrm{Re} = 300$ the confined channel ran healthily — $C_d \simeq 1.6$, C_l swinging through ± 1.3 — and then went non-finite, at step 79 999 with `BGK` and 90 000 with central moments, i.e. the collision operator is not the variable. The blow-up *coordinates* pointed into the far wake, 1200 cells past the cylinder and 260 short of the outlet, and that reading was wrong: the non-finite region was a contiguous block spanning $x = 1745 \ldots 1999$ over the *full* channel height, 255 columns anchored on the boundary and walking upstream a cell a step. A location extracted from a field that is already 12.7 % destroyed is the propagation front, not the origin; the same mistake made earlier in the same session with a "first non-finite node" report at 99.1 % destroyed. `NrmOutEq` — ρ imposed, every velocity component extrapolated, populations set to equilibrium, which

is what Breuer et al. (2000) describe for their own lattice-Boltzmann runs — clears it, and the case then averages 12 shedding cycles to $C_d = 1.7113$ and $St = 0.1292$ at Ma $= 0.0866$. Halving the Mach number to 0.052 moves C_d to 1.5908 and the lift amplitude from 3.1855 to 2.9575 — both forces down about 7 % — while St goes to 0.1294, i.e. does not move at all. Compressibility scales the pressure field so the forces follow it, and the shedding frequency is kinematic and does not; anything that had altered the flow rather than the force scale would have moved St too. The 7 % by which the first run sat above the paper's LBA curve was a Mach artefact. It is bounded rather than apportioned: τ must move with $u_{\max}$ at fixed Re and D, so the two are confounded, and separating them needs τ held fixed at four times the cells. The cost is that `NrmOutEq` and `NrmOutFree` impose ρ *and* $\boldsymbol{u}$, over-determining the boundary into a mass source: measured 8.31×10^{-2} and 8.36×10^{-2}, and a drag normalised by $\rho = 1$ then read 8.30 % and 8.35 % high — matching the drift to three digits, so none of it was the flow. A proportional servo on the outlet density restores 7×10^{-8} mass conservation, after which all three outlets agree on C_d to 0.008 %. Use `NrmOutXp` for a steady outlet; it conserves mass unaided and is the more accurate closure.

48. **The magnetic volume penalisation injects $\nabla\cdot\boldsymbol{B}$, and in three dimensions it is about half the field's gradient.** `demonstrator/mhd_sphere.cpp` confines a decaying MHD flow to a sphere by penalising both fields, and it works: $|\boldsymbol{B}\cdot\boldsymbol{n}|/|\boldsymbol{B}|$ on the shell falls from 0.503 to 8×10^{-4}, the initial condition is solenoidal to 3.7×10^{-16}, and the run shows clean selective decay (E_u down 3500× over four turnovers against E_b down 84×). The cost is that $-\chi(\boldsymbol{B}\cdot\boldsymbol{n})\boldsymbol{n}/\varepsilon_m$ is not solenoidal. Measured as $\mathrm{rms}(\nabla\cdot\boldsymbol{b})/\mathrm{rms}(|\boldsymbol{j}|)$ at $N = 64$, Re $= 200$, one turnover: both penalisations 0.393, fluid only **0.0087**, magnetic only **0.314**, neither 0.011. The magnetic source carries all of it. It does not refine away (0.399 at $N = 48$ against 0.393 at $N = 64$) and it grows with time, reaching 0.502 by four turnovers. The two-dimensional disc reaches 0.073 on the same diagnostic at the same Reynolds number, so three dimensions cost about 5×.

    Two cautions follow. ε_m is *not* the knob: sweeping it $2 \to 100$ moves $|\boldsymbol{B}\cdot\boldsymbol{n}|/|\boldsymbol{B}|$ by 56× but the divergence by 17 %, because $\nabla\cdot\boldsymbol{B}$ saturates where injection balances the resistive decay of the error — a parameter sweep over a saturating quantity measures the approach, not the cause, and only switching each penalisation off separately identifies it. And $\cos(\boldsymbol{J}, \boldsymbol{B})$ rises to 0.231 at one turnover and then *falls* to 0.170, which cannot be separated from the divergence error and must not be read as a Taylor relaxation. *Untested:* any divergence cleaning. There is no projection, no Powell source and no constrained transport in this tree, and no Poisson solver with which to build the first.

49. **The BGK family's body force is second order, the moment operators' is complete, and nothing measured the gap until 2026-09-15.** `guo_source_raw` is *exactly* the second-order Hermite expansion of $-\boldsymbol{F}\cdot\nabla_{\boldsymbol{\xi}} f^{\mathrm{eq}}$ — proved symbolically in `MATLAB/d3q27_shifted_force6.py` and asserted numerically at 1.0×10^{-16} in `validation/forcing_cm.cpp`. So BGK and TRT, in *both* codebases, leave a residue in the third-order central moments of about $2u^2|\boldsymbol{F}|$: 5.2×10^{-3} at $|\boldsymbol{u}| = 0.05$ and 2.1×10^{-2} at 0.10. That is Guo (2002) working as intended rather than a defect. What was a defect is that the choice of operator therefore changes the force's fidelity with no diagnostic, and that a driver selecting BGK where it might have selected central moments drops silently from the complete expansion to the second-order one.

    The complete order is $2D$ in D dimensions — four on D2Q9, *six* on D3Q27 — because the product index set closes at exponent two per axis ($H_3 \equiv 0$ and $H_4 \equiv -H_2$ on $\{-1, 0, 1\}$). Carrying the two-dimensional "fourth order" across leaves k_{212}, k_{221}, k_{122} wrong at $O(u^4)$ and k_{222} at $O(u^5)$, measured at 2.10×10^{-4} and 1.53×10^{-5} of $|\boldsymbol{F}|$ at $|\boldsymbol{u}| \le 0.10$. `HermiteForce4` is named for the two-dimensional case and neither builds nor needs a source; the shifted operators write K_F into the first-order slot, which *is* the complete expansion, exactly.

Untested: what the second-order residue costs a *run*. Every measurement above is per-node algebra. The residue sits in ghost moments and leaves $\sum_i \boldsymbol{c}_i f_i$ exact, so the momentum balance is untouched and the hydrodynamic force cases (`forcing_3d`, `forcing_transverse`) cannot see it even in principle — a steady balance has no k_{222} in it.

50. **The regularised D3Q7 scalar is linearly unstable under advection once $|u|$ exceeds about $1.64\sqrt{D}$, and that limit falls as $D \to 0$.** Measured 2026-10-15 on the passive mixer of `validation/mixer_forward`: at $D = 10^{-3}$ ($\omega = 1.98413$) a von Neumann scan of the 7×7 one-step map gives $|u| < 0.052$ along an axis (0.087 on a body diagonal), below the empty duct's own centreline speed, and the two-grid run diverged on both grids. `ScalarRegularised` is the TRT member with $\Lambda = (\tau - \frac{1}{2})/2$, which vanishes as $D \to 0$; `ScalarTRT` at $\Lambda = \frac{1}{4}$ is stable to $|u| = 0.5$ in every direction at any D (its banner has the table, and `tests/test_scalar_trt.cpp` the differential test). Every case below the limit is unaffected; one that advects a scalar at small D should check it.

51. **A conservative scalar keeps a uniform field uniform only where the velocity it is given is divergence-free in *its own* stencil, and a Brinkman drag that varies over a cell breaks that.** Measured on the same mixer: next to `DarcyGuo` drag regions the D3Q7 central divergence of the fluid's velocity reached $0.145\,U$ per cell, and exceeded $|u|$ itself at near-stagnant nodes, and a *uniform* inlet $c = 1$ came out between 0.70 and 1.44 at $L = 40$. The opt-in `ScalarSolver::set_uniform_correction`, calibrated per velocity field, restores it to the calibration's accuracy (2.3×10^{-10} at $L = 16$; an error in the factor acts as a source for a residence time, so not to round-off) at the price of conservation where the correction acts — and only for *uniform* fields. Most of what it changed on the mixer was not the drag but *compressibility*: the density falls 8.5 % along the duct at $\mathrm{Ma}_{\max} = 0.175$, the fluid conserves $\rho\boldsymbol{u}$ rather than $\boldsymbol{u}$, and a scalar advected by $\boldsymbol{u}$ is diluted by the density ratio — the outlet's flux-weighted mean was 0.461 for an inlet mean of 0.5, and the mixing variance 15 % low, grid-independently. The drag part refined away (1.2 % at $L = 16$, 0.03 % at $L = 32$). With two streams the $L = 16$ case still converges to $c \in [-0.084, 1.727]$, the maximum inside a near-stagnant drag region. Any scalar advected through a penalised body inherits this; it is invisible in uniform or smooth flow.

References

[1] J. Latt, B. Chopard, O. Malaspinas, M. Deville and A. Michler, *Straight velocity boundaries in the lattice Boltzmann method*, Phys. Rev. E **77**, 056703 (2008).

[2] P. J. Dellar, *Moment-based boundary conditions for lattice Boltzmann magnetohydrodynamics*, in *Numerical Mathematics and Advanced Applications*, Springer.

[3] U. Ghia, K. N. Ghia and C. T. Shin, *High-Re solutions for incompressible flow using the Navier–Stokes equations and a multigrid method*, J. Comput. Phys. **48**, 387 (1982).

[4] M. Lehmann, *Esoteric Pull and Esoteric Push: two simple in-place streaming schemes for the lattice Boltzmann method on GPUs*, Computation **10**(6), 92 (2022).

[5] P. J. Dellar, *Lattice kinetic schemes for magnetohydrodynamics*, J. Comput. Phys. **179**, 95–126 (2002).

[6] A. De Rosis, E. Lévêque and R. Chahine, *Advanced lattice Boltzmann scheme for high-Reynolds-number magneto-hydrodynamic flows*, J. Turbulence **19**(6), 446–462 (2018).

[7] G. de Vahl Davis, *Natural convection of air in a square cavity: a bench mark numerical solution*, Int. J. Numer. Methods Fluids **3**, 249–264 (1983).

[8] S. A. Orszag and C.-M. Tang, *Small-scale structure of two-dimensional magnetohydrodynamic turbulence*, J. Fluid Mech. **90**, 129–143 (1979).

[9] I. Ginzburg, F. Verhaeghe and D. d'Humières, *Two-relaxation-time lattice Boltzmann scheme: about parametrization, velocity, pressure and mixed boundary conditions*, Commun. Comput. Phys. **3**, 427–478 (2008).

[10] Z. Guo, C. Zheng and B. Shi, *Discrete lattice effects on the forcing term in the lattice Boltzmann method*, Phys. Rev. E **65**, 046308 (2002).

[11] P. D. Mininni, A. G. Pouquet and D. C. Montgomery, *Small-scale structures in three-dimensional magnetohydrodynamic turbulence*, Phys. Rev. Lett. **97**, 244503 (2006).

[12] A. De Rosis and C. Coreixas, *Multiphysics flow simulations using D3Q19 lattice Boltzmann methods based on central moments*, Phys. Fluids **32**, 117101 (2020).

[13] A. De Rosis, *Nonorthogonal central-moments-based lattice Boltzmann scheme in three dimensions*, Phys. Rev. E **95**, 013310 (2017).

[14] S. Mohammed, D. Graham and T. Reis, *Assessing moment-based boundary conditions for the lattice Boltzmann equation: A study of dipole-wall collisions*, Comput. Fluids **176**, 79–96 (2018).

[15] H. J. H. Clercx and C.-H. Bruneau, *The normal and oblique collision of a dipole with a no-slip boundary*, Comput. Fluids **35**, 245–279 (2006).

[16] A. De Rosis and E. Enan, *A three-dimensional phase-field lattice Boltzmann method for incompressible two-components flows*, Physics of Fluids **33**, 043315 (2021).

[17] Y. Zhou, A. De Rosis and A. Revell, *Coupling finite volume-lattice Boltzmann methods for advanced heat transfer simulations*, Engineering with Computers **42**:56 (2026).

[18] H. Luan, H. Xu, L. Chen, Y. Feng, Y. He and W. Tao, *Coupling of finite volume method and thermal lattice Boltzmann method and its application to natural convection*, Int. J. Numer. Meth. Fluids **70**, 200–221 (2012).

[19] T. Cheng and W. Liu, *Effect of temperature gradient orientation on the characteristics of mixed convection flow in a lid-driven square cavity*, Comput. Fluids **39**, 965–978 (2010).

[20] S. Bettaibi, F. Kuznik and E. Sediki, *Hybrid lattice Boltzmann finite difference simulation of mixed convection flows in a lid-driven square cavity*, Phys. Lett. A **378**, 2429–2435 (2014).

[21] S. Patnaik, A. Skillen and A. De Rosis, *A central-moments-based lattice Boltzmann method for electrohydrodynamic flows*, Engineering with Computers **41**, 4977–5002 (2025).

[22] T. Fusegi, J. M. Hyun, K. Kuwahara and B. Farouk, *A numerical study of three-dimensional natural convection in a differentially heated cubical enclosure*, Int. J. Heat Mass Transf. **34**, 1543–1557 (1991).